

CHAPTER 01: GENERAL, SITE FACILITIES & SAFETY						
Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
01.1	Erection and maintenance of semi permanent site office and removal of the same after completion of work in accordance with the conditions of contract. In addition to the office required for own use, the contractor shall provide and maintain furnished site office for the use of Engineer-in-charge and his staff. The site office must have tiles floor, adequate foundation, brick walls, false ceiling of gypsum board and all windows are to be glazed, shuttered and provided with steel grill. Outside and inside wall surface are to be painted on plaster acceptable to the Engineer-in-charge. The site office shall be maintained in a secure condition by the contractor until the completion of the contract and shall be provided with electricity, water supply, wash rooms and sewerage facilities. All doors shall be fitted with approved locks and windows shall be provided with screen/blinds. Before construction the contractor shall submit plans and drawings showing proposed details and location for the site office, including foundations, access roads, shades, layout of electrical and water supply and hard standings there for the approval of Engineer-in-charge. The Engineer-in-charge may require revision of the plans prior to giving approval for construction. The contractor shall also submit detailed proposed furniture, fittings and other items of equipment and plant to the Engineer-in-charge for approval. These items shall be of the standard quality suitable for site. The office, complete with furnishings, fittings, access roads and hard standings shall be ready for occupation by the Engineer-in-charge within 28 days of the date when the contractor first occupies the site. The contractor shall provide day and night guards and an attendant for the field office. At the end of the contract all materials, equipment and plant, furniture, fittings recovered from dismantling the office and removing access road will be the property of the contractor. No interim payment shall be certified unless engineer's office with required facilities are constructed and accepted by the Engineer-in-charge. (This is a time related item; proportionate payment for this item shall be made distributing in each bill on the basis of percentage progress of the whole works under contract)					
01.1.1	Engineer's site office of minimum 10 sqm plinth area with providing necessary facilities including office furniture, consumables, stationeries etc.	job	Tk. 50,308.10	Tk. 50,308.00	Tk. 50,308.00	Tk. 50,308.00
01.1.2	Engineer's site office of minimum 15 sqm plinth area with providing necessary facilities including office furniture, consumables, stationeries, water purifier etc.	job	Tk. 80,370.26	Tk. 80,370.00	Tk. 80,370.00	Tk. 80,370.00
01.1.3	Engineer's site office of minimum 38 sqm plinth area with providing necessary facilities including office furniture, 24000 BTU air cooler, umbrella, crockeries, water purifier, PC with monitor, uninterruptible power supply (UPS), laserjet printer (minimum 20 ppm), LED flood light fittings(Halogen) for sight security etc.	job	Tk. 487,129.69	Tk. 487,130.00	Tk. 487,130.00	Tk. 487,130.00

01.1.4.1	Providing necessary facilities in construction site for maintaining site safety of 30 nos construction worker including safety helmet, safety belt, apron, gumboot, goggles etc.	set	Tk. 38,651.35	Tk. 38,651.00	Tk. 38,651.00	Tk. 38,651.00
01.1.4.2	Providing necessary facilities in construction site for maintaining site safety of 10 nos construction inspection team member including safety helmet, safety goggles, safety shoe etc.	set	Tk. 10,797.84	Tk. 10,798.00	Tk. 10,798.00	Tk. 10,798.00
01.1.4.3	Supplying and providing of first aid box with necessary materials/medicine (hygienic gown, thermometer, adhesive dressings, antiseptic solutions, bandages, cotton balls or swaps, emergency blanket, gloves, hand sanitizer, ice pack, saline etc.). All complete as per direction of Engineer-in-charge.	each	Tk. 16,215.16	Tk. 16,215.00	Tk. 16,215.00	Tk. 16,215.00
01.1.5	Hiring, installation, testing & commissioning of CCTV/Surveillance facilities for monitoring construction site including supply, fixing of 5.0 megapixel IR fixed bullet network IP camera including lens, bracket, housing with outdoor night vision facilities, full HD 1080 pixel, real time video, network video recorder (NVR), 2 TB HDD for NVR, 4/8/16/32 port PoE switch, 32"/42" Full HD CCTV monitor, cable & accessories, internet connection with operation & maintenance cost throughout contract completion period, re-fitting & re-fixing with progress of construction all complete as per direction of Engineer in charge.	job	Tk. 49,081.08	Tk. 49,081.00	Tk. 49,081.00	Tk. 49,081.00
01.1.6	Providing and maintenance one project profile signboards to be placed at a suitable place of the site including submission of proposals for the materials of the signboards and text layout containing 3D picture, safety instructions, project information with security light etc. to the Engineer-in-charge for approval which will be positioned as directed by the Engineer-in-charge and removing the same on completion of the works or as instructed by the Engineer-in-charge.	sqm	Tk. 3,300.21	Tk. 3,276.00	Tk. 3,239.00	Tk. 3,239.00
01.2.1	Providing 3 sets as-built drawings subject to Engineer's approval produced in AutoCAD software in 584.5 mm x 413.5 mm (A-2 size) standard drawing paper, and operating and maintenance manual of the equipment and plant incorporated in the works, if any, in original by the date stated in the particular conditions of contract (PCC). If the contractor does not supply the as-built drawings and operating & maintenance manuals by the date stated in the particular conditions of contract (PCC), or they do not receive the Engineer-in-charge's approval, the Engineer-in-charge shall withhold the amount stated in the PCC from the payments due to the contractor. The as-built drawings must show the permanent works as actually constructed and reflect the revision of drawings supplied to the contractor during the Contract as well as revisions of drawings supplied to the contractor during the contract. (One set of as-built drawings shall be considered for measurement and payment)	per tender	Tk. 28,256.39	Tk. 28,256.00	Tk. 28,256.00	Tk. 28,256.00
01.2.2	Providing additional copy of as-built drawing produced in AutoCAD software in 584.5 mm x 413.5 mm (A-2) standard drawing paper accepted by the Engineer-in-charge.	sqm of drawing sheet	Tk. 330.19	Tk. 331.00	Tk. 331.00	Tk. 331.00
01.3.1	Providing coloured progress picture & video of works, CCTV monitoring system operation and related data processing, regular data updating & all other relevant works regarding online monitoring system etc.	per month	Tk. 32,497.81	Tk. 32,498.00	Tk. 32,498.00	Tk. 32,498.00

01.4.1	Preparation & submission of work programme / schedule to the Engineer-in-charge for approval prior ground breaking showing the events (works) and successive sub-events of works including all the management & control parameters like Earliest Start Time (EST), Earliest Finish Time (EFT), float events on critical path, resource allocation etc. with description of general method of works, in orderly manner all in the form of network diagram prepared in computer software like MS project/ Primavera and presenting in displayable hard and soft copy minimum 3 (three) sets. (This is a time related item. Proportionate payment for this item shall be made distributing in each bill on the basis of percentage progress of the whole works under contract)	per tender	Tk. 32,076.56	Tk. 32,076.00	Tk. 32,076.00	Tk. 32,076.00
01.4.2	Updating the work programme and submission to the Engineer-in-charge at intervals no longer than 28 days showing actual physical and financial progress achieved on each activity and the effect of the progress on the timing of the remaining work, including any changes to the sequence of the activities. However, Engineer's approval of the programme shall not alter the contractor's obligations. The contractor may revise the programme and submit it to the Engineer-in-charge again at any time. A revised programme shall show the variations and events related to change of contract price. Month wise documentation shall be the basis of payments on this item.	per month	Tk. 9,499.26	Tk. 9,499.00	Tk. 9,499.00	Tk. 9,499.00
01.5	Providing all necessary logistic support to frame 'monthly progress meeting' between contractor/contractor's representatives and employer / employer's representatives at the Engineer-in-charge's office at site. Month wise documentation shall be the basis of payments on this item.	per month	Tk. 2,425.30	Tk. 2,381.00	Tk. 2,205.00	Tk. 2,205.00
01.6	Providing layout and carry over PWD bench-mark (BM) at site from nearby BM pillar, property lines, existing ground level (EGL), formation ground level (FGL), highest flood levels (HFL), plinth levels (PL), mean sea level (MSL), setting and marking all pillars, marker, pegs etc. showing and maintaining reduced levels (RL's) including locating, establishing, protecting all public utilities within the premise of work and finally all to be presented in black and white.	sqm	Tk. 32.76	Tk. 32.00	Tk. 30.00	Tk. 30.00
01.7	Mobilization and cleaning site before commencing actual physical work and during contract period and demobilization after completion of the works under contract to be accepted by the Engineer-in-charge. This work shall also cover cleaning and clearing, cutting or filling, dressing the project area on and in the ground to an extent that all the events of works of the project can be executed smoothly in a working environment with a particular attention on safety and security in all respects, and to stockpile the end outcome to a place for disposal agreed by the Engineer-in-charge, where, payments are to be based on ground area determined by the Engineer-in-charge and be proportionate to the percentage progress of work under contract as a whole in all respects and approved by the Engineer-in-charge.	sqm	Tk. 217.93	Tk. 214.00	Tk. 198.00	Tk. 198.00

CHAPTER 02: EXCAVATION, FILLING & SITE DEVELOPMENT AND PALISADING

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
02.1	Earth work in excavation in all kinds of soil for foundation trenches including layout, providing center lines, local bench-mark pillars, levelling, ramming and preparing the base, fixing bamboo spikes and marking layout with chalk powder, providing necessary tools and plants, protecting and maintaining the trench dry etc., stacking, cleaning the excavated earth at a safe distance out of the area enclosed by the layout etc. all complete and accepted by the Engineer-in-charge, subject to submit method statement of carrying out excavation work to the Engineer-in-charge for approval. However, engineer's approval shall not relieve the contractor of his responsibilities and obligations under the contract.					
02.1.1	Layout and marking for earthwork in excavation in foundation accepted by the Engineer-in-charge. [Plinth area of the structure shall be considered for measurement]	sqm	Tk. 19.18	Tk. 19.00	Tk. 18.00	Tk. 18.00
02.1.2	Earthwork in excavation in foundation trenches up to 1.5 m depth and maximum 10 m lead	cum	Tk. 168.09	Tk. 165.00	Tk. 153.00	Tk. 153.00
02.1.3	Added rate for each additional 0.5 meter depth exceeding 1.5 meter.	cum	Tk. 29.79	Tk. 29.00	Tk. 27.00	Tk. 27.00
02.1.4	Added rate for each 1 m additional lead beyond 10 m.	cum	Tk. 4.29	Tk. 4.00	Tk. 4.00	Tk. 4.00
02.2	Bailing out of trapped water caused by inundation or rain, by pump from foundation trenches.	hour	Tk. 473.63	Tk. 474.00	Tk. 458.00	Tk. 458.00
02.3	Supplying of straight and strong borrak bamboo posts of having minimum 75 mm diameter althrough and driving the same vertically in the ground up to required depth by any means shaping the top of the post, all complete and accepted by the Engineer-in-charge.	meter	Tk. 90.33	Tk. 90.00	Tk. 81.00	Tk. 81.00
02.4	Supplying of wooden ballah having an average diameter not less than 150 mm (without barks) at one third distance from larger end with a minimum end diameter of 100 mm for retaining purpose and driving to a depth as per design and drawing by any method including all arrangement for staging, hoisting, carrying etc. complete and accepted by the Engineer-in-charge.	meter	Tk. 603.88	Tk. 598.00	Tk. 571.00	Tk. 571.00
02.5	Palisading work by supplying of bitumen drum sheet walling tied with 20 BWG G.I. wire fixing the same with already driven borrak bamboo posts with half split borrak bamboo runners @ 450mm c/c horizontally with iron nails, G.I wire etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the cost of bamboo post).	sqm	Tk. 612.84	Tk. 608.00	Tk. 583.00	Tk. 583.00
02.6	Supplying and fitting-fixing horizontal bracing in palisading with half split wooden ballah having an average diameter not less than 150 mm with a minimum end diameter of 100 mm @ 450 mm including supply of iron nails, gazals etc. complete approved and accepted by the Engineer-in-charge.	meter	Tk. 225.53	Tk. 224.00	Tk. 216.00	Tk. 216.00

02.7.1	Shore protection work during excavation in foundation trenches up to 1.5 m depth to protect loss due to damage of property by palisading accepted by the Engineer-in-charge. [The rate is including the cost of vertical post]	sqm	Tk. 711.46	Tk. 706.00	Tk. 673.00	Tk. 673.00
02.7.2	Shore protection work during excavation in foundation trenches beyond 1.5 m depth to protect loss due to damage of property by palisading accepted by the Engineer-in-charge, [The rate is including the cost of vertical post]	sqm	Tk. 1,042.97	Tk. 1,045.00	Tk. 992.00	Tk. 992.00
02.8	Supply of 2 mm thick best quality geo-textile from approved manufacturer/origin and laying as per manufacturer's instructions approved and accepted by the Engineer-in-charge. Before commencement of laying of geo-textile, the contractor must submit the method statement for carrying out this work including sample with evidence of origin and compliance certificate from independent testing laboratory for approval.	sqm	Tk. 213.59	Tk. 212.00	Tk. 209.00	Tk. 209.00
02.9	Removing sludge or semi-liquid clay/substances etc not suitable for pumping out or carrying by head load from the bottom of any marshy land, lake, ditch, pond by mixing fine local sand of minimum F.M. 0.8 to a proportion such that it can be carried to an intermediate point of disposal and cutting the same including stacking and spreading properly on a suitable place to dry in the sun for a certain days and dispose the dried sludge or clay to a safe distance etc. complete and accepted by Engineer-in-charge subject to submit method statement to the engineer in charge for carrying out the works including ultimate disposal of the sludge under this item for approval. However, engineer's approval shall not relieve the Contractor of his responsibilities and obligations under the contract.	cum	Tk. 1,053.85	Tk. 1,044.00	Tk. 1,002.00	Tk. 1,002.00
02.10.1	Sand filling in foundation trenches and plinth with sand having minimum F.M. 0.5 in 150 mm layers including leveling, watering and compaction to achieve minimum dry density of 95% with optimum moisture content (Modified proctor test) by ramming each layer up to finished level as per design supplied by the design office only, all complete and accepted by the Engineer-in-charge.	cum	Tk. 1,087.93	Tk. 1,093.00	Tk. 1,069.00	Tk. 1,069.00
02.10.2	Sand filling in foundation trenches and plinth with sand having minimum F.M. 0.8 in 150 mm layers including leveling, watering and compaction to achieve minimum dry density of 95% with optimum moisture content (Modified proctor test) by ramming each layer up to finished level as per design supplied by the design office only, all complete and accepted by the Engineer-in-charge.	cum	Tk. 1,166.80	Tk. 1,169.00	Tk. 1,114.00	Tk. 1,114.00
02.10.3	Sand filling in foundation trenches and plinth with sand having min. F.M. 1.2 in 150 mm in layers including leveling, watering and compaction to achieve minimum dry density of 95% with optimum moisture content (Modified proctor test) by ramming each layer up to finished level as per design supplied by the design office only, all complete and accepted by the Engineer-in-charge.	cum	Tk. 1,296.36	Tk. 1,301.00	Tk. 1,246.00	Tk. 1,246.00
02.10.4	Sand filling in foundation trenches and plinth with sand having min. F.M. 2.2 in 150 mm layers including leveling, watering and compaction to achieve minimum dry density of 95% with optimum moisture content (Modified proctor test) by ramming each layer up to finished level as per design supplied by the design office only, all complete and accepted by the Engineer-in-charge.	cum	Tk. 3,256.72	Tk. 3,152.00	Tk. 3,176.00	Tk. 3,142.00

02.11	50 mm down graded picked jhama khoa consolidation in foundation trenches by mixing the same with best quality local sand (F.M. 1.2) in (2:1) (khoa: sand) proportion to achieve minimum dry density of 95% with optimum moisture content (Modified proctor test) including breaking and screening chips, laying and spreading in 150 mm layers uniformly and compacting etc. all complete and accepted by the Engineer-in-charge.	cum	Tk. 5,060.36	Tk. 5,045.00	Tk. 4,892.00	Tk. 4,825.00
02.12	50 mm down graded picked jhama khoa consolidation in foundation trenches by mixing in mixer machine with best quality local sand (F.M. 1.2) and cement in (1:6:12) (cement: sand: khoa) proportion to achieve minimum dry density of 95% with optimum moisture content (Modified proctor test) including breaking and screening chips, laying and spreading in 150 mm layers uniformly and compacting etc. all complete and accepted by Engineer-in-charge.	cum	Tk. 5,794.78	Tk. 5,784.00	Tk. 5,650.00	Tk. 5,584.00
02.13	Earth filling in foundation trenches and plinth in 150 mm layer with earth available within 90 m of the building site to achieve minimum dry density of 95% with optimum moisture content (Modified proctor test) including carrying watering, leveling, dressing and compacting to a specified percentage each layer up to finished level etc. all complete and accepted by Engineer-in-charge.	cum	Tk. 206.91	Tk. 202.00	Tk. 189.00	Tk. 189.00
02.14	Cleaning and disposing of excavated earth from the construction site by truck or by any other means to a place within 30 km radius of the city or town area including loading & unloading at both ends, leveling and dressing the carried earth etc. complete and accepted by the Engineer-in-charge.	cum	Tk. 755.93	Tk. 751.00	Tk. 686.00	Tk. 686.00
02.15	Earth filling in foundation trenches and plinth in 150 mm layers with carted earth carried by truck or by any other means including loading and unloading at both ends, leveling, watering and compacting to achieve minimum dry density of 95% with optimum moisture content (modified proctor test) including local carriage each layer up to finished level including cost of water and test (carried from beyond 300 m) etc. all complete and accepted by the Engineer-in-charge	cum	Tk. 709.35	Tk. 706.00	Tk. 689.00	Tk. 689.00
02.16	Site development/Improvement by carted earth or dredged sand, sandy silt (free from any organic, foreign, environmental hazardous substances) carried by head or truck or any other means including cost of cutting or by dredging of sand, sandy silt, all; including local carrying, placing the earth/sand, sandy silt in the designated area, maintaining slopes, breaking lumps, levelling and dressing in layers up to finished level etc. all complete as per direction and accepted by the engineer in charge.					
02.16.1	By other method/means than dredgeing.	cum	Tk. 759.66	Tk. 755.00	Tk. 735.00	Tk. 735.00
02.16.2	By dredging including necessary systems thereof.	cum	Tk. 510.17	Tk. 509.00	Tk. 504.00	Tk. 504.00
02.17	Mechanical compaction of earth beyond plinth area , required for pre-approved specific engineering purpose in 150 mm layers including levelling, watering and consolidation each layer with chain dozer, grader, roller etc. to achieve minimum dry density of 95% with optimum moisture content (modified proctor test) up to finished level all complete and accepted by the engineer-in-charge subjected to submission of the method statement.	cum	Tk. 222.04	Tk. 219.00	Tk. 208.00	Tk. 208.00

02.18	Earthwork in excavation of pond of any dimension in all kinds of soil including cutting up to required depth including bailing out water and throwing on the embankment, breaking clods, ramming and leveling, dressing in 225 mm layer with maintaining the side slopes and level of both pond and the embankment as per design and accepted by the Engineer-in-charge.					
02.18.1	For initial lead of 30 m and lift of 1.5 m	cum	Tk. 259.24	Tk. 255.00	Tk. 236.00	Tk. 236.00
02.18.2	Added for each additional lift of 0.3 m beyond 1.5 m and up to 3 m	cum	Tk. 8.94	Tk. 9.00	Tk. 8.00	Tk. 8.00
02.18.3	Added for each additional lift of 0.3 m beyond 3 m and up to 4.5 m	cum	Tk. 13.24	Tk. 13.00	Tk. 12.00	Tk. 12.00
02.18.4	Added for each additional lift of 0.3 m beyond 4.5 m and up to 6 m	cum	Tk. 19.07	Tk. 19.00	Tk. 17.00	Tk. 17.00
02.18.5	Added for each additional lead of 30 m or part thereof beyond initial lead 30 meter	cum	Tk. 34.56	Tk. 34.00	Tk. 31.00	Tk. 31.00
02.19.1	Bailing out water by 1 cusec diesel operated pump including hire charge of pump, fuel, operator, spares and maintenance etc. all complete and accepted by the Engineer-in-charge.	per hour	Tk. 676.79	Tk. 670.00	Tk. 644.00	Tk. 644.00
02.19.2	Bailing out water by 1 cusec electricity operated pump including hire charge of pump, fuel, operator, spares and maintenance etc. all complete and accepted by the Engineer-in-charge.	per hour	Tk. 562.40	Tk. 556.00	Tk. 530.00	Tk. 530.00

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CHAPTER 03: BRICK SOLING, CC, DPC ETC.

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
03.1.1	One layer brick flat soling in foundation or in floor with first class/picked jhama bricks (BDS 208) including preparation of bed and filling the interstices with local sand, leveling etc. complete and accepted by the Engineer-in-charge	sqm	Tk. 517.61	Tk. 515.00	Tk. 500.00	Tk. 492.00
03.1.2	One layer brick flat soling in foundation or in floor with sand cement solid block (SCSB) (240x114x70)mm including preparation of bed and filling the interstices with local sand, leveling etc. complete and accepted by the Engineer-in-charge	sqm	Tk. 596.64	Tk. 595.00	Tk. 591.00	Tk. 591.00
03.2.1	One layer Herringbone bond (HBB) brick soling in foundation or in floor with first class/picked jhama bricks (BDS 208) including preparation of bed and filling the interstices with local sand, leveling including supply of water, other charges, costs of tools and plants etc. all complete and accepted by the Engineer-in-charge	sqm	Tk. 803.10	Tk. 800.00	Tk. 779.00	Tk. 766.00
03.2.2	One layer Herringbone bond (HBB) brick soling in foundation or in floor with sand cement solid block (SCSB) (240x114x70)mm including preparation of bed and filling the interstices with local sand, leveling including supply of water, other charges, costs of tools and plants etc. all complete and accepted by the Engineer-in-charge	sqm	Tk. 999.01	Tk. 996.00	Tk. 989.00	Tk. 989.00

CHAPTER 04: BRICK WORKS, PATENT STONE IN FLOOR AND FANCY SCREEN BLOCK WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
04.1	Brick works with first class bricks with cement sand (F.M. 1.2) mortar (1:6) in foundation and plinth, filling the joints/interstices fully with mortar, racking out the joints, cleaning and soaking the bricks at least for 24 hours before use and curing at least for 7 days etc. all complete including cost of water, electricity and other charges and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)	cum	Tk. 7,529.12	Tk. 7,501.00	Tk. 7,276.00	Tk. 7,181.00
04.2	Brick works of width one brick or one and a half brick length of first class bricks with cement sand (F.M. 1.2) mortar (1:6) in superstructure including raking out joints, filling the interstices with mortar, cleaning and soaking the bricks at least for 24 hours before use and washing of sand, necessary scaffolding, curing at least for 7 days etc. all complete (measurement to given as 250 mm width for one brick length and 375 mm for one brick and a half brick length) and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	cum	Tk. 8,173.69	Tk. 8,136.00	Tk. 7,872.00	Tk. 7,776.00
04.3	Brick works with first class bricks with cement sand (F.M. 1.2) mortar (1:4) in exterior walls including filling the interstices with mortar, raking out joints, cleaning and socking the bricks at least for 24 hours before use and washing of sand, necessary scaffolding, curing at least for 7 days etc. all complete including cost of water, electricity and other charges (measurement to given as 250 mm width for one brick length and 375 mm for one brick and a half brick length) accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	cum	Tk. 8,419.60	Tk. 8,383.00	Tk. 8,122.00	Tk. 8,027.00
04.4	Exposed brick works (for exterior wall without outside plaster) with special quality sorted out first class bricks , very carefully laid in cement sand (F.M. 1.2) mortar (1:4) in superstructure with uniform width and depth of joints, true to vertical and horizontal lines including raking out joints, scaffolding, soaking the bricks at least for 24 hours before use, washing and screening of sand, curing at least for 7 days and high class flush pointing with cement sand (F.M. 1.2) mortar (1:1) including cost of water, electricity and other charges etc all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	cum	Tk. 9,197.42	Tk. 9,159.00	Tk. 8,871.00	Tk. 8,766.00

04.5	Exposed brick works (for exterior wall without outside plaster) with automatic machine made first class bricks , very carefully laid in cement sand (F.M. 1.2) mortar (1:4) in superstructure with uniform width and depth of joints, true to vertical and horizontal lines including raking out joints, scaffolding, soaking the bricks at least for 24 hours before use, washing and screening of sand, curing at least for 7 days and high class flush pointing with cement sand (F.M. 1.2) mortar (1:1) including cost of water, electricity and other charges etc all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	cum	Tk. 9,435.75	Tk. 9,397.00	Tk. 9,214.00	Tk. 9,214.00
04.6	Brick works with 10 holes machine made bricks of approved size (241 mm x 114 mm x 70 mm) having uniform colour carefully laid in cement sand (F.M. 1.2) mortar (1:4) in superstructure with uniform width and depth of joints, true to vertical and horizontal lines including raking out joints, filling the interstices with mortar, cleaning and soaking bricks at least for 24 hours before use and washing and screening of sand, necessary scaffolding, curing at least for 7 days and pointing with cement sand (F.M. 1.2) mortar (1:2) including cost of water, electricity and other charges etc. complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	cum	Tk. 13,439.67	Tk. 13,401.00	Tk. 13,218.00	Tk. 13,218.00
04.7	Brick work with 241 mm x 114 mm x 70 mm size machine made solid klinker facing bricks having uniform colour carefully laid in cement sand (F.M. 1.2) mortar (1:4) in superstructure with uniform width and depth of joints, true to vertical and horizontal lines including raking out joints, filling the interstices with mortar, cleaning and soaking bricks at least for 24 hours before use and washing and screening of sand, necessary scaffolding, curing at least for 7 days and pointing with cement sand (F.M. 1.2) mortar (1:2) including cost of water, electricity and other charges etc. complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	cum	Tk. 10,781.77	Tk. 10,746.00	Tk. 10,569.00	Tk. 10,569.00
04.8	Brick works with 241 mm x 114 mm x 70 mm size 10 hole machine made rock-face or textured klinker facing bricks having uniform colour carefully laid in cement sand (F.M. 1.2) mortar (1:4) in superstructure with uniform width and depth of joints, true to vertical and horizontal lines including raking out joints, filling the interstices with mortar, cleaning and soaking bricks at least for 24 hours before use and washing and screening of sand, necessary scaffolding, curing at least for 7 days and pointing with cement sand (F.M. 1.2) mortar (1:2) including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	cum	Tk. 13,194.84	Tk. 13,163.00	Tk. 13,001.00	Tk. 13,001.00
04.9	Brick works with 241 mm x 114 mm x 70 mm size 3 hole machine made reinforced klinker facing bricks having uniform colour carefully laid in cement sand (F.M.1.2) mortar (1:4) in superstructure with uniform width and depth of joints, true to vertical and horizontal lines including raking out joints, filling the interstices with mortar, cleaning and soaking bricks at least for 24 hours before use and washing and screening of sand, necessary scaffolding, curing at least for 7 days and pointing with cement sand (F.M. 1.2) mortar (1:2) including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	cum	Tk. 15,164.65	Tk. 15,123.00	Tk. 14,967.00	Tk. 14,964.00
04.10	Added cost for each additional floor above ground floor for brick works (item No.- 4.2 to 4.9)	cum	Tk. 179.00	Tk. 175.00	Tk. 162.00	Tk. 162.00

04.11	Added cost for brick-work for each additional meter (or part of meter) free height of brick wall beyond 4 m from the respective floor in auditorium, go-down, overhead water tank etc. as description of the item No. 4.2 to 4.9 including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Added cost shall be incorporated in the estimate only when both sides of wall is uninterruptedly free beyond 4.0 meter)	cum	Tk. 226.41	Tk. 224.00	Tk. 212.00	Tk. 212.00
04.12	Providing brick works for 200 mm x 100 mm facing in superstructure with 200 mm x 100 mm x 50 mm machine made klinker hard pressed facing bricks of approved quality with cement sand (F.M. 1.2) mortar (1:4), including racking out joints, cutting the bricks to required size wherever necessary including high class recessed pointing, cleaning and soaking the bricks at least for 24 hours before use and washing and screening of sand, necessary scaffolding, curing at least for 7 days for all floors up to 4th floor including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (100x200 mm face should be placed for facing). (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 2,004.48	Tk. 1,997.00	Tk. 1,953.00	Tk. 1,953.00
04.13	Providing brick works for 200 mm x 50 mm facing in superstructure with 200 mm x 50 mm x 50 mm machine made hard pressed klinker facing red or maroon strips of approved quality with 20 mm thick cement sand (F.M.1.2) mortar (1:4), 10 mm thick cement mortar in adjacent sides including racking out joints and cutting the bricks to required sizes wherever necessary with high class recessed pointing, cleaning and soaking the bricks for at least 24 hours before use and washing of sand, necessary scaffolding, curing at least for 7 days for all floors up to 4th floor including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 2,899.40	Tk. 2,892.00	Tk. 2,848.00	Tk. 2,848.00
04.14	Providing clay wall tile facing strips (water absorption ratio below 4.5%) in superstructure with machine made hard pressed klinker facing strips of approved quality with 20 mm thick cement sand (F.M.1.2) mortar (1:3) in adjacent side wall, column, drop wall etc. and racking out joints, cutting to required sizes wherever necessary with high class recessed pointing, cleaning and soaking at least for 24 hours before use washing and screening of sand and necessary scaffoldings, curing at least 7 days including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor.					
04.14.1	With 200 mm x 62 mm x 16.51 mm red strip	sqm	Tk. 2,627.23	Tk. 2,619.00	Tk. 2,574.00	Tk. 2,574.00
04.14.2	With 200 mm x 62 mm x 12.7 mm beige strip	sqm	Tk. 2,868.98	Tk. 2,861.00	Tk. 2,816.00	Tk. 2,816.00
04.14.3	With 200 mm x 62 mm x 8.9 mm red strip	sqm	Tk. 2,441.26	Tk. 2,433.00	Tk. 2,388.00	Tk. 2,388.00
04.14.4	With 241 mm x 70 mm x 12.7 mm red strip	sqm	Tk. 2,902.53	Tk. 2,894.00	Tk. 2,850.00	Tk. 2,850.00
04.14.5	With 242 mm x 70 mm x 12.7 mm matt glazed textured strip	sqm	Tk. 3,330.46	Tk. 3,322.00	Tk. 3,277.00	Tk. 3,277.00
04.15	125 mm brick work with first class bricks with cement sand (F.M. 1.2) mortar (1:6) and making bond with connected walls including necessary scaffolding, raking out joints, cleaning and soaking the bricks for at least 24 hours before use and washing of sand, curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 1,160.43	Tk. 1,155.00	Tk. 1,116.00	Tk. 1,103.00

04.16	125 mm brick works with first class bricks with cement sand (F.M. 1:2) mortar (1:4) and making bond with connected walls including necessary scaffolding, raking out joints, cleaning and soaking the bricks for at least 24 hours before use and washing of sand, curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 1,185.23	Tk. 1,180.00	Tk. 1,142.00	Tk. 1,129.00
04.17	Exposed 125 mm thick pointing ornamental brick work with sorted out first class bricks with cement sand (F.M. 1:2) mortar (1:4) and making bond with connected walls including necessary scaffolding, raking out joints, cleaning and soaking the bricks for at least 24 hours before use and washing of sand, curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 1,449.33	Tk. 1,443.00	Tk. 1,394.00	Tk. 1,379.00
04.18	Exposed 125 mm thick pointing ornamental brick work with automatic machine made first class bricks with cement sand (F.M. 1:2) mortar (1:4) and making bond with connected walls including necessary scaffolding, raking out joints, cleaning and soaking the bricks for at least 24 hours before use and washing of sand, curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 1,447.90	Tk. 1,441.00	Tk. 1,405.00	Tk. 1,405.00
04.19	Providing 125 mm thick brick work in superstructure (partition walls) with cement sand (F.M. 1:2) mortar (1:4) with 10 hole machine made ceramic bricks of approved size having uniform colour, carefully laid in cement mortar of uniform width & depth of joints, true to vertical and horizontal lines including racking out joints, filling the interstices with same mortar, making bond with connected walls, cleaning, soaking brick at least 24 hours and washing of sand, curing 7 days in all floors, including cost of water, electricity and other charges accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 1,890.52	Tk. 1,885.00	Tk. 1,855.00	Tk. 1,855.00
04.20	Fancy screen work with different size hollow machine made design blocks of uniform colour, carefully laid in cement sand (F.M. 1:2) mortar (1:3) in all floors in wall with uniform mortar joints, true to vertical and horizontal line including racking out joints, cleaning and soaking the blocks for minimum 24 hours before use including washing and screening of sand, scaffolding, curing etc. complete in all respect including supply of all necessary materials, cost of water, electricity and other charges accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)					
04.20.1	With design block size: 114 mm x 114 mm x 114 mm or equivalent volume & design. In ground floor	sqm	Tk. 4,555.67	Tk. 4,545.00	Tk. 4,490.00	Tk. 4,490.00
04.20.2	With design block size: 140 mm x 140 mm x 102 mm or equivalent volume & design. In ground floor	sqm	Tk. 3,257.35	Tk. 3,247.00	Tk. 3,195.00	Tk. 3,195.00
04.20.3	With design block size: 152 mm x 152 mm x 102 mm or 152 mm x 152 mm x 114 mm or equivalent volume and design. In ground floor	sqm	Tk. 3,007.73	Tk. 2,997.00	Tk. 2,946.00	Tk. 2,946.00
04.20.4	With design block size: 203 mm x 203 mm x 108 mm or equivalent volume and design. In ground floor	sqm	Tk. 2,257.43	Tk. 2,247.00	Tk. 2,202.00	Tk. 2,202.00
04.20.5	With design block size: 203 mm x 121 mm x 102 mm or equivalent volume and design. In ground floor	sqm	Tk. 2,479.32	Tk. 2,469.00	Tk. 2,424.00	Tk. 2,424.00

04.20.6	With design block size: 152 mm x 152 mm x 152 mm or equivalent volume and design. In ground floor	sqm	Tk. 3,007.73	Tk. 2,997.00	Tk. 2,946.00	Tk. 2,946.00
04.20.7	With design block size: 241 mm x 114 mm x 70 mm or equivalent volume and design. In ground floor	sqm	Tk. 2,405.46	Tk. 2,395.00	Tk. 2,343.00	Tk. 2,343.00
04.21	Added cost for each additional floor above ground floor for brick and screen block works (item No.- 4.12 to 4.20)	sqm	Tk. 29.00	Tk. 29.00	Tk. 26.00	Tk. 26.00
04.22	Klinker paving brick works in floor or pavement with machine made pressed bricks of approved quality with cement sand (F.M. 1.2) mortar (1:4) on minimum 12 mm thick cement sand (F.M. 1.2) mortar (1:4) including raking out joints, cutting the bricks to required size, soaking the same for 24 hours before use wherever necessary including high class flush pointing in cement mortar (1:2), cleaning, curing at least for 7 days etc. including washing and screening of sand, cost of water, electricity and other charges complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)					
04.22.1	100 mm thick flat brick pavement with 200 mm x 100 mm x 50 mm size klinker facing bricks.	sqm	Tk. 3,131.22	Tk. 3,119.00	Tk. 3,073.00	Tk. 3,073.00
04.22.2	50 mm thick brick on edge pavement with 200 mm x 100 mm x 50 mm klinker paving bricks in 1:4 cement sand mortar having 200 mm x 100 mm face at top.	sqm	Tk. 1,907.56	Tk. 1,897.00	Tk. 1,860.00	Tk. 1,860.00
04.22.3	50 mm thick brick pavement with 200 mm x 50 mm x 50 mm klinker facing bricks.	sqm	Tk. 2,822.98	Tk. 2,812.00	Tk. 2,770.00	Tk. 2,770.00
04.23	125 mm thick brick pavement or flooring with first class brick on edge with cement sand (F.M. 1.2) mortar (1:4) on minimum 12 mm thick cement sand (F.M. 1.2) mortar (1:4) including cutting the bricks as per requirement, cleaning and racking out joints, finishing with flush pointing with cement sand (F.M. 1.2) mortar (1:2) and curing for at least 7 days etc. including cost of water, washing and screening of sand (F.M. 1.2), electricity and other charges complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)	sqm	Tk. 1,397.54	Tk. 1,387.00	Tk. 1,337.00	Tk. 1,324.00
04.24	125 mm thick brick pavement/flooring with 10 hole machine made ceramic bricks with cement sand (F.M. 1.2) mortar (1:4) on minimum 12 mm thick cement sand (F.M. 1.2) mortar (1:4) including cutting the bricks as per requirement, cleaning and racking out joints, finishing with flush pointing with cement sand (F.M. 1.2) mortar (1:2) and curing for at least 7 days etc. including cost of water, washing and screening of sand (F.M. 1.2), electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)	sqm	Tk. 2,041.05	Tk. 2,031.00	Tk. 1,993.00	Tk. 1,993.00
04.25	75 mm thick cement concrete (1:3:6) flooring with cement, best quality coarse sand and 19 mm down graded picked jhama brick chips including breaking of chips, screening, mixing, laying, compacting, washing and screening of sand (F.M 1.2) and curing at least for 7 days etc. including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)	sqm	Tk. 585.04	Tk. 585.00	Tk. 571.00	Tk. 566.00

04.26	25 mm thick artificial patent stone (1:2:4) flooring with cement, best quality coarse sand (50% quantity of Sylhet sand or coarse sand of equivalent F.M. 2.2 and 50% best quality local sand of FM 1.2) and 12 mm down well graded brick chips including breaking chips, screening, laying the concrete in alternate panels, compacting and finishing the top with neat cement and curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 485.76	Tk. 481.00	Tk. 463.00	Tk. 461.00
04.27	38 mm thick artificial patent stone (1:2:4) flooring with cement, best quality coarse sand (50% quantity of Sylhet sand or coarse sand of equivalent F.M. 2.2 and 50% best quality local sand of FM 1.2) and 12 mm down well graded brick chips including breaking chips, screening, laying the concrete in alternate panels, compacting and finishing the top with neat cement and curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 593.94	Tk. 589.00	Tk. 570.00	Tk. 567.00
04.28	25 mm thick artificial patent stone (1:2:4) flooring with cement, best quality coarse sand (50% quantity of Sylhet sand or coarse sand of equivalent F.M. 2.2 and 50% best quality local sand of FM 1.2) and 12 mm down well graded stone chips, laying the concrete in alternate panels, compacting and finishing the top with neat cement and curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 566.20	Tk. 563.00	Tk. 537.00	Tk. 532.00
04.29	38 mm thick artificial patent stone (1:2:4) flooring with cement, best quality coarse sand (50% quantity of Sylhet sand or coarse sand of equivalent F.M. 2.2 and 50% best quality local sand of FM 1.2) and 12 mm down well graded stone chips, laying the concrete in alternate panels, compacting and finishing the top with neat cement and curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. Cement: (CEM-II/B-M) In ground floor	sqm	Tk. 714.59	Tk. 711.00	Tk. 680.00	Tk. 673.00
04.30	25 mm thick artificial patent stone (1:1.5:3) flooring with cement, best quality coarse sand (50% quantity of Sylhet sand or coarse sand of equivalent F.M. 2.2 and 50% best quality local sand of FM 1.2) and 12 mm down well graded stone chips , laying the concrete in alternate panels, compacting and finishing the top with neat cement and curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 584.03	Tk. 581.00	Tk. 555.00	Tk. 550.00

04.31	38 mm thick artificial patent stone (1:1.5:3) flooring with cement, best quality coarse sand (50% quantity of Sylhet sand or coarse sand of equivalent F.M. 2.2 and 50% best quality local sand of FM 1.2) and 12 mm down well graded stone chips , laying the concrete in alternate panels, compacting and finishing the top with neat cement and curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge.(Cement: CEM-II/B-M) In ground floor	sqm	Tk. 741.33	Tk. 738.00	Tk. 707.00	Tk. 700.00
04.32	Added cost for each additional floor above ground floor for patent stone works.	sqm	Tk. 25.00	Tk. 25.00	Tk. 23.00	Tk. 23.00

..... END OF CHAPTER FOUR

CHAPTER 05 : MOSAIC WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
05.1	10 mm thick finished and 12.5 mm laying thickness cast in situ mosaic work on floor with one part of 10 mm machine crushed well graded dust free and properly washed marble chips and one part of mixture containing white cement / grey cement including preparation of mix and laying the same over minimum 25 mm thick artificial patent stone flooring with brick chips (1:2:4) including making suitable panel, compacting, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No.120) /cutting with mosaic cutting machine and finishing with minar stone and spreading oxalic acid including breaking of brick chips, screening, washing mosaic chips and supply of all necessary materials and water, electricity and other charges in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					
05.1.1	With Indian chips and glass strip using 100% white cement In ground floor	sqm	Tk. 2,149.93	Tk. 2,135.00	Tk. 2,076.00	Tk. 2,074.00
05.1.2	With Indian chips and glass strip using 100% grey cement In ground floor	sqm	Tk. 1,837.90	Tk. 1,823.00	Tk. 1,764.00	Tk. 1,762.00
05.1.3	With Pakistani chips and glass strip using 100% white cement In ground floor	sqm	Tk. 2,182.95	Tk. 2,168.00	Tk. 2,109.00	Tk. 2,107.00
05.1.4	With Pakistani chips without glass strip using 100% grey cement In ground floor	sqm	Tk. 1,588.61	Tk. 1,575.00	Tk. 1,525.00	Tk. 1,523.00
05.1.5	With Indian chips without glass strip using 100% grey cement In ground floor	sqm	Tk. 1,555.59	Tk. 1,542.00	Tk. 1,491.00	Tk. 1,490.00
05.1.6	With Pakistani chips without glass strip using 100% white cement In ground floor	sqm	Tk. 1,900.64	Tk. 1,887.00	Tk. 1,837.00	Tk. 1,835.00

05.2	12 mm thick finished and laying thickness 14.5 mm cast in situ mosaic work on floor with one part of 10 mm machine crushed well graded dust free and properly washed marble chips and one part of mixture containing 100% white cement including preparation of mix and laying the same over minimum 25 mm thick artificial patent stone flooring with brick-chips (1:2:4) including making suitable panel, compacting, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No. 120)/ cutting with mosaic cutting machine and finishing with minar stone including breaking of brick chips, screening, spreading oxalic acid, washing of mosaic chips and supply of all necessary materials and water, electricity and other charges in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					
05.2.1	With Indian chips and glass strip In ground floor	sqm	Tk. 2,239.09	Tk. 2,225.00	Tk. 2,165.00	Tk. 2,163.00
05.2.2	With Pakistani chips and glass strip In ground floor	sqm	Tk. 2,276.07	Tk. 2,262.00	Tk. 2,202.00	Tk. 2,200.00
05.2.3	With Indian chips without glass strip In ground floor	sqm	Tk. 1,956.77	Tk. 1,943.00	Tk. 1,893.00	Tk. 1,891.00
05.2.4	With Pakistani chips without glass strip In ground floor	sqm	Tk. 1,993.75	Tk. 1,980.00	Tk. 1,930.00	Tk. 1,928.00
05.3	10 mm thick finished and laying thickness 12.5 mm silver grey cast in situ mosaic work on floor with one part of 10 mm machine crushed well graded, dust free and properly washed marble chips and one part of mixture containing white cement and grey cement in proportion (1:1) including preparation of mix and laying the same over minimum 25 mm thick artificial patent stone flooring with brick-chips (1:2:4) including making suitable panel, compacting, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No. 120) and finishing with minar stone including breaking of brick chips, screening, washing of mosaic chips and supply of all necessary materials including water, electricity and other charges in all respect and accepted by the Engineer-in-charge.(Cement: CEM-I/OPC)					
05.3.1	With Indian chips and glass strip	sqm	Tk. 1,993.92	Tk. 1,979.00	Tk. 1,920.00	Tk. 1,918.00
05.3.2	With Pakistani chips and glass strip	sqm	Tk. 2,026.94	Tk. 2,012.00	Tk. 1,953.00	Tk. 1,951.00
05.3.3	With Indian chips without glass strip In ground floor	sqm	Tk. 1,711.60	Tk. 1,698.00	Tk. 1,647.00	Tk. 1,646.00
05.3.4	With Pakistani chips without glass strip In ground floor	sqm	Tk. 1,744.62	Tk. 1,731.00	Tk. 1,680.00	Tk. 1,679.00
05.4	12 mm thick finished and laying thickness 14.5 mm silver grey cast in situ mosaic work on floor with one part of 10 mm machine crushed well graded dust free and properly washed marble chips and one part of mixture containing white cement and grey cement in proportion (1:1) including preparation of mix and laying the same over minimum 25 mm thick artificial patent stone flooring with brick-chips (1:2:4) including making suitable panels, compacting, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No. 120) and finishing with minar stone including breaking of brick chips, screening, washing of mosaic chips and supply of all necessary materials and water, electricity and other charges in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					

05.4.1	With Indian chips and glass strip In ground floor	sqm	Tk. 2,064.35	Tk. 2,050.00	Tk. 1,990.00	Tk. 1,988.00
05.4.2	With Pakistani chips and glass strip In ground floor	sqm	Tk. 2,101.33	Tk. 2,087.00	Tk. 2,027.00	Tk. 2,025.00
05.4.3	With Indian chips without glass strip In ground floor	sqm	Tk. 1,782.03	Tk. 1,769.00	Tk. 1,718.00	Tk. 1,716.00
05.4.4	With Pakistani chips without glass strip In ground floor	sqm	Tk. 1,819.01	Tk. 1,805.00	Tk. 1,755.00	Tk. 1,753.00
05.5	10 mm thick finished and laying thickness 12.5 mm cast in situ mosaic work in wall with one part of 10 mm machine crushed well graded dust free and properly washed marble chips and one part of mixture containing 100% white cement including preparation of mix and laying the same on 12 mm thick cement plaster (1:4) to wall including compacting, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No. 120), finishing the corners and edges and also finishing with minar stone and supplying all necessary materials including water, electricity and other charges in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					
05.5.1	With Indian chips In ground floor	sqm	Tk. 2,204.92	Tk. 2,172.00	Tk. 2,096.00	Tk. 2,096.00
05.5.2	With Pakistani chips In ground floor	sqm	Tk. 2,237.94	Tk. 2,205.00	Tk. 2,129.00	Tk. 2,129.00
05.6	10 mm thick finished and laying thickness 12.5 mm silver grey cast in situ mosaic work in wall with one part of 10 mm machine crushed well graded, dust free and properly washed marble chips and one part of mixture containing white cement and grey cement in proportion (1:1) including preparation of mix. and laying the same on 12 mm thick cement plaster (1:4) to wall including compacting, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No 120) including finishing the corners and edges and also finishing with minar stone and supplying all necessary materials and water, electricity and other charges in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					
05.6.1	With Indian chips In ground floor	sqm	Tk. 2,048.90	Tk. 2,016.00	Tk. 1,940.00	Tk. 1,940.00
05.6.2	With Pakistani chips In ground floor	sqm	Tk. 2,081.92	Tk. 2,049.00	Tk. 1,973.00	Tk. 1,973.00
05.6.3	With Indian chips using 100% grey cement	sqm	Tk. 1,892.89	Tk. 1,860.00	Tk. 1,784.00	Tk. 1,784.00

05.7	10 mm thick finished laying in 12.5 mm thickness coloured cast in situ mosaic work in floor with one part of 10 mm machine crushed well graded, dust free and properly washed marble chips and one part of mixture containing white and grey cement in proportion (9:1) and necessary quantity of approved colouring materials including preparation of mix and laying the same over min 25 mm thick artificial patent stone flooring with brick-chips (1:2:4) including making suitable panel, compacting, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No. 120) and finishing with Minar stone including breaking of brick chips, screening, washing of mosaic chips and supply of all necessary materials and water, electricity and other charges in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					
05.7.1	With Indian chips and glass strip In ground floor	sqm	Tk. 2,234.46	Tk. 2,220.00	Tk. 2,160.00	Tk. 2,159.00
05.7.2	With Pakistani chips and glass strip In ground floor	sqm	Tk. 2,267.48	Tk. 2,253.00	Tk. 2,193.00	Tk. 2,192.00
05.7.3	With Indian chips without glass strip In ground floor	sqm	Tk. 1,640.12	Tk. 1,627.00	Tk. 1,576.00	Tk. 1,574.00
05.7.4	With Pakistani chips without glass strip In ground floor	sqm	Tk. 1,829.15	Tk. 1,816.00	Tk. 1,765.00	Tk. 1,763.00
05.8	12 mm thick finished and laying thickness 14 mm coloured cast in situ mosaic work on floor with one part of 10 mm machine crushed well graded, dust free and properly washed marble chips and one part of mixture containing white cement and grey cement in proportion (9:1) and necessary quantity of approved colouring materials including preparation of mix and laying the same over minimum 25 mm thick artificial patent stone flooring with brick-chips (1:2:4) including making suitable panel, compacting, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No. 120) and finishing with minar stone including breaking of brick chips, screening, washing of mosaic chips and supply of all necessary materials and water, electricity and other charges in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					
05.8.1	With Indian chips and glass strip In ground floor	sqm	Tk. 2,323.61	Tk. 2,309.00	Tk. 2,250.00	Tk. 2,248.00
05.8.2	With Pakistani chips and glass strip In ground floor	sqm	Tk. 2,360.60	Tk. 2,346.00	Tk. 2,287.00	Tk. 2,285.00
05.8.3	With Indian chips without glass strip In ground floor	sqm	Tk. 2,041.30	Tk. 2,028.00	Tk. 1,977.00	Tk. 1,975.00
05.8.4	With Pakistani chips without glass strip In ground floor	sqm	Tk. 2,078.28	Tk. 2,065.00	Tk. 2,014.00	Tk. 2,012.00

05.9	10 mm thick finished and laying thickness 12.5 mm coloured cast in situ mosaic work in walls with one part of 10 mm machine crushed well graded, dust free and properly washed marble chips and one part of mixture containing white cement and grey cement in proportion (9:1) and necessary quantity of approved colouring materials including preparation of mix and laying the same on 12 mm thick cement sand (F.M. 1:2) plaster (1:4) to walls including pressing, curing at least for 7 days, cutting with pumice stone (No. 40, No. 80, No. 120) including finishing the corners and edges and also finishing with minar stone and supply of all necessary materials etc. and water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					
05.9.1	With Indian chips In ground floor	sqm	Tk. 2,289.45	Tk. 2,256.00	Tk. 2,181.00	Tk. 2,181.00
05.9.2	With Pakistani chips In ground floor	sqm	Tk. 2,322.47	Tk. 2,289.00	Tk. 2,214.00	Tk. 2,214.00
05.10	200 mm x 200 mm x 20 mm mosaic terrazzo tiles flooring having 10 mm thick finished mosaic top with one part of 10 mm machine crushed well graded dust free and properly washed marble chips and one part of mixture containing white cement and grey cement in 9:1 proportion pressed on 12 mm thick cement sand (F.M. 1:2 and 2.5 by 50% each) mortar (1:2) base including preparing the base; finally, setting the tiles on 40 mm thick cement sand (F.M.1.2 and 2.5 by 50% each) mortar (1:3) including compacting, cutting with pumice stone (No. 40, No. 80, No. 120) and finishing the corners and edges and also finishing with minar stone including screening, washing mosaic chips and supply of all necessary materials etc. and water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-I/OPC)					
05.10.1	200 mm x 200 mm x 20 mm (finished) mosaic terrazzo tiles flooring with Indian chips In ground floor	sqm	Tk. 1,934.25	Tk. 1,925.00	Tk. 1,884.00	Tk. 1,884.00
05.10.2	200 mm x 200 mm x 20 mm (finished) mosaic terrazzo tiles flooring with Pakistani chips In ground floor	sqm	Tk. 1,965.95	Tk. 1,956.00	Tk. 1,916.00	Tk. 1,915.00
05.11	Added cost for mosaic and terrazzo tiles work above ground floor (item no.- 5.1 to 5.10).	sqm	Tk. 18.00	Tk. 18.00	Tk. 17.00	Tk. 17.00

..... **END OF CHAPTER FIVE**

CHAPTER 06: TILES, MARBLE AND GRANITE STONE WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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06.1	Supplying, fitting and fixing country made GP(Gress Porcellanto) - glazed or unglazed homogeneous floor tiles complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm², irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting and laying the tiles in proper way and finishing with care etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.1.1	GP (homogeneous) 300 mm x 300 mm floor tiles	sqm	Tk. 1,328.33	Tk. 1,321.00	Tk. 1,295.00	Tk. 1,295.00
06.1.2	GP (homogeneous) 400 mm x 400 mm floor tiles	sqm	Tk. 1,467.01	Tk. 1,460.00	Tk. 1,434.00	Tk. 1,434.00
06.1.3	GP (homogeneous) 300 mm x 600 mm floor tiles	sqm	Tk. 1,591.82	Tk. 1,585.00	Tk. 1,559.00	Tk. 1,559.00
06.1.4	GP (homogeneous) 600 mm x 600 mm floor tiles	sqm	Tk. 1,633.42	Tk. 1,626.00	Tk. 1,600.00	Tk. 1,600.00
06.1.5	GP (homogeneous) 600 mm x 1200 mm floor tiles	sqm	Tk. 3,047.97	Tk. 3,041.00	Tk. 3,015.00	Tk. 3,015.00
06.1.6	GP (homogeneous) 305 mm x 305 mm high traffic floor tiles	sqm	Tk. 1,425.40	Tk. 1,418.00	Tk. 1,392.00	Tk. 1,392.00
06.2	Supplying, fitting and fixing country made mirror polished homogeneous floor tiles complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm², irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting and laying the tiles in proper way and finishing with care etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor					
06.2.1	GP (mirror polished) 400 mm x 400 mm floor tiles	sqm	Tk. 1,633.42	Tk. 1,626.00	Tk. 1,600.00	Tk. 1,600.00
06.2.2	GP (mirror polished) 500 mm x 500 mm floor tiles	sqm	Tk. 1,772.11	Tk. 1,765.00	Tk. 1,739.00	Tk. 1,739.00
06.2.3	GP (mirror polished) 600 mm x 600 mm floor tiles	sqm	Tk. 1,952.39	Tk. 1,945.00	Tk. 1,919.00	Tk. 1,919.00
06.2.4	GP (mirror polished) 800 mm x 800 mm floor tiles	sqm	Tk. 2,091.07	Tk. 2,084.00	Tk. 2,058.00	Tk. 2,058.00
06.2.5	GP (mirror polished) 600 mm x 900 mm floor tiles	sqm	Tk. 2,354.57	Tk. 2,347.00	Tk. 2,321.00	Tk. 2,321.00
06.2.6	GP (mirror polished) 600 mm x 1200 mm floor tiles	sqm	Tk. 2,784.48	Tk. 2,777.00	Tk. 2,751.00	Tk. 2,751.00
06.2.7	GP Mirror Polished (Double Charge) 600 mm x 600 mm floor tiles	sqm	Tk. 2,174.28	Tk. 2,167.00	Tk. 2,141.00	Tk. 2,141.00
06.2.8	GP Mirror Polished (Double Charge) 800 mm x 800 mm floor tiles	sqm	Tk. 2,659.66	Tk. 2,653.00	Tk. 2,626.00	Tk. 2,626.00
06.3	Supplying, fitting and fixing country made homogeneous matt finished/ rustic floor tiles complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm², irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting and laying the tiles in proper way and finishing with care etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor					
06.3.1	Matt or rustic floor tiles of size 400 mm x 400 mm and below	sqm	Tk. 1,702.77	Tk. 1,696.00	Tk. 1,670.00	Tk. 1,670.00
06.3.2	Matt or rustic floor tiles of size 600 mm x 600 mm and below	sqm	Tk. 1,924.66	Tk. 1,918.00	Tk. 1,891.00	Tk. 1,891.00
06.3.3	Matt or rustic floor tiles of size 800 mm x 800 mm and below	sqm	Tk. 2,368.43	Tk. 2,361.00	Tk. 2,335.00	Tk. 2,335.00
06.4	Supplying, fitting and fixing foreign made polished pocelain/ mirror polished homogeneous floor tiles complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 35 N/mm², irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting and laying the tiles in proper way and finishing with care etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor					

06.4.1	Polish pocelain/ mirror polished 600 mm x 600 mm floor tiles	sqm	Tk. 2,576.46	Tk. 2,569.00	Tk. 2,543.00	Tk. 2,543.00
06.4.2	Polished pocelain/ mirror polished 800 mm x 800 mm floor tiles	sqm	Tk. 2,853.82	Tk. 2,847.00	Tk. 2,821.00	Tk. 2,821.00
06.4.3	Polished pocelain/ mirror polished 600 mm x 1200 mm floor tiles	sqm	Tk. 4,656.67	Tk. 4,650.00	Tk. 4,623.00	Tk. 4,623.00
06.4.4	Polished pocelain/ mirror polished 1000 mm x 1000 mm floor tiles	sqm	Tk. 6,043.48	Tk. 6,036.00	Tk. 6,010.00	Tk. 6,010.00
06.4.5	Polished pocelain/ mirror polished 1200 mm x 1200 mm floor tiles	sqm	Tk. 7,430.29	Tk. 7,423.00	Tk. 7,397.00	Tk. 7,397.00
06.5	Supplying, fitting and fixing foreign made homogeneous matt finished/ rustic floor tiles complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm², irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting and laying the tiles in proper way and finishing with care etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor					
06.5.1	Matt or rustic floor tiles of size 600 mm x 600 mm	sqm	Tk. 2,992.50	Tk. 2,985.00	Tk. 2,959.00	Tk. 2,959.00
06.5.2	Matt or rustic floor tiles of size 600 mm x 900 mm	sqm	Tk. 4,934.03	Tk. 4,927.00	Tk. 4,901.00	Tk. 4,901.00
06.6	Supplying, fitting and fixing country made glazed wall tiles complying BDS ISO 13006: 2015, irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:3) base and raking out the joints with white cement including cutting, laying and hire charge of machine and finishing with care etc. including cost of water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.6.1	Wall tiles less than, equal or equivalent to 250 mm x 330 mm in sizes	sqm	Tk. 1,332.74	Tk. 1,324.00	Tk. 1,288.00	Tk. 1,288.00
06.6.2	Wall tiles more than 250 mm x 330 mm & less than or equal to 250 mm x 400 mm in sizes	sqm	Tk. 1,568.50	Tk. 1,560.00	Tk. 1,523.00	Tk. 1,523.00
06.6.3	Wall tiles more than 250 mm x 400 mm & less than or equal to 300 mm x 600 mm in sizes	sqm	Tk. 1,845.86	Tk. 1,837.00	Tk. 1,801.00	Tk. 1,801.00
06.6.4	Wall tiles less than, equal or equivalent to 250 mm x 750 mm in sizes	sqm	Tk. 1,929.07	Tk. 1,920.00	Tk. 1,884.00	Tk. 1,884.00
06.6.5	Decor wall tiles of any size	sqm	Tk. 2,095.49	Tk. 2,087.00	Tk. 2,050.00	Tk. 2,050.00
06.6.6	Border wall tiles of any size	sqm	Tk. 4,411.46	Tk. 4,403.00	Tk. 4,366.00	Tk. 4,366.00
06.7	Supplying, fitting and fixing country made rustic or matt finished wall tiles complying BDS ISO 13006: 2015, irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:3) base and raking out the joints with white cement including cutting, laying and hire charge of machine and finishing with care etc. including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.7.1	Matt or rustic wall tiles less than or equal to 300 mm x 600 mm size	sqm	Tk. 1,942.94	Tk. 1,934.00	Tk. 1,898.00	Tk. 1,898.00
06.8	Supplying, fitting and fixing foreign made glazed wall tiles complying BDS ISO 13006: 2015, irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:3) base and raking out the joints with white cement including cutting, laying and hire charge of machine and finishing with care etc. including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.8.1	Glazed wall tiles 300 mm x 600 mm in sizes along with matching floor tiles, decor and border (if any)	sqm	Tk. 2,885.97	Tk. 2,877.00	Tk. 2,841.00	Tk. 2,841.00

06.9	Supplying, fitting and fixing foreign made rustic or matt finished wall tiles complying BDS ISO 13006: 2015, irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:3) base and raking out the joints with white cement including cutting, laying and hire charge of machine and finishing with care etc. including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.9.1	Matt or rustic wall tiles 300 mm x 600 mm in sizes along with matching floor tiles, decor and border tiles (if any)	sqm	Tk. 2,885.97	Tk. 2,877.00	Tk. 2,841.00	Tk. 2,841.00
06.10	Supplying, fitting and fixing country made GP homogeneous stair tiles having non skidding offsets, complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm ² , irrespective of color &/or design, with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting and laying the tiles in proper way and finishing with care etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor					
06.10.1	Glazed stair tiles having size of width 250 mm to 300 mm and length 300 mm	sqm	Tk. 1,439.27	Tk. 1,432.00	Tk. 1,406.00	Tk. 1,406.00
06.10.2	Glazed stair tiles having size of width 250 mm to 300 mm and length 325 mm to 600 mm	sqm	Tk. 1,841.45	Tk. 1,834.00	Tk. 1,808.00	Tk. 1,808.00
06.10.3	Glazed stair tiles (Double Charge) having size of width 300 mm and length 600 mm	sqm	Tk. 2,382.30	Tk. 2,375.00	Tk. 2,349.00	Tk. 2,349.00
06.11	Supplying, fitting and fixing country made rustic or matt finished stair tiles having non-skidding offsets complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm ² , irrespective of color &/or design, with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting, laying and hire charge of machine and finishing with care etc. including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.11.1	Rustic stair tiles of size 300 mm x 600 mm	sqm	Tk. 1,744.37	Tk. 1,737.00	Tk. 1,711.00	Tk. 1,711.00
06.12	Supplying, fitting and fixing foreign made GP homogeneous stair tiles having non skidding offsets, complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm ² , irrespective of color &/or design, with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting and laying the tiles in proper way and finishing with care etc. all complete and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor					
06.12.1	Glazed stair tiles of size 300 mm x 600 mm	sqm	Tk. 2,021.73	Tk. 2,015.00	Tk. 1,989.00	Tk. 1,989.00
06.13	Supplying, fitting and fixing foreign made rustic or matt finished stair tiles having non-skidding offsets complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm ² , irrespective of color &/or design, with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and raking out the joints with white cement including cutting, laying and hire charge of machine and finishing with care etc. including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.13.1	Rustic stair tiles of size 300 mm x 600 mm	sqm	Tk. 1,813.71	Tk. 1,807.00	Tk. 1,780.00	Tk. 1,780.00

06.14	Supplying, fitting and fixing country made floor tiles complying BDS ISO 13006: 2015, water absorption $\leq 0.5\%$, modulus of rupture (MOR) ≥ 27 N/mm ² , irrespective of color &/or design, with adhesives in full thickness of tiles, filler/tiles grout including cutting, shaping, placing in proper level etc. all complete and accepted by the Engineer in charge. In ground floor					
06.14.1	GP glazed / unglazed homogenous floor tiles (400 mm x 400 mm)	sqm	Tk. 1,589.14	Tk. 1,583.00	Tk. 1,552.00	Tk. 1,552.00
06.14.2	GP glazed / unglazed homogenous floor tiles (300 mm x 300 mm)	sqm	Tk. 1,450.46	Tk. 1,444.00	Tk. 1,414.00	Tk. 1,414.00
06.14.3	GP Mirror Polished floor tiles (300 mm x 300 mm)	sqm	Tk. 1,519.80	Tk. 1,513.00	Tk. 1,483.00	Tk. 1,483.00
06.14.4	GP Mirror Polished floor tiles (400 mm x 400 mm)	sqm	Tk. 1,755.55	Tk. 1,749.00	Tk. 1,719.00	Tk. 1,719.00
06.14.5	GP Mirror Polished floor tiles (600 mm x 600 mm)	sqm	Tk. 2,074.52	Tk. 2,068.00	Tk. 2,038.00	Tk. 2,038.00
06.14.6	GP Mirror Polished floor tiles (800 mm x 800 mm)	sqm	Tk. 2,213.20	Tk. 2,207.00	Tk. 2,176.00	Tk. 2,176.00
06.14.7	GP Mirror Polished floor tiles (600 mm x 900 mm)	sqm	Tk. 2,476.69	Tk. 2,470.00	Tk. 2,440.00	Tk. 2,440.00
06.14.8	GP Mirror Polished floor tiles (600 mm x 1200 mm)	sqm	Tk. 2,906.61	Tk. 2,900.00	Tk. 2,870.00	Tk. 2,870.00
06.14.9	GP Mirror Polished (Double Charge) 600 mm x 600 mm floor tiles	sqm	Tk. 2,296.41	Tk. 2,290.00	Tk. 2,260.00	Tk. 2,260.00
06.14.10	GP Mirror Polished (Double Charge) 800 mm x 800 mm floor tiles	sqm	Tk. 2,781.79	Tk. 2,775.00	Tk. 2,745.00	Tk. 2,745.00
06.15	Supplying, fitting and fixing 20mm to 25mm thick machine made cement pavement tiles having minimum compressive strength of 27 MPa, irrespective of color &/or design , with 20 mm thick cement sand (F.M. 1.2) mortar (1:4) base and making the joints carefully in true straight line including cutting, laying and hire charge of machine and finishing with care etc. including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.15.1	Pavement tiles of size 300 mm x 300 mm	sqm	Tk. 1,189.65	Tk. 1,183.00	Tk. 1,156.00	Tk. 1,156.00
06.16	Supplying, fitting and fixing machine made clay roof tiles with ridge tiles on RCC inclined slab, irrespective of color &/or design, water absorption capacity $<6\%$ with 20 mm thick cement sand (F.M. 1.2) mortar (1:3) base and making the joints carefully in true straight line including cutting, laying and hire charge of machine and finishing with care etc. including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M). In ground floor					
06.16.1	Roof tiles of size 100 mm x 200 mm or equivalent (red colour)	sqm	Tk. 2,960.20	Tk. 2,950.00	Tk. 2,900.00	Tk. 2,900.00
06.16.2	Roof tiles of size 125 mm x 170 mm to 200 mm x 200 mm or equivalent (red colour)	sqm	Tk. 3,360.29	Tk. 3,350.00	Tk. 3,300.00	Tk. 3,300.00
06.16.3	Roof tiles of size 160 mm x 268 mm or equivalent (red colour)	sqm	Tk. 3,086.40	Tk. 3,076.00	Tk. 3,026.00	Tk. 3,026.00
06.16.4	Roof tiles of size 225 mm x 300 mm or larger (red colour)	sqm	Tk. 3,169.60	Tk. 3,159.00	Tk. 3,110.00	Tk. 3,110.00
06.16.5	Flower shaped roof tiles of size 175 mm x 175 mm or equivalent (red colour)	sqm	Tk. 4,032.20	Tk. 4,022.00	Tk. 3,972.00	Tk. 3,972.00
06.16.6	Roof tiles of size 100 mm x 200 mm or equivalent (beige colour)	sqm	Tk. 2,933.85	Tk. 2,923.00	Tk. 2,874.00	Tk. 2,874.00
06.16.7	Roof tiles of size 125 mm x 170 mm to 200 mm x 200 mm or equivalent (beige colour)	sqm	Tk. 3,488.57	Tk. 3,478.00	Tk. 3,429.00	Tk. 3,429.00
06.16.8	Roof tiles of size 225 mm x 300 mm or larger (beige colour)	sqm	Tk. 3,419.23	Tk. 3,409.00	Tk. 3,359.00	Tk. 3,359.00
06.16.9	Flower shaped roof tiles of size 175 mm x 175 mm or equivalent (beige colour)	sqm	Tk. 4,195.84	Tk. 4,185.00	Tk. 4,136.00	Tk. 4,136.00
06.16.10	Roof tiles of size 125 mm x 170 mm to 200 mm x 200 mm or equivalent (grey or black colour)	sqm	Tk. 3,488.57	Tk. 3,478.00	Tk. 3,429.00	Tk. 3,429.00
06.16.11	Flower shaped roof tiles of size 175 mm x 175 mm or equivalent (grey or black colour)	sqm	Tk. 4,556.41	Tk. 4,546.00	Tk. 4,496.00	Tk. 4,496.00

06.17	Marble stone flooring with Indian standard quality (Bansowara/ Indian grey/ Indian pink or equivalent) marble stone of approved quality complying accepted National/ International standards, texture and size with 25 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including placing cement slurry below the stone, cutting the marble stone to desired size, grinding and finishing the edges, setting in position in proper level, filling the gaps with white cement and colouring pigments including supplying all necessary equipments/materials and finishing the top by pumice stone and necessary auxiliary materials and water, electricity and other charges complete in all respect and accepted by the Engineer in charge.(Cement: CEM-II/B-M).					
06.17.1	18 mm thick marble stone flooring In ground floor	sqm	Tk. 7,945.77	Tk. 7,938.00	Tk. 7,843.00	Tk. 7,842.00
06.17.2	16 mm thick marble stone flooring In ground floor	sqm	Tk. 7,417.46	Tk. 7,409.00	Tk. 7,314.00	Tk. 7,314.00
06.18	Marble stone flooring with Indian superior quality marble stone of approved quality complying accepted National/ International standards, texture and size with 25 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including placing cement slurry below the stone, cutting the marble stone to desired size, grinding and finishing the edges, setting in position in proper level, filling the gaps with white cement and colouring pigments including supplying all necessary equipments/materials and finishing the top by pumice stone and necessary auxiliary materials and water, electricity and other charges complete in all respect and accepted by the Engineer in charge. (Cement: CEM-II/B-M)					
06.18.1	18 mm thick marble stone flooring In ground floor	sqm	Tk. 8,606.16	Tk. 8,598.00	Tk. 8,503.00	Tk. 8,503.00
06.18.2	16 mm thick marble stone flooring In ground floor	sqm	Tk. 7,549.54	Tk. 7,541.00	Tk. 7,446.00	Tk. 7,446.00
06.19.1	Supplying and setting 12 mm thick Indian standard quality (Bansowara/ Indian grey/ Indian pink or equivalent) marble stone of approved quality complying accepted National/ International standards, texture and size on wall with 12 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including cutting and grinding the marble stone slab/tile, setting in position keeping horizontal and vertical alignment correct, filling the joints with white cement, cutting and grinding the top surface, finishing etc including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 6,897.46	Tk. 6,890.00	Tk. 6,795.00	Tk. 6,794.00
06.19.2	Supplying and setting 12 mm thick Indian superior quality marble stone of approved quality complying accepted National/ International standards, texture and size on wall with 12 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including cutting and grinding the marble stone slab/tile, setting in position keeping horizontal and vertical alignment correct, filling the joints with white cement, cutting and grinding the top surface, finishing etc including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 7,293.69	Tk. 7,286.00	Tk. 7,191.00	Tk. 7,191.00

06.20.1	Marble stone flooring with 18mm thick Italian or equivalent marble stone of approved quality complying accepted National/ International standards, texture and size over 25 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including. placing cement slurry below the stone, cutting the marble stone to desired size, grinding and finishing the edges, setting in position in proper level, filling the gaps with white cement and colouring pigments including supplying all necessary equipments/materials and finishing the top by pumice stone and necessary auxiliary materials and water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 10,455.24	Tk. 10,447.00	Tk. 10,352.00	Tk. 10,352.00
06.20.2	Supplying and setting 12 mm thick Italian or equivalent quality marble stone on wall of approved quality complying accepted National/ International standards, texture and size on wall with 12 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including cutting and grinding the marble stone slab/tile, setting in position keeping horizontal and vertical alignment correct, filling the joints with white cement, cutting and grinding the top surface, finishing etc including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	Sqm	Tk. 9,803.15	Tk. 9,795.00	Tk. 9,700.00	Tk. 9,700.00
06.21	Labour cost for making moulding in marble stone as per architectural drawing including cutting, grinding, finishing, supplying necessary equipment etc. complete in all respect and accepted by the Engineer in Charge. (Cement: CEM-II/B-M).					
06.21.1	18 mm thick marble stone	meter	Tk. 474.98	Tk. 473.00	Tk. 454.00	Tk. 454.00
06.21.2	16 mm thick marble stone	meter	Tk. 470.11	Tk. 468.00	Tk. 449.00	Tk. 449.00
06.21.3	12 mm thick marble stone	meter	Tk. 398.88	Tk. 397.00	Tk. 383.00	Tk. 383.00
06.22	Granite stone flooring with 18mm thick Indian or equivalent granite stone of approved quality complying accepted National/ International standards, texture and size over 25 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including. placing cement slurry below the stone, cutting the granite stone to desired size, grinding and finishing the edges, setting in position in proper level. filling the gaps with white cement and colouring pigments including supplying all necessary equipments/materials and finishing the top by pumice stone and necessary auxiliary materials and water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 9,134.46	Tk. 9,126.00	Tk. 9,031.00	Tk. 9,031.00
06.23	Granite stone flooring with 18mm thick Italian or equivalent origin granite stone of approved quality complying accepted National/ International standards, texture and size over 25 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including. placing cement slurry below the stone, cutting the granite stone to desired size, grinding and finishing the edges, setting in position in proper level, filling the gaps with white cement and colouring pigments including supplying all necessary equipments/materials and finishing the top by pumice stone and necessary auxiliary materials and water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 10,455.24	Tk. 10,447.00	Tk. 10,352.00	Tk. 10,352.00

06.24	Supplying and setting 12 mm thick Indian or equivalent granite stone of approved quality complying accepted National/ International standards, texture and size on wall with 12 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including cutting and grinding the granite stone slab, setting in position keeping horizontal and vertical alignment correct, filling the joints with white cement, cutting and grinding the top surface, finishing etc including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 7,161.61	Tk. 7,154.00	Tk. 7,059.00	Tk. 7,059.00
06.25	Supplying and setting 12 mm thick Italian or equivalent granite stone of approved quality complying accepted National/ International standards, texture and size on wall with 12 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including cutting and grinding the granite stone slab, setting in position keeping horizontal and vertical alignment correct, filling the joints with white cement, cutting and grinding the top surface, finishing etc including water, electricity and other charges complete in all respect and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) In ground floor	sqm	Tk. 8,482.38	Tk. 8,475.00	Tk. 8,380.00	Tk. 8,379.00
06.26	Kota stone flooring of approved quality complying accepted National/ International standards, texture and size with 25 mm thick cement sand (50% F.M. 1.2 and 50% F.M 2.2) mortar (1:2) including placing cement slurry below the stone, cutting the marble stone to desired size, grinding and finishing the edges, setting in position in proper level, filling the gaps with white cement and colouring pigments including supplying all necessary equipments/materials and finishing the top by pumice stone and necessary auxiliary materials and water, electricity and other charges complete in all respect and accepted by the Engineer in charge. (Cement: CEM-II/B-M)					
	14-22 mm thick Kota stone flooring In ground floor	sqm	Tk. 4,818.84	Tk. 4,814.00	Tk. 4,749.00	Tk. 4,748.00
06.27	Added rate for each additional floor above ground floor for tiles works	sqm	Tk. 15.00	Tk. 14.00	Tk. 13.00	Tk. 13.00
06.28	Added rate for each additional floor above ground floor for marble and granite stone works	sqm	Tk. 25.00	Tk. 25.00	Tk. 23.00	Tk. 23.00

..... END OF CHAPTER SIX

CHAPTER 07: REINFORCED CEMENT CONCRETE (RCC) WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
	RCC WORKS: 1:2:4 (measured on gross concrete section) ($f_c = 19$ MPa, minimum $f_{cr} = 26$ MPa in nominal mix 1:2:4), with brick-chips (Sand of F.M. 1.2 and F.M. 2.2 in equal proportion)					

07.1	Reinforced cement concrete works with minimum cement content relates to mix ratio 1:2:4 having maximum water cement ratio = 0.45 and minimum f_{cr} = 26 MPa, satisfying a specified compressive strength f_c = 19 MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality sand [50% quantity of best local sand (F.M. 1.2) and 50% quantity of Sylhet sand or coarse sand of equivalent F.M. 2.2] and 20 mm down well graded picked jhama brick chips conforming to ASTM C-33 including conducting necessary tests, breaking chips and screening, making and placing shutter in position maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; mixing in standard mixer machine with hopper fed by standard measuring boxes, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering).					
07.1.1	Individual & combined footing, pile cap, raft/mat, floor slab and foundation beam up to plinth level	cum	Tk. 8,673.00	Tk. 8,643.00	Tk. 8,521.00	Tk. 8,449.00
07.1.2	Pedestal, column, column capital, lift wall and wall up to ground floor	cum	Tk. 9,032.00	Tk. 8,996.00	Tk. 8,853.00	Tk. 8,781.00
07.1.3	Floor / roof slab, T-beam, L-beam and rectangular beam, tie beam, lintel, stair case slab and step etc. up to ground floor	cum	Tk. 8,817.00	Tk. 8,784.00	Tk. 8,654.00	Tk. 8,582.00
07.1.4	Cornice, railing, drop wall, louver, fins, sunshade, false ceiling, waffle and ribbed floor slab etc. up to ground floor	cum	Tk. 9,391.00	Tk. 9,350.00	Tk. 9,185.00	Tk. 9,113.00
	RCC WORKS: 1:2:4 (measured on gross concrete section) (f_c = 22 MPa, minimum f_{cr} = 30.5 MPa in nominal mix 1:2:4), with stone chips (100% sand of F.M. 2.2)					
07.2	Reinforced cement concrete works with minimum cement content relates to mix ratio 1:2:4 having maximum water cement ratio = 0.40 and minimum f_{cr} = 30.5 MPa, satisfying a specified compressive strength f_c = 22 MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down well graded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), conducting necessary tests, making and placing shutter in position maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; mixing in standard mixer machine with hopper fed by standard measuring boxes, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering)					
07.2.1	Individual & combined footing, pile cap, raft/mat, floor slab and foundation beam up to plinth level	cum	Tk. 13,504.00	Tk. 13,670.00	Tk. 12,724.00	Tk. 12,340.00

07.2.2	Pedestals, column, column capital, lift wall and RCC wall up to ground floor	cum	Tk. 13,863.00	Tk. 14,023.00	Tk. 13,056.00	Tk. 12,671.00
07.2.3	Floor / roof slab, T-beam, L-beam and rectangular beam, tie beam, lintel, stair case slab and steps etc. up to ground floor	cum	Tk. 13,648.00	Tk. 13,811.00	Tk. 12,857.00	Tk. 12,472.00
07.2.4	Cornice, railing, drop wall, louver, fins, sunshade, false ceiling, waffle and ribbed floor slab etc. up to ground floor	cum	Tk. 14,222.00	Tk. 14,377.00	Tk. 13,388.00	Tk. 13,003.00
	RCC WORKS: 1:1.5:3(measured on gross concrete section) (f_c = 25MPa, minimum f_{cr} = 33.5 MPa in nominal mix 1 : 1.5 : 3), with stone chips (100% sand of F.M. 2.2)					
07.3	Reinforced cement concrete works with minimum cement content relates to mix ratio 1:1.5:3 having maximum water cement ratio = 0.40 and minimum f _{cr} = 33.5 MPa, satisfying a specified compressive strength f _c = 25 MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down well graded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; mixing with standard mixer machine with hopper, fed by standard measuring boxes, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering)					
07.3.1	Individual & combined footing, pile cap, raft/mat, floor slab and foundation beam up to plinth level	cum	Tk. 13,842.00	Tk. 14,000.00	Tk. 13,095.00	Tk. 12,728.00
07.3.2	Pedestals, column, column capital, lift wall and RCC wall up to ground floor	cum	Tk. 14,201.00	Tk. 14,353.00	Tk. 13,427.00	Tk. 13,060.00
07.3.3	Floor / roof slab, T-beam, L-beam and rectangular beam, tie beam, lintel, stair case slab and steps etc. upto ground floor	cum	Tk. 13,986.00	Tk. 14,141.00	Tk. 13,228.00	Tk. 12,861.00
07.3.4	Cornice, railing, drop wall, louver, fins, sunshade, false ceiling, waffle and ribbed floor slab etc. up to ground floor	cum	Tk. 14,561.00	Tk. 14,707.00	Tk. 13,759.00	Tk. 13,392.00
	RCC WORKS : 1:1.25:2.5 (measured on gross concrete section) (f_c = 32 MPa, minimum f_{cr} = 40.5 MPa in nominal mix 1 : 1.25 : 2.5)With Stone Chips and Admixture (Type-G) (100% sand of F.M. 2.2)					

07.4	Reinforced cement concrete works with minimum cement content relates to mix ratio 1:1.25:2.5 having maximum water cement ratio = 0.40 and minimum f_{cr} = 40.5 MPa, satisfying a specified compressive strength f_c = 32 MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa)/ ASTM-C 150 Type – I, and adding approved high range water reducing admixture of complying specific type (generally be Type-G) under ASTM-C 494, best quality coarse sand [Sylhet sand or coarse sand of equivalent F.M. 2.2], 20 mm down well graded crushed stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), including conducting necessary tests, screening sand through proper sieves, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; mixing with standard mixer machine with hopper and fed by standard measuring boxes, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Doses of admixture to be fixed in consultation with design office) (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering).					
07.4.1	Individual & combined footing, pile cap, raft/mat, floor slab and foundation beam up to plinth level	cum	Tk. 14,143.00	Tk. 14,295.00	Tk. 13,421.00	Tk. 13,067.00
07.4.2	Padestal, column, column capital, lift wall and RCC wall up to ground floor	cum	Tk. 14,503.00	Tk. 14,648.00	Tk. 13,753.00	Tk. 13,399.00
07.4.3	Floor / roof slab, T-beam, L-beam and rectangular beam, tie beam, lintel, stair case slab and steps etc. up to ground floor	cum	Tk. 14,287.00	Tk. 14,436.00	Tk. 13,554.00	Tk. 13,200.00
07.4.4	Cornice, railing, drop wall, louver, fins, sunshade, false ceiling, waffle and ribbed floor slab etc. up to ground floor	cum	Tk. 14,862.00	Tk. 15,002.00	Tk. 14,085.00	Tk. 13,731.00
	RCC WORKS: 1:1:2 (measured on gross concrete section) (f_c = 40 MPa, minimum f_{cr} = 50 MPa in nominal mix 1 : 1 : 2) With stone chips and admixture (Type-G) (100% sand of F.M. 2.2)					

07.5	Reinforced cement concrete works with minimum cement content relates to mix ratio 1:1:2 having minimum $f'_{cr} = 50$ Mpa, satisfying a specified compressive strength $f'_c = 40$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, and adding approved high range water reducing admixture of complying specific type (generally be Type-G) under ASTM-C 494, best quality coarse sand [Sylhet sand or coarse sand of equivalent F.M. 2.2], 20 mm down well graded crushed stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; mixing with standard mixer machine with hopper and fed by standard measuring boxes, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete approved and accepted by the Engineer-in-charge. (Doses of admixture to be fixed in consultation with design office) (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering)					
07.5.1	Individual & combined footing, pile cap, raft/mat and foundation beam up to plinth level	cum	Tk. 14,624.00	Tk. 14,768.00	Tk. 13,935.00	Tk. 13,599.00
07.5.2	Pedestal, column, column capital, lift wall and RCC wall upto ground floor	cum	Tk. 14,984.00	Tk. 15,121.00	Tk. 14,267.00	Tk. 13,931.00
	Fair-faced RCC WORKS: 1:1.5:3 (measured on gross concrete section) ($f'_c = 25$ MPa, minimum $f'_{cr} = 33.5$ MPa in nominal mix 1:1.5:3) (100% sand of F.M. 2.2)					
07.6	Cast-in-place fair-faced concrete works with $f'_{cr}=33.5$ Mpa and $f'_c=25$ Mpa at 28 days on standard cylinders by using 75% of OPC and 25% of white cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2, 20 mm down and well graded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification) at mix ratio of 1:1.5:3 , adding coloring pigment admixture (satisfying ASTM C979) as per architectural design, conforming to the standard practice of code ACI/BNBC/ASTM including conducting necessary tests, screening sand and chips through proper sieves, washing, making and placing shutter in position maintaining true to plumb, making shutter water tight properly, placing reinforcement in position, including pouring of concrete in form, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering)					
07.6.1	For column, wall, beam, slab, drop wall, cornice etc. up to ground floor	cum	Tk. 17,381.00	Tk. 17,522.00	Tk. 16,558.00	Tk. 16,190.00
	RCC WORKS: batch mix/machine mix with approved mix design, well graded stone chips (100% sand of F.M. 2.2) and required admixture					

07.7.1	Reinforced cement concrete works from batch mix/machine mix with minimum cement content relates to approved mix ratio having minimum $f_{cr} = 33.5$ MPa, satisfying a specified compressive strength $f_c = 25$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494 (Doses to be applied as per Manufacturer's Guideline), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content as per approved design mix (minimum 385 kg/ cum), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation to 4th floor) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 14,992.00	Tk. 15,158.00	Tk. 14,290.00	Tk. 13,923.00
07.7.2	Reinforced cement concrete works from batch mix/machine mix with minimum cement content relates to approved mix ratio having minimum $f_{cr} = 40.5$ MPa, satisfying a specified compressive strength $f_c = 32$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494 (Doses to be applied as per Manufacturer's Guideline) making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content as per approved design mix (minimum 445 kg/ cum), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation to 4th floor) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 15,372.00	Tk. 15,532.00	Tk. 14,695.00	Tk. 14,341.00

07.7.3	Reinforced cement concrete works from batch mix/machine mix with minimum cement content relates to approved mix ratio having minimum $f_{cr} = 50$ MPa, satisfying a specified compressive strength $f'c = 40$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494 (Doses to be applied as per Manufacturer's Guideline), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content as per approved design mix (minimum 500 kg/ cum), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation to 4th floor) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 15,650.00	Tk. 15,803.00	Tk. 15,003.00	Tk. 14,665.00
07.7.4	Reinforced cement concrete works from batch mix/machine mix with minimum cement content relates to approved mix ratio having minimum $f_{cr} = 60$ MPa, satisfying a specified compressive strength $f'c = 50$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494 (Doses to be applied as per Manufacturer's Guideline), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content (minimum 500 kg/ cum) and micro silica (minimum 25 kg/ cum) as per approved design mix, including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation to 4th floor) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 18,111.00	Tk. 18,264.00	Tk. 17,464.00	Tk. 17,127.00
	RCC WORKS: from Ready-Mix Concrete plant with approved mix design, well graded stone chips (100% sand of F.M. 2.2) and required admixture					

07.8.1	Reinforced cement concrete works from Ready-Mix concrete with minimum cement content relates to approved mix ratio having minimum $f_{cr} = 33.5$ MPa, satisfying a specified compressive strength $f_c = 25$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in appendix), High-range water reducing chemical admixture Type-G under ASTM C494 (Doses to be applied as per Manufacturer's Guideline) making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content as per approved design mix (minimum 385 kg/ cum), carrying to the site of laying using Transit Mixer (considering setting time of concrete from plant to the site), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm (measured at site), casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation to 4th floor) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 14,267.00	Tk. 14,434.00	Tk. 13,566.00	Tk. 13,199.00
07.8.2	Reinforced cement concrete works from Ready-Mix concrete with minimum cement content relates to approved mix ratio having minimum $f_{cr} = 40.5$ MPa, satisfying a specified compressive strength $f_c = 32$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494 (Doses to be applied as per Manufacturer's Guideline), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content as per approved design mix (minimum 445 kg/ cum), carrying to the site of laying using Transit Mixer (considering setting time of concrete from plant to the site), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm (measured at site), casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation to 4th floor) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 14,647.00	Tk. 14,808.00	Tk. 13,971.00	Tk. 13,617.00

07.8.3	Reinforced cement concrete works from Ready-Mix concrete with minimum cement content relates to approved mix ratio having minimum $f_{cr} = 50$ MPa, satisfying a specified compressive strength $f'_c = 40$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494 (Doses to be applied as per Manufacturer's Guideline) making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content as per approved design mix (minimum 500 kg/ cum), carrying to the site of laying using Transit Mixer (considering setting time of concrete from plant to the site), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm (measured at site), casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation to 4th floor) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 14,926.00	Tk. 15,079.00	Tk. 14,279.00	Tk. 13,941.00
07.8.4	Reinforced cement concrete works from Ready-Mix concrete with minimum cement content relates to approved mix ratio having minimum $f_{cr} = 60$ MPa, satisfying a specified compressive strength $f'_c = 50$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5 MPa) / ASTM-C 150 Type – I, best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494 (Doses to be applied as per Manufacturer's Guideline), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content (minimum 500 kg/ cum) and micro silica (minimum 25 kg/ cum) as per approved design mix, carrying to the site of laying using Transit Mixer (considering setting time of concrete from plant to the site), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm (measured at site), casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation to 4th floor) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 17,387.00	Tk. 17,540.00	Tk. 16,740.00	Tk. 16,402.00

07.8.5	Reinforced cement concrete works for thick mat foundation from Ready-Mix concrete with minimum cement content relates to approved mix ratio having minimum $f'_{cr} = 33.5$ MPa, satisfying a specified compressive strength $f'_{c} = 25$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to CEM-II/B-M 42.5N (42.5MPa), best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494(Doses to be applied as per Manufacturer's Guideline), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content as per approved design mix (minimum 400 kg/cum), carrying to the site of laying using Transit Mixer (considering setting time of concrete from plant to the site), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm (measured at site), casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation: MAT) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 14,234.00	Tk. 14,397.00	Tk. 13,548.00	Tk. 13,189.00
07.8.6	Reinforced cement concrete works for thick mat foundation from Ready-Mix concrete with minimum cement content relates to approved mix ratio having minimum $f'_{cr} = 40.5$ MPa, satisfying a specified compressive strength $f'_{c} = 32$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to CEM-II/B-M 42.5N (42.5MPa), best quality Sylhet sand or coarse sand of equivalent F.M. 2.2 and 20 mm down wellgraded stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), High-range water reducing chemical admixture Type-G under ASTM C494(Doses to be applied as per Manufacturer's Guideline), conducting necessary tests, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; using machine batched and machine mixed concrete using cement content as per approved design mix (minimum 460 kg/cum), carrying to the site of laying using Transit Mixer (considering setting time of concrete from plant to the site), including pumping of concrete using line pump or boom placer to site of laying, maintaining allowable slump of 100mm to 150mm (measured at site), casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, cost of all materials and other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation: MAT) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office.	cum	Tk. 14,602.00	Tk. 14,759.00	Tk. 13,943.00	Tk. 13,597.00

	RCC WORKS : 1:1.25:2.5 (measured on gross concrete section) (f_c = 30 MPa, minimum f_{cr} = 38.5 MPa in nominal mix 1 : 1.25 : 2.5) With Stone Chips and Admixture (100% sand of F.M. 2.2) For Saline Zone					
07.9	Reinforced cement concrete works with minimum cement content relates to mix ratio 1:1.25:2.5 having minimum f _{cr} = 38.5 MPa, satisfying a specified compressive strength f _c = 30 MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM, Cement conforming to CEM-II/B-M 42.5N (42.5MPa), and adding approved high range water reducing admixture of complying specific type (generally be Type-G or F) under ASTM-C 494, best quality coarse sand [Sylhet sand or coarse sand of equivalent F.M. 2.2], 20 mm down well graded crushed stone chips conforming to ASTM C-33 (Aggregate grading as per table shown in technical specification), including conducting necessary tests, screening sand through proper sieves, making and placing shutter in position and maintaining true to plumb, making shutter water-tight properly, placing reinforcement in position; mixing with standard mixer machine with hopper and fed by standard measuring boxes or in batching plant, casting in forms, compacting by vibrator machine and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Doses of admixture to be fixed in consultation with design office) (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering)					
07.9.1	Individual & combined footing, pile cap, raft/mat, floor slab and foundation beam up to plinth level	cum	Tk. 14,815.00	Tk. 14,967.00	Tk. 14,093.00	Tk. 13,739.00
07.9.2	Padestal, column, column capital, lift wall and RCC wall up to ground floor	cum	Tk. 15,174.00	Tk. 15,320.00	Tk. 14,425.00	Tk. 14,071.00
07.9.3	Floor / roof slab, T-beam, L-beam and rectangular beam, tie beam, lintel, stair case slab and steps etc. up to ground floor	cum	Tk. 14,589.00	Tk. 14,739.00	Tk. 13,856.00	Tk. 13,502.00
07.9.4	Cornice, railing, drop wall, louver, fins, sunshade, false ceiling, waffle and ribbed floor slab etc. up to ground floor	cum	Tk. 15,164.00	Tk. 15,304.00	Tk. 14,387.00	Tk. 14,033.00
07.10	Added rate for RCC works/fair-faced concrete works (materials, consumables and laying/placing concrete) in column, column capital, lift wall & wall, T- beam, L-beam, rectangular beams, floor / roof slab, cantilever slab, waffle slab and drop panels where the free height of the structure exceeds 4 meter (rate is considered for each additional meter height)	cum	Tk. 340.38	Tk. 335.00	Tk. 312.00	Tk. 312.00
07.11.1	Added rate for RCC work/fair-faced work in additional floor: concrete	cum	Tk. 93.00	Tk. 93.00	Tk. 93.00	Tk. 93.00
07.11.2	Added rate for RCC work in additional floor: concrete (concrete from batching or Ready-Mix Plant)					
07.11.2.1	Add for each cum of concrete from 5th floor to 14th floor:	cum	Tk. 19.00	Tk. 19.00	Tk. 19.00	Tk. 19.00
07.11.2.2	Add for each cum of concrete above 15th floor:	cum	Tk. 47.00	Tk. 47.00	Tk. 47.00	Tk. 47.00
FORM WORK (Steel)						
07.12	Centering and shuttering, including strutting, propping etc. (The formwork must be rigid enough both in and out of plane, to make the concrete surface true to the designed shape and size by using necessary MS sheets of minimum 16 BWG, angles of minimum size 40 mm x 40 mm x 5 mm, flat bars etc.) and removal of form for:					

07.12.1	Individual and combined footing	sqm	Tk. 582.00	Tk. 579.00	Tk. 568.00	Tk. 568.00
07.12.2	In raft/mat/floor slab up to plinth level	sqm	Tk. 587.00	Tk. 584.00	Tk. 571.00	Tk. 571.00
07.12.3	Foundation beam	sqm	Tk. 493.00	Tk. 492.00	Tk. 486.00	Tk. 486.00
07.12.4	Pedestal, column, column capital, lift wall and wall up to ground floor	sqm	Tk. 522.00	Tk. 521.00	Tk. 517.00	Tk. 517.00
07.12.5	Tie beam and lintel: ground floor	sqm	Tk. 543.00	Tk. 541.00	Tk. 535.00	Tk. 535.00
07.12.6	T-beam, L-beam and rectangular beam etc up to ground floor	sqm	Tk. 574.00	Tk. 573.00	Tk. 569.00	Tk. 569.00
07.12.7	Floor and roof slab up to ground floor	sqm	Tk. 597.00	Tk. 596.00	Tk. 590.00	Tk. 590.00
07.12.8	Cornice, railing, drop walls, louver, fins etc.: ground floor	sqm	Tk. 487.00	Tk. 486.00	Tk. 479.00	Tk. 479.00
07.12.9	Sunshade, false ceiling: ground floor	sqm	Tk. 540.00	Tk. 540.00	Tk. 534.00	Tk. 534.00
07.12.10	Stair case slab and steps up to ground floor	sqm	Tk. 551.00	Tk. 550.00	Tk. 546.00	Tk. 546.00
07.12.11	Waffle or ribbed floor upto ground floor	sqm	Tk. 720.00	Tk. 719.00	Tk. 713.00	Tk. 713.00
07.12.12.1	Arches, domes, vaults up to 8m span (for single use)	sqm	Tk. 2,145.00	Tk. 2,144.00	Tk. 2,139.00	Tk. 2,139.00
07.12.12.2	Arches, domes, vaults up to 8 m span (for 2-times use)	sqm	Tk. 1,501.00	Tk. 1,501.00	Tk. 1,498.00	Tk. 1,498.00
07.12.12.3	Arches, domes, vaults up to 8 m span (for 3-times use)	sqm	Tk. 1,073.00	Tk. 1,072.00	Tk. 1,070.00	Tk. 1,070.00
07.12.12.4	Arches, domes, vaults up to 8 m span (for more than 3-times use)	sqm	Tk. 751.00	Tk. 751.00	Tk. 749.00	Tk. 749.00
07.13	Added rate for arches, domes, vaults exceeding 8 m span	sqm	Tk. 537.00	Tk. 536.00	Tk. 535.00	Tk. 535.00
07.14	Added rate for shuttering in circular work	sqm	Tk. 115.00	Tk. 115.00	Tk. 114.00	Tk. 114.00
07.15	Added rate for additional height in centering, shuttering where ever required with adequate bracing, propping etc. over a height of 4.0 m, for every additional height of 1 meter or part thereof.	sqm	Tk. 57.39	Tk. 57.00	Tk. 57.00	Tk. 57.00
07.16	Added rate for formwork/shuttering in additional floor	sqm	Tk. 18.00	Tk. 18.00	Tk. 16.00	Tk. 16.00
07.17	Formwork for making 'fair-faced' surface of the concrete as per design, drawing and direction of Engineer-in-charge. (The formwork must be rigid enough both in and out of plane, to make the concrete surface true to the designed shape and size by using necessary MS sheets of minimum 10 BWG, angles of minimum size 50 mm x 50 mm x 6 mm, flat bars, formwork releasing agent etc.)	sqm	Tk. 1,110.00	Tk. 1,108.00	Tk. 1,104.00	Tk. 1,104.00
	FORMWORK (Wooden) for boundary wall, gate, centry post etc.					
07.18	Centering and shuttering, including strutting, propping etc. and removal of form for:					
07.18.1	Individual and combined footings	sqm	Tk. 523.00	Tk. 520.00	Tk. 502.00	Tk. 502.00
07.18.2	Foundation beam	sqm	Tk. 475.00	Tk. 474.00	Tk. 458.00	Tk. 458.00
07.18.3	Padestals, column, wall	sqm	Tk. 526.00	Tk. 525.00	Tk. 508.00	Tk. 508.00
07.18.4	Tie beam & lintels	sqm	Tk. 521.00	Tk. 520.00	Tk. 502.00	Tk. 502.00
07.18.5	T-beam, L-beam and rectangular beam	sqm	Tk. 508.00	Tk. 507.00	Tk. 490.00	Tk. 490.00
07.18.6	Floor & roof slab	sqm	Tk. 561.00	Tk. 557.00	Tk. 552.00	Tk. 552.00
07.18.7	Cornice, railing, drop wall, louver, fins etc.	sqm	Tk. 459.00	Tk. 458.00	Tk. 445.00	Tk. 445.00
07.18.8	Sunshade, false ceiling	sqm	Tk. 471.00	Tk. 471.00	Tk. 459.00	Tk. 459.00
07.18.9	Stair case slab and steps	sqm	Tk. 457.00	Tk. 457.00	Tk. 445.00	Tk. 445.00

07.19	Supply and use of specified type chemical admixture delivered from an authorized local agent or manufacturer, complying with the ASTM C-494 requirements; conforming the current compliance of the admixture to specification requirements like physical properties, uniformity and equivalence in composition etc.; performance (water content, fresh concrete setting time and compressive strength) requirements, delivered in sealed water-tight containers confirming plainly marked product name & type under this specification, net weight and /or volume, manufacturing and expiry date, non aggressiveness to environment, aggregates and metals in concrete etc. and mixing the admixture in non pre-stressed cement concrete mixture in the field in accordance with manufacturers recommendation and instruction; providing safety provisions in all respects etc. all complete as per instruction and approved by the Engineer-in-charge. Dose (quantity in milli litres per 50 kg bag cement) and brand/origin/ manufacturer with respect to particular brand of cement and particular stock of aggregates and method of use to be determined by mix design / trial mix at the cost of contractor and approved by the Engineer. (Use of admixture is subjected to prior approval of design office)									
07.19.1	Supply and use of water reducing chemical admixture in concrete, complying ASTM C-494 Type – A of approved brand / origin / manufacturer and supplied by only manufacturer's authorised dealer with certificate of origin. The admixture is to reduce quantity of mixing water required for normal workability and to maintain low water-cement (W/C) ratio, for improved concrete quality assurance.	liter	Tk.	205.00	Tk.	204.00	Tk.	203.00	Tk.	203.00
07.19.2	Supply and use of retarding chemical admixture in concrete, complying ASTM C-494 Type - B of approved brand/origin/manufacturer and supplied by only manufacturer's authorised dealer with certificate of origin. The admixture for retarding setting time of concrete, there by reducing shrinkage in concrete specially for concrete laid in thick layers and intended delayed construction for improved construction joints, re-concreting etc.	liter	Tk.	209.00	Tk.	209.00	Tk.	209.00	Tk.	209.00
07.19.3	Supply and use of accelerating chemical admixture in concrete, complying ASTM C-494 Type - C of approved brand/origin/manufacturer and supplied by only manufacturer's authorised dealer with certificate of origin. An admixture that is to accelerate the setting time and early strength gain of concrete.	liter	Tk.	147.00	Tk.	147.00	Tk.	147.00	Tk.	147.00
07.19.4	Supply and use of water-reducing and retarding chemical admixture in concrete, complying ASTM C-494 Type - D of approved brand/origin/manufacturer and supplied by only manufacturer's authorised dealer with certificate of origin. The admixture to reduce the quantity of mixing water required to produce concrete of given consistency, i.e. low water-cement (W/C) ratio and intend to retard setting time of concrete, specially for layered, thick laid concrete etc.	liter	Tk.	270.00	Tk.	270.00	Tk.	270.00	Tk.	270.00
07.19.5	Supply and use of water-reducing and accelerating chemical admixture in concrete, complying ASTM C-494 Type - E of approved brand/origin/manufacturer and supplied by only manufacturer's authorised dealer with certificate of origin. The admixture to reduce the quantity of mixing water required to produce concrete of given consistency, i.e. low water-cement (W/C) ratio and intend to accelerate setting time of concrete, and early strength gain in concrete etc.	liter	Tk.	233.00	Tk.	233.00	Tk.	233.00	Tk.	233.00

07.19.6	Supply and use of water-reducing high range chemical admixture in concrete, complying ASTM C-494 Type - F of approved brand/origin/manufacture and supplied by only manufacturer's authorised dealer with certificate of origin. The admixture required to produce concrete of consistency by 12% or greater (flowing concrete) and for higher strength of concrete.	liter	Tk. 270.00	Tk. 270.00	Tk. 270.00	Tk. 270.00
07.19.7	Supply and use of water-reducing high range and retarding chemical admixture in concrete, complying ASTM C-494 Type - G of approved brand/origin/manufacture and supplied by only manufacturer's authorised dealer with certificate of origin. The admixture required to produce concrete of consistency by 12% or greater (flowing concrete) and for higher strength of concrete and intend to retard setting time of concrete.	liter	Tk. 202.00	Tk. 202.00	Tk. 202.00	Tk. 202.00
07.20	Supply and use of best quality pliable, cold applied pre-formed self adhesive rubberized/bituminous water proofing membrane of approved brand / origin / manufacture from an authorized local agent or manufacturer having minimum thickness 1.5 mm, lined with 2 ply cross laminated high density polyethylene film performances and properties all conforming to ASTM standards and laying, placing, fixing on made up firm and dry surfaces free from hollows, sharp protrusions, contaminates etc. laid over-lap by at least 75 mm along the edges and 125 mm at ends ensuring good initial bond with the surfaces all complete under the strict accordance with the manufacturer recommendations and instructions providing all safety provisions in all respects and accepted by the Engineer. (supplied by only manufacturer's authorised dealer with certificate of origin) under following specification:					
	Backing(HDPE) thickness minimum 0.10mm Membrane thickness minimum 1.40mm Total thickness minimum 1.50mm Cold flexibility greater than (-)200C: ASTM D146 Pliability: "No crack" tested in ASTM D146 Tensile strength (film only) min 3.50 MPa(35000kN/m2) testing in ASTM D882 Tensile strength (composite) at least 2.40 MPa or N/mm2 Elongation (composite) min 300% Cycling over crack at (-) 320C No effect 100 cycles ASTM C836-89a Puncture resistance greater than 240N ASTM E 154-88 Resistance to hydrostatic head at least 50m ASTM D5385 Tear resistance min 4.0 MPa (N/mm2) Adhesion to primed concrete min 4.0 N/mm : ASTM D1000 Water vapor transmission Less than 0.10g/m2 per hour : ASTM E96 Water absorption in 24 hour Less than 0.33% ASTM D570 Heat ste and curing at least for 28 days, removing centering-shutterin					

07.20.1	Water-proofing membrane on external surface of primed vertical or inclined wall /on plastered and primed back-wall and concrete cast against membrane. (Excluding cost of back fill/brick wall on back/plaster when membrane laid on back-wall and concrete cast against membrane which to be paid as per corresponding items in this schedule)	sqm	Tk. 1,084.69	Tk. 1,082.00	Tk. 1,069.00	Tk. 1,069.00
07.20.2	Water-proofing membrane on made up surface below the floor or below the horizontal surfaces with temporary protective cover using polystyrene/38 mm thick sand-cement mortar in 4:1 /or by equally effective alternative. (Rate is excluding the cost of protective cover which to be paid as per corresponding items in this schedule)	sqm	Tk. 894.81	Tk. 893.00	Tk. 886.00	Tk. 886.00
07.20.3	Water-proofing membrane on the floor or on the horizontal surfaces with permanent protective cover & wearing coarse. (Rate is excluding the cost of protective cover and wearing coarse which to be paid as per corresponding items in this schedule)	sqm	Tk. 930.39	Tk. 929.00	Tk. 921.00	Tk. 921.00
07.21	Supply of polymer based curing compound as per ASTM C 309 or BS 7542-1992 standard and apply by stirring & spraying with motorized sprayers over the surface of fresh concrete free from surface water and avoiding bleeding of concrete maintaining the spray nozzle approximately 450 mm away from the concrete surface with passed back and forth to ensure complete coverage confirming the pump pressure for fine spray and keeping the concrete surface undisturbed until it has sufficient strength to bear surface loads [The suitability of motorized sprayer should be ensured by an application trial] ensuring adequate ventilation to avoid inhalation of vapors and contact with skin and eyes all complete and accepted by Engineer-in-charge.	sqm	Tk. 218.00	Tk. 215.00	Tk. 206.00	Tk. 206.00
07.22	Supply of epoxy bonding agent following ASTM C881 / C881M - 14 or any international acceptable standard and apply over the old concrete surface after making free of standing water, removal of dust, laitance, grease, curing compounds, impregnations, waxes and any other contaminants to achieve optimum bonding in between fresh and old concrete and accepted by the Engineer-in-charge.	sqm	Tk. 1,088.00	Tk. 1,083.00	Tk. 1,068.00	Tk. 1,068.00
07.23	Supply of best quality salt guard coating on concrete surface following ASTM B117 delivered from authorized local agent of the manufacturer in a sealed container and apply by a firm experienced in applying salt guard coating or workers trained and approved by manufacturer to wall and ceiling in 2 coats elapsing specified time for drying/ recoating including cleaning and sand papering making the surface free from loose and efflorescence, flaky materials, dirt, grease, wax, polish, scraping all chalked and scaled materials, applying the paint by brush/spray following manufacturer's written instructions and necessary scaffolding etc. all complete and accepted by the Engineer-in-charge	sqm	Tk. 47.00	Tk. 46.00	Tk. 45.00	Tk. 45.00
7.24	Supplying, fitting and fixing of 0.45mm thick foreign made celulloid paper for fair face form works to add extra smoothness in surface applying in shutter top with liquid glue etc of celluloid paper and other materials, electricity, accessories, labour for installation, scaffolding, screws, nails etc. all complete as per design, approved and accepted by the engineer in charge.	sqm	Tk. 175.00	Tk. 174.00	Tk. 172.00	Tk. 172.00

..... END OF CHAPTER SEVEN						
CHAPTER 08: M.S. FABRICATION AND JOINTS IN RC WORKS						
Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
08.1	Supplying, fabrication and fixing to detail as per design : ribbed or deformed bar reinforcement (excluding laboratory test fees) for Reinforced concrete, produced and marked in accordance with BDS ISO 6935-2:2016 (or standard subsequently released from BSTI) including straightening and cleaning rust, if any, bending and binding in position with supply of G.I. wires, conducting necessary laboratory tests etc. (excluding splices or laps) complete in all respect and accepted by the Engineer-in-charge (Measurement shall be recorded only on standard mass per unit length of bars, while dia of bars exceeds its standard)					
08.1.1	Grade 300 (B300DWR: complying BDS ISO 6935-2:2016 / ASTM A615) ribbed or deformed bar produced and marked according to Bangladesh standard, with minimum yield strength, f_y (ReH)= 300 MPa but f_y not exceeding 350 MPa and whatever is the actual yield strength within allowable limit as per BNBC/ ACI 318, the ratio of ultimate tensile strength f_u to yield strength f_y , shall be at least 1.25 and minimum elongation after fracture and minimum total elongation at maximum force is 17% and 8% respectively : up to ground floor	kg	Tk. 91.17	Tk. 91.00	Tk. 90.00	Tk. 90.00
08.1.2	Grade 400 (B400DWR / B420DWR: complying BDS ISO 6935-2:2016 / ASTM A615) ribbed or deformed bar produced and marked according to Bangladesh standard, with minimum yield strength, f_y (ReH)= 400 MPa but f_y not exceeding 480 MPa and whatever is the actual yield strength within allowable limit as per BNBC/ ACI 318, the ratio of ultimate tensile strength f_u to yield strength f_y , shall be at least 1.25 and minimum elongation after fracture and minimum total elongation at maximum force is 17% and 8% respectively : up to ground floor.	kg	Tk. 99.84	Tk. 100.00	Tk. 99.00	Tk. 99.00
08.1.3	Grade 500 (B500DWR: complying BDS ISO 6935-2:2016 / ASTM A615) ribbed or deformed bar produced and marked according to Bangladesh standard, with minimum yield strength, f_y (ReH)= 500 MPa and whatever is the yield strength within allowable limit as per BNBC/ ACI 318, the ratio of ultimate tensile strength f_u to actual yield strength f_y , shall be at least 1.25 and minimum elongation after fracture and minimum total elongation at maximum force is 17% and 8% respectively : up to ground floor.(To be used for pile work, earth retaining wall, boundary wall, water reservoir & other structures as per instruction of concerned design CHAPTER)	kg	Tk. 94.89	Tk. 95.00	Tk. 94.00	Tk. 94.00
08.2	Added rate for M.S. Fabrication work in additional floor above ground floor	kg	Tk. 0.67	Tk. 0.66	Tk. 0.61	Tk. 0.61

08.3	Supplying and fixing of upset forging parallel thread rebar Couplers (excluding laboratory test fees) conforming to ACI-318 on "Building Code Requirements for Structural Concrete" and ISO: 15835 on "Steels for the reinforcement of concrete — Reinforcement couplers for mechanical splices of bars" to ribbed or deformed bars of Grade 400 (B400DWR / B420DWR : complying BDS ISO 6935-2:2016), in splicing main rebars of structural members satisfying the condition that splice shall develop at least 1.25 times greater of yield strength (fy) than that of the connecting rebars in tension and compression including threading, enlargement at connection by forging, protecting the prepared reinforcement bars and related operations as required to complete the works per direction of Engineer-in-Charge. Minimum 0.5% of the total coupled rebars are to be tested for quality assurance.					
08.3.1	Thread Coupler for 20 mm diameter Bar	each	Tk. 319.03	Tk. 319.00	Tk. 316.00	Tk. 316.00
08.3.2	Thread Coupler for 25 mm diameter Bar	each	Tk. 506.92	Tk. 506.00	Tk. 502.00	Tk. 502.00
08.3.3	Thread Coupler for 32 mm diameter Bar	each	Tk. 728.04	Tk. 727.00	Tk. 720.00	Tk. 720.00
08.4	Supplying and fixing of cold extrusion mechanical rebar Couplers (excluding laboratory test fees) conforming to ACI-318 on "Building Code Requirements for Structural Concrete" and ISO: 15835 on "Steels for the reinforcement of concrete — Reinforcement couplers for mechanical splices of bars" to ribbed or deformed bars of Grade 400 (B400DWR / B420DWR : conforming BDS ISO 6935-2:2016), in splicing main rebars of structural members satisfying the condition that splice shall develop at least 1.25 times greater of yield strength (fy) than that of the connecting rebars in tension and compression including related operations as required to complete the works as per direction of Engineer-in-Charge. Minimum 1.0% of the total coupled rebars are to be tested for quality assurance.					
08.4.1	Press Coupler for 16 mm diameter Bar (minimum Length: 100 mm, maximum Inner diameter: 21 mm, minimum Thickness: 4.5 mm)	each	Tk. 377.92	Tk. 377.00	Tk. 374.00	Tk. 374.00
08.4.2	Press Coupler for 20 mm diameter Bar (minimum Length: 120 mm, maximum Inner diameter: 25 mm, minimum Thickness: 5.5 mm)	each	Tk. 521.49	Tk. 520.00	Tk. 517.00	Tk. 517.00
08.4.3	Press Coupler for 22 mm diameter Bar (minimum Length: 132 mm, maximum Inner diameter: 28 mm, minimum Thickness: 6 mm)	each	Tk. 664.35	Tk. 663.00	Tk. 659.00	Tk. 659.00
08.4.4	Press Coupler for 25 mm diameter Bar (minimum Length: 150 mm, maximum Inner diameter: 31 mm, minimum Thickness: 7 mm)	each	Tk. 854.83	Tk. 853.00	Tk. 849.00	Tk. 849.00
08.4.5	Press Coupler for 32 mm diameter Bar (minimum Length: 192 mm, maximum Inner diameter: 38 mm, minimum Thickness: 9 mm)	each	Tk. 1,442.98	Tk. 1,441.00	Tk. 1,436.00	Tk. 1,436.00
08.5	installation of water stopper at movable and immovable RCC green joints as per following specification.					
08.5.1	Supplying, fitting and fixing ribbed PVC water stopper with non centre bulb of width approx. 250 mm and approx 9 mm thick for immovable (non working) RCC joints, having minimum tensile strength of 8.27MPa (approx. 1200 psi) min ultimate elongation 350%, approximate hardness 50, minimum tear resistance 0.0175 kN/mm (100 lbs/inch) at RCC construction green joints in basements, approved and in consultation with design office, providing 38 mm x 62 mm wooden batten at the top and bottom of water stopper, holding in position and keeping the water stopper in true horizontal or vertical by appropriate means as where necessary etc; all complete and accepted by the Engineer-in-charge.	meter	Tk. 1,021.89	Tk. 1,018.00	Tk. 994.00	Tk. 994.00

08.5.2	Supplying, fitting and fixing ribbed PVC water stopper with centre bulb of width approx. 250 mm and approx 9 mm thick for immovable (non working) RCC joints, having minimum tensile of strength 8.27 MPa (approx. 1200 psi) minimum ultimate elongation 350%, approximate hardness 50, minimum tear resistance 0.0175 kN/mm (100 lbs/inch) at RCC construction green joints in basements, approved and in consultation with design office, providing 38 mm x 62 mm wooden batten at the top and bottom of water stopper, holding in position and keeping the water stopper in true horizontal or vertical by appropriate means as where necessary etc; all complete and accepted by the Engineer-in-charge.	meter	Tk. 1,106.43	Tk. 1,103.00	Tk. 1,078.00	Tk. 1,078.00
08.6	Supplying, fitting and fixing 250 mm wide 16 SWG Steel Sheet water stopper in RCC construction joint holding in position and keeping the water stopper in true horizontal or vertical by appropriate means as where necessary etc; all complete and accepted by the Engineer-in-charge.	meter	Tk. 374.37	Tk. 374.00	Tk. 370.00	Tk. 370.00
08.7.1	Manufacturing, supplying, fitting and fixing Expansion Joint for gap upto 100mm with 125mmx50mmx6mm and 75mmx50mmx6mm galvanized MS angle and 8mm thick galvanized MS plate etc. all complete as per drawing, design and accepted by the Engineer-in-charge.	meter	Tk. 7,764.31	Tk. 7,757.00	Tk. 7,714.00	Tk. 7,714.00
08.7.2	Manufacturing, supplying, fitting and fixing Expansion Joint for gap more than 100mm up to 150mm with 150mmx50mmx6mm and 100mmx50mmx6mm galvanized MS angle and 8mm thick galvanized MS plate etc. all complete as per drawing, design and accepted by the Engineer-in-charge.	meter	Tk. 9,154.95	Tk. 9,147.00	Tk. 9,105.00	Tk. 9,105.00
08.7.3	Manufacturing, supplying, fitting and fixing Expansion Joint for gap more than 150mm up to 200mm with 175mmx50mmx6mm and 125mmx50mmx6mm galvanized MS angle and 8mm thick galvanized MS plate etc. all complete as per drawing, design and accepted by the Engineer-in-charge.	meter	Tk. 10,545.59	Tk. 10,538.00	Tk. 10,496.00	Tk. 10,496.00

..... END OF CHAPTER EIGHT

CHAPTER 09: PILE WORKS AND PILE TEST

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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09.1	Boring/drilling by wash or percussion method for cast in situ pile up to the required depth and diameter with minimum 6 m long temporary steel casing, true to vertical, providing bentonite slurry and maintaining water level in the hole, washing the hole for at least 30 minutes, cleaning the bore hole and making the bore hole ready for placing steel cage and concreting including hire charge of rig set with winch machine, tripod stand, tremie pipe, cost of fuel, lubricant, mobilization, demobilization, maintenance, spares, stand-byes, insurance coverage, water, electricity and other charges all complete, approved and accepted by the Engineer-in-charge. Before commencing boring operation, contractor shall submit the method statement of cast-in-situ pile work including sequence of boring and casting, disposal of spoils to the Engineer-in-charge for approval. However, Engineer's approval shall not relieve the contractor of his responsibilities and obligations under contract.					
09.1.1	400 mm dia	meter	Tk. 818.59	Tk. 813.00	Tk. 795.00	Tk. 795.00
09.1.2	450 mm dia	meter	Tk. 1,040.66	Tk. 1,034.00	Tk. 1,011.00	Tk. 1,011.00
09.1.3	500 mm dia	meter	Tk. 1,221.14	Tk. 1,213.00	Tk. 1,186.00	Tk. 1,186.00
09.1.4	550 mm dia	meter	Tk. 1,455.95	Tk. 1,446.00	Tk. 1,415.00	Tk. 1,415.00
09.1.5	600 mm dia	meter	Tk. 1,678.82	Tk. 1,668.00	Tk. 1,632.00	Tk. 1,632.00
09.2	Auger Boring/Rotary drilling for cast in situ pile up to the required depth and diameter with minimum 6 m long temporary steel casing, true to vertical, providing bentonite slurry and maintaining water level in the hole, washing the hole for at least 30 minutes, cleaning the bore-hole and making the bore- hole ready for placing steel cage and concreting including hire charge of rig set , tremie pipe, cost of fuel, lubricant, mobilization, demobilization, maintenance, spares, stand-byes, insurance coverage, bentonite, water, electricity and other charges etc. complete approved and accepted by the Engineer-in-charge. Before commencing boring operation, contractor shall submit the method statement of cast-in-situ pile work including sequence of boring and casting, disposal of spoils to the Engineer-in-charge for approval. However, Engineer's approval shall not relieve the contractor of his responsibilities and obligations under contract.					
09.2.1	600 mm dia	meter	Tk. 3,866.19	Tk. 3,861.00	Tk. 3,844.00	Tk. 3,844.00
09.2.2	750 mm dia	meter	Tk. 4,815.61	Tk. 4,809.00	Tk. 4,789.00	Tk. 4,789.00
09.2.3	1000 mm dia	meter	Tk. 6,984.87	Tk. 6,976.00	Tk. 6,947.00	Tk. 6,947.00
09.3	Removing of spoils / mud accumulated during boring for cast in situ pile by wash boring from working site to a safe distance by contractor's own arrangement i.e. with container set in truck or on cart including loading, unloading everything complete as per standard practice and accepted by the Engineer-in-charge. (Quantity should be given three times of solid volume of boring)	cum	Tk. 743.83	Tk. 742.00	Tk. 689.00	Tk. 689.00

09.4	Cast in situ pile with reinforced cement concrete works of high slump by adding high range water reducing admixture (ASTM C494 Type A or F) with minimum cement content relates to mix ratio 1 : 1.5 : 3 having minimum $f_{cr} = 26$ Mpa, and satisfying a specified compressive strength $f_c = 21$ Mpa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM & cement conforming to BDS EN-197-1- CEM -1,52.5 N / ASTM-C 150 Type – I, best quality coarse sand [Sylhet sand or coarse sand of equivalent F.M.2.2], 20 mm down well graded crushed stone chips conforming to ASTM C-33, including breaking chips, screening through proper sieves, making, placing re-bar cage in position, placing and removing tri-pod as per requirement, pouring the concrete in bore-hole with the help of a tremie pipe, maintaining the tremie pipe immersed in concrete by at least 1 meter throughout the period of concreting, maintaining required slump, etc. mixing the aggregates with standard mixer machine with hopper, casting in forme and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Doses of admixture to be fixed in consultation with design office) (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and th	cum	Tk. 15,715.24	Tk. 15,879.00	Tk. 14,909.00	Tk. 14,519.00
09.5	Providing and making point welding at contact point of the spiral binders at reasonable intervals with the main reinforcements by electric arc welding for construction of cast in situ bored pile carefully with highly oxidized electrodes, making the points prominent and accepted by the Engineer-in-charge (Rate is inclusive of all materials labour, tools and plants, electricity and all equipment).	point	Tk. 3.68	Tk. 4.00	Tk. 4.00	Tk. 4.00
09.6	Providing and making welded splice over two sides of contact by welding of minimum 300 mm length at the lap of main reinforcement in re-bar cage to be placed in bore-hole where necessary by electric arc welding with highly oxidized electrodes making the joint prominent all complete and accepted by the Engineer-in-charge. (Rate is inclusive of all materials labour, tools and plants, electricity and all equipment).	meter of weld	Tk. 483.08	Tk. 483.00	Tk. 483.00	Tk. 483.00
09.7	Labour for breaking head of hardened cast in situ bored pile/pre-cast pile up to a required length by any means but without damaging the rest and removing the dismantled materials such as concrete to a safe distance including scraps and cleaning concrete from steel/M.S. rods, straightening and bending of pile bars, preparation and making platform where necessary, carrying, all sorts of handling, stacking the same properly after clearing, leveling and dressing the situ and clearing the bed etc. complete in all respect and accepted by the Engineer-in-charge. (Measurement will be given for the actual pile head volume to be broken)	cum	Tk. 3,594.81	Tk. 3,518.00	Tk. 3,274.00	Tk. 3,274.00

09.8	Conducting static load test as per ASTM D1143 or equivalent standard for the cast-in-situ/pre-cast pile providing required scaffolding, bracing, jacks, pressure test gauge, loading unloading, Kentledge and other plants and equipment including staging, mobilization, demobilization, hire charge, gunny bags, sand and filling sacs/gunny bags for loading, record readings and preparation of results in standard forms and other incidental charges as per standard practice and procedures including submission of load test report, furnishing all graph and chart etc. complete in all respect approved and accepted by the Engineer-in-charge (Minimum two cyclic loading; one at service load and another cycle at double the service load then to continue loading till failure of the pile). Before commencing load test, contractor shall submit method statement for conducting load test to the Engineer-in-charge for approval. However, Engineer's approval shall not relieve the Contractor of his responsibilities and obligations under Coe and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Doses of admixture to be fixed in consultation with design					
09.8.1	Mobilization & de-mobilization of testing equipment	per set per site	Tk. 5,644.32	Tk. 5,644.00	Tk. 5,154.00	Tk. 5,154.00
09.8.2	Calibration of pressure gauge	per set per site	Tk. 6,625.95	Tk. 6,626.00	Tk. 6,626.00	Tk. 6,626.00
09.8.3	For 50 tonne load					
09.8.3.1	For first test at one site for one testing set	per test	Tk. 89,808.38	Tk. 89,565.00	Tk. 88,241.00	Tk. 88,241.00
09.8.3.2	For 2nd & additional tests at same site for same testing set	per test	Tk. 80,827.54	Tk. 80,608.00	Tk. 79,417.00	Tk. 79,417.00
09.8.4	For 100 tonne load					
09.8.4.1	For first test at one site for one testing set	per test	Tk. 149,800.01	Tk. 149,599.00	Tk. 147,786.00	Tk. 147,786.00
09.8.4.2	For 2nd & additional tests at same site for same testing set	per test	Tk. 134,820.01	Tk. 134,639.00	Tk. 133,007.00	Tk. 133,007.00
09.8.5	For 150 tonne load					
09.8.5.1	For first test at one site for one testing set	per test	Tk. 208,441.90	Tk. 208,292.00	Tk. 206,087.00	Tk. 206,087.00
09.8.5.2	For 2nd & additional tests at same site for same testing set	per test	Tk. 187,597.71	Tk. 187,463.00	Tk. 185,479.00	Tk. 185,479.00
09.8.6	For 200 tonne load					
09.8.6.1	For first test at one site for one testing set	per test	Tk. 269,169.74	Tk. 269,024.00	Tk. 266,292.00	Tk. 266,292.00
09.8.6.2	For 2nd & additional tests at same site for same testing set	per test	Tk. 242,252.77	Tk. 242,121.00	Tk. 239,663.00	Tk. 239,663.00
09.8.7	For 250 tonne load					
09.8.7.1	For first test at one site for one testing set	per test	Tk. 330,265.69	Tk. 330,171.00	Tk. 327,048.00	Tk. 327,048.00
09.8.7.2	For 2nd & additional tests at same site for same testing set	per test	Tk. 297,239.12	Tk. 297,154.00	Tk. 294,343.00	Tk. 294,343.00
09.8.8	For 300 tonne load					
09.8.8.1	For first test at one site for one testing set	per test	Tk. 390,993.54	Tk. 390,917.00	Tk. 387,267.00	Tk. 387,267.00
09.8.8.2	For 2nd & additional tests at same site for same testing set	per test	Tk. 351,894.18	Tk. 351,825.00	Tk. 348,541.00	Tk. 348,541.00
09.8.9	For 350 tonne load					
09.8.9.1	For first test at one site for one testing set	per test	Tk. 449,635.43	Tk. 449,610.00	Tk. 445,569.00	Tk. 445,569.00
09.8.9.2	For 2nd & additional tests at same site for same testing set	per test	Tk. 404,671.89	Tk. 404,649.00	Tk. 401,012.00	Tk. 401,012.00
09.8.10	For 400 tonne load					
09.8.10.1	For first test at one site for one testing set	per test	Tk. 508,277.33	Tk. 508,303.00	Tk. 503,871.00	Tk. 503,871.00
09.8.10.2	For 2nd & additional tests at same site for same testing set	per test	Tk. 457,449.59	Tk. 457,472.00	Tk. 453,484.00	Tk. 453,484.00

09.9	Rate for incremental loading					
09.9.1	For intermediate loading from 50 tonne to 100 tonne for each additional tonne					
09.9.1.1	For first test at one site for one testing set	tonne	Tk. 1,199.83	Tk. 1,201.00	Tk. 1,191.00	Tk. 1,191.00
09.9.1.2	For 2nd & additional tests at same site for same testing set	tonne	Tk. 1,079.85	Tk. 1,081.00	Tk. 1,072.00	Tk. 1,072.00
09.9.2	For intermediate loading from 100 tonne to 150 tonne for each additional tonne.					
09.9.2.1	For first test at one site for one testing set	tonne	Tk. 1,172.84	Tk. 1,174.00	Tk. 1,166.00	Tk. 1,166.00
09.9.2.2	For 2nd & additional tests at same site for same testing set	tonne	Tk. 1,055.55	Tk. 1,056.00	Tk. 1,049.00	Tk. 1,049.00
09.9.3	For intermediate loading from 150 tonne to 200 tonne for each additional tonne.					
09.9.3.1	For first test at one site for one testing set	tonne	Tk. 1,214.56	Tk. 1,215.00	Tk. 1,204.00	Tk. 1,204.00
09.9.3.2	For 2nd & additional tests at same site for same testing set	tonne	Tk. 1,093.10	Tk. 1,093.00	Tk. 1,084.00	Tk. 1,084.00
09.9.4	For intermediate loading from 200 tonne to 250 tonne for each additional tonne.					
09.9.4.1	For first test at one site for one testing set	tonne	Tk. 1,221.92	Tk. 1,223.00	Tk. 1,215.00	Tk. 1,215.00
09.9.4.2	For 2nd & additional tests at same site for same testing set	tonne	Tk. 1,099.73	Tk. 1,101.00	Tk. 1,094.00	Tk. 1,094.00
09.9.5	For intermediate loading from 250 tonne to 300 tonne for each additional tonne.					
09.9.5.1	For first test at one site for one testing set	tonne	Tk. 1,214.56	Tk. 1,215.00	Tk. 1,204.00	Tk. 1,204.00
09.9.5.2	For 2nd & additional tests at same site for same testing set	tonne	Tk. 1,093.10	Tk. 1,093.00	Tk. 1,084.00	Tk. 1,084.00
09.9.6	For intermediate loading from 300 tonne to 350 tonne for each additional tonne.					
09.9.6.1	For first test at one site for one testing set	tonne	Tk. 1,172.84	Tk. 1,174.00	Tk. 1,166.00	Tk. 1,166.00
09.9.6.2	For 2nd & additional tests at same site for same testing set	tonne	Tk. 1,055.55	Tk. 1,056.00	Tk. 1,049.00	Tk. 1,049.00
09.9.7	For intermediate loading from 350 tonne to 400 tonne for each additional tonne.					
09.9.7.1	For first test at one site for one testing set	tonne	Tk. 1,172.84	Tk. 1,174.00	Tk. 1,166.00	Tk. 1,166.00
09.9.7.2	For 2nd & additional tests at same site for same testing set	tonne	Tk. 1,055.55	Tk. 1,056.00	Tk. 1,049.00	Tk. 1,049.00
09.10	Pre-cast pile made in reinforced cement concrete with minimum cement content relates to mix ratio 1:1.25:2.5 having minimum f _{cr} = 38.5 Mpa, and satisfying specified compressive strength f _c = 30 Mpa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM & cement conforming to BDS EN-197-1- CEM-I (52.5 N) / ASTM-C 150 Type – I, best quality coarse sand (F.M.2.2), 20 mm down well graded crushed stone chips conforming to ASTM C-33, mixing in standard mixture machine and fed by standard measuring boxes, including all related works like screening through proper sieves, cleaning and washing, centering and placing reinforcement cages in position, casting, compacting by vibrators and tapered rods as where necessary, curing for 28 days etc. cost of water, electricity and other charges, providing fitting and fixing pile shoe in position, tools, plants & equipments, mobilization, demobilization, labour, test of materials and concrete etc. all complete as per design, drawing and accepted by ticular brand of cement and particular stock of aggregates and method of use to be determined by mix design / trial mix	cum	Tk. 14,286.58	Tk. 14,436.00	Tk. 13,554.00	Tk. 13,199.00
09.11	Cost of bed preparation including one layer brick flat soling with first class/picked jhama bricks including preparation of bed and filling the interstices with local sand, leveling etc. complete and accepted by the Engineer-in-charge, Minimum 12 mm thick cement sand (F.M. 1.2) plaster with neat cement finishing with cement (1:4) including washing of sand, finishing the edges and corners and curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect and accepted by the Engineer-in-charge (Cement: CEM-II/B-M) and polythene as separator between pile layers during casting concrete	sqm	Tk. 320.22	Tk. 324.00	Tk. 324.00	Tk. 324.00

09.12	Centering and shuttering, including strutting, propping etc. (The formwork must be rigid enough both in and out of plane, to make the concrete surface true to the designed shape and size by using necessary MS sheets of minimum 18 BWG, angles of minimum size 1.5 x 1.5 x 3/16" for sides, angles of minimum size 1 x 1 x 3/16" for frames, 1.5" x 3/16" flat bars etc.) and removal of form for precast pile (The formwork is considered for two sides and front shuttering of the pile)	sqm	Tk. 404.51	Tk. 409.00	Tk. 398.00	Tk. 398.00
09.13	Pre-cast pile shoe making & supplying pre-fabricated frustum of square cone having four sides and a toe made of 6 mm thick ASTM A36 complied MS steel plate jointed by line welding, 4 nos 12 mm dia deformed MS bar jointed by line welding with MS plate along the four edges of shoe aligning the corner main bar of the pile and 1-12mm dia bar to fix with and perpendicular to shoe toe plate aligning centre of pile, including cutting, shaping, sizing of MS plate and bar, all complete as per design & drawing and accepted by the Engineer-in-charge	each	Tk. 3,214.81	Tk. 3,187.00	Tk. 3,064.00	Tk. 3,064.00
09.14	Mobilization and demobilization of drop hammer type pre-cast pile driving rig.	per set per site	Tk. 11,963.51	Tk. 11,951.00	Tk. 10,921.00	Tk. 10,921.00
09.15	Driving 250 mm x 250 mm size pre-cast pile with drop hammer type rig, and maintaining driving log in prescribed format Before commencing driving operation, contractor shall submit method statement for carrying out the driving operation including sequence of driving to the Engineer-in-charge for approval. However, Engineer's approval shall not relieve the contractor of his responsibilities and obligations under contract.	meter	Tk. 513.78	Tk. 508.00	Tk. 489.00	Tk. 489.00
09.16	Driving 300 mm x 300 mm to 350 mm x 350 mm size pre-cast pile with drop hammer type rig, and maintaining driving log in prescribed format Before commencing driving operation, contractor shall submit method statement for carrying out the driving operation including sequence of driving to the Engineer-in-charge for approval. However, Engineer's approval shall not relieve the contractor of his responsibilities and obligations under contract.	meter	Tk. 817.37	Tk. 809.00	Tk. 778.00	Tk. 778.00
09.17	Mobilization and demobilization of automatic diesel hammer mounted complete rig set.	per set per site	Tk. 186,508.09	Tk. 186,508.00	Tk. 186,508.00	Tk. 186,508.00
09.18	Driving 250 mm x 250 mm size pre-cast pile by rig fitted with automatic diesel operated hammer. Before commencing driving operation by rig fitted with automatic diesel operated hammer, contractor shall submit method statement for carrying out the operation including sequence of driving to the Engineer-in-charge for approval. However, engineer's approval shall not relieve the contractor of his responsibilities and obligations under contract.	meter	Tk. 706.14	Tk. 702.00	Tk. 691.00	Tk. 691.00
09.19	Driving 300 mm x 300 mm & 350 mm x 350 mm size pre-cast pile by rig fitted with automatic diesel operated hammer. Before commencing driving operation by rig fitted with automatic diesel operated hammer, contractor shall submit method statement for carrying out the operation including sequence of driving to the Engineer-in-charge for approval. However, engineer's approval shall not relieve the contractor of his responsibilities and obligations under contract.	meter	Tk. 914.00	Tk. 909.00	Tk. 894.00	Tk. 894.00

09.20	Driving 400 mm x 400 mm size pre-cast pile by rig fitted with automatic diesel operated hammer. Before commencing driving operation by rig fitted with automatic diesel operated hammer, contractor shall submit method statement for carrying out the operation including sequence of driving to the Engineer-in-charge for approval. However, engineer's approval shall not relieve the contractor of his responsibilities and obligations under contract.	meter	Tk. 1,035.71	Tk. 1,030.00	Tk. 1,013.00	Tk. 1,013.00
09.21	Driving 450 mm x 450 mm size pre-cast pile by rig fitted with automatic diesel operated hammer. Before commencing driving operation by rig fitted with automatic diesel operated hammer, contractor shall submit method statement for carrying out the operation including sequence of driving to the Engineer-in-charge for approval. However, engineer's approval shall not relieve the contractor of his responsibilities and obligations under contract.	meter	Tk. 1,109.73	Tk. 1,104.00	Tk. 1,085.00	Tk. 1,085.00
09.22	Pre-cast driven micro-pile with reinforced cement concrete works with minimum cement content relates to mix ratio 1:1.25:2.5 having minimum $f_{cr} = 35$ MPa, and satisfying a specified compressive strength $f_c = 30$ MPa at 28 days on standard cylinders as per standard practice of Code ACI/BNBC/ASTM & cement conforming to BDS EN-197-1- CEM-I (32.5 to 52.5 N) / ASTM-C 150 Type – I, best quality coarse sand (F.M.2.5) and 20 mm down well-graded crushed stone chips conforming to ASTM C-33, mixing in standard mixture machine with hopper and fed by standard measuring boxes, including cost of breaking chips, screening through proper sieves, cleaning and washing, centering and shuttering with M.S sheet, M.S angle, F.I bar, nuts and bolts, preparation of bed, laying polythene, placing of reinforcement cage, casting, compacting by vibrators and tapered rods, curing for 28 days etc. including cost of water, electricity and other charges as per standard practice or specified method providing head cushion, all equipments including brand of cement and particular stock of aggregates and method of use to be determined by mix design / trial mix at the cost of contractor and approved by the Engineer. (Use of admixture is subjected to prior approval of design office)hine and curing at least for 28 days, removing c					
09.22.1	Casting of 180 mm x 180 mm x 7.6 m Pile with reinforced cement concrete (1:1.25:2.5) with stone chips	meter	Tk. 596.38	Tk. 600.00	Tk. 570.00	Tk. 559.00
09.22.2	Casting of 205 mm x 205 mm x 7.6 m Pile with reinforced cement concrete (1:1.25:2.5) with stone chips	meter	Tk. 745.06	Tk. 750.00	Tk. 711.00	Tk. 697.00
09.22.3	Driving of 180 mm x 180 mm x 7.6 m pre-cast pile	meter	Tk. 217.40	Tk. 215.00	Tk. 207.00	Tk. 207.00
09.22.4	Driving of 205 mm x 205 mm x 7.6 m pre-cast pile	meter	Tk. 261.11	Tk. 258.00	Tk. 248.00	Tk. 248.00

09.22.5	Supply of Cold-Drawn Wire Pre-stressed Concrete (CWPC) micro-pile which is made by using cold drawn wire minimum cement content relates to mix ratio 1:1.5:3 cement conforming to BDS EN-197-1-CEM-1, 52.5N (52.5MPa) / ASTM-C 150 type - 1, having minimum $f_{cr} = 35$ MPa and satisfying specified compressive strength $f_c = 30$ MPa at 28 days best quality coarse sand (F.M. 2.2), 20 mm down well graded stone chips conforming to ASTM C-33, mixing in standard mixture machine and fed by standard measuring boxes, including cost of all kind of reinforcement BDS ISO 6935-2:2006 including strengthening and cleaning rust if any bending and binding in position with supply of GI wires (24 BWG), centering and shuttering with M.S. sheet, M.S. angle, F.I. bar, nuts and bolts, preparation of bed, supplying & tensioning of cold drawn wire, laying polythene, placing of reinforcement cage in position, casting, compacting by vibrator and tapered rods, curing at least 28 days etc. including cost of water, electricity and other charges, pre and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Doses of admixture to be fixed in consultation with design office) (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost	meter	Tk. 1,265.93	Tk. 1,257.00	Tk. 1,216.00	Tk. 1,210.00
09.23	PIT (Pile integrity test) : Conducting low strain dynamic testing of pile as per ASTM D5882-The standard method for low-strain integrity testing of pile by a qualified and approved PIT expert, preparing pile heads and providing all relevant materials, equipment, experts, labour, mobilization, demobilization, data acquisition and analysis, preparation of results in standard forms and compiling final report with recommendations on the tests etc. complete in all respect approved and accepted by the Engineer-in-charge. (Report shall contain driving log / boring and pouring logs, graphs & statement on "Calibration Pile" (pile that tested for its capacity and PIT done earlier on it, at a place of known soil profiles), soil investigation report. (Calibration pile shall be made and tested first). Method statement for conducting PIT shall be submitted to the Engineer for approval, and shall be the part of PIT report, However, Engineer's approval shall not relieve the contractor of his responsibilities and obligations of c					
09.23.1	Mobilization and demobilization to site and PIT on 12(twelve) piles or less	upto 1st 12 piles	Tk. 34,356.75	Tk. 34,357.00	Tk. 34,357.00	Tk. 34,357.00
09.23.2	For PIT test of each additional pile after 12(Twelve) pile tested at same site	per pile	Tk. 1,472.43	Tk. 1,472.00	Tk. 1,472.00	Tk. 1,472.00

09.24	Pre-cast spliced pile jointing by supplying, fitting, fixing prefabricated square shaped two steel caps made up by 6 mm thick ASTM A36 complied M.S plate, sides formed by bent in channel shape to have only two line of joinery on faces, having min 50 mm dia hole on its web; capping by min 12 mm thick ASTM A36 complied M.S plate shaping the edges in 45°, having min 40 mm dia hole at centre to pass 25 mm M.S bar 433 mm in length; jointed at channels root by welding; caps fitted with 2 nos BDS ISO 6935-2009: B400 DWR complied d20 U-bar 1244mm in length by welding, equidistant from centre holes of cap and shall have min 75 mm clear from pile faces; and finally, the system to be fitted in correct alignments on pile head positions waiting for concreting; both the pile heads fitted with the steel caps to be welded together through v-notches in field, while one of the companion segment of piles driven in ground and other one placed over it maintaining true verticality; all complete as per direction and stepwise appoe and curing at least for 28 days, removing centering					
09.24.1	X-sec of pre cast pile segments :300 mm x 300 mm	each	Tk. 10,599.12	Tk. 10,581.00	Tk. 10,479.00	Tk. 10,466.00
09.24.2	X-sec of pre cast pile segments :350 mm x 350 mm	each	Tk. 12,020.60	Tk. 12,002.00	Tk. 11,892.00	Tk. 11,880.00
09.24.3	X-sec of pre cast pile segments :400 mm x 400 mm	each	Tk. 13,594.92	Tk. 13,577.00	Tk. 13,459.00	Tk. 13,447.00
09.24.4	X-sec of pre cast pile segments :450 mm x 450 mm	each	Tk. 13,950.85	Tk. 13,934.00	Tk. 13,819.00	Tk. 13,807.00
09.25	Pile Driving by Hydraulic Static Pile Driver					
09.25.1	Mobilization and demobilization of hydraulic static pile driver(Upto 320 metric ton capacity)	per set per site	Tk. 559,524.28	Tk. 559,524.00	Tk. 559,524.00	Tk. 559,524.00
09.25.2	Mobilization and demobilization of hydraulic static pile driver(321 Metric Ton to 500 Metric Ton Capacity)	per set per site	Tk. 671,429.14	Tk. 671,429.00	Tk. 671,429.00	Tk. 671,429.00
09.25.2.1	Driving pre-cast pile of 300 mm x 300 mm to 350 mm x 350 mm X-section made with shoe/ without shoe by Hydraulic static pile driver of upto 320 tonne (metric ton) capacity; arranged with necessary cranes, power source, maintaining driving log and record of pressure exerted on piles driven in to desired RL or capacity level ; complete in all respect including handling of piles, dressing & leveling of site, way making, surveying, layout setting etc ended with the submission of pile driving report in three copies duly signed by professional engineer and approved by the Engineer-in-charge.	meter	Tk. 886.07	Tk. 885.00	Tk. 883.00	Tk. 883.00
09.25.2.2	Driving pre-cast pile of 400 mm x 400 mm X-section made with shoe/ without shoe by hydraulic static pile driver of approx. 321 tonne to 500 tonne (metric ton) capacity; arranged with necessary cranes, power source, maintaining driving log and record of pressure exerted on piles driven in to desired RL or capacity level ; complete in all respect including handling of piles, dressing & leveling of site, way making, surveying, layout setting etc ended with the submission of pile driving report in three copies duly signed by professional engineer and approved by the Engineer-in-charge.	meter	Tk. 996.56	Tk. 996.00	Tk. 993.00	Tk. 993.00
09.25.2.3	Driving pre-cast pile of 450 mm x 450 mm X-section made with shoe/ without shoe by hydraulic static pile driver of approx. 500 tonne (metric ton) capacity; arranged with necessary cranes, power source, maintaining driving log and record of pressure exerted on piles driven in to desired RL or capacity level ; complete in all respect including handling of piles, dressing & leveling of site, way making, surveying, layout setting etc ended with the submission of pile driving report in three copies duly signed by professional engineer and approved by the Engineer-in-charge.	meter	Tk. 1,043.38	Tk. 1,042.00	Tk. 1,039.00	Tk. 1,039.00

09.26	Mobilization and demobilization of Crane (crawler) and crane mounted Vibratory Hammer complete rig set.	per set per site	Tk. 149,206.48	Tk. 149,206.00	Tk. 149,206.00	Tk. 149,206.00
09.27	Supplying U-shape hot rolled steel sheet pile of specified sections having chemical composition Phosphorus = 0.04% (Maximum), Sulphur = 0.04% (Maximum), and mechanical properties Tensile strength=> 490 N/mm ² , Yield strength => 296 N/mm ² , Elongation =15% (Minimum) (according to ASTM Standards), as specified by appropriate means to the construction site and stacking at proper place including loading, unloading, local handling as per specifications, including all taxes, freights, incidental charges etc. complete as per direction of the Engineer -in-charge U-shape, hot- rolled steel sheet pile width = 400mm to 600mm: height => 100mm, length = 12.0m, Thickness >= 10.5mm, Thickness >= 13.0mm (±0.50mm), Thickness >= 15.5mm (±0.50mm), Thickness >= 17.0mm (±0.50mm) This rate excludes the cost of bed preparation. Sheet piles have to be unloaded on a firm flat base. (Salvage materials shall be the contractors property after completion of the work, rate is excluding the cost of driving and taking out).	kg	Tk. 52.50	Tk. 52.00	Tk. 52.00	Tk. 52.00
09.28	Cutting (if necessary) charge of U-type Hot rolled steel sheet piles to design length and shape as per design and specification (thickness, length and mechanical properties) and instruction by all complete as per direction of the Engineer-in-charge.	Each	Tk. 372.00	Tk. 372.00	Tk. 358.00	Tk. 358.00
09.29	Jointing steel sheet piles (if necessary) of different thickness by welding along width (both sides) and along collars (outside) (i.e. at 4 points) to increase the length of pile as per requirement including necessary modification of the ends to receive the weld, supply of welding materials, equipments and other accessories as per specification and direction of Engineer in charge.	Each	Tk. 495.00	Tk. 493.00	Tk. 476.00	Tk. 476.00
09.30	Fabrication and construction of corner Joints in steel sheet piles including cutting, welding, shaping as per design and specification and as per requirement including supply of necessary welding materials, equipments and other accessories as per specification and direction of Engineer in charge.	Each	Tk. 225.00	Tk. 225.00	Tk. 221.00	Tk. 221.00
09.31	Driving steel sheet piles					
09.31.1	Driving steel sheet piles (up to 6.0m depth) of various sections and weights in any type of soil by mechanical device (Crawler Crane having 50 to 55 Ton capacity and Vibratory Hammer etc.) including handling and placing in position, supply of all materials and equipments related with driving of sheet piles, correcting leaning beyond tolerance and other defects etc. all complete as per direction of Engineer in charge (measurement will be taken on projected width x height).	sqm	Tk. 1,108.93	Tk. 1,115.00	Tk. 1,109.00	Tk. 1,109.00
09.31.2	Driving steel sheet piles (beyond 6.0m up to 9.0m depth) of various sections and weights in any type of soil by mechanical device (Crawler Crane having 50 to 55 Ton capacity and Vibratory Hammer etc.) including handling and placing in position, supply of all materials and equipments related with driving of sheet piles, correcting leaning beyond tolerance and other defects etc. all complete as per direction of Engineer in charge (measurement will be taken on projected width x height).	sqm	Tk. 1,023.62	Tk. 1,029.00	Tk. 1,024.00	Tk. 1,024.00

09.31.3	Driving steel sheet piles (beyond 9.0m up to 12.0m depth) of various sections and weights in any type of soil by mechanical device (Crawler Crane having 50 to 55 Ton capacity and Vibratory Hammer etc.) including handling and placing in position, supply of all materials and equipments related with driving of sheet piles, correcting leaning beyond tolerance and other defects etc. all complete as per direction of Engineer in charge (measurement will be taken on projected width x height).	sqm	Tk. 998.03	Tk. 1,003.00	Tk. 999.00	Tk. 999.00
09.32	Lifting steel sheet piles					
09.32.1	Lifting steel sheet piles (up to 6.0m depth) of any section and weights from any type of soil by mechanical device including supply of all equipments etc., handling and placing in stacks etc. all complete as per direction of Engineer in charge (measurement will be taken on projected width x height).	sqm	Tk. 909.02	Tk. 917.00	Tk. 910.00	Tk. 910.00
09.32.2	Lifting steel sheet piles (beyond 6.0m up to 9.0m depth) of any section and weights from any type of soil by mechanical device including supply of all equipments etc., handling and placing in stacks etc. all complete as per direction of Engineer in charge (measurement will be taken on projected width x height).	sqm	Tk. 942.69	Tk. 951.00	Tk. 943.00	Tk. 943.00
09.32.3	Lifting steel sheet piles (beyond 9.0m up to 12.0m depth) of any section and weights from any type of soil by mechanical device including supply of all equipments etc., handling and placing in stacks etc. all complete as per direction of Engineer in charge (measurement will be taken on projected width x height).	sqm	Tk. 1,060.53	Tk. 1,070.00	Tk. 1,061.00	Tk. 1,061.00
09.33	Mobilization & demobilization of joist driving rig	per set per site	Tk. 9,472.65	Tk. 9,473.00	Tk. 9,473.00	Tk. 9,473.00
09.34	Driving joist (up to 6.5m depth) of any sections and weights in any type of soil by mechanical device including handling and placing in position, supply of all materials and equipments related with driving of sheet piles, correcting leaning beyond tolerance and other defects etc. all complete as per direction of Engineer in charge (measurement will be taken on meter).	meter	Tk. 400.49	Tk. 396.00	Tk. 383.00	Tk. 383.00

..... END OF CHAPTER NINE

CHAPTER 10: STRUCTURAL STEEL WORKS, SHEET ROOFING AND SAFETY CANOPY

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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10.1	Supply and fixing of galvanized anchor bolts of variable dia for rigid frame conforming to ASTM F1554 Grade 55, Galvanized to A153, Class C or equivalent with minimum yield strength of 380 MPa, manual of steel construction by American Institute of Steel Construction (AISC) etc. including the cost of washer & bolts, material testing etc. all complete as per drawing, specification and direction of the Engineer-in-charge.	kg	Tk. 192.00	Tk. 192.00	Tk. 192.00	Tk. 192.00
10.2	Supply, fabrication and installation of built-up sections i.e. columns, beams, rafters, bracings, struts etc. (up to level-4) from steel plates conforming to ASTM A572, with a minimum yield strength of 345 MPa , including the cost of testing of plates, application of red/grey-oxide primer etc. all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 137.00	Tk. 137.00	Tk. 137.00	Tk. 137.00
10.3	Supply, fabrication and installation of built-up sections i.e. columns, beams, rafters, bracings, struts etc (up to level-4) from steel plates conforming to ASTM A36, with a minimum yield strength of 250 MPa , including the cost of testing of plates, application of red/grey-oxide primer etc. all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 134.00	Tk. 134.00	Tk. 134.00	Tk. 134.00
10.4	Supply, fabrication and installation of hot-rolled sections (W, H, I-shape etc.) for columns, beams, rafters, bracings etc. (up to level-4) conforming to ASTM A36, with a minimum yield strength of 250 MPa , including the cost of testing of plates, application of red/grey-oxide primer etc. all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 131.00	Tk. 131.00	Tk. 131.00	Tk. 131.00
10.5	Supply, fabrication and installation of hot rolled structural grade steel pipe and tube of any thickness conforming to ASTM A283 with a minimum yield strength of 275 MPa, including the cost of testing, application of red/grey-oxide primer etc. (up to level-4) all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 200.00	Tk. 200.00	Tk. 200.00	Tk. 200.00
10.6	Supply, fabrication and installation of hot rolled structural grade steel channel and angle in roof truss or any form of structure of any thickness conforming to ASTM A36 with a minimum yield strength of 250 MPa, including the cost of testing, application of red/grey-oxide primer etc. (up to level-4) all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 129.00	Tk. 129.00	Tk. 129.00	Tk. 129.00
10.7	Supply and fabrication of built-up sections for shore piles from steel plates conforming to ASTM A36, with a minimum yield strength of 250 MPa , including the cost of testing of plates, application of red/grey-oxide primer etc. all complete as per drawing, specification and direction of Engineer-in-charge (Salvage materials shall be the contractors property after completion of the work, rate is excluding the cost of driving and taking out).	kg	Tk. 68.00	Tk. 68.00	Tk. 68.00	Tk. 68.00

10.8	Supply and fabrication of hot rolled structural grade steel Sections of any thickness conforming to ASTM A36 with a minimum yield strength of 250 MPa for shore piling , including the cost of testing , application of red/grey-oxide primer etc. all complete as per drawing, specification and direction of Engineer-in-charge(Salvage materials shall be the contractors property after completion of work, rate is excluding the cost of driving and taking out).	kg	Tk. 53.00	Tk. 53.00	Tk. 53.00	Tk. 53.00
10.9	Supply ,fabrication, installation and de-installation of built-up sections for shore pile bracing at any level basement work from steel plates conforming to ASTM A36, with a minimum yield strength of 250 MPa , including the cost of testing of plates, application of red/grey-oxide primer etc. all complete as per drawing, specification and direction of Engineer-in-charge(Salvage materials shall be the contractors property after completion of work).	kg	Tk. 88.00	Tk. 88.00	Tk. 88.00	Tk. 88.00
10.10	Supply ,fabrication, installation and de-installation of built-up sections for shore pile bracing at any level basement work from steel plates conforming to ASTM A572, with a minimum yield strength of 345 MPa , including the cost of testing of plates, application of red/grey-oxide primer etc. all complete as per drawing, specification and direction of Engineer-in-charge(Salvage materials shall be the contractors property after completion of work).	kg	Tk. 89.00	Tk. 89.00	Tk. 89.00	Tk. 89.00
10.11	Supply and fixing of connection bolts of variable diameter with nut and washer according to " ASTM A325 Type 1 or equivalent" with $F_u = 720$ Mpa, including the cost of testing of bolts, all complete as per drawing, specification and direction of Engineer-In-Charge.	kg	Tk. 361.00	Tk. 361.00	Tk. 361.00	Tk. 361.00
10.12	Supply and fixing of connection bolts of variable diameter with nut and washer according to " ASTM A307 Type A or equivalent" with $F_u = 414$ Mpa, including the cost of testing of bolts, all complete as per drawing, specification and direction of Engineer-In-Charge.	kg	Tk. 324.00	Tk. 324.00	Tk. 324.00	Tk. 324.00
10.13	Supply and installation of GI purlin & girt of any size (up to level-4) conforming to "ASTM A653" grade 45, with a minimum yield strength of 310 MPa, including the cost of testing of materials, all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 164.00	Tk. 163.00	Tk. 164.00	Tk. 164.00
10.14	Supply and fixing of sag rod (up to level-4) conforming to grade 40, with a minimum yield strength of 275 MPa, including the cost of making threads, nuts and washers, red oxide primer etc. all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 139.00	Tk. 139.00	Tk. 139.00	Tk. 139.00
10.15	Supply and fixing of cablw/wire bracing conforming to "ASTM A475" with a minimum yield strength of 1193 MPa, with I-bolt and hill side washer, all complete as per drawing, specification and direction of Engineer-in-charge.	meter	Tk. 223.00	Tk. 223.00	Tk. 223.00	Tk. 223.00
10.16	Supplying and fixing of shear stud of variable dia conforming to ASTM A108 or equivalent with minimum yield strength of 275 MPa, including the cost testing, welding etc. all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 169.00	Tk. 169.00	Tk. 169.00	Tk. 169.00

10.17	Supply and installation of 0.70 mm thick galvanized decking panel according to ASTM A653, with minimum yield strength of 345 MPa, including all necessary accessories (sheeting screws with washers, sealants etc.), testing of materials etc. all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 186.00	Tk. 186.00	Tk. 186.00	Tk. 186.00
10.18	Supply, fabrication and installation of 6 mm thick Chequer Plate conforming to ASTM A572, with minimum yield strength of 345 MPa (up to level-4), including all necessary accessories ,testing of materials etc. all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 196.00	Tk. 196.00	Tk. 196.00	Tk. 196.00
10.19	Supply and installation of 0.47 mm thick zinc alum color coated profile sheet from local source/China with minimum warranty period of 5 years after installation for roof, wall, capping, flashing, trimming, gutter etc. (up to level-4) conforming to ASTM A792, Grade: 80, AZ150 gm/m ² , including the cost of necessary accessories (sheeting screws with washers, silicon sealant etc.), testing of materials etc. all complete as per drawing specification and direction of Engineer-in-charge.	kg	Tk. 200.00	Tk. 200.00	Tk. 200.00	Tk. 200.00
10.20	Supply and installation of 0.47 mm thick zinc alum color coated profile sheet of durability class 2 with minimum warranty period of 10 years of durability, for roof, wall, capping, flashing, trimming, gutter etc. conforming to ASTM A792, Grade: 80, AZ150 gm/m ² , including the cost of necessary accessories (sheeting screws with washers, silicon sealant etc.), testing of materials etc. all complete as per drawing specification and direction of Engineer-in-charge (supplier has to produce papers to proof the source of material).	kg	Tk. 322.00	Tk. 322.00	Tk. 322.00	Tk. 322.00
10.21	Supply and installation of 0.457 mm thick corrugated galvanized iron sheet (Bangladesh made) having min weight 63-65 kg per bundle (2'-6" width, 70 – 72 rft long) fitted and fixed on M.S. sections with 'J' hook or wooden purlin with screws, limpet washers and putty etc. (up to level-4) all complete and accepted by the Engineer-in-charge.	sqm	Tk. 578.00	Tk. 577.00	Tk. 569.00	Tk. 569.00
10.22	0.27 mm thick galvanized iron plain sheet ridging with 300 mm lap on either side fitted and fixed with galvanized bolts and nuts etc. all complete and accepted by the Engineer-in-charge.					
10.22.1	Galvanized	meter	Tk. 472.00	Tk. 471.00	Tk. 459.00	Tk. 459.00
10.22.2	Colored	meter	Tk. 523.00	Tk. 522.00	Tk. 510.00	Tk. 510.00
10.23	Supply and installation of 2.0 mm thick fiber glass sheet (transparent sheet) for roof, wall etc. on M.S. purlin, angle etc. with 'J' hook or screws with washer and putty, all complete as per drawing, specification and direction of Engineer-in-charge (up to level-4).	sqm	Tk. 1,437.00	Tk. 1,436.00	Tk. 1,428.00	Tk. 1,428.00
10.24	1.5 mm thick plain fiberglass sheet ridging with 300 mm lap on either side fitted and fixed with galvanized bolts and nuts etc. all complete and accepted by the Engineer-in-charge.	meter	Tk. 875.00	Tk. 874.00	Tk. 862.00	Tk. 862.00

10.25	Supply and installation of double bubble foil insulation at wall and roof of nominal thickness of 8 mm conforming to Class A in accordance with the ASTM-E84-09 fire test standard, including the cost of lapping, costape, adhesive and necessary accessories, all complete as per drawing, specification and direction of Engineer-in-charge.	sqm	Tk. 251.00	Tk. 251.00	Tk. 251.00	Tk. 251.00
10.26	Premium Synthetic Enamel paint of approved best quality and colour delivered from authorized local agent of the manufacturer(Berger robbialac supergloss synthetic enamel/Elite lucky seven/ Asian apcolipe premium paint or equivalent brand) in a sealed container, having highly water resistant, high bondibilty, flexiblity property; using specified brand thinner applying to metallic surface by spray in two coats over single coat anti-corrosive coating including cleaning drying, making free from dirt, grease, wax, removing all chalked and scaled materials, all complete in all floors accepted by the Engineer-in-charge. (payment shall be made on strucutral steel weight basis)	kg	Tk. 9.00	Tk. 9.00	Tk. 9.00	Tk. 9.00
10.27	Supply and application of Epoxy based corrosion protection paint to the surface of the structural steel members conforming to SA 2.5; the corrosion class shall be -C3 in accordance with BS EN ISO 12944-2 and durability class in accordance with BS EN ISO 12944-5; the Steel members to be shot blasted inside the enclosed shot blasting chamber, final coat paint must be applied on site after installation, including the cost of primer, testing and necessaey accessories, all complete as per drawing, specification and direction of Engineer-in-charge.	kg	Tk. 26.00	Tk. 26.00	Tk. 26.00	Tk. 26.00
10.28	Supply and pouring of non-shrink grout to steel column bases conforming to ASTM C 109 having compressive strength not less than 45 MPa, including the cost of shutter and necessary accessories, all complete as per drawing, specification and direction of Engineer-in-charge.	cum	Tk. 147,568.00	Tk. 147,568.00	Tk. 147,568.00	Tk. 147,568.00
10.29	Saftey Canopy: Supplying temporary safety canopy around construction work place where public safety is likely to be endangered due to construction activities; which shall be made of truss system of steel sections (main frame) at 1800 mm c/c with purlins @ 750 mm c/c, making flooring system by corrugated galvanized iron sheets of thickness 0.45 mm, laying wire mesh net on iron sheets, providing continuous gutter along the edges of the building with downpipe @ 6000 mm c/c, including fitting and fixing in position providing necessary anchors, cables, wires, ties etc. by standard anchoring and welding, nut-bolts etc, all complete and accepted by the Engineer-in-charge	sqm	Tk. 2,421.00	Tk. 2,416.00	Tk. 2,398.00	Tk. 2,398.00
10.30	Safety net with G.I Wire: Supply, fabrication, installation and taking out temporary horizontal safety net around construction work place where public safety is likely to be endangered due to construction activities; which shall be made of Chain link net with 24 BWG wire fitted at 15'-0" height horizontally of the outer periphery of the building with 75mmx40mmx5mm MS Chanel post and purlin, 20mm MS Rod and 40mmx40mmx6mm MS Angle as ties etc. by standard anchoring and welding, nut-bolts etc, all complete and accepted by the Engineer-in-charge(Salvage materials shall be the contractors property after completion of work.)	sqm	Tk. 1,515.00	Tk. 1,502.00	Tk. 1,482.00	Tk. 1,482.00

10.31	Safety shed over drive way : Supply, fabrication, installation and taking out temporary horizontal safety shed over the drive way around construction work place where public safety is likely to be endangered due to construction activities; which shall be made of 0.27 mm thick galvanized iron plain sheet fitted at 15'-0" height horizontally over the drive way or path way at the outer periphery of the building with 75mm x 40mm x 5 mm M.S Chanel post and purlin, 20mm MS Rod and 40mmx40mmx6mm MS Angle as ties etc. by standard anchoring and welding, nut-bolts etc, all complete and accepted by the Engineer-in-charge(Salvage materials shall be the contractors property after completion of work.)	sqm	Tk. 1,638.00	Tk. 1,631.00	Tk. 1,603.00	Tk. 1,603.00
10.32	Scaffolding for outside netting of building: Supplying, fitting & fixing scaffolding for outside netting of building with single layer 38 mm dia MS pipe connected each other with minimum 4'-0" centre to centre both direction with 38 mm dia swivel clamp, including fitting and fixing in position providing necessary anchors, wires, ties etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the cost of netting)	sqm	Tk. 348.00	Tk. 348.00	Tk. 348.00	Tk. 348.00
10.33	Safety net: Supplying temporary safety net (hessian cloth) around construction work place (along the height of the building) where public safety is likely to be endangered due to construction activities; which shall be supported using scaffolding around the building for brick work/ plaster; including fitting and fixing in position providing necessary anchors, wires, ties etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the cost of scaffolding)	sqm	Tk. 51.00	Tk. 51.00	Tk. 50.00	Tk. 50.00
10.34	Supply, fabrication, installation and taking out of 1.2 m high temporary safety fencing at different floor during construction work around the outer periphery of building or around the punch in the floor slab for safety of the worker during construction, which shall be made of 10mm dia MS Rod @ 450mm C/C both way welded at each point together with a 12mm Dia MS Rod as top rail drilled anchored to the floor and sides as necessary all complete and accepted by the Engineer-in-charge(Salvage materials shall be the contractors property after completion of work.).	sqm	Tk. 380.00	Tk. 380.00	Tk. 380.00	Tk. 380.00
10.35	Providing bearing joints fixed or free with 250 mm x 375 mm x 10 mm M.S. (Grade - A36) plate fitted in the truss angles with 16 mm diameter M.S. rod counter sunk rivets, welded and anchored to 250 x 375 x 10 mm. M.S. plate by two Nos. 19 mm dia M.S. "U" rod embedded in the bed of the lintel around the reinforcement, cleaning and greasing the joints for all floors etc. all complete and accepted by the Engineer-in-charge	each	Tk. 2,941.95	Tk. 2,934.00	Tk. 2,903.00	Tk. 2,903.00
10.36	Galvanization: Galvanization of finished structural steel built-up and hot rolled finished materials of any shape and size (average thickness not more than 8 mm) conforming to the requirement of the corrosion class C3 and Galvanization shall be performed in accordance with ASTM A123 all complete as per drawing, specification and direction of engineer in charge	kg	Tk. 71.00	Tk. 71.00	Tk. 71.00	Tk. 71.00
10.37	Additional Erection and installation charge for items 10.2~10.6 and 10.18 shall be 2% of the gross item rate for every four floor above level-4					

..... END OF CHAPTER TEN

CHAPTER 11 : WOOD WORKS IN DOOR AND WINDOW FRAME, ROOF TRUSS AND RAILINGS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
11.1	Supplying and making door and window frames (Chowkat) for all floors in/c wall side grove cutting, moulding by CNC machine or any other means with matured seasoned wood of required size including painting two coats of coal tar to the surface in contact with wall, fitting and fixing in position etc. all complete and accepted by the Engineer-in-charge. (All sizes of wood are finished).					
11.1.1	Frame made from seasoned Mehgoni wood (Chemically treated)					
11.1.1.1	Upto Size 0'-2.5" x 0'-6" (finished)	cum	Tk. 135,509.26	Tk. 135,148.00	Tk. 132,944.00	Tk. 132,944.00
11.1.1.2	Size greater than 0'-2.5" x 0'-6" and upto 0'-2.5" x 0'-11" (finished)	cum	Tk. 118,023.63	Tk. 117,823.00	Tk. 116,579.00	Tk. 116,579.00
11.1.2	Frame made from seasoned Garjan/Jam/Local Sal Wood					
11.1.2.1	Upto Size 0'-2.5" x 0'-6" (finished)	cum	Tk. 145,620.16	Tk. 140,204.00	Tk. 145,583.00	Tk. 145,583.00
11.1.2.2	Size greater than 0'-2.5" x 0'-6" and upto 0'-2.5" x 0'-11" (finished)	cum	Tk. 127,831.55	Tk. 122,727.00	Tk. 128,839.00	Tk. 128,839.00
11.1.3	Frame made from seasoned Chattogram Silkarai /Azobe / Okan Loha Kath					
11.1.3.1	Upto Size 0'-2.5" x 0'-6" (finished)	cum	Tk. 160,786.52	Tk. 160,425.00	Tk. 158,221.00	Tk. 158,221.00
11.1.3.2	Size greater than 0'-2.5" x 0'-6" and upto 0'-2.5" x 0'-11" (finished)	cum	Tk. 142,543.42	Tk. 142,343.00	Tk. 141,099.00	Tk. 141,099.00
11.1.4	Frame made from seasoned Local Manjuree / Jarul					
11.1.4.1	Upto Size 0'-2.5" x 0'-6" (finished)	cum	Tk. 221,451.96	Tk. 216,035.00	Tk. 221,414.00	Tk. 221,414.00
11.1.4.2	Size greater than 0'-2.5" x 0'-6" and upto 0'-2.5" x 0'-11" (finished)	cum	Tk. 201,390.92	Tk. 196,287.00	Tk. 202,398.00	Tk. 202,398.00
11.1.5	Frame made from seasoned Chattogram Teak					
11.1.5.1	Upto Size 0'-2.5" x 0'-6" (finished)	cum	Tk. 256,840.13	Tk. 251,424.00	Tk. 264,335.00	Tk. 264,335.00
11.1.5.2	Size greater than 0'-2.5" x 0'-6" and upto 0'-2.5" x 0'-11" (finished)	cum	Tk. 235,718.62	Tk. 230,614.00	Tk. 244,033.00	Tk. 244,033.00
11.1.6	Frame made from seasoned Myanmar(Burma) Teak					
11.1.6.1	Upto Size 0'-2.5" x 0'-6" (finished)	cum	Tk. 393,337.37	Tk. 387,921.00	Tk. 395,827.00	Tk. 395,827.00
11.1.6.2	Size greater than 0'-2.5" x 0'-6" and upto 0'-2.5" x 0'-11" (finished)	cum	Tk. 368,125.49	Tk. 363,021.00	Tk. 371,585.00	Tk. 371,585.00
11.1.7	Frame made from seasoned Jessore & Other Local Teak					
11.1.7.1	Upto Size 0'-2.5" x 0'-6" (finished)	cum	Tk. 186,063.79	Tk. 185,703.00	Tk. 183,499.00	Tk. 183,499.00
11.1.7.2	Size greater than 0'-2.5" x 0'-6" and upto 0'-2.5" x 0'-11" (finished)	cum	Tk. 167,063.21	Tk. 166,863.00	Tk. 165,619.00	Tk. 165,619.00
11.2.1	Supplying, fitting and fixing M.S. flat bar clamp of 225 mm x 38 mm x 6 mm size having bifurcated ends to door and window frames with necessary screws including cutting grooves in chowkat if necessary and encasing inside the wall with cement concrete (1:2:4) etc, all complete and accepted by the Engineer-in-charge.	each	Tk. 123.16	Tk. 122.00	Tk. 117.00	Tk. 117.00
11.2.2	Supplying, fitting and fixing M.S. flat bar clamp of 150 mm x 38 mm x 6 mm size having bifurcated ends to door and window frames with necessary rowel plug, screws etc. including cutting grooves in chowkat if necessary etc. all complete and accepted by the Engineer-in-charge.	each	Tk. 131.14	Tk. 130.00	Tk. 126.00	Tk. 126.00

11.3.1	Supplying, fitting and fixing double leaf door frame for all floors made with 38mm x 38 mm x 6 mm M.S. angle fitted with 150 mm long 6 nos. iron clamps of same size M.S. angle (with one end bifurcated) with the vertical members of the frame, fixing the frame in wall with cement concrete (1:2:4), mending good the damages, fixing 6 Nos. of 100 mm long iron hinges with the vertical members of the frame all complete including making holes in walls, cutting, sizing, welding, fabricating, carrying from work-shop to site and local carriage, curing etc. all complete as per plan and drawing and accepted by the Engineer-in-charge.	meter	Tk. 730.61	Tk. 730.00	Tk. 720.00	Tk. 720.00
11.3.2	Supplying, fitting and fixing single leaf door frame for all floors made with 38 mm x 38 mm x 6 mm M.S. angle, fixing 125 mm long 6 nos. iron clamps of same size (with one end bifurcated) with the vertical members of the frame, fixing the frame in wall with cement concrete (1:2:4), mending good the damages, fixing 4 nos. 100 mm size iron hinges with the vertical members of the frame, all complete including making holes in walls, cutting, sizing, welding, fabricating, carriage from workshop to site including local carriage, curing etc. as per plan and drawing etc. all complete and accepted by the Engineer-in-charge.	meter	Tk. 707.27	Tk. 707.00	Tk. 696.00	Tk. 696.00
11.4	Supplying matured seasoned wood and making roof truss of any size with wall plates as per design including supplying, fabricating and hoisting, scaffolding, fitting and fixing in position with bolts, nuts, painting two coats of coal tar etc. all complete and accepted by the Engineer-in-charge. (All sizes of wood are finished).					
11.4.1	Garjan / Jam/ Local Sal wood	cum	Tk. 125,547.13	Tk. 120,889.00	Tk. 126,617.00	Tk. 126,617.00
11.4.2	Mehgoni	cum	Tk. 116,447.31	Tk. 116,339.00	Tk. 115,243.00	Tk. 115,243.00
11.4.3	Silkarai /Chickrashi	cum	Tk. 139,196.85	Tk. 139,088.00	Tk. 137,992.00	Tk. 137,992.00
11.5	Supplying and making hand rail for stair case with matured seasoned wood including necessary grove cutting, moulding by CNC machine or any other means with proper corner, finishing as per design, fitting and fixing in position in all floors etc. all complete and accepted by the Engineer-in-charge.					
11.5.1	Mehgoni wood stair rail	cum	Tk. 121,567.79	Tk. 121,377.00	Tk. 119,180.00	Tk. 119,180.00
11.5.2	Chattogram Silkarai / Chikrashi wood stair rail	cum	Tk. 144,317.33	Tk. 144,126.00	Tk. 141,929.00	Tk. 141,929.00
11.5.3	Chattogram Teak wood stair rail	cum	Tk. 230,765.58	Tk. 226,024.00	Tk. 237,432.00	Tk. 237,432.00
11.5.4	Myanmar(Burma) Teak wood stair rail	cum	Tk. 353,613.10	Tk. 348,872.00	Tk. 355,775.00	Tk. 355,775.00
11.5.5	Jessore & other Local Teak wood stair rail	cum	Tk. 167,066.87	Tk. 166,876.00	Tk. 164,678.00	Tk. 164,678.00

..... END OF CHAPTER ELEVEN

CHAPTER 12 : DOOR-WINDOW SHUTTER & FLY PROOF WIRE-NET SHUTTER

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
12.1	Supplying, fitting and fixing of 38 mm thick well matured seasoned (moisture content should be between 12% to 16%) solid wood single leaf door shutter having stile section 100mm x 38mm, top rail section 125mm x 38mm, lock rail section 100mm x 38mm and bottom rail section 225mm x 38mm (made from minimum 250mm wide plank) with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished)					
12.1.1.1	Kathal (SS Fittings)	sqm	Tk. 9,041.49	Tk. 9,011.00	Tk. 8,824.00	Tk. 8,824.00
12.1.1.2	Kathal (Brass Fittings)	sqm	Tk. 9,410.00	Tk. 9,380.00	Tk. 9,192.00	Tk. 9,192.00
12.1.2.1	Gamari (SS Fittings)	sqm	Tk. 7,927.16	Tk. 7,674.00	Tk. 7,932.00	Tk. 7,932.00
12.1.2.2	Gamari (Brass Fittings)	sqm	Tk. 8,295.67	Tk. 8,042.00	Tk. 8,301.00	Tk. 8,301.00
12.1.3.1	Chattogram Teak (SS Fittings)	sqm	Tk. 13,498.79	Tk. 13,246.00	Tk. 13,725.00	Tk. 13,725.00
12.1.3.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 13,867.30	Tk. 13,614.00	Tk. 14,093.00	Tk. 14,093.00
12.1.4.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 19,516.15	Tk. 19,263.00	Tk. 19,521.00	Tk. 19,521.00
12.1.4.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 19,884.66	Tk. 19,631.00	Tk. 19,890.00	Tk. 19,890.00
12.2	Supplying, fitting and fixing 38 mm thick well matured seasoned (moisture content should be between 12 to 16%) wooden ornamental special design (both sides) single panel flush door shutter having stile, top rail and lock rail section 125 x 38mm and bottom rail section 225mm x 38mm (made from minimum 250mm wide plank) with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished)					
12.2.1.1	Chattogram Teak (SS Fittings)	sqm	Tk. 15,685.06	Tk. 15,410.00	Tk. 15,797.00	Tk. 15,797.00
12.2.1.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 15,270.89	Tk. 15,002.00	Tk. 15,448.00	Tk. 15,448.00
12.2.2.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 21,121.42	Tk. 20,853.00	Tk. 21,069.00	Tk. 21,069.00
12.2.2.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 21,522.55	Tk. 21,254.00	Tk. 21,470.00	Tk. 21,470.00
12.3	Supplying, fitting and fixing 38mm thick well matured seasoned (moisture content should be between 12 to 16%) wooden double leaf panel door shutters having stile and top rail section 100mm x 38mm, lock rail section 125mm x 38mm and bottom rail section 225mm x 38mm, panelling 38mm thick both sides raised (made from minimum 250mm wide plank) with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished)					
12.3.1.1	Kathal (SS Fittings)	sqm	Tk. 9,917.32	Tk. 9,898.00	Tk. 9,713.00	Tk. 9,713.00

12.3.1.2	Kathal (Brass Fittings)	sqm	Tk. 10,714.47	Tk. 10,695.00	Tk. 10,510.00	Tk. 10,510.00
12.3.2.1	Gamari (SS Fittings)	sqm	Tk. 8,686.93	Tk. 8,420.00	Tk. 8,728.00	Tk. 8,728.00
12.3.2.2	Gamari (Brass Fittings)	sqm	Tk. 9,484.08	Tk. 9,217.00	Tk. 9,525.00	Tk. 9,525.00
12.3.3.1	Chattogram Teak (SS Fittings)	sqm	Tk. 14,838.89	Tk. 14,579.00	Tk. 15,130.00	Tk. 15,130.00
12.3.3.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 15,636.04	Tk. 15,376.00	Tk. 15,927.00	Tk. 15,927.00
12.3.4.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 21,483.00	Tk. 21,230.00	Tk. 21,538.00	Tk. 21,538.00
12.3.4.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 22,280.16	Tk. 22,027.00	Tk. 22,335.00	Tk. 22,335.00
12.4	Supplying, fitting and fixing 38mm thick well matured seasoned (moisture content should be between 12 to 16%) wooden double leaf panel door shutters with 1/4th glazed and 3/4th wooden panel having stile and top rail section 100mm x 38mm, lock rail section 125mm x 38mm, and bottom rail section 225mm x 38mm, panelling 38mm thick both sides raised (made from minimum 250mm wide plank) with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished)					
12.4.1.1	Kathal (SS Fittings)	sqm	Tk. 8,787.23	Tk. 8,771.00	Tk. 8,610.00	Tk. 8,610.00
12.4.1.2	Kathal (Brass Fittings)	sqm	Tk. 8,912.83	Tk. 8,900.00	Tk. 8,753.00	Tk. 8,753.00
12.4.2.1	Gamari (SS Fittings)	sqm	Tk. 7,252.83	Tk. 7,031.00	Tk. 7,299.00	Tk. 7,299.00
12.4.2.2	Gamari (Brass Fittings)	sqm	Tk. 7,869.84	Tk. 7,648.00	Tk. 7,916.00	Tk. 7,916.00
12.4.3.1	Chattogram Teak (SS Fittings)	sqm	Tk. 12,420.62	Tk. 12,206.00	Tk. 12,679.00	Tk. 12,679.00
12.4.3.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 13,047.06	Tk. 12,833.00	Tk. 13,305.00	Tk. 13,305.00
12.4.4.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 18,001.83	Tk. 17,795.00	Tk. 18,063.00	Tk. 18,063.00
12.4.4.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 18,628.28	Tk. 18,422.00	Tk. 18,689.00	Tk. 18,689.00
12.5	Supplying, fitting and fixing 38 mm thick well matured seasoned (moisture content should be between 12 to 16%) wooden grooved single panel flush door shutter having stile, top rail and horizontal panel section 100mm x 38mm and bottom rail section 250mm x 38mm (made from min250mm wide plank) including keeping 6mm x 12mm even groove all around and minimum 12mm grooved lap to each panel with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished)					
12.5.1.1	Kathal (SS Fittings)	sqm	Tk. 9,530.13	Tk. 9,489.00	Tk. 9,287.00	Tk. 9,287.00
12.5.1.2	Kathal (Brass Fittings)	sqm	Tk. 9,898.64	Tk. 9,857.00	Tk. 9,656.00	Tk. 9,656.00
12.5.2.1	Gamari (SS Fittings)	sqm	Tk. 8,357.99	Tk. 8,084.00	Tk. 8,351.00	Tk. 8,351.00
12.5.2.2	Gamari (Brass Fittings)	sqm	Tk. 8,726.50	Tk. 8,452.00	Tk. 8,719.00	Tk. 8,719.00
12.5.3.1	Chattogram Teak (SS Fittings)	sqm	Tk. 14,218.66	Tk. 13,938.00	Tk. 14,436.00	Tk. 14,436.00
12.5.3.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 14,587.17	Tk. 14,306.00	Tk. 14,805.00	Tk. 14,805.00
12.5.4.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 20,548.18	Tk. 20,260.00	Tk. 20,527.00	Tk. 20,527.00
12.5.4.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 20,916.70	Tk. 20,628.00	Tk. 20,895.00	Tk. 20,895.00

12.6	Supplying, fitting and fixing 36 mm thick factory made solid wooden door shutter from a reputed manufacturer having manufacturer's warranty made of chemically treated (pressure treatment method) timber (moisture content should be between 12 to 16%) having stile 125mm x 36mm, top and bottom rail 140mm x 36mm and horizontal rail 120mm x 36mm including keeping 5mm x 8mm even groove all around with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (Price includes cost of polishing) (All sizes of wood are finished)					
12.6.1	Commercial solid wooden door (Normal Polish) With SS Fittings	sqm	Tk. 8,127.23	Tk. 8,121.00	Tk. 8,078.00	Tk. 8,078.00
12.6.2	Commercial solid wooden door (Normal Polish) With Brass Fittings	sqm	Tk. 8,456.61	Tk. 8,450.00	Tk. 8,408.00	Tk. 8,408.00
12.7	Supplying, fitting and fixing 36 mm thick factory made solid wooden door shutter (moisture content should be between 12 to 16%) from a reputed manufacturer with manufacturer's warranty having stile 150mm x 36mm, top rail 125mm x 36mm, bottom rail 225mm x 36mm with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished)					
12.7.1.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 30,454.88	Tk. 30,448.00	Tk. 30,406.00	Tk. 30,406.00
12.7.1.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 30,784.26	Tk. 30,778.00	Tk. 30,735.00	Tk. 30,735.00
12.7.2.1	Mahogany (Chemically treated by Pressure treatment method) (SS Fittings)	sqm	Tk. 10,138.11	Tk. 10,132.00	Tk. 10,089.00	Tk. 10,089.00
12.7.2.2	Mahogany (Chemically treated by Pressure treatment method) (Brass Fittings)	sqm	Tk. 10,467.48	Tk. 10,461.00	Tk. 10,419.00	Tk. 10,419.00
12.7.3.1	Beech Wood (SS Fittings)	sqm	Tk. 18,805.67	Tk. 18,799.00	Tk. 18,757.00	Tk. 18,757.00
12.7.3.2	Beech Wood (Brass Fittings)	sqm	Tk. 19,135.05	Tk. 19,128.00	Tk. 19,086.00	Tk. 19,086.00
12.7.4	Added Cost for both side Ultra Violet (UV) Lacquer Polish (One side of door area to be used for measurement)	sqm	Tk. 4,622.70	Tk. 4,623.00	Tk. 4,623.00	Tk. 4,623.00
12.8	Supplying, fitting and fixing of 40mm thick Wood Plastic Composite (WPC) door shutter (minimum inner wall thickness 6mm and hinge side wall thickness 15mm) having door frame section 150mm x 62mm weighing 16.7kg/sqm with 4nos 100mm long, 3mm thick hinge, 1 no round lock, 2nos 150mm long handle, hinged cleats and carrying the same to the site and local carriage etc. All complete in all floors and accepted by the Engineer-in-charge.					
12.8.1	Door Size 6'9" x 2'6"	each	Tk. 17,698.02	Tk. 17,695.00	Tk. 17,667.00	Tk. 17,667.00
12.8.2	Door Size 6'9" x 3'0"	each	Tk. 21,315.30	Tk. 21,312.00	Tk. 21,285.00	Tk. 21,285.00
12.8.3	Door Size 6'9" x 3'4"	each	Tk. 23,065.04	Tk. 23,062.00	Tk. 23,034.00	Tk. 23,034.00

12.9	Supplying, fitting and fixing 36 mm thick and having density 400 kg/cum solid particle board single leaf flush door shutter for all floors for internal use made of well matured mechanical seasoned Rubber (Malaysian timber) or other approved wooden frame having 1.2mm thick veneer pasted on each side by mechanical process with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished)					
12.9.1.1	Myanmar(Burma) Teak veneered (SS Fittings)	sqm	Tk. 5,528.98	Tk. 5,526.00	Tk. 5,504.00	Tk. 5,504.00
12.9.1.2	Myanmar(Burma) Teak veneered (Brass Fittings)	sqm	Tk. 5,858.36	Tk. 5,855.00	Tk. 5,834.00	Tk. 5,834.00
12.9.2.1	Champ veneered (SS Fittings)	sqm	Tk. 4,208.21	Tk. 4,205.00	Tk. 4,184.00	Tk. 4,184.00
12.9.2.2	Champ veneered (Brass Fittings)	sqm	Tk. 4,537.58	Tk. 4,534.00	Tk. 4,513.00	Tk. 4,513.00
12.9.3.1	Chapalish veneered (SS Fittings)	sqm	Tk. 4,208.21	Tk. 4,205.00	Tk. 4,184.00	Tk. 4,184.00
12.9.3.2	Chapalish veneered (Brass Fittings)	sqm	Tk. 4,537.58	Tk. 4,534.00	Tk. 4,513.00	Tk. 4,513.00
12.9.4.1	Garjan veneered (SS Fittings)	sqm	Tk. 4,366.70	Tk. 4,363.00	Tk. 4,342.00	Tk. 4,342.00
12.9.4.2	Garjan veneered (Brass Fittings)	sqm	Tk. 4,696.08	Tk. 4,693.00	Tk. 4,672.00	Tk. 4,672.00
12.9.5.1	For Formica laminated Door (Commercial Lipping) (SS Fittings)	sqm	Tk. 6,003.14	Tk. 6,000.00	Tk. 5,979.00	Tk. 5,979.00
12.9.5.2	For Formica laminated Door (Commercial Lipping) (Brass Fittings)	sqm	Tk. 6,332.51	Tk. 6,329.00	Tk. 6,308.00	Tk. 6,308.00
12.10	Supplying, fitting and fixing well matured seasoned (moisture content should be between 12 to 16%) wooden fixed louver shutter having 62 mm x 125 mm frame and inner horizontal wooden louver 150 mm x 25 mm (minimum 250 mm wide plank) size spaced @ 50 mm c/c fixed with frame in grooves with 4nos 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished).					
12.10.1	Gamari	sqm	Tk. 11,566.77	Tk. 11,101.00	Tk. 11,816.00	Tk. 11,816.00
12.10.2	Chapalish	sqm	Tk. 13,333.74	Tk. 12,868.00	Tk. 13,583.00	Tk. 13,583.00
12.10.3	Chattogram Teak	sqm	Tk. 22,610.30	Tk. 22,145.00	Tk. 23,297.00	Tk. 23,297.00
12.10.4	Myanmar(Burma) Teak	sqm	Tk. 34,537.32	Tk. 34,072.00	Tk. 34,786.00	Tk. 34,786.00
12.11	Supplying, fitting and fixing well matured seasoned (moisture content should be between 12 to 16%) wooden glazed ventilator louver shutter having outer frame section 62 mm x 125 mm with 62 mm x 125 mm vertical divider with inner 50mm x 38mm shutter and with 3 Nos. horizontal 38 mm x 38 mm sash bar in each part (minimum 250 mm wide plank) and inner horizontal glazed louver of size 150 mm x 5mm fixed in frame @ 154 mm c/c by cutting grooves in the frame with 4nos. 100mm long, 3mm thick hinge, 12mm dia 250mm long tower bolt, 12mm dia 200mm long socket bolt, 2nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished).					
12.11.1	Gamari	sqm	Tk. 7,906.44	Tk. 7,626.00	Tk. 8,014.00	Tk. 8,014.00
12.11.2	Chapalish	sqm	Tk. 8,946.04	Tk. 8,666.00	Tk. 9,054.00	Tk. 9,054.00
12.11.3	Chattogram Teak	sqm	Tk. 14,403.95	Tk. 14,123.00	Tk. 14,769.00	Tk. 14,769.00
12.11.4	Myanmar(Burma) Teak	sqm	Tk. 21,421.25	Tk. 21,141.00	Tk. 21,529.00	Tk. 21,529.00

12.12	Supplying fitting and fixing 38 mm thick well matured seasoned (moisture content should be between 12 to 16%) wooden double leaf glazed door shutters having stiles and rails section 150 mm x 38 mm (made from minimum 250 mm wide plank) and 5 mm glass panes fitted and fixed with wooden batten, putty and nails, 8nos 100mm long, 3mm thick hinge, 4nos 12mm dia 250mm long tower bolt, 4nos 150mm long handle, hinged cleats and buffer blocks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished).					
12.12.1.1	Chattogram Teak (SS Fittings)	sqm	Tk. 8,786.19	Tk. 8,653.00	Tk. 8,870.00	Tk. 8,870.00
12.12.1.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 9,342.14	Tk. 9,209.00	Tk. 9,426.00	Tk. 9,426.00
12.12.2.1	Silkarai (SS Fittings)	sqm	Tk. 6,586.48	Tk. 6,569.00	Tk. 6,440.00	Tk. 6,440.00
12.12.2.2	Silkarai (Brass Fittings)	sqm	Tk. 7,142.43	Tk. 7,125.00	Tk. 6,996.00	Tk. 6,996.00
12.12.3.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 11,912.09	Tk. 11,779.00	Tk. 11,882.00	Tk. 11,882.00
12.12.3.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 12,468.04	Tk. 12,335.00	Tk. 12,437.00	Tk. 12,437.00
12.13	Supplying, fitting and fixing 38 mm thick well matured seasoned (moisture content should be between 12 to 16%) wooden glazed window shutters having horizontal and vertical sections of 87 mm x 38 mm and sash bar 38 mm x 38 mm (made from minimum 250 mm wide plank) fitted with 3 mm glass panes with putty and nails and provided with 6nos 75mm long hinges, 2nos 150mm tower bolt, 2 nos handle and 2nos 225mm catch hooks. All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished).					
12.13.1.1	Kathal (SS Fittings)	sqm	Tk. 7,755.68	Tk. 7,734.00	Tk. 7,570.00	Tk. 7,570.00
12.13.1.2	Kathal (Brass Fittings)	sqm	Tk. 8,545.74	Tk. 8,524.00	Tk. 8,360.00	Tk. 8,360.00
12.13.2.1	Gamari (SS Fittings)	sqm	Tk. 7,003.08	Tk. 6,830.00	Tk. 6,968.00	Tk. 6,968.00
12.13.2.2	Gamari (Brass Fittings)	sqm	Tk. 7,793.14	Tk. 7,620.00	Tk. 7,758.00	Tk. 7,758.00
12.13.3.1	Chattogram Teak (SS Fittings)	sqm	Tk. 10,766.08	Tk. 10,593.00	Tk. 10,880.00	Tk. 10,880.00
12.13.3.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 11,556.14	Tk. 11,383.00	Tk. 11,670.00	Tk. 11,670.00
12.13.4.1	Chapalish (SS Fittings)	sqm	Tk. 7,605.16	Tk. 7,433.00	Tk. 7,570.00	Tk. 7,570.00
12.13.4.2	Chapalish (Brass Fittings)	sqm	Tk. 8,395.22	Tk. 8,223.00	Tk. 8,360.00	Tk. 8,360.00
12.13.5.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 14,830.12	Tk. 14,657.00	Tk. 14,795.00	Tk. 14,795.00
12.13.5.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 15,620.18	Tk. 15,448.00	Tk. 15,585.00	Tk. 15,585.00
12.14	Supplying, fitting and fixing 38 mm thick well matured seasoned (moisture content should be between 12 to 16%) wooden window shutters one third glazed and two third raised paneling on both sides having horizontal and vertical sections of 87 mm x 38 mm and sash bars 38 mm x 38 mm (made from minimum 250 mm wide plank) fitted with 3 mm glass panes with putty and nails and provided with 6nos 75mm long hinges, 2nos 150mm tower bolt, 2 nos 100mm long handle and 2nos 225mm long catch hooks, All complete in all floors and accepted by the Engineer-in-charge. (All sizes of wood are finished).					
12.14.1.1	Kathal (SS Fittings)	sqm	Tk. 11,390.03	Tk. 11,359.00	Tk. 11,143.00	Tk. 11,143.00
12.14.1.2	Kathal (Brass Fittings)	sqm	Tk. 12,236.08	Tk. 12,205.00	Tk. 11,989.00	Tk. 11,989.00
12.14.2.1	Gamari (SS Fittings)	sqm	Tk. 10,038.91	Tk. 9,738.00	Tk. 10,062.00	Tk. 10,062.00
12.14.2.2	Gamari (Brass Fittings)	sqm	Tk. 10,884.97	Tk. 10,584.00	Tk. 10,908.00	Tk. 10,908.00
12.14.3.1	Chattogram Teak (SS Fittings)	sqm	Tk. 16,794.50	Tk. 16,493.00	Tk. 17,085.00	Tk. 17,085.00

12.14.3.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 17,640.55	Tk. 17,339.00	Tk. 17,932.00	Tk. 17,932.00
12.14.4.1	Chapalish (SS Fittings)	sqm	Tk. 11,119.81	Tk. 10,819.00	Tk. 11,143.00	Tk. 11,143.00
12.14.4.2	Chapalish (Brass Fittings)	sqm	Tk. 11,965.86	Tk. 11,665.00	Tk. 11,989.00	Tk. 11,989.00
12.14.5.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 24,090.53	Tk. 23,789.00	Tk. 24,114.00	Tk. 24,114.00
12.14.5.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 24,936.58	Tk. 24,635.00	Tk. 24,960.00	Tk. 24,960.00
12.15.1	Supplying, fitting and fixing steel glazed window shutters for all floors with frames as per design having 25 mm x 25 mm x 5 mm angle section for Chowkat and shutters and 50 mm x 25 mm x 5 mm T-section used for vertical member and 25 mm x 5 mm F.I. bar for making box (combining with angle) and 25 mm x 19 mm x 5 mm T-section used for glass divider including all charges for welding, riveting etc. including cost of supplying all essential fittings like iron hinges, plate handle, pin-stopper, adjustable cleat and clamps including supplying, fitting, fixing 3 mm glass panes by pucca putty, for fitting and fixing the frame, making necessary holes in brick walls and R.C.C works and fitting with C.C. (1:2:4) and mending good the damages including all cost of carriage, workshop charges, electricity and labour for fittings, fixing, curing etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the costs of painting).	sqm	Tk. 6,767.32	Tk. 6,768.00	Tk. 6,760.00	Tk. 6,759.00
12.15.2	Supplying, fitting and fixing M.S. door shutters made with 38 mm x 38 mm x 6 mm angle outer frame, inner member 25 mm x 6 mm F.I. bar, cladding with 16 BWG M.S. sheet, 04 nos 75 mm long hinge, 02 nos. locking arrangement with 50 mm x 50 mm x 6 mm M.S. angle chowkat including all charges for welding, riveting etc. including cost of supplying all essential fittings like iron hinges, socket bolts, clamps, for fitting and fixing the frame, making necessary holes in brick walls and R.C.C works and fitting with C.C. (1:2:4) and mending good the damages including all cost of carriage, workshop charges, electricity and labour for fittings, fixing, curing etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the costs of painting).	sqm	Tk. 7,630.29	Tk. 7,597.00	Tk. 7,380.00	Tk. 7,380.00
12.16	Supplying, fitting and fixing steel glazed window shutters for all floors with frames as per design having 19mm x 19mm x 19mm x 3mm Z-section in frame and 19mm x 3mm F.I. bar for making T-section for horizontal dividers of shutters and 25mm x 3mm F.I. bar for middle vertical members of frame including all cost of charges for fabrication and manufacture by welding, riveting, etc. Supplying all essential fittings like 2 nos stopper handle, 300mm long 3 or 2 nos adjustable iron cleats, 6 or 4 nos 50mm long iron pin hinges, 150mm long 4 nos clamps including supplying, fitting, fixing 3mm glass panes by pucca putty, for fitting and fixing frame making necessary holes in brick and R.C.C. wall or member as necessary filling with C.C. (1:2:4) and mending good the damages including all cost of carriage, electricity, workshop charges and labour for fitting, fixing, etc. all complete as per drawing and design and accepted by the Engineer-in-charge. (Rate is excluding the costs of painting)					
12.16.1	With provision for fan-light by providing necessary additional materials, 03 nos adjustable 300 mm long iron cleats and 6 nos 50 mm long iron pin hinges (total wt of grill without glass panes is min 28.13 kg for 15 sft)	sqm	Tk. 4,400.87	Tk. 4,401.00	Tk. 4,394.00	Tk. 4,392.00
12.16.2	Without provision for fan-light by providing 02 nos adjustable 300 mm long iron cleats and 4 nos 50 mm long iron pin hinges (total wt of grill without glass panes is min 24.84 kg for 15 sft)	sqm	Tk. 4,112.47	Tk. 4,112.00	Tk. 4,105.00	Tk. 4,104.00

12.17	Manufacturing, supplying fitting and fixing of fixed glazed frame for all floors with outer member of 19mm x 19mm x 19mm x 3mm Z-section and vertical and horizontal dividers of 19mm x 19mm x 3mm T-section including all cost of fabrication and welding, supplying and fixing of 19mm x 19mm x 3mm angle clamps of 150mm long (with one end bifurcated) of required Nos with the frame, fixing the frame in wall with (1:2:4) cement concrete in the hole for clamp after making necessary grooves, supplying and fixing of 3mm glass panes with pucca putty, mending good all damages, painting iron face with two coats of synthetic enamel paint over a coat of anti-corrosive priming of approved colour and quality etc. cost of electricity, workshop charges, carriage etc. all complete as per drawing and accepted by the Engineer-in-charge. [Rate is excluding the cost of painting] (Scope of work of this item is in staircase and over partition wall etc.)	sqm	Tk. 4,044.91	Tk. 4,045.00	Tk. 4,036.00	Tk. 4,034.00
12.18	Supplying, fitting and fixing 25 mm thick wooden double leaf fly proof wire mesh window shutters for all floors with 75 mm x 25 mm vertical and horizontal stile and sash bar of 38 mm x 25 mm section provided with 24 B.W.G. G.I wire mesh netting of 20 mm mesh fitted and fixed with shutter by 12 mm x 12 mm wooden bit, screws, nails, etc. including supply of all necessary fittings such as 2 Nos 150 mm long tower and socket bolt of best quality 4 Nos 75mm long hinges, 2 Nos 150 mm long catch hook and 2 Nos. 100 mm long handle including fitting and fixing the same in position with frame with supply of necessary screws, nails, etc. complete and accepted by the Engineer-in-charge. (All sizes of wood are finished)					
12.18.1.1	Kathal (SS Fittings)	sqm	Tk. 5,840.56	Tk. 5,822.00	Tk. 5,683.00	Tk. 5,683.00
12.18.1.2	Kathal (Brass Fittings)	sqm	Tk. 6,979.77	Tk. 6,962.00	Tk. 6,822.00	Tk. 6,822.00
12.18.2.1	Chattogram Teak (SS Fittings)	sqm	Tk. 7,770.10	Tk. 7,655.00	Tk. 7,804.00	Tk. 7,804.00
12.18.2.2	Chattogram Teak (Brass Fittings)	sqm	Tk. 8,909.31	Tk. 8,795.00	Tk. 8,943.00	Tk. 8,943.00
12.18.3.1	Myanmar(Burma) Teak (SS Fittings)	sqm	Tk. 10,374.97	Tk. 10,260.00	Tk. 10,314.00	Tk. 10,314.00
12.18.3.2	Myanmar(Burma) Teak (Brass Fittings)	sqm	Tk. 11,514.18	Tk. 11,400.00	Tk. 11,453.00	Tk. 11,453.00
12.19	Supplying, fitting and fixing of well matured wooden louver shutter (Chattogram Teak) having 75.50 mm x 38 mm size all around frame and inner horizontal wooden louver blade 65 mm x 20 mm spaced @ 38 mm c/c, fitted and fixed with 2 nos hinges, 1 no handle and 2 nos magnetic catch. All complete in all floors and accepted by the Engineer-in-charge.					
12.19.1	With SS Fittings	sqm	Tk. 16,628.52	Tk. 16,331.00	Tk. 16,846.00	Tk. 16,846.00
12.19.2	With Brass Fittings	sqm	Tk. 16,919.09	Tk. 16,621.00	Tk. 17,136.00	Tk. 17,136.00
12.20	Supplying, fitting and fixing hydraulic door closer (big size) of approved quality including all necessary tools and accessories etc. all complete and accepted by the Engineer-in-charge.					
12.20.1	Standard Quality	each	Tk. 2,684.12	Tk. 2,682.00	Tk. 2,666.00	Tk. 2,666.00
12.21	Supplying, fitting and fixing mortice door lock with lever handle of approved quality including all necessary tools and accessories etc. all complete approved and accepted by the Engineer-in-charge.					
12.21.1	Standard Quality	each	Tk. 1,242.36	Tk. 1,239.00	Tk. 1,212.00	Tk. 1,212.00
12.22	Supplying, fitting and fixing rim/round door lock of approved quality including all necessary tools and accessories etc. all complete and accepted by the Engineer-in-charge.					

12.22.1	Standard Quality	each	Tk. 1,058.31	Tk. 1,055.00	Tk. 1,028.00	Tk. 1,028.00
12.23	Supplying, fitting and fixing best quality heavy type 19 mm dia and 300 mm long hasp bolt including cutting grooves in door shutter and frames, screws etc. all complete approved and accepted by the Engineer-in-charge.					
12.23.1	SS Hasp Bolt	each	Tk. 343.57	Tk. 344.00	Tk. 337.00	Tk. 337.00
12.23.2	Brass Hasp Bolt	each	Tk. 552.16	Tk. 552.00	Tk. 546.00	Tk. 546.00
12.24	Supplying, fitting and fixing of best quality magnetic door holder of approved quality including all necessary tools and accessories all complete and accepted by the Engineer-in-charge.					
12.24.1	Standard Quality	each	Tk. 374.24	Tk. 373.00	Tk. 362.00	Tk. 362.00
12.25	Supplying, fitting and fixing approved best quality foreign made stainless steel/brass door viewer/eye viewer from 12 mm to 20 mm diameter with adjustable thickness suitable for fitting from 37 mm to 50 mm thick door shutter in/c all necessary work as per direction and accepted by the Engineer-in-charge.	each	Tk. 613.51	Tk. 611.00	Tk. 589.00	Tk. 589.00

..... END OF CHAPTER TWELVE

CHAPTER 13 : WINDOW GRILL, VERANDAH GRILL & NETTING

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
13.1	Supplying, fitting and fixing window grills made of 16 mm dia M.S. rod @ 100 mm c/c fitted with 3 (three) nos. horizontal M.S. flat bar (38 mm x 6 mm) including fabrication, welding, riveting, cost of electricity, workshop charges, carriage, fixing with clamps in wall/RCC member for all floors, finished with anti-corrosive painting (Red-Oxide) etc. complete and accepted by the Engineer-in-charge. (Total weight per sqm should be approx. 19 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,609.63	Tk. 2,608.00	Tk. 2,589.00	Tk. 2,589.00
13.2	Supplying, fitting and fixing window grills of any design made of 20 mm x 3 mm mild steel section @ 100 mm c/c with outer frame of 25 mm x 6 mm F.I. bar including fabrication, welding, cost of electricity workshop charges, carriage, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-Oxide) etc complete for all floors and accepted by the Engineer-in-charge. (Total weight per sqm should be approx 10.00 kg. and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 1,959.30	Tk. 1,957.00	Tk. 1,932.00	Tk. 1,932.00

13.3.1	Supplying, fitting and fixing window grills of any design made with 20 mm x 5 mm F.I. bar @ 100 mm c/c as inner and outer section ; including fabricating, welding, cost of electricity and tools and plants, finished with anti-corrosive painting (Red-Oxide), carriage, cutting grooves, mending good the damages etc. complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be approx. 11 kg. and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,075.04	Tk. 2,072.00	Tk. 2,048.00	Tk. 2,048.00
13.3.2	Supplying, fitting and fixing window grills of any design made with 20 mm x 5 mm F.I. bar @ 100 mm c/c as inner members but no outer frame ; including fabrication, welding, cost of electricity, workshop charges, carriage, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-Oxide) etc. complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be approx. 9 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 1,902.45	Tk. 1,900.00	Tk. 1,875.00	Tk. 1,875.00
13.4.1	Supplying, fitting, fixing window grills of any design made with 25 mm x 6 mm F.I. bar @ 100 mm c/c as both outer and inner section ; including fabrication, welding, cost of electricity workshop charges, carriage, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-Oxide) etc. complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be approx. 19 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,798.22	Tk. 2,796.00	Tk. 2,771.00	Tk. 2,771.00
13.4.2	Supplying, fitting, fixing window grills of any design made with 25 mm x 6 mm F.I. bar @ 100 mm c/c as inner section but no outer frame ; including fabrication, welding, cost of electricity workshop charges, labour charges for fitting and fixing grills in position, local carriage charges, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-oxide) etc. complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be approx 16 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,491.32	Tk. 2,489.00	Tk. 2,464.00	Tk. 2,464.00
13.4.3	Supplying, fitting, fixing window grills of any design made with 25 mm x 6 mm F.I. bar as outer frame and 20 mm x 5 mm F.I. inner member @ 100 mm c/c ; including fabrication, welding, cost of electricity workshop charges, labour charges for fitting and fixing grills in position, local carriage charges, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-Oxide) etc. complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be approx 12 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,209.35	Tk. 2,207.00	Tk. 2,182.00	Tk. 2,182.00
13.5.1	Supplying, fitting and fixing window grill made of 10 mm x 10 mm M.S. solid bar @ 100mm c/c with outer frame of 38 mm x 6mm F.I. bar and 25 mm x 6 mm F.I. bar for clamp as per design including fabrication, welding, cost of electricity workshop charges, carriage, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-Oxide) etc. complete for all floors approved and accepted by the Engineer-in-charge. (Total weight per sqm should be approx 17 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,682.63	Tk. 2,679.00	Tk. 2,649.00	Tk. 2,649.00

13.5.2	Supplying, fitting and fixing window grill made of 12 mm x 12 mm M.S. solid bar @ 100mm c/c with outer frame of 38 mm x 6mm F.I. bar and 25 mm x 6 mm F.I. bar for clamp as per design including fabrication, welding, cost of electricity workshop charges, labour charges for fitting and fixing grills in position, local carriage charges, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-Oxide) etc. complete for all floors approved and accepted by the Engineer-in-charge. (Total weight per sqm should be approx 19 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,825.26	Tk. 2,822.00	Tk. 2,792.00	Tk. 2,792.00
13.5.3	Supply, fitting & fixing window grills made of 10 mm x 10 mm M.S. solid bar @ 100 mm c/c with 25 x 6 mm F.I. Bar for Clamp and no outer frame; including fabrication, welding, cost of electricity workshop charges, carriage, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-Oxide) etc. complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be approx 14 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,269.83	Tk. 2,268.00	Tk. 2,247.00	Tk. 2,247.00
13.5.4	Supply, fitting & fixing window grills made of 12 mm x 12 mm M.S. solid bar @100 mm c/c as inner members but no outer frame , with 25 x 6 mm F.I. Bar for Clamp;including fabrication, welding, cost of electricity workshop charges, carriage, cutting grooves, mending good the damages, tools and plants, finished with anti-corrosive painting (Red-Oxide) etc. all complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be approx 15 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,399.95	Tk. 2,398.00	Tk. 2,378.00	Tk. 2,378.00
13.5.5	Supply, fitting & fixing window grills made of 12 mm x 12 mm stainless steel (SS) bar with wall thickness 1.5 mm @ 100 mm c/c as inner frame but no outer frame; including fabrication, welding, cost of electricity workshop charges, carriage, cutting grooves, mending good the damages, tools and plants, clamps and other necessary accesories for fitting, fixings grills in position, polishing etc. complete for all floors accepted by the Engineer-in-charge.	sqm	Tk. 3,361.42	Tk. 3,359.00	Tk. 3,339.00	Tk. 3,339.00
13.6	Supplying, fitting and fixing veranda grill made of 10 mm dia M.S. rod provided @ 100 mm c/c in any position both ways as per design with outside frame 25 mm x 25 mm x 6 mm M.S. angle including fabricating, welding in each point, painting a coat of anti-corrosive painting (Red-Oxide), including cost of electricity, tools and plants, workshop charges, carriage of the same including cutting groves in the R.C.C. or brick work, mending good the damages with C.C. (1:1.5:3) etc. complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be aprox. 17 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,473.67	Tk. 2,472.00	Tk. 2,453.00	Tk. 2,453.00

13.7	Supplying, fitting and fixing verandah grill made of 10 mm dia M.S. rod provided @ 75 mm c/c in any position both ways as per design with outside frame of 25 mm x 25 mm x 6 mm M.S. angle and F.I. bar clamps including fabricating, welding in each point, painting a coat of anti-corrosive painting (Red-Oxide), including cost of electricity, tools and plants, workshop charges, carriage of the same including cutting grooves in the R.C.C. or brick work, mending good the damages with C.C. (1:1.5:3) etc, all complete for all floors accepted by the Engineer-in-charge. (Total weight per sqm should be aprox. 21 kg and add or deduct @ Tk. 100.00 for each kg/sqm excess or less respectively)	sqm	Tk. 2,781.09	Tk. 2,779.00	Tk. 2,761.00	Tk. 2,761.00
13.8	Supplying, fitting and fixing MS verandah grill as per design and shape with 75 mm x 50 mm MS box with 3 mm wall thickness middle rail (horizontal), one pair 50 mm x 25 mm MS hollow box with 3 mm wall thickness @ 750 mm c/c vertical and 12 mm x 12 mm MS solid bar @ 75 mm c/c horizontal and providing approx. 750 mm x 750 mm imergency exit with proper locking arrangement (if necessary) as per required measurement including cutting grooves in wall up to 150 mm depth and mending good the damages with C.C. (1:1.5:3), finished with anti-corrosive paint (Red-Oxide painting) etc. complete as per drawing and design and accepted by the Engineer.	sqm	Tk. 4,735.85	Tk. 4,734.00	Tk. 4,714.00	Tk. 4,714.00
13.9	Supplying, fitting and fixing MS verandah grill as per design and shape with 75 mm x 50 mm MS box with 3mm wall thickness middle rail (horizontal), 50 mm x 25 mm vertical rail with 3mm wall thickness @ approx. 750 mm c/c, 25x25 mm MS box with 2 mm wall thickness horizontally @100mm c/c as per required measurement including cutting grooves in wall up to 150 mm depth and mending good the damages with C.C. (1:1.5:3), finished with anti-corrosive paint (Red-Oxide painting) etc. complete as per drawing and design and accepted by the Engineer.	sqm	Tk. 4,626.49	Tk. 4,624.00	Tk. 4,607.00	Tk. 4,607.00
13.10	Supplying, fitting, fixing Jail patern M.S window grill made of 25 mm dia M.S Plain Rod @ 75 mm C/C fitted and fixed with M.S Angle frame 65 mm x 65 mm x 6 mm size placed 5 Nos. Horizontally and 2 Nos. Vertically including fabricating, welding, drilling, cutting and riveting the F.I bar and M.S angle clamp 10 Nos. including making holes in RCC Cloumn/Brick wall, fixing the clamps and mending good the damages with C.C (1:1.5:3), i finished with anti-corrosive paint (Red-Oxide painting) etc. all complete as per direction by Engineer-in-Charge.	sqm	Tk. 9,572.54	Tk. 9,568.00	Tk. 9,526.00	Tk. 9,524.00
13.11	Supplying, fitting, fixing Jail patern M.S Verandah grill made of 20 mm dia M.S Plain Rod @ 100 mm C/C fitted and fixed with M.S Angle frame 50 mm x 50 mm x 6 mm size placed 5 Nos. Horizontally and 2 Nos. Vertically including fabricating, welding, drilling, cutting and riveting the F.I bar and M.S angle clamp 10 Nos. including making holes in RCC Cloumn/Brick wall fixing the clamps and mending good the damaged with C.C (1:1.5:3), finished with anti-corrosive paint (Red-Oxide painting) etc. all complete as per direction by Engineer-in-Charge.	sqm	Tk. 5,830.10	Tk. 5,828.00	Tk. 5,807.00	Tk. 5,805.00

13.12	Supplying, Fitting, Fixing Jail patern M.S grill door made of 25mm dia M.S Plain Rod @ 75 mm C/C fitted and fixed with M.S Angle frame 65 mm x 65 mm x 6 mm size 7 Nos. placed Horizontally and 2 Nos. Vertically providing 25 mm dia M.S rod of 1200 mm long and 38 thick M.S plate up to a size of 1'-6"x1'-0" for locking arrangement, 38 mm dia G.I pipe of 1'-0" long for locking) including fabricating, welding, drilling, cutting and riveting the F.I bar and M.S angle clamp 8 Nos. including making holes in wall fixing the clamps and mending good the damaged with C.C (1:1.5:3) including anticorrosive paint etc. all complete as per direction by the Engineer-in-Charge.	sqm	Tk. 12,695.13	Tk. 12,689.00	Tk. 12,637.00	Tk. 12,635.00
13.13	Supplying, Fitting, Fixing Steel window shutter with frames top portion ventilation (2'-6") and bottom portion (3'-8") outer frame of the window made with 38mm x 38mm x 6mm M.S Angle and middle vertical member of frame made with 38mm x 38mm x 6mm M.S- T and shutter frame made of M.S Angle of same section, ventilation portion paired with 1.5 mm thick M.S sheet fixed @ 38mm C/C (20 Nos.) and bottom 3'-8" panel portion divided into 3 equal panes with 1.50 mm thick M.S sheet,including fabricating, welding, reveting, provided with necessary fittings like 12mm dia 150mm long socket bolt, 300mm long 25mm x 6mm size F.I bar adjustable cleats, 125mm long 50mm x 50mm x 6mm size M.S angle handle , 300mm long M.S angle clamp (8 Nos. hinge) including making holes in wall for fixing the clamps and mending good the damages with C.C (1:1.5:3) and a cot of anticorrosive paint etc. all complete as per direction by the Engineer-in-Charge.	sqm	Tk. 9,331.02	Tk. 9,316.00	Tk. 9,254.00	Tk. 9,253.00

..... END OF CHAPTER THIRTEEN

CHAPTER 14: ALUMINIUM DOOR, WINDOW FRAME & GLASS WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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14.1	Supplying fitting and fixing of aluminium swing door with spandrel as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 1.5 mm thick wall frame (size 101.60 mm, 44.45 mm), 2.0 mm mm thick shutter side (size 54 mm, 46 mm), 0.99 mm thick door glass bit (size 16.54 mm, 15.49 mm, 0.115 kg/m), 1.8 mm thick clousure section (size 101.60 mm, 42.93 mm),1.5 mm thick 106.60 mm clousure cover (0.392 kg/m), 4 mm thick floor bottom (size 101.60 mm, 12.70 mm, 1 kg/m), 1.8 mm thick shutter bottom (size 82.6 mm, 43.99 mm, 0.60 kg/m), 1.8 mm thick shutter top (size 51 mm, 43.99 mm, 1.88 kg/m) and 2.3 mm to 4.01 mm thick handle (size 101.60 mm, 38.10 mm, 25.40 mm short, 1.35 kg/m), 1.5 mm thick 31.75 mm shutter divider(0.535 kg/m), 1.5 mm thick 7.14 mm spandrel section of all aluminum members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 micrones in thickness or powder coated to any colour with a coat not less thane and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Doses of admixture to be fixed in consultation with design office) (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and th					
14.1.1	Anodized to bronze colour	sqm	Tk. 8,300.00	Tk. 8,289.00	Tk. 8,228.00	Tk. 8,228.00
14.1.2	Powder coated to any colour	sqm	Tk. 9,159.00	Tk. 9,149.00	Tk. 9,088.00	Tk. 9,088.00
14.2	Supplying fitting and fixing of aluminium swing door as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 1.5 mm thick wall frame (size 101.60 mm, 44.45 mm, 83.21 mm), 2.0 mm thick shutter side (size 54 mm, 46 mm), 0.99 mm thick door glass bit (size 16.54 mm, 15.49 mm, 0.115 kg/m), 1.8 mm thick clousure section (size 101.60 mm, 42.93 mm),1.5 mm thick 106.60 mm clousure cover (0.392 kg/m), 4 mm thick floor bottom (size 101.60 mm, 12.70 mm, 1 kg/m), 1.8 mm thick shutter bottom (size 82.6 mm, 43.99 mm, 0.60 kg/m), 1.8 mm thick shutter top (size 51 mm, 43.99 mm, 1.88 kg/m) and 2.3 mm to 4.01 mm thick handle (size 101.60 mm, 38.10 mm, 25.40 mm short, 1.35 kg/m) section of all aluminum members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 micrones in thickness or powder coated to any colour with a coat not less than 25 micrones in thickness and density of 4 mg per square cm etc. including all accessories icular brand of cement and particular stock of aggregates and method of use to be determined by mix design / trial mix at the cost of contractor and approved by the Engineer. (Use of admixture is subjected to prior approval of design office)hine and curing at least for 28 days, removing centering-shutterin					
14.2.1	Anodized to any colour	sqm	Tk. 7,588.00	Tk. 7,580.00	Tk. 7,527.00	Tk. 7,527.00
14.2.2	Powder coated to any colour	sqm	Tk. 8,274.00	Tk. 8,266.00	Tk. 8,213.00	Tk. 8,213.00

14.3	Supplying, fitting and fixing of aluminium composite swing doors with fixed partition as per the U. S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 1.5 mm thick wall frame (size 101.60, 44.45 mm, 0.86 kg/m), 1.8 mm thick shutter side (size 54 mm, 46 mm, 1.2 kg/m), 0.99 mm thick door glass bit (size 16.54 mm, 14.49 mm, 0.115 kg/m), 1.8 mm thick closure section (size 101.54 mm, 42.93 mm, 1.20 kg/m), 1.5 thick closure cover (101.60 mm, 0.45 kg/m), 4 mm thick floor bottom (size 101.60 mm, 12.70 mm, 1 kg/m), 1.8 mm thick shutter bottom (size 86.60 mm, 43.99 mm, 0.60 kg/m), 1.8 mm thick shutter top (size 51 mm, 43.99 mm, 1.88 kg/m) 2.3mm to 4.01mm thick handle (short) (size 101.60 mm, 38.10 mm, 25.40 mm, 1.35 kg/m), minimum 1.2 mm thick top and side (size 101.60 mm, 44.45 mm, 0.934 kg/m), minimum 1.2 mm thick bottom (size 101.35 mm, 44.45 mm, 0.734 kg/m), minimum 1.2 mm thick bottom cover (44.32 mm, 19.61mm, 0.333 kg/m), minimum 1.2 mm thick groove and curing at least for 28 days, removing centering-shuttering after specified time approved; including cost of water, electricity, other charges etc. all complete, approved and accepted by the Engineer-in-charge. (Doses of admixture to be fixed in consultation with design office) (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering) approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation: MAT) Note: The mix design shall be done from PWD testing laboratory or other recognized laboratories and be consulted with concerned design office. Design office and removing access road will be the property of the contractor					
14.3.1	Anodized to any colour	sqm	Tk. 3,911.00	Tk. 3,906.00	Tk. 3,876.00	Tk. 3,876.00
14.3.2	Powder coated to any colour	sqm	Tk. 4,319.00	Tk. 4,314.00	Tk. 4,284.00	Tk. 4,284.00
14.4	Supplying, fitting and fixing of aluminium sliding doors as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick outer bottom (size 75.50 mm, 17.79 mm, 0.47 kg/m), minimum 1.2 mm thick outer top (size 75.50 mm, 28.50 mm, 0.705 kg/m), minimum 1.2 mm thick shutter top (size 33 mm, 26.80 mm, 0.42 kg/m), minimum 1.2 mm shutter bottom (size 60 mm, 24 mm, 0.589 kg/m), minimum 1.2 mm thick outer side (size 75.50 mm, 19.90 mm, 0.52 kg/m), minimum 1.2 mm thick shutter lock (size 49.20 mm, 25.80 mm, 0.543 kg/m), minimum 1.2 mm thick inter lock (size 34.40 mm, 32.13 mm, 0.562 kg/m) sections all aluminum members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc. including all accessories like handle, sliding door key lock, sliding door wheel, sliding door mohar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5mm thick glass including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge. Size up to: 2100 mm x 2100 mm	OK				
14.4.1	Anodized to any colour	sqm	Tk. 3,269.00	Tk. 3,265.00	Tk. 3,222.00	Tk. 3,222.00
14.4.2	Powder coated to any colour	sqm	Tk. 3,794.00	Tk. 3,789.00	Tk. 3,746.00	Tk. 3,746.00

14.5	Supplying, fitting and fixing of aluminium sliding doors with mosquito mesh as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick outer bottom (size 95 mm, 22.39 mm, 0.476 kg/m), minimum 1.2 mm thick outer top (size 95.00 mm, 28.50 mm, 0.749 kg/m), minimum 1.2 mm thick shutter top (size 33 mm, 26.80 mm, 0.42 kg/m), minimum 1.2 mm thick shutter bottom (size 60 mm, 24.40mm, 0.589 kg/m), minimum 1.2 mm thick outer side(size 95.00 mm, 19.50 mm, 0.591kg/m), minimum 1.2 mm thick shutter lock (size 49.20 mm, 25.80 mm, 0.543 kg/m), minimum 1.2 mm thick inter lock (size 34.40 mm, 32.13 mm 0.562 kg/m), minimum 1.2 mm thick shutter divider (size 31.75 mm, 0.535 kg/m) sections all aluminum members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc. including all accessories like handle, sliding door key lock, sliding door wheel, sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5mm thick glass including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge. Size up to: 2100 mm x 2100 mm"	OK				
14.5.1	Anodized to any colour	sqm	Tk. 4,610.00	Tk. 4,605.00	Tk. 4,555.00	Tk. 4,555.00
14.5.2	Powder coated to any colour	sqm	Tk. 5,435.00	Tk. 5,431.00	Tk. 5,380.00	Tk. 5,380.00
14.6	Supplying, fitting and fixing of aluminium sliding doors with spandrel as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick outer bottom (size 75.50 mm, 17.79 mm, 0.605 kg/m), minimum 1.2 mm thick outer top (size 75.50 mm, 28.50 mm, 0.705 kg/m), minimum 1.2 mm thick shutter top (size 33 mm, 26.80 mm, 0.42 kg/m), minimum 1.2 mm thick shutter bottom (size 60 mm, 24 mm, 0.589 kg/m), minimum 1.2 mm thick outer side(size 75.50 mm, 19.90 mm, 0.52 kg/m), minimum 1.2 mm thick shutter lock (size 49.20 mm, 25.80 mm, 0.543 kg/m), minimum 1.2 mm thick inter lock (size 34.40 mm, 32.13 mm, 0.562 kg/m), minimum 1.2 mm thick shutter divider (size 31.75 mm, 0.535 kg/m) and minimum 1.5 mm thick spandrel (7.14 mm) sections all aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc. including all accessories like sliding door key lock, sliding door wheel, sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5mm thick glass including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge. Size up to: 1800 mm x 2100 mm	OK				
14.6.1	Anodized to any colour	sqm	Tk. 4,976.00	Tk. 4,971.00	Tk. 4,921.00	Tk. 4,921.00
14.6.2	Powder coated to any colour	sqm	Tk. 5,795.00	Tk. 5,790.00	Tk. 5,740.00	Tk. 5,740.00

14.7	Supplying, fitting and fixing of aluminium sliding window as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick outer bottom (size 75.50 mm, 32mm,0.605 kg/m), minimum 1.2 mm thick outer top (size 75.50 mm, 28.50 mm,0.705 kg/m), minimum 1.2 mm thick shutter top (size 33 mm,26.80 mm,0.42 kg/m), minimum 1.2 mm thick shutter bottom (size 60mm, 24.40 mm,0.589 kg/m), minimum 1.2 mm thick outer side (size 75.50 mm,19.90 mm,0.52 kg/m), minimum 1.2 mm thick shutter lock (size 49.20 mm 25.80 mm,0.543 kg/m) and minimum 1.2 mm thick inter lock (size 34.40 mm, 32.13 mm,0.562 kg/m) sections all aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc. including all accessories like sliding door key lock, sliding door wheel, sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5 mm thick glass including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge.	OK				
14.7.1	Size up to: 1500 mm x 1400 mm.					
14.7.1.1	Anodized to any colour	sqm	Tk. 3,814.00	Tk. 3,807.00	Tk. 3,764.00	Tk. 3,764.00
14.7.1.2	Powder coated to any colour	sqm	Tk. 4,583.00	Tk. 4,576.00	Tk. 4,534.00	Tk. 4,534.00
14.7.2	Size up to: 900 mm x 1400 mm					
14.7.2.1	Anodized to any colour	sqm	Tk. 5,190.00	Tk. 5,180.00	Tk. 5,124.00	Tk. 5,124.00
14.7.2.2	Powder coated to any colour	sqm	Tk. 6,523.00	Tk. 6,514.00	Tk. 6,458.00	Tk. 6,458.00
14.8	Supplying, fitting and fixing of aluminium sliding window with mosquito mesh as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick outer bottom (95.00 mm, 38.00 mm outer bottom,0.72 kg/m), minimum 1.2 mm thick outer top (95.00 mm, 28.5 mm outer top, 0.75 kg/m), minimum 1.2 mm thick shutter top (size 33 mm,26.80 mm,0.42 kg/m), minimum 1.2 mm thick shutter bottom (size 60mm, 24.40 mm,0.589 kg/m), minimum 1.2 mm thick outer side (95.00 mm, 19.50 mm outer side ,0.591 kg/m), minimum 1.2 mm thick shutter lock (size 49.20 mm 25.80 mm,0.543 kg/m) and minimum 1.2 mm thick inter lock (size 34.40 mm, 32.13 mm,0.562 kg/m) sections all aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc. including all accessories like sliding door key lock, density of 4 mg per square cm etc. including all accessories like sliding door key lock, sliding door wheel , sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5mm thick glass including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and	OK				
14.8.1	Size up to: 1524 mm x 1372 mm					
14.8.1.1	Anodized to any colour	sqm	Tk. 5,119.00	Tk. 5,113.00	Tk. 5,068.00	Tk. 5,068.00
14.8.1.2	Powder coated to any colour	sqm	Tk. 6,222.00	Tk. 6,216.00	Tk. 6,171.00	Tk. 6,171.00
14.8.2	Size up to: 1280 mm x 1372 mm					
14.8.2.1	Anodized to any colour	sqm	Tk. 5,828.00	Tk. 5,821.00	Tk. 5,766.00	Tk. 5,766.00
14.8.2.2	Powder coated to any colour	sqm	Tk. 7,054.00	Tk. 7,047.00	Tk. 6,992.00	Tk. 6,992.00
14.8.3	Size up to: 762 mm x 610 mm					

14.8.3.1	Anodized to any colour	sqm	Tk. 12,106.00	Tk. 12,080.00	Tk. 11,930.00	Tk. 11,930.00
14.8.3.2	Powder coated to any colour	sqm	Tk. 14,430.00	Tk. 14,404.00	Tk. 14,254.00	Tk. 14,254.00
14.9	Supplying, fitting and fixing of equal dimension frame(EDF) section aluminium sliding window with mosquito mesh as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 1.5 mm thick outer bottom (size 91.50 mm, 38.00 mm), 1.5 mm thick outer top (size 91.00 mm, 28.70 mm), 1.5 mm thick shutter top (size 65.00 mm, 27.30 mm), 1.5 mm thick shutter bottom (size 72.20 mm, 24.65 mm), 1.5 mm thick outer side (size 91.50 mm, 28.50 mm), 1.5 mm thick shutter lock (size 65.30 mm, 26.00 mm), 1.5 mm thick shutter inter lock (size 55.20 mm, 26.00 mm), 1.5 mm thick mesh section (size 55.20 mm, 13.20 mm), all aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc. including all accessories like sliding door key lock, sliding door wheel, sliding door mohiar, sliding door n density of 4 mg per square cm etc. including all accessories like sliding door key lock, sliding door whee , sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5mm thick glass including labour charge for fitting of accessori					
14.9.1	Size up to: 1524 mm x 1372 mm					
14.9.1.1	Anodized to any colour	sqm	Tk. 6,482.00	Tk. 6,471.00	Tk. 6,402.00	Tk. 6,402.00
14.9.1.2	Powder coated to any colour	sqm	Tk. 6,767.00	Tk. 6,761.00	Tk. 6,715.00	Tk. 6,715.00
14.9.2	Size up to: 1280 mm x 1372 mm					
14.9.2.1	Anodized to any colour	sqm	Tk. 7,025.00	Tk. 7,018.00	Tk. 6,964.00	Tk. 6,964.00
14.9.2.2	Powder coated to any colour	sqm	Tk. 7,704.00	Tk. 7,697.00	Tk. 7,642.00	Tk. 7,642.00
14.9.3	Size up to: 762 mm x 610 mm					
14.9.3.1	Anodized to any colour	sqm	Tk. 14,305.00	Tk. 14,279.00	Tk. 14,129.00	Tk. 14,129.00
14.9.3.2	Powder coated to any colour	sqm	Tk. 15,579.00	Tk. 15,552.00	Tk. 15,402.00	Tk. 15,402.00

14.10	Supplying, fitting and fixing of aluminium fixed composite window as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick outer bottom (size 75.50 mm, 32 mm, 0.605 kg/m), minimum 1.2 mm thick outer top (size 75.50mm, 28.50 mm, 0.705 kg/m), minimum 1.2 mm thick shutter top (size 33 mm, 26.80 mm, 0.42 kg/m), minimum 1.2 mm thick shutter bottom (size 60 mm, 24.40 mm, 0.589 kg/m), minimum 1.2 mm thick outer side (size 75.50 mm, 19.90 mm, 0.52 kg/m), minimum 1.2 mm thick shutter lock (size 49.20 mm, 25.80 mm, 0.543 kg/m), minimum 1.2 mm thick inter lock (size 34.40 mm, 32.13 mm, 0.562 kg/m), minimum 1.2 mm thick bottom cover (size 17.40 mm, 31.78 mm, 0.25 kg/m), minimum 1.2 mm thick fixed bottom (size 76.20 mm, 38.10 mm, 0.640 kg/m), minimum 1.2 mm thick groove cover (size 57.15 mm, 15.88 mm, 0.362 kg/m) and minimum 1.2 mm thick top and side (size 76.20 mm, 38.10 mm, 0.769 kg/m) sections all aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc.including all accessories like sliding door key lock, sliding door wheel, sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5 mm thick glass including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge. Size up to: 1500 mm x 2100 mm.	OK				
14.10.1	Anodized to any colour	sqm	Tk. 4,086.00	Tk. 4,082.00	Tk. 4,043.00	Tk. 4,043.00
14.10.2	Powder coated to any colour	sqm	Tk. 4,786.00	Tk. 4,782.00	Tk. 4,743.00	Tk. 4,743.00
14.11	Supplying, fitting and fixing of aluminium top hung/casement windows as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 1.5 mm thick casement outer (size 38.86 mm, 36.36 mm, 0.395kg/m), 1.9 mm thick casement shutter (size 47.62 mm, 28.57 mm, 0.692 kg/m), and 1.245 mm thick shutter glass clip (size 15.87 mm, 27 mm, 0.188 kg/m) sections of all aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 1.5 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc.including all accessories 4-bar hinge, casement handle, bolts and nuts keeping provision for fitting 5 mm thick glass including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge.	OK				
14.11.1	Anodized to any colour	sqm	Tk. 11,311.00	Tk. 11,300.00	Tk. 11,159.00	Tk. 11,159.00
14.11.2	Powder coated to any colour	sqm	Tk. 11,524.00	Tk. 11,513.00	Tk. 11,387.00	Tk. 11,387.00

14.12	Supplying fitting and fixing of aluminium corrugated curtain wall as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick bottom cover (size 17.40 mm 31.78 mm, 0.250 kg/m), minimum 1.2 mm thick fixed bottom (size 76.20 m 38.10 mm, 0.640 kg/m), minimum 1.2 mm thick grouve cover(size 57.15 mm, 15.80 mm 0.362 kg/m), minimum 1.2 mm thick top and side (size 76.2 mm, 57.15 mm, 0.769 kg/m) and 1.5 mm thick spandrel (7.14 mm) sections of will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 micrones in thickness or powder coated to any colour with a coat not less than 25 micrones in thickness and density of 4 mg per square cm etc. including all accessories like rivet, screws fixed neoprene etc. complete in all respect as per drawing and accepted by the Engineer-in-charge.	OK				
14.12.1	Size up to: 3000 mm x 2100 mm					
14.12.1.1	Anodized to any colour	sqm	Tk. 7,516.00	Tk. 7,512.00	Tk. 7,471.00	Tk. 7,471.00
14.12.1.2	Powder coated to any colour	sqm	Tk. 8,604.00	Tk. 8,600.00	Tk. 8,559.00	Tk. 8,559.00
14.12.2	Size up to: 3000 mm x 750 mm					
14.12.2.1	Anodized to any colour	sqm	Tk. 8,924.00	Tk. 8,914.00	Tk. 8,845.00	Tk. 8,845.00
14.12.2.2	Powder coated to any colour	sqm	Tk. 10,203.00	Tk. 10,193.00	Tk. 10,124.00	Tk. 10,124.00
14.13	Supplying fitting and fixing of aluminium fixed fan light as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick fixed bottom (size 76.2 mm, 38.10 mm, 0.640 kg/m), minimum 1.2 mm thick (17.40 mm, 31.78 mm bottom cover 0.250 kg/m) and minimum 1.2 mm thick grouve cover (size 57.15mm, 15.80 mm, 0.362 kg/m) and minimum 1.2 mm thick top and side section (76.20 mm, 16 mm, 38.10 mm, 0.769 kg/m) fixed with above mentioned frames and all aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 micrones in thickness or powder coated to any colour with a coat not less than 25 micrones in thickness and density of 4 mg per square cm etc. including all accessories, neoprene, sealant, nut and bolt etc. complete in all respect as per drawing and accepted by the Engineer-in-charge.	OK				
14.13.1	Anodized to any colour	sqm	Tk. 5,913.00	Tk. 5,887.00	Tk. 5,751.00	Tk. 5,751.00
14.13.2	Powder coated to any colour	sqm	Tk. 6,377.00	Tk. 6,350.00	Tk. 6,215.00	Tk. 6,215.00
14.14	Supplying fitting and fixing of aluminium fixed louver as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having minimum 1.2 mm thick wall frame (size 76.20 mm, 38.10 mm, 0.564 kg/m), 1.5 mm thick louver (size 55.02 mm, 37.39 mm, 0.281 kg/m) and minimum 1.2 mm thick louver outer section (size 39.67 mm, 15.06 mm) and inner inclined aluminium louver section fixed with above mentioned frame and all aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 micrones in thickness or powder coated to any colour with a coat not less than 25 micrones in thickness and density of 4 mg per square cm etc. including all accessories, neoprene, sealant, nuts and bolts etc.	OK				
14.14.1	Anodized to any colour	sqm	Tk. 10,881.00	Tk. 10,845.00	Tk. 10,680.00	Tk. 10,680.00
14.14.2	Powder coated to any colour	sqm	Tk. 12,300.00	Tk. 12,265.00	Tk. 12,100.00	Tk. 12,100.00

14.15	Supplying, fitting, fixing of aluminium sliding fly proof Netting shutter as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 2.0 mm thick outer frame (size 38mm x 38mm), minimum 1.2 mm thick mosquito net shutter frame mesh section (42.2 mm x13.2mm). All aluminium members will be anodized to aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in thickness or powder coated to any colour with a coat not less than 25 microns in thickness and density of 4 mg per square cm etc. including all accessories like wheel, sealant, rubber, bolts and nuts keeping provision for fitting fiber mesh including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and electricity, all complete in all respect as per drawing and accepted by the Engineer-in-charge.	OK				
14.15.1	Anodized to any colour	sqm	Tk. 3,798.00	Tk. 3,796.00	Tk. 3,782.00	Tk. 3,782.00
14.15.2	Powder coated to any colour	sqm	Tk. 4,372.00	Tk. 4,370.00	Tk. 4,356.00	Tk. 4,356.00
14.16	Supplying, fitting and fixing in Aluminium door frames, windows, partitions and curtain wall distortion free glass of approved quality and shade including cost of fitting fixing all necessary accessories etc. complete in all respect as per drawing and accepted by the Engineer-in-charge.					
14.16.1.1.1	5 mm thick tinted glass	sqm	Tk. 1,492.00	Tk. 1,489.00	Tk. 1,473.00	Tk. 1,473.00
14.16.1.1.2	5 mm thick tinted tempered glass	sqm	Tk. 2,394.00	Tk. 2,390.00	Tk. 2,374.00	Tk. 2,374.00
14.16.1.2.1	5 mm thick clear glass	sqm	Tk. 1,229.00	Tk. 1,225.00	Tk. 1,209.00	Tk. 1,209.00
14.16.1.2.2	5 mm thick clear tempered glass	sqm	Tk. 2,491.00	Tk. 2,487.00	Tk. 2,471.00	Tk. 2,471.00
14.16.1.3	5 mm thick ambushed glass	sqm	Tk. 1,215.00	Tk. 1,211.00	Tk. 1,195.00	Tk. 1,195.00
14.16.1.4.1	5 mm thick reflective glass	sqm	Tk. 1,659.00	Tk. 1,655.00	Tk. 1,639.00	Tk. 1,639.00
14.16.1.4.2	5 mm reflective tempered glass	sqm	Tk. 2,879.00	Tk. 2,875.00	Tk. 2,859.00	Tk. 2,859.00
14.16.2.1.1	6 mm thick clear glass	sqm	Tk. 1,492.00	Tk. 1,489.00	Tk. 1,473.00	Tk. 1,473.00
14.16.2.1.2	6 mm thick clear tempered glass	sqm	Tk. 2,879.00	Tk. 2,875.00	Tk. 2,859.00	Tk. 2,859.00
14.16.2.2	6 mm thick ambushed glass	sqm	Tk. 2,602.00	Tk. 2,598.00	Tk. 2,582.00	Tk. 2,582.00
14.16.2.3.1	6 mm thick reflecting glass	sqm	Tk. 1,839.00	Tk. 1,835.00	Tk. 1,819.00	Tk. 1,819.00
14.16.2.3.2	6 mm thick reflecting tempered glass	sqm	Tk. 2,838.00	Tk. 2,834.00	Tk. 2,818.00	Tk. 2,818.00
14.16.3.1	8 mm thick clear glass	sqm	Tk. 2,120.00	Tk. 2,115.00	Tk. 2,093.00	Tk. 2,093.00
14.16.3.2	8 mm thick clear tempered glass	sqm	Tk. 3,853.00	Tk. 3,848.00	Tk. 3,827.00	Tk. 3,827.00
14.16.4.1	10 mm thick clear glass	sqm	Tk. 2,359.00	Tk. 2,352.00	Tk. 2,326.00	Tk. 2,326.00
14.16.4.2	10 mm thick clear tempered glass	sqm	Tk. 4,273.00	Tk. 4,266.00	Tk. 4,240.00	Tk. 4,240.00
14.16.5.1	12 mm thick clear glass	sqm	Tk. 3,094.00	Tk. 3,087.00	Tk. 3,061.00	Tk. 3,061.00
14.16.5.2	12 mm thick clear tempered glass	sqm	Tk. 5,937.00	Tk. 5,930.00	Tk. 5,904.00	Tk. 5,904.00
14.16.6.1	13.52mm (6mm tempered glass+1.52mm PVB lamination+6mm tempered glass) foreign made tempered lamination glass	sqm	Tk. 9,080.00	Tk. 9,071.00	Tk. 9,034.00	Tk. 9,034.00
14.16.6.2	13.52 mm(6mm low-E glass+1.52mm PVB lamination+6mm glass) foreign made tempered lamination glass	sqm	Tk. 11,369.00	Tk. 11,359.00	Tk. 11,322.00	Tk. 11,322.00
14.16.7.1	17.52 mm (8mm tempered glass+1.52mm PVB lamination+8mm tempered glass) foreign made tempered double glazing glass	sqm	Tk. 11,065.00	Tk. 11,052.00	Tk. 11,002.00	Tk. 11,002.00

14.16.7.2	17.52 mm(8mm tempered Low-E glass+1.52mm PVB lamination+8mm clear tempered glass) foreign made tempered double glazing glass	sqm	Tk. 12,452.00	Tk. 12,439.00	Tk. 12,389.00	Tk. 12,389.00
14.16.8.1	24mm (6mm tempered glass+12mm argon gas filled gap+6mm tempered glass) foreign made tempered double glazing glass	sqm	Tk. 7,977.00	Tk. 7,964.00	Tk. 7,911.00	Tk. 7,911.00
14.16.8.2	24 mm(6mm clear tempered Low-E glass+12mm argon gas filled gap+6mm clear tempered glass) foreign made tempered double glazing glass	sqm	Tk. 9,226.00	Tk. 9,212.00	Tk. 9,160.00	Tk. 9,160.00
14.17	Supplying, fabrication and Installation of approved colour 4 mm thick Fire Rated Aluminium Composite Panel (Alucobond/Alcopanel/Alpolc or equivalent) of 0.5 mm thick aluminum skin cover both side with 3 coat PVDF on the front face panel sheet and non-combustible mineral filled core/ aluminum hydroxide filled (fr) core, skin cover sheet in Peralumam -100, ENAW -5005 (AlMg1) on the both side(Total weight 7.60kg/sqm), as per US Architectural Aluminium Manufacturer's Association (AAMA) in column, beam and wall surface providing with 44 mm x 44 mm x 1.5 mm aluminum section including supply of necessary hardware, consumables, scaffolding etc fitted with necessary accessories like rivet, screw, rowel bolt etc. all complete in all respect as per drawing and direction of the Engineer-in-charge	sqm	Tk. 13,051.00	Tk. 13,048.00	Tk. 12,994.00	Tk. 12,994.00
14.18	Supplying, fitting and fixing of aluminium curtain wall with aluminium top hung/casement windows as per the US Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 134mm , 70mm, 3.5mm curtain wall sub-mullim outer and inner frame, 3.0 mm thick casement shutter (size 58.40 mm, 37.20 mm, 57.40 mm) 1.2 mm thick shutter glass strips, all aluminium member will be powder coated with coat not less than 1.5 micron in thickness including all accessories 4 bar hinge casement handle, 50 mm x 50 mm x 6 mm aluminium angle, steel rowel bolt, screws , rivets norton tape masking tape, structural sealant, gum bracket rod etc. all complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 13,409.00	Tk. 13,404.00	Tk. 13,320.00	Tk. 13,320.00
14.19	Supplying, fitting and fixing of aluminium curtain wall with aluminium top hung/casement windows as per the US Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 100 mm , 50mm, 2mm curtain wall sub-mullim outer and inner frame, 3.0 mm thick casement shutter (size 58.40 mm, 37.20 mm, 57.40 mm), 1.2 mm thick shutter glass strips, all aluminium member will be powder coated with coat not less than 1.5 micron in thickness including all accessories 4 bar hinge casement handle, 50 mm x 50 mm x 6 mm aluminium angle, steel rowel bolt, screws , rivets norton tape masking tape, structural sealant, gum bracket rod etc. all complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 10,815.00	Tk. 10,803.00	Tk. 10,696.00	Tk. 10,696.00
14.20	Supplying, fitting and fixing of aluminium curtain wall and aluminium top hung/casement windows as per the US Architectural Aluminium Manufacturer's Association (AAMA) standard specification and BDS 1879:2014 having 50 mm , 50 mm, 2 mm curtain wall sub-mullim outer and inner frame, 3.0 mm thick casement shutter (size 58.40 mm, 37.20 mm, 57.40 mm) , 1.2 mm thick shutter glass strips, all aluminium member will be powder coated with coat not less than 1.5 micron in thickness including all accessories 4 bar hinge casement handle, 50 mm x 50 mm x 6 mm aluminium angle, steel rowel bolt, screws , rivets norton tape masking tape, structural sealant, gum bracket rod etc. all complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 9,378.00	Tk. 9,366.00	Tk. 9,259.00	Tk. 9,259.00

14.21	Supplying, fitting and fixing of aluminium curtain wall along with 10 mm thick clear tempered glass as per the US Architectural aluminium Manufacturer's Association (AAMA) Standard Specification and BDS 1879:2014 having 115 mm , 64 mm, 3 mm curtain wall sub-mullion outer and inner frame, 3.0 mm thick casement shutter (size 58.40 mm, 37.20 mm, 57.40 mm) , 1.2 mm thick shutter glass strips, all aluminium member will be powder coated with coat not less than 1.5 micron in thickness including all accessories 4 bar hinge casement handle, 50 mm x 50mm x 6mm aluminium angle, steel rowel bolt, screws , rivets norton tape masking tape, structural sealant, gum bracket rod etc. all complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 11,770.00	Tk. 11,764.00	Tk. 11,694.00	Tk. 11,694.00
14.22	Supplying, fitting and fixing of 12 mm thick clear tempered glass wall upto 3.0 m height with vertical fin glass support of same thickness and support shall be at least 1.2 m c/c fixed properly with glass by silicon glue with supply and fittings of all required accessories such as SS u channel, nut bolts, aluminium angle, steel rowel bolt, screws , rivets norton tape masking tape, structural sealant, gum bracket rod etc. all complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 21,510.00	Tk. 21,500.00	Tk. 21,414.00	Tk. 21,414.00
14.23	Supplying, fitting and fixing of 12 mm thick tempered glass wall from 3.0 m to 6.0 m height with spider point fixing joint and vertical fin glass support of same thickness, support shall be at least 1.2 m c/c fixed properly with main glass by silicon glue with supply and fittings of all required accessories such as SS u channel, nut bolts, aluminium angle, steel rowel bolt, screws , rivets norton tape masking tape, structural sealant, gum bracket rod etc. all complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 22,108.00	Tk. 22,103.00	Tk. 22,042.00	Tk. 22,042.00
14.24	Supplying, fitting and fixing of 12 mm thick clear tempered glass wall from 6.0 m to 8.2 m height with spider point fixing joint and vertical/ horizontal fin glass support of double thickness, support shall be at least 2 m c/c fixed properly with main glass by silicon glue with supply and fittings of all required accessories such as SS u channel, nut bolts, aluminium angle, steel rowel bolt, screws , rivets norton tape masking tape, structural sealant, gum bracket rod etc. all complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 22,488.00	Tk. 22,485.00	Tk. 22,428.00	Tk. 22,428.00
14.25	Supplying fitting and fixing of aluminium box fixed louver as per the U.S. Architectural Aluminium Manufacturer's Association (AAMA) standard specification having 50mm x 25 mm, 1.5 mm thick outer section and horizontally @ 75 mm c/c fixed with above mentioned frame and all aluminium members will be anodized to aluminium with a coat not less than 15 micron in thickness and density of 4 mg per square cm etc. including all accessories, neoprene, sealant, nuts and bolts etc. complete in all respect as per drawing and accepted by the Engineer.	OK				
14.25.1	Anodized to any colour	sqm	Tk. 8,760.00	Tk. 8,756.00	Tk. 8,726.00	Tk. 8,726.00
14.25.2	Powder coated to any colour	sqm	Tk. 9,870.00	Tk. 9,865.00	Tk. 9,836.00	Tk. 9,836.00

14.26	Supplying fitting and fixing of aluminium ventilation louver as per the U.S Architectural Aluminium Manufacturer's Association (AAMA) standard specification having 38 mmx 38mm, 2 mm thick outer frame, 39.67 mm x15.06 mm louver outer of minimum 1.2 mm thick and horizontal louver 55.02 mm x 37.39 mm of 1.50 mm thick @ 25 mm c/c fixed with above mentioned frame and aluminium members will be Powder coated colour etc. including all accessories, neoprene, sealant, nuts and bolts etc. complete in all respect as per drawing and accepted by the Engineer.	OK				
14.26.1	Anodized to any colour	sqm	Tk. 15,662.00	Tk. 15,644.00	Tk. 15,547.00	Tk. 15,547.00
14.26.2	Powder coated to any colour	sqm	Tk. 17,589.00	Tk. 17,571.00	Tk. 17,473.00	Tk. 17,473.00

..... END OF CHAPTER FOURTEEN

CHAPTER 15 : CEMENT PLASTER, FAIR-FACE PLASTER AND POINTING WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
15.1.1	Minimum 12 mm thick cement sand (F.M. 1.2) plaster (1:4) with fresh cement to both inner-and outer surface of wall, finishing the corner and edges including washing of sand, cleaning the surface, curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 299.37	Tk. 297.00	Tk. 281.00	Tk. 281.00
15.1.2	Minimum 12 mm thick cement sand (F.M. 1.2) plaster (1:4) with fresh cement to outer surface of wall , finishing the corner and edges including washing of sand, cleaning the surface, curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 315.88	Tk. 314.00	Tk. 298.00	Tk. 298.00
15.2	Minimum 12 mm thick cement sand (F.M. 1.2) plaster with neat cement finishing to plinth wall with cement (1:4) up to 150 mm below ground level including washing of sand, finishing the edges and corners and curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 361.89	Tk. 360.00	Tk. 340.00	Tk. 340.00

15.3	Minimum 12 mm thick cement sand (F.M. 1.2) plaster with neat cement finishing to dado with cement (1:4) up to 150 mm including washing of sand, finishing the edges and corners and curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 372.90	Tk. 371.00	Tk. 351.00	Tk. 351.00
15.4	Minimum 12 mm thick cement sand (F.M. 1.2) plaster (1:6) having with fresh cement to both inner and outer surface of wall, finishing the edges and corners including washing of sand, cleaning the surface, curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 291.45	Tk. 289.00	Tk. 273.00	Tk. 273.00
15.5	Minimum 6 mm thick cement sand (F.M. 1.2) plaster (1:4) with fresh cement to ceiling, R.C.C. columns, beams, surface of stair case, sunshades, cornices, railings, drop wall, louvers, fins and finishing the corners and edges including washing of sand, cleaning the surface, curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 293.29	Tk. 291.00	Tk. 272.00	Tk. 272.00
15.6	Minimum 6 mm thick cement sand (F.M. 1.2) plaster (1:3) with fresh cement to ceiling, R.C.C. columns, beams, surface of stair case, sunshades, cornices, railings, drop wall, louvers, fins and finishing the corners and edges including washing of sand, cleaning the surface, curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 284.96	Tk. 283.00	Tk. 267.00	Tk. 267.00
15.7	Flush pointing to brick wall with cement sand (F.M. 1.2) mortar (1:2) with cement including raking out the joints, curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 196.19	Tk. 194.00	Tk. 182.00	Tk. 182.00
15.8	Rule pointing to brick wall with cement sand (F.M. 1.2) mortar (1:2) with fresh cement and raking out the joints, curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 232.88	Tk. 231.00	Tk. 217.00	Tk. 217.00
15.9	Raised or tack pointing work to brick wall with cement sand (F.M. 1.2) mortar (1:2) with fresh cement of minimum 12 mm x 20 mm in size including raking out the joints and necessary scaffolding, curing at least for 7 days, cleaning each and every brick, cutting the bricks with pumice stone where necessary etc. all complete in all floors, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M) ground floor.	sqm	Tk. 608.90	Tk. 600.00	Tk. 565.00	Tk. 565.00

15.10	Providing drip course or nosing at the edge of sunshade or cornice with cement sand (F.M. 1:2) mortar (1:2) with fresh cement, curing at least for 7 days, cost of water, electricity, scaffolding and other charges etc. all complete in all respect as per drawing and accepted by the Engineer-in-charge.	meter	Tk. 169.72	Tk. 169.00	Tk. 157.00	Tk. 157.00
15.11.1	Making groove in plaster on outside wall upto 50 mm X 6 mm size including scaffolding, curing etc. complete as per drawing and accepted by the Engineer-in-charge.	meter	Tk. 29.39	Tk. 29.00	Tk. 27.00	Tk. 27.00
15.11.2	Making groove in plaster on outside wall upto 38 mm X 6 mm size including scaffolding, curing etc. complete as per drawing and accepted by the Engineer-in-charge.	meter	Tk. 45.06	Tk. 44.00	Tk. 42.00	Tk. 42.00
15.12	Minimum 10 mm thick fair face plaster using premix cement plaster (CP2 LGT/Stucco base premix or equivalent); initial background should be 19 mm rough cement plaster (1:4); then after passing 7 days curing time surface of newly constructed rough plaster should be thoroughly cleaned using a stiff brush to remove any dust, loose particles, wood pieces, mould oil etc, then the background needs to be thoroughly neutralized by applying potable water; Mixing premix cement plaster with water at water/powder ratio according to the specification of the manufacturer to form paste and apply on rough plaster surface; powder for making paste must be mixed with potable water only; apply the paste on the rough plaster surface by trowel to get smooth and shiny surface and cure the finished plaster surface for 2-3 days(3-4 times daily); finally water repellent coating to protect concrete surface; all complete with necessary scaffolding etc. accepted by the Engineer-in-charge.(Rate is excluding the cost of 19 mm thick rough cemen aluminium bronze/silver/ss/black colour with a coat not less than 15 microns in	sqm	Tk. 2,046.54	Tk. 2,040.00	Tk. 1,991.00	Tk. 1,991.00
15.13.1	Making groove in fair face plaster on outside wall upto 50 mm X 6 mm size including scaffolding, curing etc. complete as per drawing and accepted by the Engineer-in-charge.	meter	Tk. 55.20	Tk. 55.00	Tk. 53.00	Tk. 53.00
15.13.2	Making groove in fair face plaster on outside wall upto 38 mm X 6 mm size including scaffolding, curing etc. complete as per drawing and accepted by the Engineer-in-charge.	meter	Tk. 67.43	Tk. 67.00	Tk. 64.00	Tk. 64.00
15.14	Minimum 1mm thick fair face coating using Master Emaco N 303 Rendroc FC/Top Crete 220 or equivalent product micro topping skim coat, a high performance single component polymer fortified, colored, cementitious micro topping that can be applied on a variety of horizontal and vertical plaster surfaces; initially background should be thoroughly cleaned using a stiff brush to remove any dust, loose particles, mould oil etc.; then the background needs to be thoroughly neutralized by applying potable water ; The ratio of fair face coating and water is 1.5:1 to 2:1 or as required for workability ; powder will be mixed with potable water in a drum by blending machine; the mixture can be spread to any horizontal and vertical plaster surface using trowel to get smooth and shiny surface; all complete with necessary scaffolding etc. accepted by the Engineer-in-charge. (Rate is excluding the cost of 19 mm thick rough cement plaster) Ground floor.	sqm	Tk. 1,508.83	Tk. 1,508.00	Tk. 1,496.00	Tk. 1,496.00

15.15.1	Minimum 12mm thick cement sand (F.M-1.2) water proof, damp proof, dry and breathable plaster (1:4) with water proof Izonil Cement (STN-EN -1015-11 , Compressive Strength 34 MPa ,Max depth of water penetration into hardened plaster is < 1 mm) or equivalent compound to wall surface, finishing the corner and edges including washing of sand, cleaning the surface, scaffolding and curing at least for 3 days, cost of water, electricity and other charges etc. all complete in all respect as per drawing and accepted by the Engineer –in-charge. (Izonil Cement/equivalent compound: Water proof, damp proof, dry and breathable cement).	sqm	Tk. 645.30	Tk. 643.00	Tk. 626.00	Tk. 626.00
15.15.2	Minimum 19mm thick cement sand (F.M-1.2) water proof, damp proof, dry and breathable plaster (1:4) with water proof Izonil Cement (STN-EN -1015-11 , Compressive Strength 34 MPa ,Max depth of water penetration into hardened plaster is < 1 mm) /equivalent compound to wall surface, finishing the corner and edges including washing of sand, cleaning the surface, scaffolding and curing at least for 3 days, cost of water, electricity and other charges etc. all complete in all respect as per drawing and accepted by the Engineer –in-charge. (Izonil Cement/equivalent compound: Water proof, damp proof, dry and breathable cement).	sqm	Tk. 830.21	Tk. 828.00	Tk. 810.00	Tk. 810.00
15.16	Added rate for plaster and all types of pointing for additional floor above ground floor	sqm	Tk. 15.00	Tk. 14.00	Tk. 13.00	Tk. 13.00
15.17	Pre-cast ventilators of minimum 25 mm thick of any design with cement sand (F.M.1.2) mortar (1:4) fitted and fixed in position, finishing with cement sand (F.M. 1.2) plaster (1:6) including necessary scaffolding and curing, in all floors accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)	sqm	Tk. 2,895.24	Tk. 2,879.00	Tk. 2,702.00	Tk. 2,702.00
15.18	75 mm x 75 mm cornice of brick masonry in cement sand (F.M. 1.2) mortar (1:6) having minimum 12 mm thick cement sand (F.M. 1.2) plaster (1:6) including scaffolding and curing at least for 7 days in all floors accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)	meter	Tk. 267.12	Tk. 264.00	Tk. 247.00	Tk. 247.00
15.19	Supplying, fitting, fixing of broken and sharp glass 19 mm to 32 mm height, av. 19 mm width placed at a distance approximately 25 mm center to center, embedding the same to 12 mm to 19 mm thick cement plaster with neat cement finishing including necessary scaffolding approved and accepted by the Engineer-in-charge. (Cement: CEM-II/B-M)	sqm	Tk. 301.14	Tk. 300.00	Tk. 294.00	Tk. 294.00

..... END OF CHAPTER FIFTEEN

CHAPTER 16 : PAINTING AND POLISHING

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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16.1.1	Exterior standard acrylic emulsion paint of approved best quality and color having water resisting properties and resistance properties against fungi, fading & flaking delivered from authorized local agent of the manufacturer in a sealed container; applying to exterior surface with surface preparation including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying necessary exterior sealer of specified brand on prepared surface; then applying necessary exterior putty of specified brand for levelling, spot filling, crack filling and cutting by sand paper/zero water paper; finally applying 2 coats of exterior emulsion paint spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 274.05	Tk. 273.00	Tk. 261.00	Tk. 261.00
16.1.2	Exterior premium acrylic emulsion paint of approved best quality and color with high performance against dirt picking tendency and efflorescence resistance properties along with water resisting properties and resistance properties against fungi, fading and flaking from authorized local agent of the manufacturer in a sealed container; applying to exterior surface with surface preparation including cleaning drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying necessary exterior sealer of specified brand on prepared surface; then applying necessary exterior putty of specified brand for levelling, spot filling, crack filling and cutting by sand paper/zero water paper; finally applying 2 coats of exterior emulsion paint spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 298.46	Tk. 297.00	Tk. 286.00	Tk. 286.00
16.2.1	Interior standard acrylic emulsion paint (plastic or matt finish) of approved best quality and colour delivered from authorized local agent of the manufacturer in a sealed container; applying to interior wall and ceiling with surface preparation including cleaning drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying necessary interior sealer of specified brand on prepared surface; then applying necessary interior putty of specified brand for levelling, spot filling, crack filling and cutting by sand paper/zero water paper; finally applying 2 coats of interior emulsion paint spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 258.06	Tk. 257.00	Tk. 245.00	Tk. 245.00

16.2.2	Interior premium acrylic emulsion painting (silky finish) of approved best quality and colour delivered from authorized local agent of the manufacturer in a sealed container; applying to interior wall and ceiling with surface preparation including cleaning drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying necessary interior sealer of specified brand on prepared surface; then applying necessary interior putty of specified brand for levelling, spot filling, crack filling and cutting by sand paper/zero water paper; finally applying 2 coats of interior emulsion paint spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors and accepted by the Engineer-in-charge	sqm	Tk. 283.42	Tk. 282.00	Tk. 270.00	Tk. 270.00
16.2.3	Interior super premium acrylic emulsion painting (odorless) of approved best quality and colour delivered from authorized local agent of the manufacturer in a sealed container; applying to interior wall and ceiling with surface preparation including cleaning drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying necessary interior sealer of specified brand on prepared surface; then applying necessary interior putty of specified brand for levelling, spot filling, crack filling and cutting by sand paper/zero water paper, finally applying 2 coats of interior emulsion paint spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors accepted by the Engineer-in-charge.	sqm	Tk. 314.06	Tk. 313.00	Tk. 301.00	Tk. 301.00
16.3.1	Standard synthetic enamel paint of approved best quality and colour delivered from authorized local agent of the manufacturer in a sealed container, having high water resistance, high bondability, flexibility property; using specified brand thinner applying to metallic or wooden surface by brass/roller/spray in 2 coats over single coat anti-corrosive coating including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 229.09	Tk. 228.00	Tk. 216.00	Tk. 216.00
16.3.2	Premium synthetic enamel paint of approved best quality and colour delivered from authorized local agent of the manufacturer in a sealed container, having high water resistance, high bondability, flexibility property; using specified brand thinner applying to metallic or wooden surface by brass/roller/spray in two coats over single coat anti-corrosive coating including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 238.33	Tk. 237.00	Tk. 225.00	Tk. 225.00

16.4	Interior design paint of approved best quality and colour delivered from authorized local agent of the manufacturer in a sealed container; applying to interior wall and ceiling with surface preparation including cleaning drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying necessary interior sealer of specified brand on prepared surface; then applying necessary interior putty of specified brand for levelling, spot filling, crack filling and cutting by sand paper/zero water paper; applying 3 coats of interior design paint; 1st coat on prepared surface, then applying 1 design coat on the 1st coat and finally applying 1 protection coat on the design coat for protecting the design spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 861.33	Tk. 856.00	Tk. 819.00	Tk. 819.00
16.5	Textured paint of approved best quality and colour delivered from authorized local agent of the manufacturer in a sealed container; applying to interior wall and ceiling including surface preparation including cleaning drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus if required must be treated with fungicidal solution, mending good the surface defects using sand paper and necessary scaffolding; applying necessary sealer of specified brand on prepared surface; then applying necessary interior putty of specified brand for levelling, spot filling, crack filling and cutting by sand paper/zero water paper; applying 3 coats of texture paint; 1st coat on prepared surface, then applying 1 texture coat on the 1st coat and finally applying 1 top coat on the texture coat upto desired finishing by brush/roller/spray etc. with necessary scaffolding; all complete in all floors and accepted by the Engineer-in-charge	sqm	Tk. 752.38	Tk. 749.00	Tk. 730.00	Tk. 730.00
16.6	Chlorinated rubber paint of approved quality and colour delivered from authorized local agent of the manufacturer in a sealed container; surface preparation including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus (if required must be treated with fungicidal solution) flaking paints, washing the surface with soap water/caustic soda, mending good the surface defects, smoothening the surface by putty, sand papering the surface; applying to concrete surfaces and walls in three coats by brush/roller/spray etc. upto desired finishing, elapsing specified time for drying and re-coating; with necessary scaffolding all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 380.04	Tk. 378.00	Tk. 364.00	Tk. 364.00
16.7.1	Epoxy enamel paint of approved quality and colour delivered from authorized local agent of the manufacturer in a sealed container; surface preparation including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus (if required must be treated with fungicidal solution) flaking paints, washing the surface with soap water/caustic soda, mending good the surface defects, smoothening the surface by putty, sand papering the surface; applying to Interior concrete surfaces and walls in three coats by brush/roller/spray etc. upto desired finishing;elapsing specified time for drying and recoating; with necessary scaffolding all complete in all floors and accepted by the Engineerin- charge.	sqm	Tk. 460.95	Tk. 459.00	Tk. 441.00	Tk. 441.00

16.7.2	Epoxy enamel paint of approved quality and colour delivered from authorized local agent of the manufacturer in a sealed container; surface preparation including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus (if required must be treated with fungicidal solution) flaking paints, washing the surface with soap water/caustic soda, mending good the surface defects, smoothening the surface by putty, sand papering the surface; applying to Exterior concrete surfaces and walls in three coats by brush/roller/spray etc. upto desired finishing;elapsing specified time for drying and recoating; with necessary scaffolding all complete in all floors and accepted by the Engineerin- charge.	sqm	Tk. 452.36	Tk. 450.00	Tk. 433.00	Tk. 433.00
16.8.1	Silicon based water repellent of approved quality delivered from authorized local agent of the manufacturer in a sealed container; surface preparation including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying 3 coats of silicon based water repellent on exposed brick surface/fair face surface spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors and accepted by the Engineer-in-charge	sqm	Tk. 257.22	Tk. 256.00	Tk. 244.00	Tk. 244.00
16.8.2	Solvent based water repellent of approved quality delivered from authorized local agent of the manufacturer in a sealed container; surface preparation including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying 3 coats of silicon based water repellent on exposed brick surface/fair face surface spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors and accepted by the Engineer-in-charge	sqm	Tk. 448.73	Tk. 447.00	Tk. 433.00	Tk. 433.00
16.9.1	Standard French polishing to Wooden board surface by three coats over a coat of priming including putty, cleaning, finishing and polishing with sand paper etc. all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 431.55	Tk. 429.00	Tk. 407.00	Tk. 407.00
16.9.2	Premium French polishing to door and window frames, shutter and any other wooden surface by three coats over a coat of priming including putty, cleaning, finishing and polishing with sand paper etc. all complete in all floors and accepted by the Engineer-in-charge.	sqm	Tk. 518.62	Tk. 515.00	Tk. 487.00	Tk. 487.00
16.10.0	Wood varnishing of approved quality and colour delivered from authorized local agent of the manufacturer in a sealed container using brand specified thinner to door and window frame and shutter in three coats, applied with brush including cleaning oil, grease, dirt, making dry, sand papering, if required washing with weak detergent solution, stripping out the colour straps, finishing the surface, etc. all complete in all floors and accepted by the Engineer-in-charge.					
16.10.1	Yacht	sqm	Tk. 207.59	Tk. 206.00	Tk. 196.00	Tk. 196.00
16.10.2	Egg-shell	sqm	Tk. 209.93	Tk. 209.00	Tk. 198.00	Tk. 198.00
16.10.3	Matt	sqm	Tk. 210.71	Tk. 209.00	Tk. 199.00	Tk. 199.00

16.11	Approved best quality and colour synthetic polyvinyl distemper delivered from authorized local agent of the manufacturer in a sealed container, applying to wall and ceiling with surface preparation including cleaning drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending good the surface defects using sand paper and necessary scaffolding; applying 1 coat of interior sealer of specified brand on prepared surface; then applying 1 coat of interior putty of specified brand for levelling, spot filling, crack filling and cutting by sand paper/zero water paper; finally applying 2 coats of synthetic polyvinyl distemper spreading by brush/roller/spray & necessary scaffolding etc. upto desired finishing, elapsing specified time for drying or recoating; all complete in all floors and accepted by the Engineer-in-charge.	sqm.	Tk. 213.01	Tk. 212.00	Tk. 202.00	Tk. 202.00
16.12	White washing by three coats, lime mixture prepared at least 12 hours before use, slacking stone lime, supplying of gums, blue, stirring thoroughly, removing the floating materials from the mixer, surface cleaning to free from all foreign materials before application of each coat, applying one vertical and one horizontal wash for each coat and successive coat is to be applied after drying up of previous coat including hair brass, providing necessary scaffolding and cleaning plinth, floors, doors, windows, portions and ventilators by washing, rubbing, oiling if necessary after white wash for all floors including cost of water, electricity and other charges etc. complete in all respect in all floors and accepted by the Engineer-in-charge.	sqm.	Tk. 24.60	Tk. 24.00	Tk. 23.00	Tk. 23.00
16.13	Colour wash with yellow orchard/any other colour pigment by two coats over a prime coat of white wash, lime mixture prepared at least 12 hours before use, slacking stone lime, supplying of gums, blue, stirring thoroughly, removing the floating materials from the mixer, surface cleaning to free from all foreign materials before application of each coat, applying one vertical and one horizontal wash for each coat and successive coat is to be applied after drying up of previous coat including hair brass, providing necessary scaffolding and cleaning plinth, floors, doors, windows, portions and ventilators by washing, rubbing, oiling if necessary after white wash for all floors including cost of water, electricity and other charges etc. complete in all respect in all floors and accepted by the Engineer-in-charge.	sqm.	Tk. 28.45	Tk. 28.00	Tk. 27.00	Tk. 27.00
16.14.1	Application of 02 coats of self leveling epoxy compound (3 mm thick) as base coat & finishing coat over a single layer of screet coat (primer) for making smooth floor surface; all chemicals delivered from authorized local agent of the manufacturer in sealed container; materials having high water resistance, high bondability, flexibility property; applying by applicator machine after being mixed by force action mixer elapsing time for drying; surface preparation including surface cleaning from dust, oil or dirt, preparing underlying surface having minimum moisture content, smoothening, finishing & polishing by grinding machine & necessary tools, repairing significant ups/downs in floor before application with scaffolding, testing charges etc. all complete in all floors approved & accepted by the Engineer-in-charge.	sqm.	Tk. 1,979.09	Tk. 1,976.00	Tk. 1,962.00	Tk. 1,962.00

16.14.2	Application of 02 coats of self leveling epoxy compound (4 mm thick) as base coat & finishing coat over a single layer of screet coat (primer) for making smooth hygienic floor surface; all chemicals delivered from authorized local agent of the manufacturer in sealed container; materials having high water resistance, high bondability, flexibility property; applying by applicator machine after being mixed by force action mixer elapsing time for drying; surface preparation including surface cleaning from dust, oil or dirt, preparing underlying surface having minimum moisture content, smoothening, finishing & polishing by grinding machine & necessary tools, repairing significant ups/downs in floor before application with scaffolding, testing charges etc. all complete in all floors approved & accepted by the Engineer-in-charge.	sqm.	Tk. 2,363.77	Tk. 2,361.00	Tk. 2,344.00	Tk. 2,344.00
16.14.3	Application of 02 coats of self leveling epoxy compound (5 mm thick) as base coat & finishing coat over a single layer of screet coat (primer) for making smooth floor surface; all chemicals delivered from authorized local agent of the manufacturer in sealed container; materials having high water resistance, high bondability, flexibility property; applying by applicator machine after being mixed by force action mixer elapsing time for drying; surface preparation including surface cleaning from dust, oil or dirt, preparing underlying surface having minimum moisture content, smoothening, finishing & polishing by grinding machine & necessary tools, repairing significant ups/downs in floor before application with scaffolding, testing charges etc. all complete in all floors approved & accepted by the Engineer-in-charge.	sqm.	Tk. 3,250.34	Tk. 3,247.00	Tk. 3,227.00	Tk. 3,227.00
16.15	Standard Doco paint with approved best quality and colour delivered from authorized local agent of the manufacturer in a sealed container, having high water resistance, high bondability, flexibility property; using specified brand thinner applying to metallic or wooden surface by brass/roller/spray machine in 2 coats over single coat primer coating including cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, all complete in all floors and accepted by the Engineer-in-charge.	sqm.	Tk. 2,098.71	Tk. 2,096.00	Tk. 2,059.00	Tk. 2,059.00
16.16	Lacquer Polish (China or equivalent) of Standard design of wooden surface Top coat, Thinner, Clear sealer, Color pigment etc. complete from authorized local agent of the manufacturer in a sealed container, including paint spray machine etc. including surface cleaning from dust, oil or dirt, smoothening, finishing and polishing with sand paper and necessary tools, scaffolding, es etc. all complete in all floors approved and accepted by the Engineer.	sqm.	Tk. 2,143.27	Tk. 2,135.00	Tk. 2,064.00	Tk. 2,064.00
16.17.1	Providing painting top of the road surface as road marking in two coats with approved best quality and colour of synthetic enamel approved quality of paint from authorized local agent of the manufacturer in a sealed container having highly water resistant, high bondability, flexible, using specific brand thinner applied by brass/ roller or others over a coat of priming elapsing time for drying including surface cleaning from dust, oil or dirt, smoothening, finishing necessary tools etc. all complete in approved and accepted by the Engineer.	sqm.	Tk. 489.88	Tk. 486.00	Tk. 457.00	Tk. 457.00

16.17.2	Providing painting top of the road surface as road marking in two coats with approved best quality and colour of Florocent paint, approved quality of paint from authorized local agent of the manufacturer in a sealed container having highly water resistant, high bondability, flexible, using specific brand thinner applied by brass/ roller or others over a coat of priming elapsing time for drying including surface cleaning from dust, oil or dirt, smoothening, finishing necessary tools etc. all complete in approved and accepted by the Engineer.	sqm.	Tk. 999.70	Tk. 996.00	Tk. 967.00	Tk. 967.00

..... END OF CHAPTER SIXTEEN

CHAPTER 17: LIME TERRACING, ROOF TOP WATER PROOFING FOR GARDEN & SWIMMING POOL

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
17.1.1	Supply and application of non-toxic two components acrylic polymer modified cementitious coating of minimum 1.5 mm thickness for water proofing of roof/ roof garden/ swimming pool which consists of powder and liquid acrylic emulsion; under a protective cover of plaster/ cement concrete/ tiles etc. as per standard specification and accepted by the Engineer-in-charge. (Rate is excluding the cost of protective cover)	sqm	Tk. 780.91	Tk. 776.00	Tk. 766.00	Tk. 766.00
17.1.2	Supply and application of waterproofing treatment to roofs and terraces using three coats of acrylic based fiber reinforced elastomeric liquid water proofing membrane conforming to relevant ASTM standards product available in market. Surface preparation shall include cleaning, drying, making free from dirt, grease, wax, removing all chalked and scaled materials, fungus, mending the surface defects using sandpaper, filling cracks using suitable Crack Seal as necessary. First coat shall be diluted by 30% water and used as self-priming coat for the following two coats, achieving force coverage of 10sf/lt. In case of greater surface area, fiber mesh can be used in between the priming coat and second coat, followed by two more coats for better durability. All shall be completed as per manufacturer's specifications & recommendations and accepted by the Engineer in charge.					
17.1.2.1	Rate for cover area less than or equal to 500 sqm(without glass fiber mesh)	sqm	Tk. 1,080.39	Tk. 1,076.00	Tk. 1,066.00	Tk. 1,066.00
17.1.2.2	Rate for cover area greter than 500 sqm(with glass fiber mesh of 10 mm X 10 mm grid and weight 110 gm/sqm placed in between first two layers)	sqm	Tk. 1,204.21	Tk. 1,199.00	Tk. 1,188.00	Tk. 1,188.00

17.2	Preparation and making gravel pack for roof top gardening with supply of 25 mm down graded pea gravels including sieving, washing, compacting etc. complete as per standard specification and accepted by the Engineer-in-charge. Ground floor	cum	Tk. 5,052.54	Tk. 5,034.00	Tk. 4,597.00	Tk. 4,597.00
17.2.1	Added rate for gravel pack work in additional floor:					
17.2.1.1	Add for each additional floor above ground floor	cum	Tk. 60.00	Tk. 58.00	Tk. 54.00	Tk. 54.00
17.3	Supply and application of 19mm thick (9.5 mm X 2 layer) water proof, damp proof, dry and breathable plaster on roof slab/ underground water reservoir/ overhead water reservoir/ basement/retaining wall with water proof, damp proof, dry and breathable Izonil cement (STN-EN - 1015-11 , compressive strength 34 MPa ,max depth of water penetration into hardened plaster is < 1 mm) or equivalent compound in a proportion of 1:2.4 (Izonil cement or equivalent compound 1: sand 2.4) after cement grouting on existing concrete surface including finishing the corner and edges, washing sand, cleaning the surface scaffolding and curing at least 3 days etc. all completed in all respects as per direction of Engineering-in-charge. Above mentioned plaster includes glass fiber mesh of 10 mm X 10 mm grid and weight 110 gm/sqm placed in between two layers.	sqm	Tk. 1,236.05	Tk. 1,233.00	Tk. 1,215.00	Tk. 1,215.00
17.4	Average 100 mm thick finished lime terracing with 20 mm down graded first class brick chips (khoa), surki from 1st class bricks and minimum lime content 500 kg per 2.83 cubic meter (stone lime brought at site, not being powdered in open air and to be slaked in presence of engineer-in-charge and to be measured in volume three days after slaking for using in the mix) in the proportion 7:2:2 (brick chips : surki : lime) including preparation of the mix on the ground by making a suitable platform under proper polythene cover. Cutting the mix twice daily with limewater (1:10) at least for 7 days until the mix attain desirable consistency. Laying the mix in proper slope, beating the same with standard 'koppa' for minimum 7 days to gain maximum consolidation, making ghoondy and neat finishing with lime Surki mortar (1:2) and curing for 21 days providing polythene cover after each day work and cleaning etc. complete in all respect accepted by the Engineer-in-charge. 1st Floor	cum	Tk. 17,763.98	Tk. 17,639.00	Tk. 16,977.00	Tk. 16,882.00
17.4.1	Added rate for lime terracing work in additional floor:					
17.4.1.1	Add for each additional floor above ground floor	cum	Tk. 119.00	Tk. 117.00	Tk. 108.00	Tk. 108.00
17.5.1	25 mm thick ferrocement treatment work for water proofing on roof having minimum cement content relates to mix ratio 1:2 cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ASTM-C 150 type-1, and best quality coarse sand (F.M. 2.2) including, the supply of all materials, cutting, binding of one layer of 20 BWG galvanized wire mesh having minimum yield strength $f_y = 450$ MPa & having 2 mesh per 25 mm in both ways, clear cover 12 mm at the middle of the ferrocement lining, with washing of sand, chipping, cleaning the surface, watering, grouting, casting, laying on proper level of roof top or floor in panels (1.25 m x 1.25 m). Compaction including leveling, curing at least for 21 days including cost of water, electricity, in all respect as per drawing, design and accepted by the Engineer-in-charge.	sqm	Tk. 1,485.21	Tk. 1,477.00	Tk. 1,438.00	Tk. 1,430.00

17.5.2	25 mm thick ferrocement treatment work for heat and water proofing on roof having minimum cement content relates to mix ratio 1:2 cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ASTM-C 150 type-1, and best quality coarse sand (F.M. 2.2) including, the supply of all materials, cutting, binding of one layer of 20 BWG galvanized wire mesh having minimum yield strength $f_y = 450 \text{ Mpa}$ & having 2 mesh per 25mm in both ways, clear cover 12mm at the middle of the ferrocement lining, applying 62 mm thick EPS sheet having density 15 kg per cum, with washing of sand, chipping, cleaning the surface, watering, grouting, casting, laying on proper level of roof top or floor in panels (1.25 m x 1.25 m). Compaction including leveling, curing at least for 21 days including cost of water, electricity, in all respect as per drawing, design and accepted by the Engineer-in-charge.	sqm	Tk. 1,874.52	Tk. 1,865.00	Tk. 1,824.00	Tk. 1,816.00

..... **END OF CHAPTER SEVENTEEN**

CHAPTER 18 : FALSE CEILING, WALL PANELING & ACOUSTIC ITEMS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
18.1.1	Supplying, fitting and fixing plain Particle board ceiling (Drop/Plain) (complying unit wt 499 to 550 kg/m ³ , bending strength 0.35 N/mm ² , max swelling 8%, moisture content not more than 10%), of 12 mm thick with best quality and well seasoned Garjan wood frame of section 50 mm x 38 mm at 600 mm x 600 mm in grid suspended from ceiling or roof or beam by 12 SWG double ply G.I. wire fixed to the ceiling by rowel plug, screws, hooks, nails etc, maintaining straight lines and desired finished level at bottom face including vertical strut as required, cutting holes in slabs or beams by electric drill machine and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost and carriage of all materials, accessories, labour for installation, electricity charge, scaffolding, screws, nails, with best quality and approved colour etc. All complete, as per drawing design and accepted by the Engineer-in-charge. All sizes of wood are finished. All Garjan wood frames must be protected from termites, cockroach and other insects. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 2,831.13	Tk. 2,776.00	Tk. 2,764.00	Tk. 2,764.00

18.1.2	Supplying, fitting and fixing of Melamine/ Plain gypsum/ veneered board false ceiling (Drop/Plain) laminated by mechanical hot press milk white PVC membrane , framing by aluminium/ powder coated aluminium T-bar of any colour and of natural anodized finish suspended in 600 mm x 600 mm grid from ceiling by 12 SWG double ply wire, fixed to the ceiling by rowel plug, screws, hooks, nails etc., maintaining straight lines and desired finished level at bottom face including vertical wooden strut as required, making holes in slabs or beams by electric drill machine and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails, etc. all complete as per drawing, design and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area (Rate is excluding cost of polish/ paint etc. of any type where necessary).					
18.1.2.1	12 mm thick melamine board ceiling	sqm	Tk. 1,997.77	Tk. 1,993.00	Tk. 1,949.00	Tk. 1,949.00
18.1.2.2	8 mm thick melamine board ceiling		Tk. 1,585.85	Tk. 1,581.00	Tk. 1,536.00	Tk. 1,536.00
18.1.2.3	9 mm thick plain gypsum board ceiling	sqm	Tk. 2,271.73	Tk. 2,266.00	Tk. 2,219.00	Tk. 2,219.00
18.1.2.4	12 mm thick plain gypsum board ceiling		Tk. 2,350.97	Tk. 2,346.00	Tk. 2,298.00	Tk. 2,298.00
18.1.2.5	12 mm thick veneered board ceiling	sqm	Tk. 3,143.44	Tk. 3,124.00	Tk. 3,018.00	Tk. 3,018.00
18.1.3	Supplying, fitting & fixing of 12 mm thick BT ply board/ garjan ply board ceiling (Drop/Plain) screwed up with best quality and well seasoned 50 mm x 38 mm Garjan Wooden Frame at 600mm x 600 mm in grid suspended from ceiling roof or beam by 12 SWG double ply G.I wire fixed to the ceiling by M.S clumps, screws, hooks, nails etc. maintaining straight lines and desired finished level at bottom face including vertical strut as required, cutting holes in slabs or beams by electric drill machine and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost and carriage of all materials, accessories, installation, electricity charge, scaffolding, labour for installation with and necessary hardware all complete, as per drawing design and accepted by the Engineer-in-charge. All sizes of wood are finished. All Garjan wood frames must be protected from termites, cockroach and other insects. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type).					
18.1.3.1	12 mm thick BT Ply board ceiling	sqm	Tk. 3,063.81	Tk. 3,018.00	Tk. 3,054.00	Tk. 3,054.00
18.1.3.2	12 mm thick Garjan Ply board ceiling	sqm	Tk. 2,525.04	Tk. 2,479.00	Tk. 2,515.00	Tk. 2,515.00
18.1.4	Supplying, fitting and fixing 12 mm thick plain fire rated gypsum board ceiling (Drop/Plain) (size 1200mm x2400mm), framing by aluminium/ powder coated aluminium any colour T-bar of natural anodized finish suspended in 600 mm x 600 mm grid from ceiling by 12 SWG double ply wire, fixed to the ceiling by rowel plug, screws, hooks, nails etc., maintaining straight lines and desired finished level at bottom face including vertical wooden strut as required, making holes in slabs or beams by electric drill machine and mending good the damages, if any during execution of the work, including making provisions for lighting arrangement, also including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails, etc. All complete as per drawing, design and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area.	sqm	Tk. 2,763.68	Tk. 2,748.00	Tk. 2,704.00	Tk. 2,704.00

18.1.5	Supplying, fitting and fixing of Perforated gypsum board false ceiling (Drop/Plain) , framing by aluminium/ powder coated aluminium any colour T-bar of natural anodized finish suspended in (600 mm x 600 mm)/ (600 mm x 1200 mm) grid from ceiling by 12 SWG double ply wire, fixed to the ceiling by rowel plug, screws, hooks, nails etc., maintaining straight lines and desired finished level at bottom face including vertical wooden strut as required, making holes in slabs or beams by electric drill machine and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails, etc. all complete as per drawing, design and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area.					
18.1.5.1	9 mm thick perforated gypsum board ceiling (600 mm x 600 mm grid)	sqm	Tk. 2,311.35	Tk. 2,306.00	Tk. 2,259.00	Tk. 2,259.00
18.1.5.2	8 mm thick perforated gypsum board ceiling (600 mm x 600 mm grid)	sqm	Tk. 2,192.48	Tk. 2,187.00	Tk. 2,140.00	Tk. 2,140.00
18.1.5.3	8 mm thick perforated gypsum board ceiling (600 mm x 1200 mm grid)	sqm	Tk. 2,309.13	Tk. 2,304.00	Tk. 2,256.00	Tk. 2,256.00
18.1.6	Supplying, fitting and fixing of best quality PVC board false ceiling (Drop/Plain) of size 600 mm x 600 mm of approved design, minimum PVC content of 35%, framing by aluminium/ powder coated aluminium T-bar of any colour and of natural anodized finish suspended in 600 mm x 600 mm grid from ceiling by 12 SWG double ply wire, fixed to the ceiling by rowel plug, screws, hooks, nails etc., maintaining straight lines and desired finished level at bottom face including vertical wooden strut as required, making holes in slabs or beams by electric drill machine and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails, etc. all complete as per drawing, design and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area.					
18.1.6.1	6 mm thick PVC board false ceiling	sqm	Tk. 2,489.65	Tk. 2,470.00	Tk. 2,424.00	Tk. 2,424.00
18.1.6.2	8 mm thick PVC board ceiling	sqm	Tk. 2,621.73	Tk. 2,602.00	Tk. 2,556.00	Tk. 2,556.00
18.1.7	Supplying, fitting and fixing of 0.6 mm thick best quality solid aluminium metal board false ceiling (drop/ plain) of size 600 mm x 600 mm, powder coated of approved design, framing by aluminium T-bar of natural anodized finish suspended in 600 mm x 600 mm grid from ceiling by 12 SWG double ply wire, fixed to the ceiling by rowel plug, screws, hooks, nails etc., maintaining straight lines and desired finished level at bottom face including vertical wooden strut as required, making holes in slabs or beams by electric drill machine and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails, etc. all complete as per drawing, design and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area.	sqm	Tk. 3,241.17	Tk. 3,225.00	Tk. 3,188.00	Tk. 3,188.00

18.1.8	Supplying, fitting and fixing of 0.6 mm thick best quality perforated aluminium metal board (including tissue/ fabric) false ceiling (Drop/Plain) of size 600 mm x 600 mm, powder coated of approved design, inner side is covered by black tissue/ fabric to protect the hole from air circulation, framing by aluminium/ powder coated aluminium T-bar of any colour and of natural anodized finish suspended in 600 mm x 600 mm grid from ceiling by 12 SWG double ply wire, fixed to the ceiling by rowel plug, screws, hooks, nails etc., maintaining straight lines and desired finished level at bottom face including vertical wooden strut as required, making holes in slabs or beams by electric drill machine and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails, etc. all complete as per drawing, design and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area.	sqm	Tk. 2,937.40	Tk. 2,922.00	Tk. 2,885.00	Tk. 2,885.00
18.1.9	Supplying, fitting & fixing of designed seamless false ceiling (Drop/Plain) with 6 mm thick fiber cement board, framing by furring channel (38mm x 12.5 mm x 0.5 mm) G. I. angle (19 mm x 19 mm x 0.5 mm) suspended in 600 mm x 600 mm grid from ceiling using G.I fitting angle, locking clips etc. including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails etc. all complete as per drawing, design and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area. (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 3,409.77	Tk. 3,389.00	Tk. 3,330.00	Tk. 3,330.00
18.2.1	Supplying, fitting and fixing 12 mm thick burma teak (BT) veneered board/ ply board/ particle board in walling with best quality and well seasoned garjan wood frame of section of 50 mm x 25 mm at 600 mm x 600 mm in grid, including groove where necessary, fitted and fixed to wall by plugs, nails, screws etc. maintaining leveled and finished exposed faces including. making holes in wall and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost and carriage of all materials, electricity, accessories, labour for installation, scaffolding, screws, nails etc. All complete as per design, approved sample and accepted by the Engineer-in-charge. All sizes of wood are finished. All Garjan wood frames must be protected from termites, cockroach and other insects. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type).					
18.2.1.1	12 mm thick Myanmar(Burma) teak (BT) veneered board in walling	sqm	Tk. 2,720.86	Tk. 2,689.00	Tk. 2,667.00	Tk. 2,667.00
18.2.1.2	12 mm thick BT ply board walling	sqm	Tk. 3,098.10	Tk. 3,088.00	Tk. 3,032.00	Tk. 3,032.00
18.2.1.3	12 mm thick particle board walling	sqm	Tk. 2,347.19	Tk. 2,308.00	Tk. 2,277.00	Tk. 2,277.00

18.2.2	Supply fitting and fixing of 0.8 mm HPL sheet lamination (one surface only) over 12mm thick MDF board wall panneling with best quality and well seasoned garjan wood frame of section 38 mm x 38mm at 600mmx 600mm in grid, including grove where necessary, fitted and fixed to wall by royal plugs, nails, screws etc. maintaining leveled and finished exposed faces including. making holes in wall and mending good the damages, if any during execution of the work, including the provisions of lighting arrangement, also including cost and carriage of all materials, electricity, accessories, labour for installation, scaffolding, screws, nails etc. All complete as per design, approved sample and accepted by the Engineer-in-charge. (All sizes of wood are finished). All Garjan wood frames must be protected from termites, cockroach and other insects. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof.	Sqm	Tk. 4,324.22	Tk. 4,284.00	Tk. 4,239.00	Tk. 4,239.00
18.3.1	Supplying, fitting and fixing best quality wall paper along with 12 mm thick PVC solid board in walling with furring chanel frame (38mm x 12.5 mm x 0.5 mm) at 600 mm x 600 mm in grid, including grove where necessary, fitted and fixed to wall by plugs, nails, screws etc. including treatment of inner surface with termite and damp proofing agent maintaining leveled and finished exposed faces including making holes in wall and mending good the damages, if any during execution of the work, also including cost and carriage of wall paper and other materials, electricity, accessories, labour for installation, scaffolding, screws, nails etc. all complete as per design, approved and accepted by the Engineer-in-charge.	sqm	Tk. 3,477.10	Tk. 3,474.00	Tk. 3,464.00	Tk. 3,464.00
18.3.2	Supplying, fitting and fixing best quality wall paper on finished plaster surface including wall preparation cost and carriage of wall paper and other materials, electricity, accessories, labour for installation, scaffolding, screws, nails etc. all complete as per design, approved and accepted by the Engineer-in-charge.	sqm	Tk. 1,143.92	Tk. 1,140.00	Tk. 1,134.00	Tk. 1,134.00
18.4.1	Supplying, fitting and fixing of wooden louver in ceiling made of 12 mm Burma teak (BT) ply board , louver of any size and spacing framing with MS "U" bar/ channel, fixed to the ceiling by thread rod, nails, plugs, screws maintaining parallel louver with straight line/ desired finishing level at bottom face, necessary hardwares, including making holes in slabs and/ beams by electric drill machine and mending good the damages, if any during execution of the work, also including cost of all materials, electricity, accessories, scaffoldings, labour for installation etc. all complete as per drawing, design and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area. All Garjan wood frames must be protected from termites, cockroach and other insects. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type).	Sqm	Tk. 7,179.02	Tk. 7,173.00	Tk. 7,159.00	Tk. 7,159.00

18.4.2	Supplying fitting and fixing Ornamental design Louver in Ceiling with 18mm Burma Teak (BT) Veneer board fitted in drop Ceiling with garjan wood framing of size 50 mm x 50mm and 50 mm x 25 mm and with 50 mm ctg. teak wood as longitudinal rib, 18mm BT Veneer board as cross rib of any spacing the design louver fixed in drop ceiling wall by plugs, nails, screws, gum, etc. maintaining leveled and finished exposed faces including, 18mm PVC edging, making holes in wall including provisions for lighting arrangement and mending good the damages, if any, during execution of the work including all materials electricity, accessories, labour for installation, Scaffolding, etc. as per drawing design & approved sample and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area. All Garjan wood frames must be protected from termites, cockroach and other insects. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 9,202.22	Tk. 9,018.00	Tk. 9,352.00	Tk. 9,352.00
18.4.3	Supplying fitting and fixing of M.S. fixed louver made of 25mm x 25mm x 5mm M.S. Angle as outer frame, 2mm thick 25mm x 18mm MS sheet box section at 75mm c/c as internal horizontal member fixing with outer frame including scaffolding, royal bolt, anchoring, welding, grinding, welder, helper and all accessories complete in all respect as per drawing and accepted by the Engineer-in-charge. (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 8,580.25	Tk. 8,582.00	Tk. 8,582.00	Tk. 8,582.00
18.4.4	Supplying fitting fixing of louver wall panel made of 18 mm Burma teak (BT) veneered board of any size, including garjan wood frame of section 50 mm x 38mm at 250 mm spacing, fitted & fixed to wall by nails, iron screw, 38mm brass screw, gum, glue etc., maintaining vertical parallel louver and finished exposed faces, including making holes in walls and mending good the damages, if any during execution of the work, including provisions for lighting arrangement, in/c groove cutting of any size, also including cost of carriage of all materials, electricity, accessories, labour of installation, scaffolding etc. all complete as per design, approved sample and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area. All Garjan wood frames must be protected from termites, cockroach and other insects. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 6,726.16	Tk. 6,687.00	Tk. 6,725.00	Tk. 6,725.00

18.4.5	Supplying fitting fixing of louver wall panel made of 18 mm Burma teak (BT) veneered board of any size, including garjan wood frame of section 50 mm x 38 mm at 250 mm spacing, fitted & fixed to wall by nails, iron screw, 38mm brass screw, gum, glue etc., maintaining vertical paralal louver and finished exposed faces, including making holes in walls and mending good the damages, if any during execution of the work, including provisions for lighting arrangement, in/c groove cutting of any size, also including cost of carriage of all materials, electricity, accessories, labour of instalation, scaffolding, including CNC jali cutting and boxes, punch, grooving, ornamental bit or any type of special design of walling as per architectural requierment etc. all complete as per design, approved sample and excepted by the Engineer -in- charge. Measurement will be taken as per finished surface area. All Garjan wood frames must be protected from termites, cockroach and other insects. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 7,782.78	Tk. 7,744.00	Tk. 7,782.00	Tk. 7,782.00
18.4.6	Supplying fitting and fixing of fixed glass louver with Aluminium hollow box as per the U.s Architectural Aluminium Fanufactures Association (AAMA) Standard specification having 1.20 mm thick wall frame (size 100 mm x 25mm) and 10mm thick, 100 mm wide tempered glass louver section inner inclined glass louver section fixed with above mentioned frame and all aluminium members will be anodized to aluminium of any colour with a coat not less than 15 micron in thickness and density of 4mg per square cm etc. including all accessories, neoprene, sealant, nuts and bolts etc. complete in all respect as per drawing and accepted by the Engineer-in-charge.	sqm	Tk. 16,399.58	Tk. 16,394.00	Tk. 16,370.00	Tk. 16,370.00
18.4.7	Supplying fitting and fixing double leaf silding SS lovuer door shutter made with 16 BWG thick 50mm x 38mm SS hollow box as outer frame, 18 BWG thick 50mm x 12mm SS hollow box as louver with 18 BWG thick 38mm x38mm SS V Channel for bottom rail, 25mm dia SS solid bar, 18 BWG thick 62mm x 38mm SS U channel as top rail, 450 mm long 25mm dia SS handle, 75 mm SS wheel, 50 mm x 50 mm size 3 mm thick plate for locking arrangement with pad lock including grooving, making hole all necessary accesories and accepted by the Engineering -in-charge. (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 25,674.46	Tk. 25,671.00	Tk. 25,633.00	Tk. 25,633.00
18.4.8	Supplying, fitting and fixing of Aluminium fins fixed vertically BDS 1879 : 2014 having 1.2 mm thick wall frame (Size 125mmx 50mm box section @ 75 mm c/c), fitted with 3 nos horizontal M.S. Angle (20 mm x 20 mm x 3 mm) will be anodized to aluminium of any colour with a coat not less than 15 micrones in thickness or powder coated to any colour with a coat not less than 25 micrones in thickness and density of 4 mg per square cm etc. including all accessories, rawl bolt,55 mm M6 royal bolt,screws, washer etc all complete and accepted by engineer in charge. (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 21,092.52	Tk. 21,089.00	Tk. 21,073.00	Tk. 21,073.00

18.5.1	Supply and making wooden skirting with well matured seasoned wood including necessary moulding with garjan wood frame of 50mm x 38mm, groove with proper corner, finishing as per design, fitting and fixing in position by using all necessary accessories, royal plugs, screws, including making holes by electric drill machine and mending good the damages, if any during execution of the work, also including cost of all materials, electricity, labour for installation in all floors etc. all complete as per drawing, design and accepted by the Engineer-in-charge. All Garjan wood frames must be protected from termites, cockroach and other insects. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type).					
18.5.1.1	12 mm thick Myanmar(Burma) Teak Wooden skirting	sqm	Tk. 5,500.02	Tk. 5,413.00	Tk. 5,484.00	Tk. 5,484.00
18.5.1.2	12 mm thick Ctg. Teak. Wooden skirting	sqm	Tk. 4,171.00	Tk. 4,084.00	Tk. 4,196.00	Tk. 4,196.00
18.5.2	Supply and making 25 mm thick wooden Design bit/ Moulding with well matured seasoned wood including necessary moulding/ groove with proper corner, finishing as per design, fitting and fixing in position by using all necessary accessories in all floors etc all complete and accepted by engineer in charge. (Rate is excluding cost of polish/ paint etc. of any type).					
18.5.2.1	Myanmar(Burma) Teak wooden Design bit/ Moulding	sqm	Tk. 16,522.13	Tk. 16,313.00	Tk. 15,983.00	Tk. 15,983.00
18.5.2.2	Ctg Teak wooden Design bit/Moulding	sqm	Tk. 11,240.77	Tk. 11,054.00	Tk. 10,989.00	Tk. 10,989.00
18.6.1	Supplying, fitting and fixing Wall Almirah with one side attires cabinet including locker and another side shelf of Depth-1'-8" made by 18 mm B.T Veneered/ B.T. ply/ B.T marine ply board on 2.50"x2.50" (Finished) Ctg. Teak Wooden Outer Frame, 1.50" x 1.50" (Finished) Ctg. Teak wooden frame for shelf, 1.25" x 1.25" (Finished) Ctg. Teak wooden edge bit around the shutter, backside covered by the 18mm garjan Ply board, 2 nos standard handle for shutter & 3 nos standard lock for shutter and drawer, 25mm SS pipe 1.5 mm thick and including all necessary accessories for fitting & fixing all complete and accepted by the engineer-in-charge. All Ctg. teak wood frames must be protected from termites, cockroach and other insects. (Rate is excluding cost of polish/ paint etc. of any type).					
18.6.1.1	Wall Almirah with one side attire cabinet including locker and another side shelf made of 18 mm B.T Veneered board	sqm	Tk. 18,300.19	Tk. 18,152.00	Tk. 18,528.00	Tk. 18,528.00
18.6.1.2	Wall Almirah with one side attire cabinet including locker and another side shelf made of 18 mm B.T ply board	sqm	Tk. 19,766.75	Tk. 19,619.00	Tk. 19,994.00	Tk. 19,994.00
18.6.1.3	Wall Almirah with one side attire cabinet including locker and another side shelf made of 18 mm B.T Marine ply board	sqm	Tk. 21,652.31	Tk. 21,505.00	Tk. 21,880.00	Tk. 21,880.00

18.6.2	Supplying, fitting and fixing Wall Almirah with one side drawer and Another side shelf of Depth-1'-8" made by 18 mm B.T Veneered/ B.T. ply/ B.T marine ply board on 2.50"x2.50" (Finished) Ctg. Teak Wooden Outer Frame, 1.50" x 1.50" (Finished) Ctg. Teak wooden frame for shelf, 1.5" x 1.5" (Finished) Ctg. Teak wooden edge bit around the shutter, backside covered by the 18mm garjan Ply board, 2 nos standard handle for shutter & 4 nos standard lock for shutter and drawer, and including all necessary accessories for fitting & fixing all complete and accepted by the engineer-in-charge. All Ctg. teak wood frames must be protected from termites, cockroach and other insects. (Rate is excluding cost of polish/ paint etc. of any type).					
18.6.2.1	Wall Almirah with one side drawer and Another side shelf made of 18 mm B.T Veneered board	sqm	Tk. 28,167.14	Tk. 27,925.00	Tk. 28,601.00	Tk. 28,601.00
18.6.2.2	Wall Almirah with one side drawer and Another side shelf made of 18 mm B.T ply board	sqm	Tk. 31,280.24	Tk. 31,039.00	Tk. 31,714.00	Tk. 31,714.00
18.6.2.3	Wall Almirah with one side drawer and Another side shelf made of 18 mm B.T Marine ply board	sqm	Tk. 35,282.80	Tk. 35,041.00	Tk. 35,717.00	Tk. 35,717.00
18.6.3	Supplying, fitting and fixing 25mm Ctg. Teak wooden Cabinet of size 8'-0" x4'-9" x1'-8" (finished) , 18mm BT veneered/ BT ply/ BT marine ply board for shelf, 3" (Finished) Ctg. Teak wood moulding bit at bottom, and 1" moulding bit at top and middle, backside covered by the 18mm garjan Ply board, 12 nos standard handle for shutter & 6 nos standard lock, and including screw, SS chain hinge, nails, gums etc. and all necessary accessories for fitting and fixing all complete and accepted by the engineer-in-charge. All Ctg. teak wood frames must be protected from termites, cockroach and other insects. (Rate is excluding cost of polish/ paint etc. of any type).					
18.6.3.1	wooden Cabinet with B.T. veneered board shutter	sqm	Tk. 23,990.28	Tk. 23,684.00	Tk. 24,512.00	Tk. 24,512.00
18.6.3.2	wooden Cabinet with B.T. ply board shutter	sqm	Tk. 24,873.77	Tk. 24,567.00	Tk. 25,395.00	Tk. 25,395.00
18.6.3.3	wooden Cabinet with B.T. marine ply shutter	sqm	Tk. 26,009.67	Tk. 25,703.00	Tk. 26,531.00	Tk. 26,531.00
18.6.4	Supplying, fitting and fixing Kitchen Cabinet (Below Counter) Depth-1'-8" made by 18 mm B.T Veneered board/ B.T. ply board/ B.T. Marine ply board/ HPL board on 2"x1.5" (Finished) Ctg. Teak Wooden Frame, making 1" x 1" (Finished) Ctg. Teak edge bit around the shutter and Shelf made of 12mm laminated vinyle board including all necessary accessories for fitting & fixing all complete as per design and the direction of the engineer-in-charge. All Ctg.teak wood frames must be protected from termites, cockroach and other insects. (Rate is excluding cost of polish/ paint etc. of any type).					
18.6.4.1	Kitchen Cabinet (Below Counter) made by 18 mm B.T Veneered board	sqm	Tk. 11,158.37	Tk. 11,060.00	Tk. 11,115.00	Tk. 11,115.00
18.6.4.2	Kitchen Cabinet (Below Counter) made by 18 mm B.T Ply board	sqm	Tk. 11,772.03	Tk. 11,674.00	Tk. 11,729.00	Tk. 11,729.00
18.6.4.3	Kitchen Cabinet (Below Counter) made by 18 mm B.T Marine Ply board	sqm	Tk. 12,561.03	Tk. 12,463.00	Tk. 12,518.00	Tk. 12,518.00
18.6.4.4	Kitchen Cabinet (Below Counter) made by 18 mm HPL board	sqm	Tk. 15,804.68	Tk. 15,708.00	Tk. 15,763.00	Tk. 15,763.00
18.7.1	Supplying, fitting and fixing of tufted, textured loop/ cut and loop piled or similar Poly-Propylene (PP) fiber carpet tiles of custom color and any pattern interface/ carpet Inter/ equivalent 50 cm x 50 cm or any approved size, with 5 mm to 8 mm thickness (Made in Thailand/ China/ India/ Malaysia or equivalent quality product), of type 6.6 nylon Stain Repel / Stain Resist /soil protection property, including antimicrobial protection and finished pile height of 3 mm to 4 mm, with recycled glass/ cushion backing with synthetic binder and with flammability property of radiant panel (ASTM E 648), fixing with adhesives including cutting, placing in proper level etc. all complete in all respect as finish schedule, drawing and direction of Engineer-in-Charge.	sqm	Tk. 5,888.00	Tk. 5,887.00	Tk. 5,885.00	Tk. 5,885.00

18.7.2.1	Supply, Delivery and installation of Imported Fire retardant Wood Acoustic Panel (WAP) in walling WAP should be 15mm thick x 128mm high x 2440mm long made from Pine Wood granules, Grooved and perforated as per selected pattern and color (in melamine coated finish) including Non-Woven fabric inlay having of NRC >0.75, the black non-woven fabric inlay, Galvanized steel keels & clips. Panels will be fixed on 50 mm x 25mm Garjan wooden frame spacing 750 mm to 800 mm as per drawing, approved architectural and acoustic design (NRC > 0.75) and accepted by the Engineer in charge.(Made in Thailand/ China/ India/ Malayasia or equivalent quality product). All Garjan wood frames must be protected from termites, cockroach and other insects. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type if necessary).	sqm	Tk. 8,853.31	Tk. 8,813.00	Tk. 8,796.00	Tk. 8,796.00
18.7.2.2	Supply, Delivery and Installation Custom Size Fabric Wrapped Acoustic Panel (FWAP) in walling as per design, made with peripheral of 12mm Thick Commercial Veneer / MDF Boards of 50 mm height with back side of 5mm thick Garjan Plyboard and filled with 50mm thick 100% Polyester insulation / Fiberglass Insulation of NRC > 0.75. (Made in Thailand/ China/ India/ Malayasia or equivalent quality product). Wrapped fabric must be at least 600 -700 GSM in order to achieve desired Acoustic performance all complete as per drawing and architectural and acoustic design (NRC > 0.75) and accepted by the Engineer-in-charge. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type if necessary).	sqm	Tk. 3,924.69	Tk. 3,916.00	Tk. 3,861.00	Tk. 3,861.00
18.7.2.3	Supplying ,fitting and fixing of ACOUSTIC WALLING by 12mm thick B.T veneer board with best quality well seasoned garjan frame of section 50 mm x 38 mm at 600mm x 600 mm in grid, all Garjan wood frames must be protected from termites, cockroach and other insects, fitted and fixed to wall by plugs, nails, screws, gums, etc. all necessary accessories, including 50 mm imported GLASSWOOL (density 16-20 kg/m3) (Made in Thailand/ China/ India/ Malayasia or equivalent quality product), heat, sound and noise reduction quality, made of grade A1 incombustible materials with one side aluminium foil quoted/ pasted) is attached with garjan frame with black/ grey airtight fabric cloth bearded & jute hessian, cloth, garments jut including making required number and sizes holes/ groove in B.T. veneered board, scaffolding, mending good the damages, fitting and fixing as per architectrual and acoustic design (NRC 0.60-0.75), etc. all complete as per design, approved sample & accepted by the Engineer-in-charge. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type if necessary). (This item is for Hall room/ conference room/ meeting room or similar purpose uses.)	sqm	Tk. 8,894.25	Tk. 8,832.00	Tk. 8,840.00	Tk. 8,840.00

18.7.2.4	Supplying, fitting and fixing of 15mm thick Standard Acoustic panel for walling work for straight/curve area (Made in Thailand/ China/ India/ Malayasia or equivalent quality product) with high quality MDF Board with finished top Melamine/Natural wooden veneer/HPL or any other customized texture or colour finish with sound absorbing, fire proof back felt fabric, board size 133mmx2440mm to 197mm x2440mm, tongue and groove edge. Echo should be E1 standard, Density 750-800 kg/m3, perforation range 7.5%-12.8%, NRC value 0.75-0.90, including substructure of garjan wooden framing 50mm x 38mm size @ 600mm grid both way; all Garjan wood frames must be protected from termites, cockroach and other insects, backing with acoustic lining materials of 50 mm thick Glass wool (density 16-20 kg/m3) (Made in Thailand/ China/ India/ Malayasia or equivalent quality product) with air tight fabric cover in/c heat stitching etc all complete as per approved drawing and architectural and acoustic design (NRC 0.75-0.90) and accepted by the Engineer in-charge. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type if necessary).	sqm	Tk. 12,309.73	Tk. 12,240.00	Tk. 12,285.00	Tk. 12,285.00
18.7.2.5	Supplying, fitting and fixing of 15mm thick Moisture proof Acoustic panel for walling work for straight/curve area (Made in Thailand/ China/ India/ Malaysia or equivalent quality product) with high quality MDF Board with finished top Melamine/Natural wooden veneer/HPL or any other customized texture or colour finish with sound absorbing, fire proof back felt fabric, board size 133mmx2440mm to 197mm x2440mm, tongue and groove edge. Echo should be E1 standard, Density 750-800 kg/m3, perforation range 7.5%-12.8%, NRC value 0.75-0.90, including substructure of garjan wooden framing 50mm x 38mm size @ 600mm grid both way; all Garjan wood frames must be protected from termites, cockroach and other insects, backing with acoustic lining materials of 50 mm thick Glass wool (density 16-20 kg/m3) (Made in Thailand/ China/ India/ Malayasia or equivalent quality product) with air tight fabric cover in/c heat stitching etc all complete as per approved drawing and architectural and acoustic design (NRC 0.75-0.90) and accepted by the Engineer in-charge. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type if necessary).	sqm	Tk. 12,614.82	Tk. 12,545.00	Tk. 12,590.00	Tk. 12,590.00

18.7.2.6	Supplying, fitting and fixing of 15mm thick Fire Retardant Acoustic panel for walling work for straight/curve area (Made in Thailand/ China/ India/ Malayasia or equivalent quality product) with high quality MDF Board with finished top Melamine/Natural wooden veneer/HPL or any other customized texture or colour finish with sound absorbing, fire proof back felt fabric, board size 133mmx2440mm to 197mm x2440mm, tongue and groove edge. Panel materials should be incombustible, Echo should be E1 standard, Density 750-800 kg/m3, perforation range 7.5%-12.8%, NRC value 0.75-0.90, including substructure of garjan wooden framing 50mm x 38mm size @ 600mm grid both way; all Garjan wood frames must be protected from termites, cockroach and other insects, backing with acoustic lining materials of 50 mm thick Glass wool (density 16-20 kg/m3) (Made in Thailand/ China/ India/ Malayasia or equivalent quality product) with air tight fabric cover in/c heat stitching etc all complete as per approved drawing and architectural and acoustic design (NRC 0.75-0.90) and accepted by the Engineer in-charge. The brick/ RCC surface shall be dry and damp-proof. (Rate is excluding cost of polish/ paint etc. of any type if necessary).	sqm	Tk. 12,767.37	Tk. 12,697.00	Tk. 12,743.00	Tk. 12,743.00
18.7.2.7	Supplying, fitting and fixing of 15mm thick Perforated/Micro perforated wooden acoustic panel for walling work, Straight/ Curve (Made in Thailand/ China/ India/ Malayasia or equivalent quality product) with high quality MDF Board with finished top Melamine/Natural Wooden veneer/ Technology veneer/HPL or any other customized texture or colour finish with sound absorbing, fire proof back felt fabric, board size 1200mm x2400mm of incombustible material, Echo should be E1 standard, density 750-800 kg/m3, Perforation range 12%-13% (normal) / 1%-6.5% (micro), NRC value 0.75-0.90 including substructure of garjan wooden framing 50mm x 38mm size @ 600mm grid both way; all Garjan wood frames must be protected from termites, cockroach and other insects, backing with acoustic lining materials of 50 mm thick Glass wool (density 16-20 kg/m3) (Made in Thailand/ China/ India/ Malayasia or equivalent quality product) with air tight fabric cover in/c heat stitching etc all complete as per approved drawing and architectural and acoustic design (NRC 0.75-0.90) and accepted by the Engineer in-charge. The brick/ RCC surface shall be dry and damp-proof. (This item is for Auditorium/ convention hall or similar purpose uses.) (Rate is excluding cost of polish/ paint etc. of any type if necessary).	sqm	Tk. 12,462.27	Tk. 12,392.00	Tk. 12,438.00	Tk. 12,438.00

18.7.3.1	Supplying, fitting and fixing of 15 mm thick Fire Retardent, heat Insulated, Acoustic Mineral fiber board (any colour) ceiling (drop/ plain/ straight/ curve) (Made in Thailand/ China/ India/ Malaysia or equivalent quality product) with high grade mineral wool, surface finish with white ultra vinyl latex paint, surface treatment-film coating, anti-fungal & anti bacterial, humidity resistant greater than 90%, light reflectivity more than 90%, NRC value 0.60- 0.90, density-290-315 kg/m3, size of board: 600mm x 600mm, of incombustible materials, framing by aluminium T-bar of natural anodized finish suspended in 600 mm x 600 mm grid from ceiling using butterfly hanger with adjustable clip/ locking clips etc. including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails, rivet etc. all complete as per drawing, architectural and acoustic design (NRC 0.6-0.9) and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof. (This item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 2,166.07	Tk. 2,154.00	Tk. 2,126.00	Tk. 2,126.00
18.7.3.2	Supplying, fitting and fixing of 15 mm thick Fire retardent, heat Insulated, Acoustic Mineral fiber board ceiling , (drop/ plain/ straight/ curve) (Made in UK/Germany/ Italy/ France or equivalent quality product) with high grade mineral wool, surface finish with white ultra vinyl latex paint, surface treatment-film coating, anti-fungal & anti bacterial humidity resistant greater than 90%, light reflectivity more than 90%, NRC value 0.60- 0.90, density-290-315 kg/m3, size of board: 600mm x 600mm, of incombustible materials, framing by aluminium T-bar of natural anodized finish suspended in 600 mm x 600 mm grid from ceiling using butterfly hanger with adjustable clip/ locking clips etc. including cost of all materials, electricity, accessories, scaffoldings, labour for installation, screws, nails, rivet etc. all complete as per drawing, architectural and acoustic design (NRC 0.6-0.9) and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof. (This item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 4,384.96	Tk. 4,373.00	Tk. 4,345.00	Tk. 4,345.00
18.7.3.3	Supplying, fitting and fixing of 20mm thick Mineral Fiber Acoustic ceiling, (drop/ plain/ straight/ curve) (Made in UK/Germany/ Italy/ France or equivalent quality product) with high grade mineral wool, surface finish with white ultra vinyl latex paint, surface treatment-film coating, anti-fungal & anti bacterial humidity resistant greater than 90%, light reflectivity more than 90%, NRC value 0.90 -1.00, medium density, size of board: 600mm x 600mm, fire rating B1 standard, framing with 600mm x 600 mm grid suspended from ceiling using screw, nut, adjustable hanger hook, anchor, T-Grid, Wall angle, Main runner, thread rod, 38mm carrier bracket, connector, sleeve anchor, suspension bracket etc. all complete as per drawing, architectural and acoustic design (NRC 0.9-1.0) and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof.(This item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 9,929.08	Tk. 9,907.00	Tk. 9,873.00	Tk. 9,873.00

18.7.3.4	Supplying, fitting and fixing of 20mm thick Mineral Fiber Acoustic ceiling, (drop/ plain/ straight/ curve) (Made in China/Malaysia/Thailand/India or equivalent quality product) with high grade mineral wool, surface finish with white ultra vinyl latex paint, surface treatment-film coating, anti-fungal & anti bacterial humidity resistant greater than 90%, light reflectivity more than 90%, NRC value 0.90 - 1.00, medium density, size of board: 600mm x 600mm, fire rating B1 standard, framing with 600mm x 600 mm grid suspended from ceiling using T-Grid, Wall angle, Main runner, 38mm carrier bracket, adjustable hanger hook, thread rod, anchor, screw, nut, connector, sleeve anchor, suspension bracket etc. all complete as per drawing architectural and acoustic design (NRC 0.9-1.0) and accepted by the Engineer-in-charge. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof.(This item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 6,341.69	Tk. 6,319.00	Tk. 6,286.00	Tk. 6,286.00
18.7.4	Supplying, fitting and fixing of Ply board C-section or box type work on wall/Ceiling. Using C-section of 150mm x 200mm x 150mm or any size of thickness 18mm with facing 50mm, in/c substruction of garjan wooden framing (50mm x 38mm @600mm c/c) and screwed, for additional load support in/c special made full thread 8mm dia rod fixed to the ceiling or roof or beam by rowel bolt, all complete as per drawing, design and accepted by the Engineer-in-charge. All Garjan wood frames must be protected from termites, cockroach and other insects. Measurement will be taken as per finished surface area. The brick/ RCC surface shall be dry and damp-proof. (This item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type).	sqm	Tk. 5,479.80	Tk. 5,371.00	Tk. 5,491.00	Tk. 5,491.00
18.8	Additional Rate of scaffolding for walling or ceiling work (from Item no. 18.1 to 18.7)					
18.8.1	From 4m up to 6 m	Sqm	Tk. 132.00	Tk. 132.00	Tk. 132.00	Tk. 132.00
18.8.2	Beyond 6m to above height	Sqm	Tk. 264.00	Tk. 264.00	Tk. 264.00	Tk. 264.00
18.9	Supplying, fitting, fixing CNC Cutted facade work made of 5 mm(+3%) thick MS Plate , pre-treated & painted with industrial paint/ powder coated glossy paint (thickness not less than 70 micron) covered by water colour weather proof paint, supported by 50mm x 50 mm x 5mm MS angle at Top & Side of outer frame according to drawing & design, fitted and fixed in wall/ RCC members with necessary number of clamps, intermediate struts, royal bolt with epoxy. Segments of MS plate will be smoothly joined by rivets or welding, aligned and levelled. Cost is inclusive electricity, workshop charges, loading - unloading, lifting, carrying and placing to the required place & position etc. all complete as per approved design and drawing and accepted by the Engineer-in-charge.(Upto a height of 20m)	sqm	Tk. 7,967.14	Tk. 7,963.00	Tk. 7,936.00	Tk. 7,936.00
18.10	Additional Rate of scaffolding for CNC Cutted facade work					
18.10.1	Above 20m and upto 30m	Sqm	Tk. 132.00	Tk. 132.00	Tk. 132.00	Tk. 132.00
18.10.2	Beyond 30m	Sqm	Tk. 264.00	Tk. 264.00	Tk. 264.00	Tk. 264.00
..... END OF CHAPTER EIGHTEEN						

CHAPTER 19 : COLLAPSIBLE GATE, M.S. GATE, ROLLING SHUTTER ETC.

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
19.1.1	Manufacturing, supplying, fitting and fixing S.S. collapsible gate of any design and shape made of 25 mm x 25 mm x 3 mm S.S. angle placed @ 112 mm c/c vertically and connecting the same with each other by 25 mm x 3 mm S.S flat bar scissors 525 mm, 600 mm long provided in 3 rows including cutting the different S.S. members to required sizes, fabricating, welding, riveting with required size rivets, providing required size wheels, pulling handles on both sides, suitable locking arrangement, electrodes, grease and finally placing the same in position in between 2 (two) nos. 50 mm x 50 mm x 6 mm S.S. tee rail made by welding 2 nos. 50 mm x 6 mm S.S. flat bar fitted and fixed at top and bottom with R.C.C. lintel/ roof slab, floors and side wall with required nos. 150 mm to 225 mm long 38 mm x 6 mm S.S. flat bar clamps one end welded with the gate member and the other end bifurcated and embedded in C.C. at the respective point including cutting holes and mending good the damages by pouring concrete (1:2:4) into thluding cost of polish/ paint etc. of any type). work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type).tion hall o					
19.1.1.1	for S.S. Grade A304	sqm	Tk. 19,344.97	Tk. 19,327.00	Tk. 19,203.00	Tk. 19,203.00
19.1.1.2	for S.S. Grade A201	sqm	Tk. 15,416.74	Tk. 15,399.00	Tk. 15,275.00	Tk. 15,275.00
19.1.2	Manufacturing, supplying, fitting and fixing S.S. collapsible gate of any design and shape made of 20 mm x 20 mm x 3 mm S.S. angle placed @ 112 mm c/c vertically and connecting the same with each other by 20 mm x 3 mm S.S flat bar scissors 525 mm, 600 mm long provided in 3 rows including cutting the different S.S. members to required sizes, fabricating, welding, riveting with required size rivets, providing required size wheels, pulling handles on both sides, suitable locking arrangement, electrodes, grease and finally placing the same in position in between 2 (two) nos. 50 mm x 50 mm x 6 mm S.S. tee rail made by welding 2 nos. 50 mm x 6 mm S.S. flat bar fitted and fixed at top and bottom with R.C.C. lintel/ roof slab, floors and side wall with required Nos. 150 mm to 225 mm long 38 mm x 6 mm S.S. flat bar clamps one end welded with the gate member and the other end bifurcated and embedded in C.C. at the respective point including cutting holes and mending good the damages by pouring concrete (1:2:4) into thuding cost of polish/ paint etc. of any type). item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of					
19.1.2.1	for S.S. Grade A304	sqm	Tk. 16,145.21	Tk. 16,128.00	Tk. 16,003.00	Tk. 16,003.00
19.1.2.2	for S.S. Grade A201	sqm	Tk. 12,701.64	Tk. 12,684.00	Tk. 12,560.00	Tk. 12,560.00

19.1.3	Manufacturing, supplying, fitting and fixing M.S. collapsible gate of any design and shape made of 25 mm x 25 mm x 3 mm M.S. angle placed @ 112 mm c/c vertically and connecting the same with each other by 25 mm x 3 mm M.S flat bar scissors 525 mm, 600 mm long provided in 3 rows including cutting the different M.S. members to required sizes, fabricating, welding, riveting with required size rivets, providing required size wheels, pulling handles on both sides, suitable locking arrangement, electrodes, grease and finally placing the same in position in between 2 (two) Nos. 50 mm x 50 mm x 6 mm M.S. tee rail made by welding 2 nos. 50 mm x 6 mm M.S. flat bar fitted and fixed at top and bottom with R.C.C. lintel/ roof slab, floors and side wall with required nos. 150 mm to 225 mm long 38 mm x 6 mm M.S. flat bar clamps one end welded with the gate member and the other end bifurcated and embedded in C.C. at the respective point including cutting holes and mending good the damages by pouring concrete (1:2:4) into thluding cost of polish/ paint etc. of any type). work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type).tion hall or similar purpose uses.) (Rate is excluding cost of polish/ paint etc. of any type if necessary). d by the Engineer-in-charge.and electricity complete in	sqm	Tk. 5,981.64	Tk. 5,963.00	Tk. 5,830.00	Tk. 5,829.00
19.1.4	Manufacturing, supplying, fitting and fixing M.S. collapsible gate of any design and shape made of 20 mm x 20 mm x 3 mm M.S. angle placed @ 112 mm c/c vertically and connecting the same with each other by 20 mm x 3 mm M.S flat bar scissors 525 mm, 600 mm long provided in 3 rows including cutting the different M.S. members to required sizes, fabricating, welding, riveting with required size rivets, providing required size wheels, pulling handles on both sides, suitable locking arrangement, electrodes, grease and finally placing the same in position in between 2 (two) nos. 50 mm x 50 mm x 6 mm M.S. tee rail made by welding 2 nos. 50 mm x 6 mm M.S. flat bar fitted and fixed at top and bottom with R.C.C. lintel/roof slab, floors and side wall with required Nos. 150 mm to 225 mm long 38 mm x 6 mm M.S. flat bar clamps one end welded with the gate member and the other end bifurcated and embedded in C.C. at the respective point including cutting holes and mending good the damages by pouring concrete (1:2:4) into theuding cost of polish/ paint etc. of any type). item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type). f necessary).	sqm	Tk. 5,170.57	Tk. 5,152.00	Tk. 5,018.00	Tk. 5,018.00

19.2.1	Supplying, fitting, fixing, and installation of ordinary type M.S. gate (double leaf) of any design and shape with 38 mm x 38 mm x 6 mm M.S. Angle outer frame having 16 mm dia M.S. rod placed vertically @ 75 mm c/c. at bottom part and top part (200 mm long arrows) and also placed horizontally at bottom for interconnection of 16 mm dia M.S. rod placed vertically as per design, the top part covered with 18 BWG M.S. sheet and fixed with 4 nos. 38 mm x 6 mm F.I. bars placed diagonally and 4 Nos. 38 mm x 6 mm F.I. bars placed horizontally and vertically all passing through the center, providing locking arrangements on 3 mm thick M.S. plate one for main gate and one for minimum 600 mm x 1025 mm pocket gate with 38 mm x 38 mm x 6 mm M.S. angle and 16 mm dia M.S. rod placed vertically @ 75 mm c/c, providing 38 mm x 38 mm x 6 mm M.S. angle clamps fitted and fixed with the outer frame of the main gate, the clamp being embedded in R.C.C. or R.C.C. core pillar with cement concrete (1:2:4) in masonry including cutting holluding cost of polish/ paint etc. of any type). work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type).tion hall or similar purpose uses.) (Rate is excluding cost of polish/ paint etc. of any type if necessary). d by the Engineer-in-charge.and electricity complete in all respect	sqm	Tk. 6,665.97	Tk. 6,646.00	Tk. 6,496.00	Tk. 6,495.00
19.2.2	Manufacturing, supplying and fitting-fixing security door with 25 mm dia M.S. rod and outer frame with 50 mm x 50 mm x 6 mm M.S. angle including locking arrangement as per approved design and drawing providing 38 mm x 38 mm x 6 mm M.S. angle clamps fitted and fixed with the outer frame of the main gate, the clamp being embedded in R.C.C. or R.C.C. core pillar with cement concrete (1:2:4) in masonry including cutting holes and mending good the damages, finishing, curing etc. including riveting, welding as and where necessary, painting two coats with approved quality of synthetic enamel paint over a coat of priming of anti-corrosive paint etc. all complete as per drawing, design and accepted by the Engineer-in-charge. (Rate is excluding the cost of paint).	sqm	Tk. 13,268.55	Tk. 13,215.00	Tk. 12,905.00	Tk. 12,903.00
19.3	Supplying, fitting, fixing, and installation of ordinary type M.S. security gate of any design and shape with 50 mm x 50 mm x 6 mm M.S. angle outer frame and 02 (two) 50 mm x 50 mm x 6 mm angle placed diagonally having 22 mm dia M.S. rod placed vertically @ 115 mm c/c. and also 38 mm x 6 mm F.I bar placed horizontally, the bottom part covered with 3 mm thick M.S. plate providing locking arrangements on 3 mm thick M.S. plate providing heavy type clamps fitted and fixed with the outer frame of the main gate, the clamp being embedded in R.C.C. or R.C. core pillar with cement concrete (1:2:4) in masonry including cutting holes and mending good the damages, finishing, curing etc. including fabrication, riveting, welding as and where necessary, painting 2 (two) coats with approved best quality synthetic enamel paint over a coat of anticorrosive painting etc. all complete as per drawing, design and accepted by the Engineer-in-charge. (Rate is excluding the cost of paint).	sqm	Tk. 7,848.76	Tk. 7,833.00	Tk. 7,700.00	Tk. 7,699.00

19.4	Supplying, fitting, fixing and installation of ordinary type M.S. gate (double leaf) of any design and shape with 38 mm x 38 mm x 6 mm M.S. angle box (made by welding 2 nos. 38 mm x 38 mm x 6 mm angle) outer frame having 25 mm x 50 mm x 25 mm x 5 mm M.S. channel (made by welding 2 nos. of channel) placed part diagonally after cutting and shaping as per requirement, part horizontally @ 75 mm c/c, the two part of each leaf being separated by a vertical member of 38 mm x 38 mm x 6 mm M.S. box and welded the each ends of diagonal and horizontal members properly with the box frame as per architectural drawing providing full locking arrangement on 3 mm thick M.S. plates providing 38 mm x 38 mm x 6 mm M.S. angle clamps, fitting fixing with the outer frame of the gate, the clamp being embedded in R.C.C. pillars with cement concrete (1:2:4) including cutting holes and mending good the damages, finishing, curing and where necessary painting two coats with approved quality of synthetic enamel paint over a coat of priming cost of polish/ paint etc. of any type). item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type). f necessary).	sqm	Tk. 8,865.56	Tk. 8,850.00	Tk. 8,724.00	Tk. 8,723.00
19.5	Manufacturing, supplying, fitting and fixing G.I. pipe gate of any design and shape having 38 mm dia G.I. pipe outer frame and 19 mm dia. G.I. pipe vertical member placed @ 75 mm c/c by welding at top and bottom of G.I. pipe frame, cutting the pipes in proper shape and size including making semicircular band at the corners of the outer frame without damaging the pipe, covering 50% of the gate area with 18 BWG M.S. sheet, providing also two extra horizontal 38 mm dia G.I. pipes welded with vertical post, providing 6 nos. huskle domney with R.C.C. or R.C.C core pillar with cement concrete (1:2:4) in masonry including cutting holes, mending good the damages, making provision for minimum 0.61 m x 1.4 m pocket gate having its outer frame and inner vertical members made with 19 mm dia G.I. pipe including necessary locking arrangements. Painting the gate with 2 (two) coats of synthetic enamel paint over a coat of anti-corrosive priming, welding as and where necessary including necessary locking arrangement and providing cost of polish/ paint etc. of any type). work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type)	sqm	Tk. 10,063.37	Tk. 10,048.00	Tk. 9,936.00	Tk. 9,936.00

19.6	Supplying, fitting and fixing of rolling shutter of any design and shape made of 24 SWG G.P. sheet unit minimum 69 mm width having 10 mm and 12 mm dia circular folding at ends, rolled in machine, locked together properly to form rolling shutter, top of the shutter to be fitted with circular spring box made of 26 BWG sheet, containing best quality spring made in China, nutted with 38 mm G.I. pipe at one end and the other end nutted to shutter including fitting the G.I. pipe again with 375 mm x 300 mm x 6 mm pankha at ends with side guide channel (side channel consist of 50 mm x 100 mm x 6 mm M.S. plate with 2 mm x 19 mm x 3 mm F.I. bar, middle channel consist of 2 mm x 150 mm x 6 mm M.S. plate with 2 mm x 19 mm x 6 mm fixed to wall or column with clamps embedded in R.C.C. part by cement concrete (1:2:4) including cutting holes (minimum 3 mm x 25 mm x 6 mm) mending the damages, finishing, curing, bottom end of shutter fitted to folded G.P. sheet (min 125 mm width in two layers), which is again fitted to 25 mm x using cost of polish/ paint etc. of any type). item is for acoustic work in auditorium, Hall Room, Seminar Room etc.) (Rate is excluding cost of polish/ paint etc. of any type). f necessary). quare cm etc.including all accessories like sliding door key lock, sliding door wheel, sliding door mohiar, s	sqm	Tk. 6,405.26	Tk. 6,387.00	Tk. 6,276.00	Tk. 6,275.00
19.7	Manufacturing, supplying, fitting, fixing of ordinary type M.S. folding gate with 12mm x 12mm MS square solid bar and 25 mm x 25mm x 6 mm MS angle box frame and two nos outer side fixed box post made with 40 mm x 40 mm x 6 mm angle at both ends attached to column/wall side including locking arrangement providing 38 mm x 38 mm x 6 mm M.S. angle clamps fitted and fixed with the outer frame of the main gate, the clamp being embedded in R.C.C. or R.C.C. core pillar with cement concrete (1:2:4) in masonry including cutting holes and mending good the damages, finishing, curing etc. including riveting, welding as and where necessary, painting two coats with approved quality of synthetic enamel paint over a coat of priming of anti-corrosive paint etc. all complete as per drawing, design and accepted by the Engineer-in-charge. (Rate is excluding the cost of paint)	sqm	Tk. 5,630.01	Tk. 5,611.00	Tk. 5,489.00	Tk. 5,489.00
19.8	Manufacturing, supplying, fitting and fixing SS folding gate of any design and shape made of 37 mm x 1.5 mm round pipe section as vertical section spacing 450 mm c/c, 25 x 25 x 1.5 mm box section criss-cross connected with vertical section through 12 mm solid pipe section, 12 mm x 38 mm x 1.5 mm rectangular SS section fitted & fixed vertically & connected through 12 mm solid pipe section with vertical section, placing required number of SS wheel, SS Handle, SS Locking clamp etc, fitting-fixing the gate by cutting hole and mending good the damages by pouring concrete(1:2:4) into the holes and hoisting the gate in position with necessary carriage,electrodes, curing, polishing, finishing, etc.complete as per drawing and design and accepted by the Engineer-in-charge.					
19.8.1	for S.S. Grade A304	sqm	Tk. 17,043.07	Tk. 17,029.00	Tk. 16,913.00	Tk. 16,911.00
19.8.2	for S.S. Grade A201	sqm	Tk. 14,932.62	Tk. 14,919.00	Tk. 14,802.00	Tk. 14,801.00

19.9	Manufacturing, supplying, fitting and fixing SS Swing front gate of any design and shape made of 50 mm x 50mm x 1.5 mm S.S. box section as outer frame, 37mm x 37mm x 1.5 mm SS box section as vertical section @ 100 mm c/c fitted with outer frame, 25 mm x 25 mm x 1.5 mm SS box section as horizontal section , 37 mm x 1.5 mm SS round section fitted with vertical outer section for SS lettering, 25 mm dia SS solid round pipe, placing required number of solid hinges, SS wheel, SS locking arrangement, hasp bolt, tower bolt etc, fitting-fixing the gate by cutting hole and mending good the damages by pouring concrete(1:2:4) into the holes and hoisting the gate in position with necessary carriage,electrodes, curing, polishing, finishing, etc.complete as per drawing and design and accepted by the Engineer-in-charge .					
19.9.1	for S.S. Grade A304	sqm	Tk. 18,131.00	Tk. 18,119.00	Tk. 18,015.00	Tk. 18,014.00
19.9.2	for S.S. Grade A201	sqm	Tk. 14,493.22	Tk. 14,481.00	Tk. 14,377.00	Tk. 14,376.00
19.10	Manufacturing, supplying, fitting and fixing SS Sliding front gate of any design and shape made of 50 mm x 50mm x 1.5 mm S.S. box section as outer frame, 37mm x 37mm x 1.5 mm SS box section as vertical section @ 100 mm c/c fitted with outer frame, 25 mm x 25 mm x 1.5 mm SS box section as horizontal section , 37 mm x 1.5 mm SS round section fitted with vertical outer section for SS lettering, 25 mm dia SS solid round pipe & 3mm SS channel (62 mm x 62 mm) placed at bottom & top respectively, placing required number of solid hinges, SS wheel, SS locking arrangement, hasp bolt, tower bolt etc, fitting-fixing the gate by cutting hole and mending good the damages by pouring concrete(1:2:4) into the holes and hoisting the gate in position with necessary carriage,electrodes, curing, polishing, finishing, etc.complete as per drawing and design and accepted by the Engineer-in-charge .					
19.10.1	for S.S. Grade A304	sqm	Tk. 18,477.70	Tk. 18,465.00	Tk. 18,362.00	Tk. 18,361.00
19.10.2	for S.S. Grade A201	sqm	Tk. 14,839.92	Tk. 14,828.00	Tk. 14,724.00	Tk. 14,723.00
9.10.3	Manufacturing, supplying, fitting and fixing double leaf SS Main gate made with 2mm thick 75mm x 75mm ss hollow box as outer frame, 1.5 mm thick 50mm X 25mm ss hollow box as inner frame with 1.5 mm thick, 19 mm x 19 mm SS hollow box for fencing @75mm c/c, 900 mm Long 25mm dia SS box handle, 350 mm long 20mm dia SS hatch bolt, 150mm x 38mm hinge base plate, 50 mm x 8 mm S.S plate for bottom rail, 75 mm dia SS wheel, 6 mm thick 12" x 13" box SS 16 gauge plate for locking arrangement, 100 mm x 10 mm dia S.S. Rowel bolt, SS pad lock, fitting-fixing the gate by cutting hole and mending good the damages by pouring concrete(1:2:4) into the holes and hoisting the gate in position with necessary carriage,electrodes, curing, polishing, finishing, etc.complete as per drawing and design and accepted by the Engineer-in-charge.					
19.10.3.1	for S.S. Grade A304	sqm	Tk. 29,685.86	Tk. 29,667.00	Tk. 29,508.00	Tk. 29,506.00
19.10.3.2	for S.S. Grade A201	sqm	Tk. 24,564.16	Tk. 24,545.00	Tk. 24,386.00	Tk. 24,384.00

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CHAPTER 20 : STAIR & VERANDAH RAILING AND STAIR NOSING

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
20.1	Manufacturing, supplying, fitting and fixing of stair railing of standard height of any design and shape with square box (2 nos.in each tread) made by through welding of 2 nos. 19 mm x 19 mm x 3 mm M.S. angle to provide hand-rail, engraving each box 150mm in the concrete by 50 mm x 50 mm x 6 mm anchor plate at base, welding 38 mm x 6mm F.I. bar inclined plain with the top of the box to fit in the grooved wooden rail by necessary screws including cutting grooves in concrete, mending good the damages with cement concrete (1:2:4), applying french polish of approved quality and colour to wooden hand rail, 2 (two) coats of synthetic enamel paint of approved quality over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish).	sqm	Tk. 4,249.95	Tk. 4,223.00	Tk. 4,086.00	Tk. 4,086.00
20.2	Manufacturing, supplying, fitting and fixing of stair railing of standard height of any design and shape with square box (2 nos. box in each tread) made by through welding of 2nos. 25 mm x 25 mm x 5 mm M.S. angle , engraving each box 150 mm in the concrete by 50 mm x 50 mm x 6 mm anchor plate at base, welding 50 mm x 6 mm F.I. bar in inclined plain with the top of the box fitted in the grooved wooden rail and 2 Nos. 225 mm x25 mm finished section of wooden strip fitted and fixed at mid height of railing at both side of M.S. angle box in inclined plain parallel to the hand rail etc. fitted and fixed by necessary screws, nails including cutting grooves in concrete, mending good the damages with cement concrete (1:2:4), applying french polish of approved quality and colour to wooden hand rail and inclined wooden strip and painting to F.I bar and square box with 2 (two) coats of synthetic enamel paint of approved quality and colour over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail, wooden strip and cost of painting, polish).	sqm	Tk. 5,481.27	Tk. 5,453.00	Tk. 5,316.00	Tk. 5,316.00
20.3	Supplying, fitting and fixing 12 mm dia of 2.65 mm thick G.I pipe stair railing of standard height of any design and shape with 50 mm x 50 mm x 6 mm M.S. plate at the base of 12 mm dia G.I. Pipe and fitted and fixed by welding. Placing the pipes vertically @ 250 mm c/c (1 nos. in each steps) 150 mm embedded into the R.C.C tread of stair case after cutting grooves and mending good the damages with C.C. and providing 38 mm x 6 mm F.I. bar on the top of G.I. pipe to provide wooden hand rail of any design applying french polish of approved quality and colour to wooden hand rail and painting with 2 (two) coats of synthetic enamel paint of approved quality and colour over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish).					
20.3.1	With the provision of hand rail of size: 150 mm x 62 mm (finished)	sqm	Tk. 2,886.81	Tk. 2,873.00	Tk. 2,792.00	Tk. 2,792.00
20.3.2	With the provision of hand rail of size: 112 mm x 62 mm (finished)	sqm	Tk. 3,511.64	Tk. 3,488.00	Tk. 3,373.00	Tk. 3,373.00

20.4.1	Supplying, fitting and fixing of 12 mm dia of 2.65 mm thick G.I pipe stair railing of standard height and design and shape with 50 mm x 50 mm x 6 mm M.S. plate at the base of 12 mm dia G.I. Pipe and fitting & fixing by welding. Placing the pipes vertically @ 125 mm c/c (2 nos. in each steps) 150 mm embedded into the R.C.C tread of stair case after cutting grooves and mending good the damages with C.C. and providing 40 mm dia of 3.20 mm thick G.I. pipe hand rail on the top of vertical G.I. pipe, applying 2 (two) coats of synthetic enamel paint of approved quality and color over an anti-corrosive prime coat etc. all complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area of railing will be considered for measurement, rate is excluding the cost of paint).	sqm	Tk. 4,680.67	Tk. 4,666.00	Tk. 4,584.00	Tk. 4,584.00
20.4.2	Supplying, fitting and fixing of 12 mm dia of 2.65 mm thick G.I pipe stair railing of standard height and design and shape with 50 mm x 50 mm x 6 mm M.S. plate at the base of 12 mm dia G.I. Pipe and fitting & fixing by welding. Placing the pipes vertically @ 125 mm c/c (2 nos. in each steps) 150 mm embedded into the R.C.C tread of stair case after cutting grooves and mending good the damages with C.C. and providing 50 mm dia of 3.20 mm thick G.I. pipe hand rail on the top of vertical G.I. pipe, applying 2 (two) coats of synthetic enamel paint of approved quality and color over an anti-corrosive prime coat etc. all complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area of railing will be considered for measurement, rate is excluding the cost of paint).	sqm	Tk. 5,072.45	Tk. 5,054.00	Tk. 4,955.00	Tk. 4,955.00
20.5	Manufacturing, supplying, fitting and fixing of stair railing of standard height of any design and shape with 75mm x 50mm x 3mm MS box for hand rail and 03 nos vertical post of 75mm x 50mm x 3mm MS box in each flight , welding the vertical post box with 100 mm x 100 mm x 6 mm anchor base plate and fixing with concrete of stair by royal bolt, welding MS box or MS F.I bar of any shape and design horizontally between the vertical post @ 125 mm c/c , applying 2 (two) coats of synthetic enamel paint of approved quality and color over an anti-corrosive prime coat etc. finished in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area of railing will be considered for measurement, rate is excluding the cost of painting).					
20.5.1	With F.I bar of size: 38 mm x 6 mm	sqm	Tk. 5,099.00	Tk. 5,078.00	Tk. 4,959.00	Tk. 4,959.00
20.5.2	With MS box of size: 20 mm x 20 mm of 1.5 mm wall thickness	sqm	Tk. 4,622.00	Tk. 4,601.00	Tk. 4,482.00	Tk. 4,482.00
20.6.1	Manufacturing, supplying, fitting and fixing of stair railing of standard height of any design and shape with 75 mm x 75 mm finished wood for hand rail and vertical post , fitting with 02 Nos. frames (150 mm space between the two frames and vertical wooden post) of 50 mm x 25 mm 1.5 mm MS box in each flight and welding 50 mm x 6 mm F.I. bar horizontally in each frame @ 150 mm c/c, fixing the rail, post and frames by necessary screws, nails, bolts with concrete of stair, applying french polish of approved quality and colour to wooden hand rail, 2 (two) coats of synthetic enamel paint of approved quality over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish).	sqm	Tk. 4,343.00	Tk. 4,322.00	Tk. 4,216.00	Tk. 4,216.00

20.6.2	Manufacturing, supplying, fitting and fixing of stair railing of standard height of any design and shape with 100mm x 50mm finished wood for hand rail , 75mm x 50mm x 3mm MS box for vertical post (total 4 nos. post per flight) fitting by welding on 100 mm x 100 mm x 6 mm anchor plate at base, welding MS box or MS F.I bar of any shape and design horizontally between the vertical post @ 125 mm c/c , welding 50 mm x 6 mm F.I. bar in inclined fitted in the grooved wooden rail by necessary screws, nails, mending good the damages with cement concrete (1:2:4), applying french polish of approved quality and colour to wooden hand rail, 2 (two) coats of synthetic enamel paint of approved quality over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish).					
20.6.2.1	With MS box of size: 38 mm x 12 mm of 1.5 mm wall thickness	sqm	Tk. 4,726.00	Tk. 4,705.00	Tk. 4,599.00	Tk. 4,599.00
20.6.2.2	With F.I bar of size: 38 mm x 6 mm	sqm	Tk. 5,169.00	Tk. 5,148.00	Tk. 5,042.00	Tk. 5,042.00
20.6.3	Manufacturing, supplying, fitting and fixing of stair railing of standard height of any design and shape with 100mm x 62mm finished wood for hand rail, fixing with 75mm x 38mm x 3mm MS box of 400mm length which fitting between 2 nos. 75mm x 38mm x 3mm MS box (total 3 nos. post per flight) welding on 150 mm x 150 mm x 6 mm anchor plate at base, 01 no 75mm x 38mm x 3mm MS box (200 mm space from top wooden rail) inclined with top and 01 no 75mm x 38mm x 3mm MS box (fitted with plain concrete surface of stair by royal bolt) inclined with bottom, both welding with vertical post, welding 1.5 mm thick 20mm dia MS pipe @ 125mm c/c horizontally between the MS vertical post, welding 50 mm x 6 mm F.I. bar in inclined fitted in the grooved wooden rail by necessary screws, nails, mending good the damages with cement concrete (1:2:4), applying french polish of approved quality and colour to wooden hand rail, 2 (two) coats of synthetic enamel paint of approved quality over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish).	sqm	Tk. 7,334.00	Tk. 7,314.00	Tk. 7,212.00	Tk. 7,212.00
20.7	Manufacturing, supplying, fitting and fixing of stair railing of standard height of any design and shape with 100mm x 62mm finished wood for hand rail, fixed with 50mm x 25mm x 2mm SS box of 400mm length which fitting between 2 nos. 50mm x 25mm x 2mm SS box (total 3 nos. post per flight) welding on 100 mm x 100 mm x 6 mm anchor plate at base, 01 no 50mm x 25mm x 2mm SS box (200 mm space from wooden rail) inclined with top and 01 no 50mm x 25mm x 2mm SS box (fitted with plain concrete surface of stair by royal bolt) inclined with bottom, both welding with vertical post, welding 1.5 mm thick 20mm dia SS pipe 125mm c/c horizontally between the SS vertical post, welding 50 mm x 6 mm F.I. bar in inclined fitted in the grooved wooden rail by necessary screws, nails, mending good the damages with cement concrete (1:2:4) including polishing of SS box and pipe, applying french polish of approved quality and colour to wooden hand rail, 2 (two) coats of synthetic enamel paint of approved quality and color over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish of wood).	sqm	Tk. 10,614.00	Tk. 10,594.00	Tk. 10,496.00	Tk. 10,496.00

20.8.1	Supplying, fitting and fixing stainless steel (SS) stair railing of standard height with 2 mm thick 62 mm dia SS pipe for hand-rail, 2 mm thick 40 mm dia 5 nos vertical SS pipes in each flight, 1.5 mm thick 20 mm dia 5 nos horizontal SS pipes as per drawing, design including carrying, polishing, fabricating, welding and fixing with tread by 25 mm long royal bolt etc.all complete and accepted by the Engineer-in-charge.	sqm	Tk. 7,786.20	Tk. 7,770.00	Tk. 7,690.00	Tk. 7,690.00
20.8.2	Supplying, fitting and fixing stainless steel (SS) stair railing of standard height with 2 mm thick 62 mm x 25 mm SS Box for hand-rail fixing with 2 nos of consecutive SS vertical post of 2 mm thick 50 mm x 18 mm SS Box (4 nos vertical support in each flight, each vertical support containing 2 nos of consecutive SS vertical box of 125 mm space) fixing with tread by 75 mm long royal bolt, welding 1.5 mm thick 38 mm x 12 mm 5 nos horizontal SS Box between vertical support as per drawing, design including carrying, polishing, fabricating, welding etc.all complete and accepted by the Engineer-in-charge.	sqm	Tk. 10,471.00	Tk. 10,449.00	Tk. 10,331.00	Tk. 10,331.00
20.8.3	Supplying, fitting and fixing stainless steel (SS) stair railing of standard height with 2 mm thick 62 mm dia SS pipe for hand-rail, 50mm x 6mm SS double plate SS newel post having SS pipe holder at the top to provide hand rail and having SS knob between double plate to hold the horizontal pipe (4 nos. post in each flight) fitting by welding on 150 mm x 100 mm x 6 mm SS anchor plate at base, base plate fitting with the tread by 50mm long royal bolt, 1.5 mm thick 20 mm dia 5 nos horizontal pipes fitting in the SS knob including carrying, polishing fabricating etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 11,663.00	Tk. 11,643.00	Tk. 11,533.00	Tk. 11,533.00
20.8.4	Supplying, fitting and fixing stainless steel (SS) stair railing of standard height with 2 mm thick 62mm dia pipe for hand -rail, 2 mm thick 50mm dia SS vertical post having pipe holder at the top to provide hand rail and having SS knob to hold the horizontal pipe (4 nos. post in each flight) fitting by welding on 150 mm x 100 mm x 6 mm anchor plate at base , base plate fitting with the tread by 50 mm long royal bolt, 1.5 mm thick 20 mm dia 8 nos horizontal pipes fitting in the SS knob including carrying , polishing fabricating etc.all complete and accepted by the Engineer-in-charge.	sqm	Tk. 11,033.00	Tk. 11,014.00	Tk. 10,913.00	Tk. 10,913.00
20.9.1	Supplying, fitting and fixing 10 mm thick tempered glass railing in stair with 62 mm dia 2 mm thick SS pipe for hand rail, 04 nos 62 mm x 50 mm x 2 mm SS vertical box post (having 4 nos. SS glass holder and 1 no rail holder) in each flight fitting and fixing with 65 mm x 55 mm x 6 mm SS base plate including welding, bending, fabricating, polishing all complete as per drawing, desing and specification etc in all respect and accepted by Engineer-in-charge.	sqm	Tk. 12,230.79	Tk. 12,212.00	Tk. 12,097.00	Tk. 12,097.00
20.9.2	Supplying, fitting and fixing 10 mm thick tempered glass railing in stair of standard height with 2 mm thick 62 mm dia SS pipe for hand-rail, 2 mm thick 62 mm dia SS vertical post (4 nos. post in each flight) having 1 no. pipe holder at the top to provide top rail and 4 nos. SS glass holder to fit the tempered glass, welding vertical post on 100 mm x 100 mm x 6 mm SS base plate and fitting base plate with the tread by 50mm long royal bolt including welding, bending, fabricating, polishing all complete as per drawing, desing and specification etc in all respect and accepted by Engineer-in-charge.	sqm	Tk. 22,005.00	Tk. 21,982.00	Tk. 21,859.00	Tk. 21,859.00

20.9.3	Supplying, fitting and fixing 10 mm thick tempered glass railing in stair of standard height with 2 mm thick 62 mm dia SS pipe for hand-rail, fixing rail with the glass by SS holder, fitting the tempered glass with 150mm/200 mm long SS glass clamp (4 nos. clamp in each flight), welding the SS glass clamp on 100 mm x 100 mm x 6 mm SS base plate and fitting base plate with the tread by 50mm long royal bolt including welding, bending, fabricating, polishing all complete as per drawing, desing and specification etc in all respect and accepted by Engineer-in-charge.	sqm	Tk. 12,096.00	Tk. 12,078.00	Tk. 11,970.00	Tk. 11,970.00
20.9.4	Supplying, fitting and fixing 10 mm thick tempered glass railing in stair of standard height with 2 mm thick 62 mm dia SS pipe for hand-rail, 2 mm thick 50 mm x 25 mm SS vertical post (4 nos. post in each flight) having 1 no. pipe holder at the top to provide top rail and 2 nos. SS glass holder to fit the tempered glass, welding vertical post on 100 mm x 100 mm x 6 mm SS base plate and fitting base plate on outer side of the tread by 75mm long royal bolt including welding, bending, fabricating, polishing all complete as per drawing, desing and specification etc in all respect and accepted by Engineer-in-charge.	sqm	Tk. 11,918.00	Tk. 11,899.00	Tk. 11,787.00	Tk. 11,787.00
20.10.1	Supplying, fitting and fixing 10 mm thick tempered glass railing in stair of standard height with 62mm dia finished wood with 10mm x 18mm groove for hand rail, fitting 150mm x 50mm finished wood with 10mm x 50mm groove with the plain RCC surface at bottom by 50mm long royal bolt, 10 mm thick tempered glass fitting in the groove between the hand rail and bottom wooden section by glass protector bit/rubber gasket, silicon including carrying, fabricating etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding cost of wood and cost of polish).	sqm	Tk. 6,672.00	Tk. 6,662.00	Tk. 6,584.00	Tk. 6,584.00
20.10.2	Supplying, fitting and fixing 10 mm thick tempered glass railing in stair of standard height with 75 mm x 50 mm finished wood for hand rail and vertical post of both sides of stair railing having 10mm x 38mm groove for fitting the tempered glass, 1.5mm thick 62 mm x 25 mm alluminium U channel fitting with the RCC plain surface at bottom by 50mm long royal bolt, 10 mm thick Tempered glass fitting between the wooden groove and alluminium U channel through glass protector bit/rubber gasket, silicon as per drawing, design including carrying, fabricating etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding cost of wood and cost of polish).	sqm	Tk. 7,578.00	Tk. 7,568.00	Tk. 7,489.00	Tk. 7,489.00
20.10.3	Supplying, fitting and fixing 10 mm thick tempered glass railing in stair of standard height with 75 mm x 50 mm finished wood for hand rail and vertical post of both sides of stair tread, fixing the rail with the tempered glass by SS holder, fitting the bottom edge of the tempered glass with 1.5mm thick 38mm x 18 mm alluminium U channel and the top edge of the tempered glass with 1.5 mm thick 12mm x 12 mm alluminium U channel through glass protector bit/rubber gasket, silicon and fixing with the tread of stair by 50mm long royal bolt, fitting 38 mm x 6 mm F.I. bar at the bottom of the grooved wooden hand rail by necessary screws, nails including fitting, fixing, carrying, fabricating etc. as per drawing, design all complete and accepted by the Engineer-in-charge. (Rate is excluding cost of wood and cost of polish).	sqm	Tk. 8,478.00	Tk. 8,467.00	Tk. 8,389.00	Tk. 8,389.00

20.10.4	Supplying, fitting and fixing 10 mm thick tempered glass railing in stair of standard height with 2 mm thick 62 mm dia SS pipe or 2 mm thick 62 mm x 38 mm SS box for hand rail and vertical post of both sides of stair tread, fixing the rail with the tempered glass by SS holder, fitting the bottom edge of the tempered glass with 1.5mm thick 38mm x 18 mm alluminium U channel and the top edge of the tempered glass with 1.5 mm thick 12mm x 12 mm alluminium U channel through glass protector bit/rubber gasket, silicon and fixing with the tread of stair by 50 mm long royal bolt, welding vertical post on 100 mm x 100 mm x 6 mm SS base plate and fitting base plate with the concrete of stair by 50mm long royal bolt including fitting, fixing,carrying, fabricating, polishing etc. as per drawing, design all complete and accepted by the Engineer-in-charge.	sqm	Tk. 10,961.00	Tk. 10,948.00	Tk. 10,851.00	Tk. 10,851.00
20.11	Supplying, fitting and fixing stainless steel railing of standard height (approximately 225 mm above RCC or Brick wall) with 2 mm thick 62 mm dia SS pipe for hand-rail, 2 mm thick 40 mm dia vertical SS pipes @ 1200 mm c/c fitting and fixing with 65 mm x 55 mm x 6 mm SS base plate and fitting base plate with the surface by 50mm long royal bolt including welding, bending, fabricating, polishing all complete as per drawing, desing and specification etc all complete in all respect and accepted by Engineer-in-charge.	meter	Tk. 2,236.00	Tk. 2,236.00	Tk. 2,236.00	Tk. 2,236.00
20.12.1	Supplying, fitting and fixing stainless steel railing on inner side of stair with 2 mm thick 62 mm dia SS pipe for hand-rail (50mm space from wall), fitting and fixing by a SS pipe holder @ 1200 mm c/c welding with 65 mm x 55 mm x 6 mm SS base plate and fitting base plate on RCC or Brick wall by SS Royel Bolt including welding, bending, fabricating, polishing all complete as per drawing, desing and specification etc all complete in all respect and accepted by Engineer-in-charge.	meter	Tk. 2,288.36	Tk. 2,287.00	Tk. 2,276.00	Tk. 2,276.00
20.12.2	Supplying, fitting and fixing MS railing on inner side of stair with 3 mm thick 62 mm x 38 mm MS box for hand-rail (50mm space from wall), fitting and fixing by a MS box holder @ 1200 mm c/c welding with 65 mm x 55 mm x 6 mm MS base plate and fitting base plate on RCC or Brick wall by Royel Bolt including welding, bending, fabricating, all complete as per drawing, desing and specification etc all complete in all respect and accepted by Engineer-in-charge.	meter	Tk. 1,329.21	Tk. 1,328.00	Tk. 1,317.00	Tk. 1,317.00
20.12.3	Supplying, fitting and fixing 50 mm dia finished Wooden railing on inner side of stair (50mm space from wall), fitting both sides by 2 mm thick 50 mm dia SS pipe caping one end of 250 mm length, fitting and fixing by a SS holder @ 1200 mm c/c welding with 65 mm x 55 mm x 6 mm SS base plate and fitting base plate on RCC or Brick wall by Royel Bolt including welding, bending, fabricating, all complete as per drawing, desing and specification etc all complete in all respect and accepted by Engineer-in-charge	meter	Tk. 2,264.53	Tk. 2,245.00	Tk. 2,258.00	Tk. 2,258.00
20.13	Supplying, fitting and fixing SS Swimming Pool Ladder of 3000 mm x 600 mm size in which outer frame of 2 mm thick 50 mm dia SS pipe and horizontal bar for lower part (approximately 1800 mm) of 2 mm thick 40 mm dia SS pipe @ 200 mm c/c and fitting and fixing the whole with 100 mm x 100 mm x 6 mm SS base plate on both side and fitting base plate on RCC or Brick wall by Royel Bolt including welding, bending, fabricating, all complete as per drawing, desing and specification etc all complete in all respect and accepted by Engineer-in-charge.	sqm	Tk. 8,154.88	Tk. 8,135.00	Tk. 8,031.00	Tk. 8,031.00

20.14.1	Supplying, fitting and fixing verandah railing with 2 mm thick 50 mm dia SS hand rail, 1.5 mm thick 20 mm dia 5 nos SS horizontal pipes, 2 mm thick 50 mm x 25 mm SS box vertical post @ 450 mm c/c. Among the vertical box post, 5 nos box posts will be fitted with floor bed by 65 mm x 55 mm x 6 mm SS plate, remaining SS box post will be fitted and fixed with 50 mm x 25 mm x 2 mm SS horizontal box at bottom (50 mm above from floor) including all fittings, fixtures, polishing, carrying etc. all complete as per approved design and drawing and accepted by the Engineer-in-charge.	sqm	Tk. 9,532.99	Tk. 9,504.00	Tk. 9,349.00	Tk. 9,349.00
20.14.2	Supplying, fitting and fixing verandah railing with 2 mm thick 62 mm dia SS hand rail, 1.5 mm thick 25 mm square SS box @150mm c/c vertically, 2 mm thick 50 mm dia SS vertical post @ 900 mm c/c, 2 nos 1.5 mm thick 38mm x 38mm square SS box fitting horizontally at 125mm below from top rail and 125mm above from floor, vertical box posts will be fitted with floor bed by 65 mm x 55 mm x 6 mm SS plate including all fittings, fixtures, polishing, carrying etc. all complete as per approved design and drawing and accepted by the Engineer-in-charge.	sqm	Tk. 10,685.00	Tk. 10,671.00	Tk. 10,595.00	Tk. 10,595.00
20.14.3	Supplying, fitting and fixing verandah railing of standard height with 2 mm thick 62 mm dia SS pipe for hand-rail, 2mm thick 75mm x 75mm SS square vertical post having pipe holder to provide hand rail and SS knob for horizontal pipe @ 900mm c/c fitting by welding on 150 mm x 150 mm x 6 mm SS anchor plate at base, base plate fitting with the floor by 50mm long royal bolt, 1.5 mm thick 25 mm dia 8 nos horizontal SS pipes fitting with SS knob of vertical post including carrying, polishing fabricating etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 12,001.00	Tk. 11,988.00	Tk. 11,918.00	Tk. 11,918.00
20.14.4	Supplying, fitting and fixing verandah railing of standard height with 2 mm thick 62 mm dia SS pipe for hand-rail, 2 mm thick 50mm dia vertical SS post having pipe holder to provide hand rail and SS knob for horizontal pipe @ 900mm c/c fitting by welding on 150 mm x 150 mm x 6 mm SS anchor plate at base, base plate fitting with the floor by 50mm long royal bolt, 1.5 mm thick 25 mm dia 8 nos horizontal SS pipes fitting with SS knob of vertical post including carrying, polishing fabricating etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 11,946.00	Tk. 11,932.00	Tk. 11,856.00	Tk. 11,856.00
20.15.1	Supplying, fitting and fixing 10 mm thick tempered glass railing in verandah with 62mm dia 2 mm thick SS pipe for hand rail, 62 mm x 50 mm x 2 mm SS vertical box fitted with concrete slab by 100 mm x 55 mm x 6 mm SS plate @ 600 mm c/c and base plate fitting with the floor by 50mm long royal bolt including all fittings, fixtures as per drawing, desing and specification etc all complete in all respect and accepted by Engineer-in-charge.	sqm	Tk. 11,949.27	Tk. 11,937.00	Tk. 11,867.00	Tk. 11,867.00
20.15.2	Supplying, fitting and fixing 10 mm thick tempered glass railing in verandah of standard height with 2 mm thick 62 mm dia SS pipe for hand-rail, 2 mm thick 50 mm dia SS vertical post having pipe holder to provide hand rail and SS glass holder @ 900mm c/c fitting by welding on 100 mm x 100 mm x 6 mm SS anchor plate at base, base plate fitting with the floor by 50mm long royal bolt, 10 mm thick Tempered glass fitting between the post through glass holder as per drawing, design including carrying, polishing fabricating etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 11,779.00	Tk. 11,768.00	Tk. 11,700.00	Tk. 11,700.00

20.15.3	Supplying, fitting and fixing 10 mm thick tempered glass railing in verandah of standard height with 1.5 mm thick 50mm x 15mm SS U channel at top of rail, fitting the tempered glass with 150mm/200 mm long SS glass clamp @ 900mm c/c, welding the SS glass clamp on 100 mm x 100 mm x 6 mm SS base plate and fitting base plate with the floor by 50mm long royal bolt including welding, bending, fabricating, polishing all complete as per drawing, desing and specification etc in all respect and accepted by Engineer-in-charge.	sqm	Tk. 10,735.00	Tk. 10,725.00	Tk. 10,663.00	Tk. 10,663.00
20.16	Supplying, fitting and fixing 10 mm thick tempered glass railing in verandah of standard height with 3mm thick 50mm x 50mm square MS box for hand rail, 3mm thick 75mm x 38mm mm MS vertical post box @ 900 mm c/c fitting by welding on 150 mm x 100 mm x 6 mm anchor plate at base, base plate fitting with the floor by 50mm long royal bolt, fitting 10mm thick tempered glass between the vertical post through glass holder, applying 2 (two) coats of synthetic enamel paint of approved quality over an anti-corrosive prime coat including all other accessories and carrying, fabricating etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the cost of painting).	sqm	Tk. 9,091.00	Tk. 9,078.00	Tk. 8,997.00	Tk. 8,997.00
20.17.1	Supplying, fitting and fixing verandah railing of standard height with 3mm thick 75mm x 50mm MS box for vertical post box @ 900 mm c/c fitting by welding on 150 mm x 100 mm x 6 mm anchor plate at base, base plate fitting with the floor by 50mm long royal bolt, 3mm thick 50mm x 38mm MS horizontal box fitting with the vertical post by welding @ 150 mm c/c including all other accessories and carrying, febrication, applying 2 (two) coats of synthetic enamel paint of approved quality over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area of railing will be considered for measurement, rate is excluding the cost of painting).	sqm	Tk. 5,857.00	Tk. 5,841.00	Tk. 5,753.00	Tk. 5,753.00
20.17.2	Supplying, fitting and fixing verandah railing of standard height with 5mm thick 75mm x 75mm MS box for vertical post box @ 900 mm c/c fitting by welding on 150 mm x 150 mm x 6 mm anchor plate at base, base plate fitting with the floor by 50mm long royal bolt, fitting 5mm thick 50mm x 50mm MS horizontal box with the vertical post by welding at top of rail and 125mm above from the floor, 3mm thick 25mm x 25mm MS square box fitting by welding vertically @ 150 mm c/c between the top and bottom rail including all other accessories and carrying, febrication, applying 2 (two) coats of synthetic enamel paint of approved quality over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area of railing will be considered for measurement, rate is excluding the cost of painting).	sqm	Tk. 6,531.00	Tk. 6,515.00	Tk. 6,426.00	Tk. 6,426.00

20.17.3	Supplying, fitting and fixing verandah railing of standard height with 5mm thick 75mm x 50mm MS box for vertical post box @ 900 mm c/c fitting by welding on 150 mm x 150 mm x 6 mm anchor plate at base, base plate fitting with the floor by 50mm long royal bolt, 5mm thick 75mm x 50mm MS horizontal box fitting by welding with the vertical post for hand rail, 2 nos. 5mm thick 50mm x 50mm MS horizontal box fitting with the vertical post by welding 125 mm below the top rail and 125 mm above the floor, 3mm thick 25mm x 25mm MS square box fitting by welding vertically @ 150 mm c/c between the 2 nos. 50mm x 50mm horizontal MS box including all other accessories and carrying, fabrication, applying 2 (two) coats of synthetic enamel paint of approved quality over an anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area of railing will be considered for measurement, rate is excluding the cost of painting).	sqm	Tk. 6,958.00	Tk. 6,941.00	Tk. 6,853.00	Tk. 6,853.00
20.18	Providing nosing to the nose of tread of stair including supplying 3 nos. of 75 mm long 12 mm x 3 mm F.I./SS bar clamps with bifurcated ends in each treads including fitting and fixing with the angle by welding and fitting and fixing the angle by cutting grooves in R.C.C, fixing and finishing and the same with C.C. (1:2:4) etc. cost of electricity and other charges all complete and accepted by the Engineer-in-charge					
20.18.1	Providing 38 mm x 38 mm x 6 mm M.S. angle to the nosing of steps of staircase	meter	Tk. 478.91	Tk. 477.00	Tk. 470.00	Tk. 470.00
20.18.2	Providing nosing 32 mm x 32 mm x 6 mm M.S. angle to the nosing of steps of stair case.	meter	Tk. 421.21	Tk. 420.00	Tk. 413.00	Tk. 413.00
20.18.3	Providing 25 mm x 25 mm x 6 mm size M.S. angle to the nosing of stair case.	meter	Tk. 364.85	Tk. 363.00	Tk. 356.00	Tk. 356.00
20.18.4	Providing 38 mm x 38 mm x 6 mm brass angle to the nosing of steps of stair case	meter	Tk. 1,517.59	Tk. 1,516.00	Tk. 1,509.00	Tk. 1,509.00
20.18.5	Providing 38 mm x 38 mm x 6 mm S.S. angle to the nosing of steps of staircase	meter	Tk. 1,475.32	Tk. 1,474.00	Tk. 1,467.00	Tk. 1,467.00
20.18.6	Providing 32 mm x 32 mm x 6 mm S.S. angle to the nosing of steps of stair case	meter	Tk. 1,179.42	Tk. 1,178.00	Tk. 1,171.00	Tk. 1,171.00
20.18.7	Providing 25 mm x 25 mm x 6 mm size S.S. angle to the nosing of stair case.	meter	Tk. 1,116.01	Tk. 1,114.00	Tk. 1,107.00	Tk. 1,107.00
20.19	Supplying, fitting & fixing of column guard made of 50 mm x 50 mm x 6 mm M.S. angle around column & 50 mm x 6 mm F.I. bar for Tie at the top including fitting and fixing with the angle by welding, fitting and fixing the angle by cutting grooves in R.C.C, fixing and finishing the same with C.C. (1:2:4) etc. cost of electricity and other charges all complete and accepted by the Engineer-in-charge	meter	Tk. 629.85	Tk. 628.00	Tk. 617.00	Tk. 617.00
20.20.1	Supplying, fitting & fixing of column guard made of 75 mm x 75 mm x 3 mm S.S. angle at corner of column including supplying 3 Nos. of 75 mm long 12 mm x 3 mm SS bar clamp with bifurcated ends, fitting and fixing with the angle by welding, fitting and fixing the angle by cutting grooves in R.C.C, fixing and finishing the same with C.C. (1:2:4) etc. cost of electricity and other charges all complete and accepted by the Engineer-in-charge.	meter	Tk. 1,824.08	Tk. 1,823.00	Tk. 1,812.00	Tk. 1,812.00
20.20.2	For column guard made of 50 mm x 50 mm x 3 mm S.S. angle	meter	Tk. 1,743.76	Tk. 1,742.00	Tk. 1,731.00	Tk. 1,731.00
..... END OF CHAPTER TWENTY						

CHAPTER 21 : GRILL FENCING AND BARBED WIRE FENCING

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
21.1	Supplying, fitting and fixing MS grill fencing of any design and shape in boundary wall with 31 mm x 6 mm F.I. bar @ 150 mm c/c placed vertically after cutting as per required measurement. Welding the said F.I. bar with 2 nos. horizontal 30 mm x 6 mm F.I. bar placed one at top and another at bottom including embedding/fixing the horizontal F.I. bar in position with cutting grooves in wall up to 150 mm depth and mending good the damages with C.C. (1:2:4), finishing with two coats of painting with synthetic enamel paint over a coat of anti-corrosive paint etc. complete as per drawing and design and accepted by the Engineer-in-charge. (Rate is excluding the cost of painting)	sqm	Tk. 2,614.52	Tk. 2,602.00	Tk. 2,522.00	Tk. 2,522.00
21.2	Supplying, fitting and fixing MS grill fencing of any design and shape made with 19 mm x 19 mm x 3 mm box (box made with 2 nos. 19 mm x 19 mm x 3 mm angle by continuous welding) placed vertically @ 150 mm c/c and welding the said vertical boxes with two Nos. horizontal 25 x 25 x 6 mm angle box (made with 2 Nos. 25 x 25 x 6 mm angle by welding), placing one at bottom and another at top and again fitting & fixing by welding the said horizontal tube box with 38 mm x 38 mm x 6 mm angle box vertical post made with 2 nos. 38 mm x 38 mm x 6 mm angle by welding) provided @ 1800 mm c/c including embedding the tube box post in brick R.C.C. pillar or core wall up to 375 mm depth after cutting grooves in wall/pillar including mending good the damages with C.C. (1:2:4) and finishing, curing, 2 coats of approved synthetic enamel paint over a coat of anti-corrosive paint etc. all complete as per design, drawing and accepted by the Engineer-in-charge. (Rate is excluding the cost of painting)	sqm	Tk. 5,253.87	Tk. 5,232.00	Tk. 5,087.00	Tk. 5,087.00
21.3	Manufacturing, supplying, fitting and fixing M.S. grill fencing as per design with outer frames having 38 mm x 38 mm x 6 mm M.S. angle and inner members having 10 mm dia M.S. rod placed @ 100 mm c/c diagonally on both direction, welding each criss-cross end of rod with corners of outer frame including cutting rods and size angles to required shapes and size and setting the entire fence with the previously installed box tube (box tube made with 2 nos. angle) post including thorough and full welding the frame with the angle box posts, painting 2 coats of synthetic enamel paint over a coat of anti-corrosive priming etc. all complete as per drawing and accepted by the Engineer-in-charge. (Rate is excluding the cost of paint)	sqm	Tk. 2,535.85	Tk. 2,522.00	Tk. 2,445.00	Tk. 2,445.00
21.4	Manufacturing, supplying, fitting and fixing M.S. box tube (made with 2 nos angle) post (bottom end bifurcated) made by thorough welding of two Nos. 38 x 38 x 6 mm M.S. angle and vertically placing the post into C.C. or R.C.C. foundation including cutting angle to required shape and size including covering the top of the post with 6 mm thick M.S. plate etc. all complete as per drawing and accepted by the Engineer-in-charge. (Rate is excluding the cost of C.C./R.C.C. foundation and paint)	meter	Tk. 1,105.36	Tk. 1,102.00	Tk. 1,078.00	Tk. 1,078.00

21.5	Providing barbed wire fencing with R.C.C (1:2:4) pillars @ 2.43 m c/c and of section 100 mm x 100 mm at top and 150 mm x 150 mm at bottom of 2.13 m total height (1.37 m above G.L. and 0.76 m below G.L.) including 150 mm thick, 450 mm square spread footings (pillars reinforced with 4 Nos. 10 mm dia main rod and 6 mm dia stirrups @ 150 mm c/c, footing reinforced with 5 Nos. 10 mm dia rod both ways) supplying, fitting and fixing 8 lines of barbed wire horizontally (fixed with the post through 6 mm dia rods embedded in to the post) and 2 lines diagonally from post to post with 12 BWG 2 ply barbed wire, with 4 points barbs @ at least 112 mm c/c including 6 mm thick (1:4) cement plaster up to 1500 mm length of the pillars etc. complete and accepted by the Engineer-in-charge. (Rate is excluding the cost of concrete, reinforcement & plaster which is to be paid as per corresponding items in the schedule)	sqm	Tk. 156.83	Tk. 156.00	Tk. 154.00	Tk. 154.00
21.6	Providing barbed wire fencing with average 100 mm dia best quality local ballah post of 1.98 m total height (1.37 m above G.L. and min. 0.60 m embedded in the ground) placed at 1800 mm c/c with 9 nos. horizontal lines and 2 nos. diagonal lines of 12 BWG 2 ply barbed wire with 4 points barbs @ at least 112 mm c/c in each way of 1.8 m including embedding the post tightly in ground and providing a cement concrete base of 200 mm x 200 mm x 300 mm around ballah at E.G.L, stretching the wires and fixing with the post with the help of minimum 62 mm long G.I. nails including supply of all materials etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the cost of ballah and its driving/fixing and concrete which is to be paid as per corresponding items in the schedule)	sqm	Tk. 279.77	Tk. 278.00	Tk. 275.00	Tk. 275.00
21.7	Supplying, fitting and fixing 12 BWG barbed wire (2 ply, 4 points) in fencing work @ 150 mm c/c in both horizontally and vertically, supported by 38 mm x 38 mm x 6 mm M.S. L-shape angle post (300 mm embedded in R.C.C. or in brick work with a cement concrete base of 75 mm x 75 mm x 300 mm) 600 mm vertical and 450 mm inclined or as per requirement @ 2.4 m c/c including straightening, binding the joints with 18 BWG wire, making holes in the angle etc. including supplying all necessary materials complete in all respect and accepted by the Engineer-in-charge. (Rate is excluding the cost of paint, R.C.C /brick work or C.C. which is to be paid as per corresponding items in the schedule).	sqm	Tk. 959.64	Tk. 953.00	Tk. 933.00	Tk. 933.00
21.8	Supplying, fitting and fixing 12 BWG barbed wire (2 ply, 4 points) in fencing work @ 150 mm c/c both horizontally and vertically, supported by 38 mm x 38 mm x 6 mm M.S. Y-shape angle post (300 mm embedded in R.C.C. or in brick work with a cement concrete base of 75 mm x 75 mm x 300 mm) 600 mm vertical and 450 mm inclined or as per requirement @ 2.4 m c/c including straightening, binding the joints with 18 BWG wire, making holes in the angle etc. including supplying all necessary materials complete in all respect and accepted by the Engineer-in-charge. (Rate is excluding the cost of paint, R.C.C /brick work or C.C. which is to be paid as per corresponding items in the schedule).	sqm	Tk. 1,033.23	Tk. 1,026.00	Tk. 1,006.00	Tk. 1,006.00
21.9	Supplying, fitting and fixing 12 BWG barbed wire fencing in a circular shape of 600 mm dia and 100 mm pitch fitted with MS rod casing (made by 8 nos. 10 mm dia plain bar in a circular pattern and placed at equal interval), 38 x 38 x 6 mm M.S. angle post (300 mm embedded in R.C.C. or in brick work with a cement concrete base of 75 mm x 75 mm x 300 mm) and 600 mm vertical and 450 mm inclined placed @ 3.0 m c/c including straightening, binding the joints with 18 BWG wire, making holes in the angle etc. including supply of all necessary materials complete in all respect and accepted by the Engineer-in-charge. (Rate is excluding the cost of Painting).	meter	Tk. 2,427.51	Tk. 2,430.00	Tk. 2,382.00	Tk. 2,382.00

21.10	Supplying, fitting and fixing Fency grill in Boundary wall made of 10 mm dia M.S. rod provided @ 75 mm c/c in any position both ways as per design with outside frame of 50 mm x 50 mm x 6 mm M.S. tube and two inner frame made of 25 mm x 25 mm x 6 mm M.S. tube including fabricating, welding in each point, painting 2 coats of synthetic enamel paint over a coat of anti-corrosive priming including cost of electricity, tools and plants, workshop charges, carriage of the same including cutting groves in the R.C.C. or brick work, mending good the damages with C.C. (1:2:4) etc, all complete for all floors accepted by the Engineer-in-charge. [Rate is excluding the cost of painting]	sqm	Tk. 4,123.86	Tk. 4,119.00	Tk. 4,082.00	Tk. 4,082.00
21.11.1	Manufacturing, supplying, fitting and fixing S.S. grill fencing(Grade 304) as per design in boundary wall with 50mm x 50mm x1.5mm box horizontal section for top & bottom part & 25mmx1.5mm ss round horizontal section @ 4" c/c between top & bottom which are attached with 4 mm thickness ss base plate of required size & ss royal bolt with side column/wall, 37 mm x 1.5 mm ss round vertical section 3'-0" c/c section attached with bottom RCC section and top SS vertical section with required welding, mending good the damages with concrete(1:2:4) required polishing, finishing etc. all complete as per drawing and accepted by the Engineer-in-charge.	sqm	Tk. 13,716.83	Tk. 13,709.00	Tk. 13,675.00	Tk. 13,675.00
21.11.2	Manufacturing, supplying, fitting and fixing S.S. grill fencing(Grade 201) as per design in boundary wall with 50mm x 50mm x1.5mm box horizontal section for top & bottom part & 25mmx1.5mm ss round horizontal section @ 4" c/c between top & bottom which are attached with 4 mm thickness ss base plate of required size & ss royal bolt with side column/wall, 37 mm x 1.5 mm ss round vertical section 3'-0" c/c section attached with bottom RCC section and top SS vertical section with required welding, mending good the damages with concrete(1:2:4) required polishing, finishing etc. all complete as per drawing and accepted by the Engineer-in-charge.	sqm	Tk. 9,404.20	Tk. 9,396.00	Tk. 9,362.00	Tk. 9,362.00
21.12	Supplying, fitting and fixing fency grill in Boundary wall made of 38 mm x 38 mm x 6 mm M.S. tube vertical section @ 150 mm c/c and 19 mm x 19 mm x 3 mm M.S. Tube horizontal section @ 900 mm c/c fabricating, welding in each point, including cost of electricity, tools and plants, workshop charges, carriage of the same including cutting groves in the R.C.C. or brick work, mending good the damages with C.C. (1:2:4) etc, all complete for all floors accepted by the Engineer-in-charge. (Rate is excluding the cost of painting)	sqm	Tk. 2,942.92	Tk. 2,938.00	Tk. 2,912.00	Tk. 2,912.00
21.13	Supplying, fitting and fixing SS galvanized barbed wire (dia 2.5 mm) fencing in a circular shape of 100 mm pitch fitted with SS rod casing (made by 6 nos. 10 mm dia SS solid bar(304 grade) in a circular pattern and placed at equal interval), 38 x 38 x 6 mm S.S. angle(304 grade) Y-post 600 mm vertical and 450 mm inclined placed @ 1.5 m c/c (300 mm embedded in R.C.C. or in brick work with a cement concrete base of 75 mm x 75 mm x 300 mm) and including straightening, binding the joints with 18 BWG wire, making holes in the angle etc. including supply of all necessary materials complete in all respect and accepted by the Engineer-in-charge.					
21.13.1	600 mm circular dia	meter	Tk. 11,486.91	Tk. 11,480.00	Tk. 11,444.00	Tk. 11,444.00
21.13.2	450 mm circular dia	meter	Tk. 10,118.12	Tk. 10,109.00	Tk. 10,073.00	Tk. 10,072.00

21.14	Supplying, fitting and fixing SS galvanized barbed wire(dia 2.5 mm) fencing in a circular shape of 100 mm pitch fitted with SS rod casing (made by 6 nos. 10 mm dia SS solid bar(201 grade) in a circular pattern and placed at equal interval), 38 x 38 x 6 mm S.S. angle(201 grade) Y- post 600 mm vertical and 450 mm inclined placed @ 1.5 m c/c (225 mm embedded in R.C.C. or in brick work with a cement concrete base of 75 mm x 75 mm x 225 mm) including straightening, binding the joints with 18 BWG wire, making holes in the angle etc. including supply of all necessary materials complete in all respect and accepted by the Engineer-in-charge.					
21.14.1	600 mm circular dia	meter	Tk. 10,512.19	Tk. 10,504.00	Tk. 10,468.00	Tk. 10,468.00
21.14.2	450 mm circular dia	meter	Tk. 8,395.66	Tk. 8,390.00	Tk. 8,354.00	Tk. 8,354.00
21.15	Manufacturing, supplying, fitting and fixing steel grill fencing in boundary wall with 38 mm dia MS Pipe with thickness 6 mm placed @ 150 mm c/c vertically and mending good the damages with C.C. (1:2:4), finishing with two coats of painting with synthetic enamel paint over a coat of anti-corrosive paint etc. complete as per drawing and design and accepted by the Engineer-in-charge. (Rate is excluding the cost of painting)	sqm	Tk. 4,329.06	Tk. 4,319.00	Tk. 4,272.00	Tk. 4,272.00

..... END OF CHAPTER TWENTY ONE

CHAPTER 22 : MISCELLANEOUS ITEMS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
22.1	Supplying, fitting and fixing of wooden wedge (12 mm base x 12 mm height) instead of putty for fixing glass panes with wooden frame by nails or by any other approved method including the cost of necessary wood (Garjan/ Jam/ Local sal), nails etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the cost of wooden frame, painting and 5 mm thick glass panes)	meter	Tk. 78.67	Tk. 78.00	Tk. 75.00	Tk. 75.00
22.2	Supplying, fitting and fixing of best quality of different dia foreign made aluminium pipe curtain rail for door and window including cutting and fitting-fixing as per required measurement and supplying necessary screws, SS clamp, electric drill and rowel plug including charge of electricity etc. all complete and accepted by the Engineer-in-charge.					
22.2.1	20 mm dia Aluminium pipe curtain rail	meter	Tk. 613.78	Tk. 717.00	Tk. 709.00	Tk. 709.00
22.2.2	25 mm dia Aluminium pipe curtain rail	meter	Tk. 791.84	Tk. 791.00	Tk. 788.00	Tk. 788.00

22.3	Supplying, fitting and fixing different dia best quality Stainless steel (SS) pipe as curtain rail for door and windows including cutting and fitting fixing as per required measurement and supplying heavy type SS Clamps & 2 nos SS End Caps, necessary screws, electric drill and rowel plug including charge of electricity etc. all complete and accepted by the Engineer-in-charge.					
22.3.1	20 mm dia Stainless steel (SS) pipe with wall thickness 1.00 mm .	meter	Tk. 563.04	Tk. 563.00	Tk. 559.00	Tk. 559.00
22.3.2	25 mm dia Stainless steel (SS) pipe with wall thickness 1.20 mm .	meter	Tk. 689.85	Tk. 689.00	Tk. 686.00	Tk. 686.00
22.3.3	32 mm dia Stainless steel (SS) pipe with wall thickness 1.20 mm .	meter	Tk. 858.94	Tk. 859.00	Tk. 855.00	Tk. 855.00
22.4	Supplying, fitting and fixing special quality 10 mm thick clear tempered glass door including all accessories, 1 set floor mounted auto door closure, 2 nos. clamping devices, 1 set locking device, top hinge and handle etc. complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 15,816.25	Tk. 15,813.00	Tk. 15,786.00	Tk. 15,786.00
22.5	Supplying, fitting and fixing best quality of different thick clear glass partition with 44 mm, 44 mm, 1.5 mm aluminium box, protector bit (single), protector bit (double), Silicon including all accessories etc. all complete and as per drawing and direction of the Engineer-in-charge.					
22.5.1	8 mm thick clear glass partition with aluminium box	sqm	Tk. 2,998.98	Tk. 2,988.00	Tk. 2,942.00	Tk. 2,942.00
22.5.2	10 mm thick clear glass partition with aluminium box	sqm	Tk. 3,162.23	Tk. 3,151.00	Tk. 3,106.00	Tk. 3,106.00
22.6	Supplying, fitting and fixing of Chattogram Teak wood strip (62 mm x 25 mm x 20 mm) for fixing of aluminium curtain rails as per design and drawing, fitting and fixing coat of priming etc. all complete and accepted by the Engineer-in-charge.(Rate is excluding the cost of polishing etc.)	meter	Tk. 912.14	Tk. 900.00	Tk. 879.00	Tk. 879.00
22.7	Supplying, fitting and fixing vertical venetian blinds made with glass fiber fabrics (foreign made) with supply of all the best quality materials (foreign made) such as head rail (aluminium), end cap-set, rod tilt, sigger (stopper), runner, spacer (link), handling cord, ball chain, link chain, cord-weight weight-plate plastic washer, glass fiber fabrics, and head cap-set etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 2,029.15	Tk. 2,029.00	Tk. 2,029.00	Tk. 2,029.00
22.8	Supplying, fitting and fixing horizontal venetian blinds of 25 mm width made with aluminium foils/ strips (foreign made) with supply of all materials (foreign made) such as head rail, bottom rail, rod tilt, head cap, bottom cap, gear box, stick, support and cap, cord drum and support, bracket, cord lock, cord equalizer, top holder, ladder, cord, handling cord and aluminium foils etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 1,566.63	Tk. 1,568.00	Tk. 1,567.00	Tk. 1,567.00
22.9	Supplying, fitting and fixing of horizontal zebra blind made with polyester fabrics, synthetic net with 3 inch vinyl/upvc coated strips (foreign made) with supply of all materials (foreign made) such as head rail, bottom rail, rod tilt, head cap, bottom cap, gear box, stick, support and cap, cord drum and support, bracket, cord lock, cord equalizer, top holder, ladder, cord, handling cord and aluminium foils etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 3,455.33	Tk. 3,456.00	Tk. 3,455.00	Tk. 3,455.00
22.10	Supplying, fitting and fixing of blackout/ translucent roller blind of PVC/ vinyl laminated fiberglass fabrics (foreign made) with supply of all materials (foreign made) such as head rail (aluminium), end cap-set, rod tilt, sigger (stopper), runner, spacer (link), handling cord, ball chain, link chain, cord-weight weight-plate plastic washer, glass fiber fabrics, and head cap-set etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 2,844.13	Tk. 2,807.00	Tk. 2,807.00	Tk. 2,807.00

22.11	Supplying, fitting and fixing of steel iron rack made of 50 mm x 50 mm x 6.35 mm vertical and horizontal angle post, welding 63.5 mm x 6.35 mm F.I. bar placed horizontally as bracing for the posts, 38 mm x 6.35 mm cross F.I. bars welded with vertical post, 7.81 mm thick M.S plate welded as cup shape at the bottom of post, supplying and fitting rivets, charge for welding, drilling, carrying, cutting, straightening in required sizes etc complete in all respect as per drawing, design and accepted by the Engineer-in-charge.	sqm	Tk. 11,246.05	Tk. 11,072.00	Tk. 10,809.00	Tk. 10,809.00
22.12	Supplying, fitting and fixing of the best quality any sizes uPVC plastic door shutter with frame having specific gravity of 1.35 - 1.45, panel wall thickness 1.7 mm-2.2 mm, shutter/panel thickness 37.5 mm & weighing 17.25 kg/m2 and other physical, chemical, thermal, fire resistivity properties etc. as per BSTI approved manufacturer standards and ASTM, BS/ISO/IS standards of different sizes uPVC plastic door shutter with uPVC plastic frame (frame size:150mm x 62.50mm) fitting - fixing in brick wall/ R.C.C wall with at least 3 Nos. SS hinges by min 64 Nos. Ø 3.17 mm and 3.97 mm 12.7 mm long rivets, 12 nos. 25.4 mm SS screws, Ø 9.38 mm, 150 mm long SS tower bolts 2 nos., 146 mm SS handle by rivet 6 Nos., G.I inner joint, 234.95 mm x 127 mm clamp, 76.2 mm x 57.15 mm, 6 Nos. GI clamp, 2 nos. outer GI joint clamp making necessary grooves and mending good the damages, finishing, curing, carrying the same to the site and local carriage etc. complete in all respect and accepted by the Engineer-in-charge	sqm	Tk. 3,773.63	Tk. 3,774.00	Tk. 3,774.00	Tk. 3,774.00
22.13	Supplying, fitting and fixing of the best quality any size uPVC plastic door shutter with frame having specific gravity of 1.35 - 1.45, panel wall thickness 2.4 mm-2.76 mm, shutter/panel thickness 37.5 mm & weighing 27.00 kg/m2 and other physical, chemical, thermal, fire resistivity properties etc. as per BSTI approved manufacturer standards or ASTM, BS/ISO/IS standards and cosmic /Foam /aristocrat or equivalent standards of different sizes uPVC plastic door shutter with uPVC plastic frame (frame size:150mm x 62.50mm) fitting - fixing in brick wall/ R.C.C wall with at least 3 Nos. SS hinges by min 64 Nos. Ø 3.17 mm and 3.97 mm 12.7 mm long rivets, 24 nos. 25.4 mm SS screws, Ø 9.38 mm, 150 mm long SS tower bolts 2 nos., 146 mm SS handle by rivet ,6 Nos. GI clamp, 6 Nos. G.I inner joint, 234.95 mm x 127 mm clamp, 76.2 mm x 57.15 mm, including making necessary grooves and mending good the damages, finishing, curing, local carriage etc . complete in all respect and accepted by the Engineer-in-charge.	sqm	Tk. 6,866.54	Tk. 6,852.00	Tk. 6,791.00	Tk. 6,791.00
22.14	Supplying, fitting & fixing of uPVC plastic or glass partition in plastic frame fabricated by 70 mm x 50 mm x 2 mm T and 50 mm x 40 mm x 2 mm / 50 mm x 30 mm x 2 mm L-sections, providing partition with 156 mm x 25 mm uPVC hollow section having wall thickness of 2.5 mm / partition with 5 mm thick glass. physical, chemical, thermal, fire resistivity properties etc of the materials as per BSTI approved manufacturer standards or ASTM, BS/ISO/IS standards, fitted and fixed in brick wall or R.C.C wall with 6 nos. GI clamps, 4 nos. inner joint GI clamps, 2 nos. outer GI joint clamps, 16 nos. rivets, making necessary grooves and mending good the damages, finishing, curing, carriage etc complete in all respect and accepted by the Engineer-in-charge.					
22.14.1	With partition of uPVC hollow section	sqm	Tk. 2,793.37	Tk. 2,786.00	Tk. 2,742.00	Tk. 2,742.00
22.14.2	With 5 mm glass	sqm	Tk. 2,664.33	Tk. 2,660.00	Tk. 2,630.00	Tk. 2,630.00

22.15	Supplying, fitting & fixing of uPVC Profile for door, window and fixed partition frames applications having specific gravity of 1.36-1.43, and other physical, chemical, thermal, fire resistivity properties etc. as per BSTI approved manufacturer standards or ASTM, BS/ISO/IS standards including all accessories, fittings with 6 nos. steel clamps complete in all respect as per drawing and accepted by the Engineer-in-charge.	meter	Tk. 618.03	Tk. 633.00	Tk. 617.00	Tk. 617.00
22.16	Supplying, fitting & fixing of swing door shutter made of low density free foam (LDFF) sheet having standard size 1200 mm x 3200 mm. Physical properties: density: 0.55-0.68 g/cm ³ , weighing 1.3 kg/sft, providing new sash (100 mm), sliding glass bead, rubber, screws, 75 mm hinges 4 Nos., 100 mm tower bolt 2 Nos., 150 mm handle 2 nos. etc. including all accessories complete in all respect as per drawing and accepted by the Engineer-in-charge.					
22.16.1	1950 mm x 750 mm size 8 mm thick uPVC swing door shutter	each	Tk. 11,098.30	Tk. 11,261.00	Tk. 11,034.00	Tk. 11,034.00
22.16.2	2100 mm x 750 mm size 8 mm thick uPVC swing door shutter	each	Tk. 11,408.74	Tk. 11,584.00	Tk. 11,357.00	Tk. 11,357.00
22.16.3	2100 mm x 900 mm size 8 mm thick uPVC swing door shutter	each	Tk. 12,102.62	Tk. 12,288.00	Tk. 12,033.00	Tk. 12,033.00
22.17	Supplying, fitting & fixing of plastic wall paneling with 3 mm sheet, weighing 0.188 kg/sft, fitting and mending good the damages, carriage to the site including glass bead, partition frame, rubber, 5 mm clear glass fitted and fixed with screws, steel clamp etc all complete and accepted by the Engineer-in-charge.	sqm	Tk. 1,358.48	Tk. 1,353.00	Tk. 1,309.00	Tk. 1,309.00
22.18	Supplying, fitting & fixing of uPVC profile fixed partition with 5 mm clear glass, weighing 1.032 kg/sft, fitting and mending good the damages, carriage etc. complete in all respect and accepted by the Engineer-in-charge.	sqm	Tk. 3,752.48	Tk. 3,792.00	Tk. 3,715.00	Tk. 3,715.00
22.19	Supplying, fitting & fixing of uPVC profile sliding window having gross weight of 14.00 kg/sqm (including 5 mm glass) fitting and mending good the damages, carries etc complete in all respect accepted by the Engineer-in-charge.					
22.19.1	uPVC sliding window with 5 mm glass (1200 mm x 1200 mm size)	sqm	Tk. 5,681.38	Tk. 5,705.00	Tk. 5,601.00	Tk. 5,601.00
22.19.2	uPVC sliding window without glass (1200 mm x 1200 mm size)	sqm	Tk. 4,802.24	Tk. 4,826.00	Tk. 4,722.00	Tk. 4,722.00
22.20	Supplying, fitting & fixing of plastic wood paneling in false ceiling with 6 mm sheet, weighing 3.85 kg/sqm, fitting and mending good the damages, carries etc complete in all respect and accepted by the Engineer-in-charge	sqm	Tk. 1,451.26	Tk. 1,445.00	Tk. 1,398.00	Tk. 1,398.00
22.21	Supplying, fitting and fixing of approved quality 190 mm x 190 mm size glass bricks in round stair wall including all accessories & charges, complete in all respect and accepted by the Engineer-in-charge.	sqm	Tk. 16,383.12	Tk. 16,381.00	Tk. 16,243.00	Tk. 16,243.00
22.22	Supplying, fitting & fixing 50 mm thick rock or glass wool padding of approved quality and accepted by the Engineer-in-charge.	sqm	Tk. 4,351.61	Tk. 4,328.00	Tk. 4,292.00	Tk. 4,292.00

22.23	Manufacturing supplying, fitting and fixing of single leaf louver door shutter made with 16 gauge thick 50 mm x 50 mm SS hollow box as outer frame, 16 gauge thick 50 mm x 38 mm SS hollow box as pannel frame and 18 gauge thick 50 mm x 12 mm SS hollow box as louver with 150 mm long 12 mm dia SS handle, 300 mm long 20 mm dia ss hasp bolt, 100 mm hinges etc, comple in all respect and accepted by the Engineering-in charge.	sqm	Tk. 24,636.00	Tk. 24,619.00	Tk. 24,469.00	Tk. 24,469.00
22.24.1	Supplying, laying, fitting & fixing best quality 225 mm dia (internal) machine made R.C.C. pipe over 100 mm thick cement concrete (1:3:6) at base and sides including single brick flat soling and gasket with cement mortar (1:4) joints, cutting and filling trenches up to required depth including excavation of soil etc. all complete in all respect as per type plan and accepted by the Engineer-in-charge (average depth of trench is 1050 mm). (Cement: CEM-II/B-M)	meter	Tk. 1,424.71	Tk. 1,423.00	Tk. 1,410.00	Tk. 1,408.00
22.24.2	Supplying, laying, fitting & fixing best quality 300 mm dia (internal) machine made R.C.C. pipe over 100 mm thick cement concrete (1:3:6) at base and sides including single brick flat soling and gasket with cement mortar (1:4) joints, cutting and filling trenches up to required depth including excavation of soil etc. all complete in all respect as per type plan and accepted by the Engineer-in-charge (average depth of trench is 1050 mm).(Cement: CEM-II/B-M)	meter	Tk. 2,277.35	Tk. 2,274.00	Tk. 2,258.00	Tk. 2,257.00
22.24.3	Supplying, laying, fitting and fixing best quality 900 mm dia (internal) machine made R.C.C. pipe with socket over 100 mm thick cement concrete (1:3:6) at base and sides including single layer brick flat soling , cutting and filling trenches up to required depth including excavation of soil etc. complete in all respect as per type plan and accepted by the Engineer-in-charge (Cement: CEM-II/B-M) . [Rate is excluding cost of excavation, back-fill, materials for back-fill and bedding]	meter	Tk. 11,961.08	Tk. 11,944.00	Tk. 11,853.00	Tk. 11,852.00
22.25	Supplying, making, fitting & fixing of R.C.C. door/window frame of 155 mm x 65 mm which is made by using M.S. rod (fy = 400 MPa) fabrication work with 24 BWG G.I. wire, steel formworks with minimum cement concrete relates mix ratio (1:2:4) having minimum f'cr = 27 MPa & satisfying a specified compressive strength, f'c= 22 MPa at 28 days and cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ ASTM-C 150 type-1, best quality 50% quantity of local sand (F.M. 1.2) and 50% quantity of coarse sand (F.M. 2.2), 6 mm down well graded stone chips, screening, making, placing in position, making shutter/formworks water-tight properly, jute hassian, polythene sheet, placing of 100 mm long welded iron hinges with 6 mm dia M.S. rod clamp in position and mixing with mixer machine, casting in forms, compacting properly and curing at least for 21 days, removing shuttering and demoulding etc. all complete including the cost of water, electricity and other charges as per design, drawing and accepted by the Engineer-in-cha	meter	Tk. 503.66	Tk. 501.00	Tk. 485.00	Tk. 485.00
22.26	Supply & installation of fire rated door complete & accepted by the Engineer-in-charge with the following specification:	sqm	Tk. 49,059.22	Tk. 49,059.00	Tk. 49,059.00	Tk. 49,059.00
	Standard(s) for safety : UL Certified, NFPA 80 standard					
	ANSI / UL 10C: Positive pressure of fire tests of door assemblies.					
	ANSI / UL 10B: Fire tests of door assemblies.					
	UL Subject 63: Outline of investigation for fire frames.					
	Shutter materials: Mild steel.					
	Fire resistivity (Door & frame): Two hours.					

	Fire resistivity (accessories): Three hours (panic bar, pul handle, ceramic fibre tape for sealing smoke, hinge, bolt and screw, mortice lock, door closer					
	Door frame: 2 vertical fire resisting jamb and 1 fire resisting horizontal jamb made with single rabbet hollow steel channel frame made of 2.3 mm imported cold rolled mild steel sheet. Four SS hinges (size: 112 mm x 100 mm x 3.4mm) are used on each leaf screwd with frame					
	Door leaf: Door leaf faces consists of 1.6 mm thick imported cold rolled MS sheet attached together using interlocking mechanism and tack welding which is filled with imported 100kg/m ³ rock wool cake. Stiffeners should be used inside to strengthen structure.					
	Vision Panel: Made of UL certified 5 mm glass. It has clear vision area of 150 mm x 100 mm. Its frame is mae of 1.6 mm imported cold rolled MS sheet and SS pop rivet used for fixing. Ceramic fibre tape is used for sealing smoke.					
	Internal insulation: Rock wool.					
	Surface treatment: All MS sheets are chemically treated coated with zinc phosphate before assembly in highly equipped chemical treatment plant. All chemicals are imported.					
	Paint Finish: Epoxy powder coated paint finish and curing in temperature controlled gas oven.					
22.27	Supplying, fitting and fixing special quality 10 mm thick clear tempered glass digital Honour board including all accessories, heavy type 6 nos. SS clamping devices/SS knob, digital print name plate including making hole etc. complete in all respect as per drawing and direction of the Engineer-in-charge.	sqm	Tk. 10,955.80	Tk. 10,934.00	Tk. 10,845.00	Tk. 10,845.00

..... END OF CHAPTER TWENTY TWO

CHAPTER 23 : FERROCEMENT WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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23.1	Supplying, fitting & fixing of pre-cast ferrocement roof panel which is made by using vibration table & steel shutter with minimum cement content relates to mix ratio 1:2 for flange having minimum compressive strength 30 MPa (28 days standard cube strength) and cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ASTM-C150 type- 1, best quality coarse sand of equivalent F.M. 2.2, 2 layers of 18 BWG galvanized/epoxy coated wire mesh having minimum yield strength $f_y = 450$ MPa & having 2 meshes per 25 mm and reinforced cement concrete works for rib with minimum cement content relates to mix ratio 1:1.5:3 having minimum $f_{cr} = 30$ MPa and satisfying a specified compressive strength $f'_c = 25$ MPa, best quality coarse sand of equivalent F.M. 2.2 and 12 mm down well graded stone chips conforming to ASTM C-33, supplying, fabrication and fixing to details as per design, deformed bar reinforcement BDS ISO 6935-2:2006, including straightening and cleaning rust, if any, bending and binding in position with supply of 24 nosl respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish). accessories like sliding door key lock, sliding door wheel, sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5 mm thick glass including labour charge for fitting of accessories, making grooves and mending good damages, carriage, and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge. Size up to: 1500 mm x 2100 mm					
23.1.1	For ground floor, span up to 3810 mm for floor/roof slab.	sqm	Tk. 2,833.00	Tk. 2,814.00	Tk. 2,750.00	Tk. 2,742.00
23.1.2	Add for each additional span length of 150 mm long direction (3810 mm < span $\leq$ 4800 mm)	sqm	Tk. 9.00	Tk. 9.00	Tk. 9.00	Tk. 9.00
23.1.3	Add for each additional floor above ground floor	sqm	Tk. 40.00	Tk. 40.00	Tk. 40.00	Tk. 40.00
23.2	Supplying, fitting & fixing of 20 mm thick ferrocement folded plate roof which is made by using vibration table & steel shutter with minimum cement content relates to mix ratio 1:2, having minimum compressive strength of 30 MPa (28 days standard cube strength) and cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ASTM-C 150 type- 1, best quality coarse sand of equivalent F.M. 2.2, 1 layer of 20 BWG galvanized/epoxy coated wire mesh having minimum yield strength $f_y = 450$ MPa & having 2 meshes per 25 mm, fabrication and fixing to details as per design, deformed bar reinforcement BDS ISO 6935-2:2006, including straightening and cleaning the rust, if any, bending and binding in position with supply of 24 BWG GI wires, removing the shutter, lifting, keeping on proper position, curing at least for 7 days including the cost of water, electricity and other charges all complete as per design, drawing and accepted by the Engineer-in-charge.					
23.2.1	For ground floor, span up to 3810 mm	sqm	Tk. 2,271.00	Tk. 2,256.00	Tk. 2,214.00	Tk. 2,213.00
23.2.2	Add for each additional span length of 150 mm long direction (3810 mm < span $\leq$ 4800 mm)	sqm	Tk. 9.00	Tk. 9.00	Tk. 9.00	Tk. 9.00
23.2.3	Add for each additional floor above ground floor	sqm	Tk. 40.00	Tk. 40.00	Tk. 40.00	Tk. 40.00

23.3	Supplying, fitting & fixing of 20 mm thick ferrocement corrugated sheet roof which is made by using vibration table & steel shutter with minimum cement content relates to mix ratio 1:2, having minimum compressive strength of 30 MPa (28 days standard cube strength) and cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ASTM-C 150 type-1, best quality coarse sand of equivalent F.M. 2.2, 1 layer of 20 BWG galvanized/epoxy coated wire mesh having minimum yield strength $f_y=450$ Mpa & having 2 meshes per 25 mm and supplying, fabrication and fixing to details as per design, deformed bar reinforcement BDS ISO 6935-2:2006 including straightening and cleaning rust, if any, bending and binding in position with supply of 24 nos GI wires, removing the shutter, lifting on ground floor roof, keeping on proper position, curing at least for 28 days including the cost of water, electricity and other charges all complete as per design, drawing and accepted by the Engineer-in-charge.					
23.3.1	For ground floor, span up to 3810 mm for floor/roof slab.	sqm	Tk. 1,849.00	Tk. 1,840.00	Tk. 1,817.00	Tk. 1,816.00
23.3.2	Add for each additional floor above ground floor	sqm	Tk. 40.00	Tk. 40.00	Tk. 40.00	Tk. 40.00
23.4	Supplying, fitting & fixing of pre-cast Ferrocement L-Panel which is made by using vibration table & steel shutter with minimum cement content relates to mix ratio 1:2 for flange having minimum compressive strength of 30 MPa (28 days standard cube strength) and cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ASTM-C 150 type- 1, best quality coarse sand of equivalent F.M. 2.2, 2 layers of 20 BWG galvanized/epoxy coated wire mesh having minimum yield strength $f_y=450$ Mpa & having 2 meshes per 25 mm and reinforced cement concrete works for rib with minimum cement content relates to mix ratio 1:1.5:3 having minimum $f_{cr} = 30$ MPa and satisfying a specified compressive strength $f_c = 25$ MPa, best quality coarse sand of equivalent F.M. 2.2 and 12 mm down well graded stone chips conforming to ASTM C-33, supplying, fabrication and fixing to details as per design, deformed bar reinforcement BDS ISO 6935-2:2006 including straightening and cleaning rust, if any, bending and binding in position, curing at least for 28 anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish					
23.4.1	For ground floor, span up to 3810 mm for floor/roof slab.	sqm	Tk. 1,612.00	Tk. 1,605.00	Tk. 1,576.00	Tk. 1,572.00
23.4.2	Add for each additional floor above ground floor	sqm	Tk. 40.00	Tk. 40.00	Tk. 40.00	Tk. 40.00
23.5	Supplying, making and fitting fixing of ferrocement door/window frame 155 mm x 65 mm which is made by using M.S. rod ($f_y = 400$ MPa) fabrication work minimum cement content relates to mix ratio 1:2 cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ ASTM-C 150 type-1, coarse sand (F.M. 2.2), cutting, bending, binding of one layer 20 BWG galvanized / epoxy coated wire mesh having minimum yield strength $f_y=450$ MPa & 12 mm c/c gap in both directions with 24 BWG. G.I. wire, jute hassain, polythene sheet, placing of 100 mm long welded iron hinges including 6 mm dia M.S. rod clamp fittings, fixing in position, casting with neat cement finishing demoulding and lean cement sand (1:8) cement, sand (F.M. 1:2) and curing at least for 28 days etc. all complete including the cost of water, electricity, testing and other charge as per design, drawing and accepted by the Engineer-in-charge. (Considering x- section of the frame is 153 mm x 64 mm).	meter	Tk. 706.82	Tk. 703.00	Tk. 680.00	Tk. 679.00

23.6	25mm thick cast-in-situ ferrocement wall with minimum cement content relates to mix ratio 1:2, cement conforming to BDS EN-197-1-CEM-I, 52.5N (52.5MPa)/ASTM-C 150 type-1, and best quality coarse sand (F.M. 2.2) M.S. rod fabrication in accordance with BDS, ISO 6935-2:2006 straightening and cleaning rust if any, bending and binding in position with supply of 8 mm dia M.S. rod 250 mm c/c vertically, 6 mm dia 250 mm c/c horizontally, 24 nos G.I. wire splices laps etc. 2 layers of 20 BWG galvanized wire mesh having minimum yield strength $f_y=450$ Mpa & 2 mesh per 25 mm, casting and finishing the both surfaces with cement sand (F.M.-1.2) mortar (1:3) and curing at least for 21 days, including the cost of water, electricity and other charges in all complete as per drawing, design and accepted by the Engineer-in-charge.	sqm	Tk. 2,066.01	Tk. 2,050.00	Tk. 2,004.00	Tk. 2,003.00
23.7	Supplying, fitting & fixing of 25 mm thick pre-cast ferrocement wall panel with minimum cement content relates to mix ratio 1:2, cement conforming to BDS EN-197-1-CEM-1, 52.5N (52.5MPa)/ ASTM-C 150 type-1 and best quality coarse sand (F.M. 2.2) and 2 layers of 20 BWG galvanized wire mesh having minimum yield strength $f_y=450$ Mpa & 2 mesh per 25 mm, including cutting, binding, bending, with 24 B.W.G. G.I. wire, casting and finishing the both surfaces with cement sand (F.M.-1.2) Mortar (1:3) and curing at least for 21 days before lifting, fitting & fixing the Ferrocement plate in proper position with making rib, supply of 8 mm dia M.S. rod 250 mm c/c vertically, 6 mm dia 250 mm c/c horizontally fabrication in accordance with BDS, ISO 6935-2:2006 straightening and cleaning rust if any, bending and binding in position with supply of G.I. wire etc. the cost of water, electricity and other charges, complete in all respect and accepted by the Engineer-in-charge					
23.7.1	Ground Floor	sqm	Tk. 1,805.00	Tk. 1,795.00	Tk. 1,768.00	Tk. 1,767.00
23.7.2	Add for each additional floor above ground floor	sqm	Tk. 20.00	Tk. 20.00	Tk. 20.00	Tk. 20.00
23.8	90 mm thick partition wall with ferrocement sandwich panel wall (cast in situ) using mortar having compressive strength not less than 15.0 N/mm^2 and fitting-fixing of EPS sheet size (3000 mm x 1200 mm x 63 mm), 8 mm M.S. bar and its fabrication in both side with 20 BWG galvanized wire mesh having minimum yield strength $f_y=450$ Mpa & 2 mesh per 25 mm, in compliance with ASTM C129 fairly uniform thickness and water absorption not more than 7% including necessary scaffolding, cleaning and curing at least for 7 days including cost of water, electricity and other charges etc. in all complete accepted by the Engineer-in-charge.					
23.8.1	Ground Floor	sqm	Tk. 2,400.00	Tk. 2,386.00	Tk. 2,345.00	Tk. 2,344.00
23.8.2	Add for each additional floor above ground floor	sqm	Tk. 15.00	Tk. 15.00	Tk. 15.00	Tk. 15.00
23.9	40 mm ferro-cement slab on soil-cement stabilized sub-grade in ground floor etc. with minimum cement content relates to mix ratio 1:2 having minimum compressive strength of 30 MPa (28 days standard cube strength) and cement conforming to BDS EN-197-1-CEM-1, best quality coarse (F.M. 2.2) sand, 2 layer of 20 BWG galvanized/epoxy coated wire mesh having minimum yield strength $f_y=450$ Mpa & having 2 mesh per 25 mm with minimum clear cover 12 mm both top & bottom, true to desired level, placing, compacting, curing at least for 7 days in all floors including cost of water, electricity and other charges etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 1,393.75	Tk. 1,386.00	Tk. 1,370.00	Tk. 1,369.00

..... END OF CHAPTER TWENTY THREE						
CHAPTER 24: ROAD AND PAVEMENT WORKS						
Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
24.1	Earth work in road embankment up to a lift of 1.5 m in all kinds of soil including cutting and throwing earth in layers not exceeding 150 mm in thickness, breaking clods up to 40 mm size in each layer, ramming, leveling and dressing as per required side slope and alignment with earth borrowed from the road side govt. acquired land for any leads and maintaining the embankment true to level and side slopes as per profiles for one year from the date of completion including making dug, bailing, profiling etc. complete and accepted by the Engineer-in-charge.					
24.1.1	For initial lift of 1.5 m	cum	Tk. 200.63	Tk. 197.00	Tk. 183.00	Tk. 183.00
24.1.2	For additional lift of 1 m beyond 1.5 m up to 2.5 m	cum	Tk. 229.23	Tk. 225.00	Tk. 209.00	Tk. 209.00
24.1.3	For additional lift of 1 m beyond 2.5 m up to 3.5 m	cum	Tk. 257.83	Tk. 253.00	Tk. 234.00	Tk. 234.00
24.1.4	For additional lift of 1 m beyond 3.5 m up to 4.5 m	cum	Tk. 286.43	Tk. 281.00	Tk. 261.00	Tk. 261.00
24.1.5	For additional lift of 1 m beyond 4.5 m up to 5.5 m	cum	Tk. 315.03	Tk. 309.00	Tk. 286.00	Tk. 286.00
24.1.6	For additional lift of 1 m beyond 5.5 m up to 6.5 m	cum	Tk. 343.63	Tk. 338.00	Tk. 313.00	Tk. 313.00
24.2	Earthwork in box cutting in all types of soil. Removing soil to a safe distance, maintaining proper alignment, camber and grade including leveling, dressing and compacting the sub grade as per drawing and accepted by the Engineer-in-charge.					
24.2.1	Depth up to 300 mm	sqm	Tk. 66.04	Tk. 65.00	Tk. 60.00	Tk. 60.00
24.2.2	Depth up to 450 mm	sqm	Tk. 98.73	Tk. 97.00	Tk. 90.00	Tk. 90.00
24.3	Creating turf on the side slopes and top of embankment with good quality turf not less than 225 mm square chunk, watering till the grass grown including all leads and lifts etc. complete and accepted by the Engineer-in-charge.	sqm	Tk. 22.45	Tk. 22.00	Tk. 21.00	Tk. 21.00
24.4.1	Single layer brick flat soling in road work with first class or picked jhama bricks as per alignment, camber and grade including filling joints with sand (F.M. 0.80) etc. complete including cost of all materials and accepted by the Engineer-in-charge.	sqm	Tk. 552.54	Tk. 549.00	Tk. 532.00	Tk. 524.00

24.4.2	One layer of Concrete Solid Block Soling having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm in road work including preparation of bed and filling the interstices with fine sand (F.M.= 0.8), leveling etc. all completed by the contractor and accepted by the Engineer-in-charge	Sqm	Tk. 661.94	Tk. 659.00	Tk. 649.00	Tk. 649.00
24.5.1	Brick on edges pavement in herring bone bond (HBB) with first class or picked jhama bricks as per alignment, camber and grade over 12 mm thick sand cushion (F.M. 0.80) including filling the joints with the same sand including cost of all materials and accepted by the Engineer-in-charge.	sqm	Tk. 892.12	Tk. 889.00	Tk. 861.00	Tk. 848.00
24.5.2	Brick on edges pavement in herring bone bond (HBB) with Concrete Solid Block (240 x 114 x 70 mm) having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm as per alignment, camber and grade over 12 mm thick sand cushion (F.M. 0.80) including filling the joints with the same sand including cost of all materials and accepted by the Engineer-in-charge.	Sqm	Tk. 1,090.23	Tk. 1,087.00	Tk. 1,073.00	Tk. 1,073.00
24.6	Double brick flat soling in roadwork with first class or picked jhama bricks as per alignment, camber and grade including filling the joints with fine sand (F.M. 0.80) and also providing 12 mm thick sand cushion in between the two layers of brick flat soling with the same quality of sand including cost of all materials and accepted by the Engineer-in-charge.	sqm	Tk. 1,112.80	Tk. 1,107.00	Tk. 1,073.00	Tk. 1,057.00
24.7.1	Brick on end edging (75 mm across the road) with first class or picked jhama bricks and filling the gaps with fine sand (F.M. 0.80) including cutting trenches, true to level and grade, removing earth, refilling and ramming the sides properly including cost of all materials and accepted by the Engineer-in-charge.	meter	Tk. 139.83	Tk. 139.00	Tk. 135.00	Tk. 133.00
24.7.2	Brick on end edging (75 mm across the road) with Concrete Solid Block(240 x 114 x 70 mm) having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm and filling the gaps with fine sand (F.M. 0.80) including cutting trenches, true to level and grade, removing earth, refilling and ramming the sides properly including cost of all materials and accepted by the Engineer-in-charge.	meter	Tk. 170.03	Tk. 169.00	Tk. 167.00	Tk. 167.00
24.8	Breaking picked jhama, first class bricks or bats including cost of bricks and bats and stacking in measurable sizes at site accepted by the Engineer-in-charge.					
24.8.1	For 62 mm to 40 mm size	cum	Tk. 4,517.41	Tk. 4,502.00	Tk. 4,368.00	Tk. 4,294.00
24.8.2	For 40 mm to 25 mm size	cum	Tk. 4,755.74	Tk. 4,736.00	Tk. 4,585.00	Tk. 4,511.00

24.9	Providing compacted water bound macadam base course with graded material of crushed well burnt, picked jhama or first class bricks and bats (50 mm downgraded brick chips having LAA value not exceeding 40%) including supplying of 12 mm downgraded chips made of same quality bricks, having compacted thickness made as per specifications including local handling and spreading uniformly to proper grade, camber and super elevation, hand packing watering, dry rolling followed by wet rolling in layers of specified thickness with 8 to 10 ton capacity power driven road roller to attain minimum soaked CBR of 80% or design CBR at minimum degree of compaction of 98% of MDD (Modified Proctor), including supplying and blinding with sand (F.M. 0.80) @ 0.012 m ³ per sqm or as required etc. all complete as per direction of the Engineer-in-charge. After adequate dry rolling, spreading of sand on the surface, sprinkling of water and rolling is to be continued until all the voids are filled. thickness of each layer should not anti-corrosive prime coat etc. complete in all respect for all floors and	cum	Tk. 5,850.65	Tk. 5,840.00	Tk. 5,712.00	Tk. 5,619.00
24.10.1	Providing compacted sub-base course with well graded material of crushed well burnt picked jhama / 1st class brick bats and sand (FM 0.8) with mixing proportion 2 : 1 having compacted thickness made as per specification including local handling, spreading uniformly to proper grade camber and super elevation, packing, rolling properly with 8 to 10 ton capacity power driven road roller and watering properly for compaction of 100% MDD (standard) blinding with sand including cost of fuel, lubricants, spares, maintenance, driver etc. all complete and accepted by the Engineer-in-charge.	cum	Tk. 5,448.88	Tk. 5,416.00	Tk. 5,199.00	Tk. 5,136.00
24.10.2	Providing improved sub-grade with sand F.M.>0.80 having compacted thickness as per specification including cost of sand, carrying, local handling, spreading uniformly in layers of 150mm to proper grade camber, super elevation, rolling properly with 7-10 tone vibratory roller and watering profusely for compaction 95% MDD (Modified Proctor Test) including cost of fuel, lubricants, spares, maintenance, driver etc. all complete and accepted by the Engineer-in-charge.	Cum	Tk. 919.12	Tk. 923.00	Tk. 887.00	Tk. 887.00
24.10.3	Providing compacted aggregate base type-I with graded metal of Crushed boulder<9mm, Crushed boulder<25mm, Crushed boulder <38mm and Sylhet sand F.M>1.00 @ ratio of (2:1) (Crushed boulder : Sand) having compacted thickness of 150mm including local handling spreading uniformly to proper grade and camber and super elevation, hand packing, rolling properly with 7-10 tone vibratory roller and watering profusely for proper consolidation with blending materials etc. in/c. cost of all materials complete as per direction of the Engineer-in-charge (Payment will be made on compacted volume and compaction of 98% MDD in Modified Proctor Test).	Cum	Tk. 10,062.72	Tk. 10,220.00	Tk. 9,451.00	Tk. 9,159.00
24.10.4	Providing compacted aggregate base type-I (With Pever) with graded metal of Crushed boulder<9mm, Crushed boulder<25mm, Crushed boulder <38mm and Sylhet sand F.M>1.00 @ ratio of (2:1) (Brick Khoa : Sand) having compacted thickness of 150mm including local handling spreading uniformly to proper grade and camber and super elevation, hand packing, rolling properly with 7-10 tone vibratory roller and watering profusely for proper consolidation with blending materials etc. in/c. cost of all materials complete as per direction of the Engineer-in-charge (Payment will be made on compacted volume and compaction of 98% MDD in Modified Proctor Test).	Cum	Tk. 10,308.23	Tk. 10,465.00	Tk. 9,696.00	Tk. 9,404.00

24.10.5	Providing compacted aggregate base type-II with graded metal of crushed well burnt picked jhama, 1st class brick chips 25mm to 38mm size and Sylhet sand F.M>1.00 @ ratio of (2:1) (Brick Khoa : Sand) having compacted thickness of 150mm including local handling spreading uniformly to proper grade and camber and super elevation, hand packing, rolling properly with 7-10 tone vibratory roller and watering profusely for proper consolidation with blending materials etc. in/c. cost of all materials complete as per direction of the Engineer-in-charge (Payment will be made on compacted volume and compaction of 98% MDD in Modified Proctor Test).	Cum	Tk. 6,409.97	Tk. 6,406.00	Tk. 6,290.00	Tk. 6,213.00
24.10.6	Providing compacted aggregate base type-II (With Pever) with graded metal of crushed well burnt picked jhama, 1st class brick chips 25mm to 38mm size and Sylhet sand F.M>1.00 @ ratio of (2:1) (Brick Khoa : Sand) having compacted thickness of 150mm including local handling spreading uniformly to proper grade and camber and super elevation, hand packing, rolling properly with 7-10 tone vibratory roller and watering profusely for proper consolidation with blending materials etc. in/c. cost of all materials complete as per direction of the Engineer-in-charge (Payment will be made on compacted volume and compaction of 98% MDD in Modified Proctor Test).	Cum	Tk. 6,649.34	Tk. 6,646.00	Tk. 6,529.00	Tk. 6,452.00
24.11	50 mm thick compacted pre-mixed bituminous carpeting with 20 mm down graded stone-chips (having LAA value <= 30%, water absorption not greater than 2%, flakiness index not greater than 35) of gradation as specified @ 0.68 cum of stonechips mixed with 46 kg of heated bitumen of 60/70 penetration grade heated straight run bitumen satisfying the requirements of ASTM/AASHTO. Bitumen content % by wt. of total is 4.5% to 5.50% or as determined by job mix design. The bitumen and stone-chips shall be separately heated to a temperature of 140°C – 155°C and 150°C – 170°C respectively before mixing. The mixing shall be done at temperature between 140°C – 160°C at a separate place away from the fire. The bitumen and stone-chips mixture shall be laid uniformly over 10 sqm of surface in proper camber, grade and super-elevation only on the prepared and accepted base or surface in a single appropriate layer to give the specified compacted thickness. The mixture should be rolled with appropriate steel drum roller (3-5 tons) &I respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish). accessories like sliding door key lock, sliding door wheel, sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5 mm thick glass including labour charge for fitting of accessories, making grooves and	sqm	Tk. 979.05	Tk. 978.00	Tk. 951.00	Tk. 935.00

24.12	38 mm thick compacted pre-mixed bituminous carpeting with 20 mm down graded stone-chips (having LAA value $\leq 30\%$, water absorption not greater than 2%, flakiness index not greater than 35) of gradation as specified @ 0.63 cum of stonechips mixed with 37 kg heated bitumen of 60/70 penetration grade heated straight run bitumen satisfying the requirements of ASTM/AASHTO. Bitumen content % by wt. of total is 4.5% to 5.50% or as determined by job mix design. The bitumen and stone-chips shall be separately heated to a temperature of 140°C – 155°C and 150°C – 170°C respectively before mixing. The mixing shall be done at temperature between 140°C – 160°C at a separate place away from the fire. The bitumen and stone-chips mixture shall be laid uniformly over 10 sqm of surface in proper camber, grade and super-elevation only on the prepared and accepted base or surface in a single appropriate layer to give the specified compacted thickness. The mixture should be rolled with appropriate steel drum roller (3-5 tons) & pnanti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish of wood).the Engineer-in-charge.and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge. Size up to: 2100 mm x 2100 mm" ring)te approved and accented by the Engineer-in-	sqm	Tk. 791.49	Tk. 791.00	Tk. 770.00	Tk. 757.00
24.13	25 mm thick compacted pre-mixed bituminous carpeting with 12 mm down graded stone-chips (having LAA value $\leq 30\%$, water absorption not greater than 2%, flakiness index not greater than 35) of gradation as specified @ 0.34 cum of stone chips mixed with 23 kg heated bitumen of 60/70 penetration grade heated straight run bitumen satisfying the requirements of ASTM/AASHTO. Bitumen content % by wt. of total is 4.5% to 5.50% or as determined by job mix design. The bitumen and stone-chips shall be separately heated to a temperature of 140°C – 155°C and 150°C – 170°C respectively before mixing. The mixing shall be done at temperature between 140°C – 160°C at a separate place away from the fire. The bitumen and stone-chips mixture shall be laid uniformly over 10 sqm of surface in proper camber, grade and super-elevation only on the prepared and accepted base or surface in a single appropriate layer to give the specified compacted thickness. The mixture should be rolled with appropriate steel drum roller (3-5 tons) & pl respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish). accessories like sliding door key lock, sliding door wheel, sliding door mohiar, sliding door neoprene, bolts and nuts including sealants, keeping provision for fitting 5 mm thick glass including labour charge for fitting of accessories, making grooves a	sqm	Tk. 504.66	Tk. 502.00	Tk. 493.00	Tk. 487.00

24.14	38 mm thick compacted pre-mixed dense bituminous carpeting with 20 mm down graded stone-chips (having LAA value $\leq 30\%$, water absorption not greater than 2%, flakiness index not greater than 35) of gradation as specified @ 0.56 cum of stone chips mixed with 45 kg heated bitumen of 60/70 penetration grade heated straight run bitumen satisfying the requirements of ASTM/ AASHTO. Bitumen content % by wt. of total is 4.5% to 5.50% or as determined by job mix design. The bitumen and stone-chips shall be separately heated to a temperature of $140^{\circ}\text{C} - 155^{\circ}\text{C}$ and $150^{\circ}\text{C} - 170^{\circ}\text{C}$ respectively before mixing. The mixing shall be done at temperature between $140^{\circ}\text{C} - 160^{\circ}\text{C}$ at a separate place away from the fire. The bitumen and stone-chips mixture shall be laid uniformly over 10 sqm of surface in proper camber, grade and super-elevation only on the prepared and accepted base or surface in a single appropriate layer to give the specified compacted thickness. The mixture should be rolled with appropriate steel drum roller (3-5 to anti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish of wood).the Engineer-in-charge.and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge. Size up to: 2100 mm x 2100 mm" ring)te, approved and accepted by the Engineer-in-charge.	sqm	Tk. 845.01	Tk. 843.00	Tk. 826.00	Tk. 814.00
24.15	25 mm thick compacted pre-mixed dense bituminous carpeting with 12 mm down graded stone-chips	sqm	Tk. 538.34	Tk. 535.00	Tk. 529.00	Tk. 524.00
24.16	Providing and laying pre-mixed dense bituminous carpeting with hot mix plant using coarse aggregate, fine aggregate, filler and bituminous binder as per design Job Mix Formula conforming Marshall Method as per specification, LAA value of aggregate should be $\leq 30\%$, water absorption not $> 2\%$, flakiness index not > 35 mixed with 60/70 penetration grade straight run bitumen satisfying the requirements of ASTM/AASHTO. including screening, cleaning of chips and preparing a uniform and quality mix in hot mix plant and ensuring a homogenous mix, in which all particles of the mineral aggregates are coated uniformly, carrying the hot premixed materials by means of dump/tipper truck, spreading the mixed materials at specified laying temperature with a hydrostatic paver finisher with sensor control to the required grade, level and alignment over the prepared surface, tamping and finishing the mix at specified compacted thickness, maintaining specified camber, grade, super-elevation and cross section, through rolling wanti-corrosive prime coat etc. complete in all respect for all floors and accepted by the Engineer-in-charge. (Exposed area excluding hand rail of railing will be considered for measurement, rate is excluding the cost of wood for hand-rail and cost of painting, polish of wood).the Engineer-in-charge.and electricity complete in all respect as per drawing and accepted by the Engineer-in-charge. Size up to: 2100 mm x 2100 mm" ring)te, approved and accepted by the Engineer-in-charge. (Rate is excluding laboratory test fees, the cost of reinforcement and its fabrication, placing, binding etc. and the cost of shuttering & centering). (Foundation: MAT)	cum	Tk. 21,693.79	Tk. 21,668.00	Tk. 21,345.00	Tk. 21,049.00
24.17	Providing tack coat @ 7.50 kg of bitumen (60/70 penetration grade) per 10 sqm including heating bitumen up to 175°C to 185°C etc. all complete (cost of bitumen included) and accepted by the Engineer-in-charge.	sqm	Tk. 88.95	Tk. 87.00	Tk. 88.00	Tk. 88.00

24.18.1	Providing seal coat with 4 cft of Sylhet sand (FM 2.2) mixed with @ 7.32 kg of bitumen (60/70 penetration grade) per 10 sqm of road surface including supplying and spreading of dry sand (F.M. 0.80) @ 0.09 cum per 10 sqm, including heating bitumen upto 176°C to 190°C, and spreading, rolling at a temperature not below 90°C celsius with appropriate steel drum roller (3 -5 ton) and pneumatic multiple tire roller (8-10 tons) to the desired compaction etc. all complete (cost of bitumen included) and accepted by the Engineer-in-charge.	sqm	Tk. 131.66	Tk. 129.00	Tk. 131.00	Tk. 130.00
24.18.2	Premixed bituminous seal coat with 0.12 cum of pea gravels mixed with 80 kg of bitumen (60/70 penetration grade) per cum of pea gravels and laid over 10 sqm road surface, spreading with proper camber, grade, surface blinding with dry sand (F.M. 1.2) @ 0.09 cum per 10 sqm, rolling at a temperature not below 90°C with appropriate steel drum roller (3-5 ton) and pneumatic multiple tire roller (8-10 tons) to the desired compaction etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 178.79	Tk. 177.00	Tk. 174.00	Tk. 174.00
24.19.1	Providing prime coat @ 1.2 liter/sqm with cutback bitumen to be prepared by cutting back 60/70 penetration grade bitumen in the ratio of 100 parts by volume of bitumen to 40-60 parts by volume of kerosene depending on the porosity of surface and will be decided by field trials, carefully cleaning of the surface of base material to be primed and spraying cutback bitumen at a temperature from 100 degree celcius to 120 degree celcius by appropriate hand device etc.all complete as per direction of the Engineer-in-charge.	sqm	Tk. 110.32	Tk. 109.00	Tk. 111.00	Tk. 111.00
24.19.2	Providing prime coat @1.2 liter/sqm with cutback bitumen to be prepared by cutting back 60/70 penetration grade bitumen in the ratio of 100 parts by volume of bitumen to 40-60 parts by volume of kerosene depending on the porosity of surface and will be decided by field trials, carefully cleaning of the surface of base material to be primed and spraying cutback bitumen at a temperature from 100°C to 120°C by mechanical distributor etc. all complete as per direction of the Engineer-in-charge.	sqm	Tk. 111.40	Tk. 110.00	Tk. 112.00	Tk. 112.00
24.20	Providing painting at the top of the road surface as road marking in two coats with approved best quality and colour of synthetic paint from authorized local agent of the manufacturer in a sealed container having high water resistance, high bondability & flexibility, using specific brand thinner applied by brass/ roller or others over a coat of priming, providing elapsing time for drying including surface cleaning from dust, oil or dirt, smoothening, finishing necessary tools etc. all complete & approved and accepted by the Engineer-in-charge.	sqm	Tk. 370.72	Tk. 368.00	Tk. 349.00	Tk. 349.00
24.21	Providing expansion joints across the R.C.C. road pavement with 20 mm gap between the adjacent panels. Supplying dowel bars of size 12 mm dia and 600 mm long M.S rod placed @ 750 mm c/c. 300 mm of the bar being fixed within one slab keeping remaining portion the bar free within the adjacent slab being inserted in 20 mm dia and 300 mm long P.V.C. pipe embedded in the slab, inside of pipe fully greased, filling the gap between the slabs (with admixture) of hot bitumen and coarse sand (F.M. 2.2) in proportion of 2:1, finishing the top with bitumen, spreading coarse sand over the joint, all complete as per plan, drawing and accepted by the Engineer-in-charge.	meter	Tk. 415.10	Tk. 410.00	Tk. 395.00	Tk. 395.00

24.22	Providing Stamped Concrete or decorative concrete flooring on RCC Casting in colored and textured form accomplished with the application of high Abrasion and UV resistant, non-fading pigment and long-wearing surface color hardener in three shakes, antiquing by using of a powdered Release Agent in a contrasting color and sealed by a Solvent-Borne Acrylic Sealer for chemical resistant according to ASTM D 543 : 95 and moisture resistant according to ASTM C 156 and stain resistant according to ASTM D 1308, providing a deep penetrating, scruff-resistance finish.	sqm	Tk. 2,317.88	Tk. 2,318.00	Tk. 2,305.00	Tk. 2,305.00

..... END OF CHAPTER TWENTY FOUR

CHAPTER 25 : ARBORICULTURE WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
25.1	Labour charge for scraping lawn surface up to 38 mm depth by spade, picking up all sorts of brick bats, rubbish by suitable tools, removing the same by head load or by any other means and stacking the same to a safe distance beyond 300 m including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 11.80	Tk. 12.00	Tk. 11.00	Tk. 11.00
25.2	Preparation of ground to make ready for plantation by spading the ground to a depth of 150 mm to 230 mm beyond 38 mm deep scrapped ground by spade, breaking earth clods to powder by hammers, picking up all sorts of rubbish, unwanted grasses by suitable tools, carrying and spreading the surplus earth into low lying area including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 23.61	Tk. 23.00	Tk. 21.00	Tk. 21.00
25.3	Leveling and dressing of lawn area to proper slope and grade by spading the same including supplying tools and plants etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 8.32	Tk. 8.00	Tk. 8.00	Tk. 8.00
25.4	Supply of best and approved quality alluvial loamy silty soil including loading, unloading at both ends, properly stacking at site including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge.	cum	Tk. 912.39	Tk. 910.00	Tk. 900.00	Tk. 900.00
25.5	Supply well decomposed cow dung carried by trucks or any other means including loading, unloading at both ends, stacking the same at site including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge	cum	Tk. 1,703.21	Tk. 1,702.00	Tk. 1,649.00	Tk. 1,649.00

25.6	Labour charge for spreading the alluvial loamy silty soil from the stacks at site on the lawn surface, leveling, dressing the same including supply all necessary tools and plants etc. all complete and accepted by the Engineer-in-charge.	cum	Tk. 193.64	Tk. 190.00	Tk. 176.00	Tk. 176.00
25.7	Added rate for lifting alluvial loamy silty soil/ decomposed cow dung work in additional floor:					
25.7.1	Add for each additional floor up to 5th floor	cum	Tk. 83.00	Tk. 82.00	Tk. 76.00	Tk. 76.00
25.7.2	Add for each additional floor from 6th floor to 9th floor	cum	Tk. 119.00	Tk. 117.00	Tk. 108.00	Tk. 108.00
25.7.3	Add for each additional floor for 10th floor & above	cum	Tk. 155.00	Tk. 152.00	Tk. 141.00	Tk. 141.00
25.8	Supply of lawn grass of approved quality by truck or by any other means, sorting the grass to proper size, washing the grass, dibbling the grass 6 mm to 50 mm apart, irrigation of lawn area till the grass grown at least for two months after plantation, weeding the undesirable grass, mowing the lawn grass by lawn mower up to two months after plantation, applying urea fertilizer on the lawn surface @ 1 kg per 9.29 sqm including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 66.93	Tk. 66.00	Tk. 62.00	Tk. 62.00
25.9	Labour charge for mixing well decomposed cow dung with alluvial loamy silty soil and excavated earth, removing the excess earth to safe distance including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge.	cum	Tk. 174.77	Tk. 172.00	Tk. 159.00	Tk. 159.00
25.10	Supplying and providing, fitting & fixing bamboo jafry fencing by best quality muli bamboo split of 20 mm in width, having 75 mm x 75 mm mesh placed diagonally doing framing a square shape in both ways and both sides tightened with 2 x 4 Nos. same split horizontally and 2 x 6 Nos. same split vertically and with G.I wire fitted with Borrak bamboo post of 100 mm dia and placed firmly in position at 1800 mm apart including supply of tools and plants, digging hole and filling back etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 549.59	Tk. 550.00	Tk. 535.00	Tk. 535.00
25.11	Supplying and providing, fitting & fixing muli bamboo tree guard of 1200 mm length and 500 mm dia best quality muli bamboo split 20 mm in size having 75 mm x 75 mm a square holes with 20 mm wide bamboo splits strengthen with 4 Nos. 20 mm wide bamboo split in both sides tightened with G.I wire fitted with 3 Nos. 63 mm dia borrak bamboo post of 1800 mm long of which 600 mm will be driven into earth by digging hole followed by fill back including supply of tools and plants all complete as per direction of the Engineer-in-charge.	each	Tk. 782.58	Tk. 782.00	Tk. 770.00	Tk. 770.00
25.12	Earth work in excavation to make a pit of 600 x 600 x 450 mm (depth) for plantation, applying cow dung mixing with loamy silty soil and excavated earth, planting the plants, tightened with 1800 mm long borrak bamboo post with jute rope and filled up the hole with mixed soil including supply of tools and plants all complete and accepted by the Engineer-in-charge.	each	Tk. 170.44	Tk. 169.00	Tk. 164.00	Tk. 164.00
25.13	Spading for preparation of flower beds in pits up to 600 mm depth, re-molding, picking up and placing and removing the spaded earth to a safe distance, mixing @ 1 cft well decomposed cow dung with 20% loamy silty soil and 6 cft excavated soil and fill back the pit including supply of tools and plants, all complete and accepted by the Engineer-in-charge.	sqm	Tk. 567.01	Tk. 562.00	Tk. 534.00	Tk. 534.00

25.14	Preparation of hedge beds by excavation of earth up to 300 mm depth, filling up the bed by a mixture of cow dung, loamy silty soil, cutted earth @ 1 cft cow dung manure per 6 cft of cutted earth and 20% loamy silty soil including supply of tools and plant etc. all complete and accepted by the Engineer-in-charge.	sqm	Tk. 280.47	Tk. 278.00	Tk. 264.00	Tk. 264.00
25.15	Carrying of rubbish , taking up all sorts of bricks, brick-bats and rubbish gravel chips, loading and unloading at both ends including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge.	cum	Tk. 177.95	Tk. 175.00	Tk. 162.00	Tk. 162.00
25.16	Supply and carrying of 12.20 m long 100 mm dia borrak bamboo at site, loading and unloading at both end, stacking the same at site including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge.	each	Tk. 537.44	Tk. 537.00	Tk. 534.00	Tk. 534.00
25.17	Supply and carrying of 7.62 m long 50 mm to 62.5 mm dia mulu bamboo at site, loading and unloading at both end, stacking the same at site including supply of tools and plants etc. all complete and accepted by the Engineer-in-charge.	each	Tk. 284.06	Tk. 283.00	Tk. 281.00	Tk. 281.00
25.18	Supply and carrying of 4.5 h.p., 550 mm blade 4 stroke, finger trip 3 position height adjuster 200 mm x 38 mm star tread wheels grass cutting (Power Lawn Mower Machine): YARDMAN No. 060A	each	Tk. 54,335.00	Tk. 54,335.00	Tk. 54,335.00	Tk. 54,335.00
25.19	Supply and carrying of 6.5 h.p., 550 mm blade 4 stroke, finger trip 3 position height adjuster 200 mm x 38 mm star tread wheels grass cutting (Power Lawn Mower Machine): YARDMAN No. 060A	each	Tk. 72,447.00	Tk. 72,447.00	Tk. 72,447.00	Tk. 72,447.00
25.20	Supply and carrying of 22 h.p. 1150 mm cutting blade power riding lawn yardman mower machine made in USA. Model No.13 BT 793G678 or equivalent.	each	Tk. 591,081.88	Tk. 591,044.00	Tk. 590,890.00	Tk. 590,890.00
25.21	Supply and carrying of 450 mm blade Hand Lawn Moar, Model No.1403-106(USA)	each	Tk. 15,093.00	Tk. 15,093.00	Tk. 15,093.00	Tk. 15,093.00

..... END OF CHAPTER TWENTY FIVE

CHAPTER 26 : SANITARY & WATER SUPPLY WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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26.01	Supplying, fitting and fixing of European type country made best quality glazed porcelain combi closet (two piece), including plastic seat cover with soft closing ,cistern system, water consumption 6 litre, siphon wash down/siphon jet single or dual flushing system which reduce water consumption, hygienic glaze in toilet bowl ,glaze in inner waste line, round bowl, outlet range 280~305 mm. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position preparing the base with cement concrete and with wire mesh or rods if necessary, in all floors including making holes wherever required and mending good the damages and fitting, fixing, finishing, complete with all necessary fittings and connection approved and accepted by the Engineer- in- charge.					
26.01.1	Approx. 670~690 X 360~ 362 mm size, minimum 27.5 kg of weight; Equivalent to Stella:Victoria /Stella: Imola / RAK: Amy / RAK: Karla or similar brand	each	Tk. 8,951.16	Tk. 8,921.00	Tk. 8,843.00	Tk. 8,843.00
26.01.2	Approx. 665~ 704 X 360~ 390 mm size, minimum 31.00 kg of weight : Equivalent to CHARU :CM18/ CHARU: CM15 / Stella :Marcela / RAK :Tampa / Stella: Marcela or similar brand	each	Tk. 10,454.00	Tk. 10,424.00	Tk. 10,347.00	Tk. 10,347.00
26.01.3	Approx. 639~670 X 370 ~399 mm size, minimum 32.5 kg of weight, Equivalent to RAK :Orient Standard /Stella: Verso or similar brand	each	Tk. 11,436.00	Tk. 11,405.00	Tk. 11,328.00	Tk. 11,328.00
26.01.4	Approx. 670~670 X 350~390 mm size, minimum 25.2 kg of weight, Equivalent to COTTO :C1480 /COTTO: C13960 / RAK Series 600 or similar brand	each	Tk. 13,252.00	Tk. 13,221.00	Tk. 13,144.00	Tk. 13,144.00
26.01.5	Approx. 668~750 x 356 ~430 mm size, minimum 37 .0 kg of weight, Equivalent to COTTO :C1390/Stella:Isabella /RAK: Washington / RAK: Jumairah or similar brand	each	Tk. 15,857.00	Tk. 15,826.00	Tk. 15,749.00	Tk. 15,749.00
26.01.6	Approx. 700~725 X 360~ 380 mm size, minimum 34.1 kg of weight; Equivalent to COTTO - C1470 / RAK: Venic or similar brand	each	Tk. 17,816.00	Tk. 17,786.00	Tk. 17,709.00	Tk. 17,709.00
26.01.7	Approx. 702~ 750x 380~412 mm size, minimum 40.0 kg of weight, Equivalent to COTTO: C118 / RAK: Shino or similar brand	each	Tk. 22,075.00	Tk. 22,045.00	Tk. 21,968.00	Tk. 21,968.00
26.01.8	Approx. 730~ 735x 430~440 mm size, minimum 42.0 kg of weight, Equivalent to COTTO: 109 / COTTO: 129 or similar brand	each	Tk. 24,810.00	Tk. 24,780.00	Tk. 24,703.00	Tk. 24,703.00

26.02	Supplying, fitting and fixing of country made European type special quality combi closet (one piece) including inbuilt flushing unit, of white color, including plastic seat cover with soft closing ,cistern system, water consumption 6 litre, siphon jet single flushing system which reduce water consumption, hygienic glaze in toilet bowl ,glaze in inner waste line, round bowl, outlet range 280~305 mm. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position preparing the base with cement concrete and with wire mesh or rods if necessary, in all floors including making holes wherever required and mending good the damages and fitting, fixing, finishing, complete with all necessary fittings and connection approved and accepted by the Engineer- in- charge.					
26.02.1	Approx. 695~719 X 340~395 mm size, minimum 39.5 kg of weight; Equivalent to RAK :Metropolitan / Stella: Vivana or similar brand	each	Tk. 17,202.92	Tk. 17,185.00	Tk. 17,126.00	Tk. 17,126.00
26.02.2	Approx. 685~729 X 360~406 mm size, minimum 36.0 kg of weight; Equivalent RAK: Oasis/Stella: Evana/ Stella: Marlina or equivalent model	each	Tk. 21,375.00	Tk. 21,356.00	Tk. 21,298.00	Tk. 21,298.00
26.02.3	Approx. 715~735 X 280~480 mm size, minimum 39.5 kg of weight; Equivalent to COTTO - C1053 / COTTO: 1141 / CHARU: CM 950 / RAK: Galaxy or similar brand	each	Tk. 25,669.00	Tk. 25,651.00	Tk. 25,592.00	Tk. 25,592.00
26.02.4	Approx. 760~735 X 430~490 mm size, minimum 40 kg of weight, Equivalent to Stella: Viola / COTTO: C1026 or similar brand	each	Tk. 31,805.00	Tk. 31,786.00	Tk. 31,727.00	Tk. 31,727.00

26.03	Supplying, fitting and fixing of country made European type glazed combi closet (Wall Hung) with concealed cistern, of any color including plastic seat cover with soft closing, water consumption 6 litre, Wash Down Daul flushing system which reduce water consumption, hygienic glaze in toilet bowl, glaze in inner waste line, round bowl, outlet range 225~230 mm. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position preparing the base with cement concrete and with wire mesh or rods if necessary, in all floors including making holes wherever required and mending good the damages and fitting, fixing, finishing, complete with all necessary fittings and connection approved and accepted by the Engineer Engineer- in- charge.					
26.03.1	Approx. 540 X 360 mm size, minimum 16.00 kg of weight	each	Tk. 22,233.73	Tk. 22,215.00	Tk. 22,157.00	Tk. 22,157.00
26.03.2	Approx. 570 X 375 mm size, minimum 22.00 kg of weight	each	Tk. 29,841.00	Tk. 29,823.00	Tk. 29,764.00	Tk. 29,764.00
26.03.3	Approx. 560 X 360 mm size, minimum 26 kg of weight	each	Tk. 39,167.00	Tk. 39,148.00	Tk. 39,090.00	Tk. 39,090.00

26.04	Supplying, fitting and fixing of European type country made glazed combi closet (Flush Valve) including flush valve, of any color, Approx. 380 mm x 710 mm x 800 mm size, minimum 28.00 kg of weight, excluding flush valve, including plastic seat cover with soft closing, Siphon jet single flushing system which reduce water consumption, hygienic glaze in toilet bowl, glaze in inner waste line, round bowl, outlet range 305 mm. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position preparing the base with cement concrete and with wire mesh or rods if necessary, in all floors including making holes wherever required and mending good the damages and fitting, fixing, finishing, complete with all necessary fittings and connection approved and accepted by the Engineer- in- charge.	each	Tk. 26,411.76	Tk. 26,393.00	Tk. 26,335.00	Tk. 26,335.00
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26.05	Supplying, fitting and fixing of country made glazed vitreous standing bowl urinal. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position after making holes in walls and floors, providing 32 mm dia plastic waste pipe with brass coupling up to grading below, 12 mm dia plastic connection pipe with brass coupling, 12 mm dia brass stop cock including mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer- in- charge.					
26.05.1	Approx. 330 X 355 X 445 mm size, minimum 10.0 kg of weight	each	Tk. 3,482.30	Tk. 3,472.00	Tk. 3,436.00	Tk. 3,436.00
26.05.2	Approx. 300 X 370 X 510 mm size, minimum 11.0 kg of weight.	each	Tk. 4,286.00	Tk. 4,276.00	Tk. 4,240.00	Tk. 4,240.00
26.05.3	Approx. 315 X 330 X 600 mm size, minimum 17.0 kg of weight.	each	Tk. 7,157.00	Tk. 7,147.00	Tk. 7,111.00	Tk. 7,111.00
26.05.4	Approx. 300 X 350 X 690 mm size, minimum 18.0 kg of weight	each	Tk. 13,206.00	Tk. 13,197.00	Tk. 13,160.00	Tk. 13,160.00

26.06	Supplying, fitting and fixing of country made glazed vitreous china flat or squatting urinal approx. 510 mm x 380 mm x 85 mm in size, minimum 9.5 kg of weight. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position with flushing inlet fitted in cement mortar (1:4) with painted cast iron body: one gallon capacity, automatic flushing cistern in each group etc. including 12 mm brass controlling valve, 32 mm dia PVC flush pipe with brass coupling, 12 mm dia plastic lead (connection) pipe including mending good the damages with 1:4 cement mortar etc. all complete approved and accepted by the Engineer- in- charge.	each	Tk. 3,310.39	Tk. 3,298.00	Tk. 3,252.00	Tk. 3,252.00
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26.07	Supplying, fitting and fixing of Bangladesh pattern, long pan with foot-rest. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position preparing the base of pan with cement mortar (1:4) and with wire mesh or rods, if necessary in all floors including making holes wherever required and mending good the damages and fitting, fixing, finishing etc. complete with all necessary fittings and connections approved and accepted by the Engineer- in- charge.					
26.07.1	Approx. 520 X 406.4 X 223.52 mm size, minimum 10.0 kg of weight	each	Tk. 2,036.86	Tk. 2,019.00	Tk. 1,960.00	Tk. 1,960.00
26.07.2	Approx. 530 X 400 X 230 mm size, minimum 11.0 kg of weight	each	Tk. 2,227.00	Tk. 2,209.00	Tk. 2,150.00	Tk. 2,150.00
26.07.3	Approx. 554 X 439. X 249 mm size, minimum 12.5 kg of weight	each	Tk. 2,687.00	Tk. 2,669.00	Tk. 2,610.00	Tk. 2,610.00
26.07.4	Approx. 540 X 460 X 290 mm size, minimum 13.5 kg of weight	each	Tk. 2,724.00	Tk. 2,706.00	Tk. 2,647.00	Tk. 2,647.00
26.07.5	Approx. 540 X 470 X 290 mm size, minimum 16 kg of weight	each	Tk. 2,847.00	Tk. 2,828.00	Tk. 2,770.00	Tk. 2,770.00

26.08	Supplying, fitting and fixing of European type glazed porcelain low-down. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed on walls or directly over water closet by necessary accessories, rowel plugs, making holes wherever required and mending good the damages and fitting, fixing & finishing etc. complete with all necessary fittings and connections approved and accepted by the Engineer- in-charge.					
26.08.1	Approx. 365 X 185 X 325 mm size, minimum 10.00 kg of weight	each	Tk. 3,608.41	Tk. 3,596.00	Tk. 3,562.00	Tk. 3,562.00
26.08.2	Approx. 490 X 190 X 345 mm size, minimum 11.00 kg of weight	each	Tk. 4,017.00	Tk. 4,005.00	Tk. 3,971.00	Tk. 3,971.00
26.08.3	Approx. 490 X 190 X 350 mm size, minimum 14 kg of weight	each	Tk. 4,069.00	Tk. 4,056.00	Tk. 4,023.00	Tk. 4,023.00
26.09	Supplying, fitting and fixing of plastic low-down of any color, on walls or directly over water closet with necessary accessories, making holes wherever required and mending good the damages and fitting, fixing & finishing etc. complete with all necessary fittings and connections approved and accepted by the Engineer- in- charge.	each	Tk. 2,429.51	Tk. 2,417.00	Tk. 2,384.00	Tk. 2,384.00
26.10	Supplying, fitting and fixing of flush valve for water closet on walls or directly over water closet with necessary accessories, making holes wherever required and mending good the damages and fitting, fixing & finishing etc. all complete with all necessary fittings and connections approved and accepted by the Engineer- in- charge.					
26.10.1	Flush valve (local)	each	Tk. 14,638.42	Tk. 14,629.00	Tk. 14,602.00	Tk. 14,602.00
26.10.2	Flush Valve (foreign)	each	Tk. 24,485.00	Tk. 24,475.00	Tk. 24,448.00	Tk. 24,448.00
26.11	Supplying, fitting and fixing of flush valve for urinal on walls or directly over water closet with necessary accessories,, rowel plugs, making holes wherever required and mending good the damages and fitting, fixing & finishing etc. all complete with all necessary fittings and connections approved and accepted by the Engineer- in- charge.	each	Tk. 5,834.51	Tk. 5,825.00	Tk. 5,798.00	Tk. 5,798.00

26.12	Supplying, fitting and fixing of country made glazed vitreous W/H wash basin excluding pedestal. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position with heavy type C.I. Brackets. 30 mm dia PVC waste water pipe with brass coupling (not exceeding 750 mm in length), basin waste with chain plug including making holes in walls and floors and fitting with wall, screws and mending good the damages, finishing etc. all complete approved and accepted by the Engineer-in-charge.					
26.12.1	Over Counter: Approx. 400X 200 mm, minimum weight 6.0 kg	each	Tk. 3,527.70	Tk. 3,509.00	Tk. 3,457.00	Tk. 3,457.00
26.12.2	Over Counter: Approx. 445X 345 mm, minimum weight 7.0 kg	each	Tk. 3,810.00	Tk. 3,792.00	Tk. 3,739.00	Tk. 3,739.00
26.12.3	Over Counter: Approx. 535X 435 mm, minimum weight 7.5 kg	each	Tk. 4,350.00	Tk. 4,331.00	Tk. 4,279.00	Tk. 4,279.00
26.12.4	Over Counter: Approx. 560 X 475 mm size, minimum weight 10.0 kg	each	Tk. 4,448.00	Tk. 4,430.00	Tk. 4,377.00	Tk. 4,377.00
26.12.5	Under Counter: Approx. 570 X 420 mm size, minimum weight 7.5 kg	each	Tk. 4,270.00	Tk. 4,252.00	Tk. 4,199.00	Tk. 4,199.00
26.12.6	Under Counter: Approx. 515 X 365 mm size, minimum weight 9.5 kg	each	Tk. 5,890.00	Tk. 5,871.00	Tk. 5,819.00	Tk. 5,819.00
26.12.7	Table Top: Approx. 400X 400 mm size, minimum weight 6.0 kg	each	Tk. 5,515.00	Tk. 5,497.00	Tk. 5,445.00	Tk. 5,445.00
26.12.8	Table Top: Approx. 450X 450 mm size :minimum weight 10.0 kg	each	Tk. 6,546.00	Tk. 6,528.00	Tk. 6,476.00	Tk. 6,476.00
26.12.9	Table Top: Approx. 480X 370 mm size, minimum weight 10.0 kg	each	Tk. 6,767.00	Tk. 6,749.00	Tk. 6,696.00	Tk. 6,696.00
26.12.10	Table Top: Approx. 485X 380 mm size, minimum weight 11.0 kg	each	Tk. 7,884.00	Tk. 7,865.00	Tk. 7,813.00	Tk. 7,813.00
26.12.11	Corner Basin: Approx. 540X 400 mm size, minimum weight 7.5 kg	each	Tk. 3,589.00	Tk. 3,571.00	Tk. 3,518.00	Tk. 3,518.00
26.12.12	Corner Basin: Approx. 460X 390 mm size, minimum weight 15.0 kg	each	Tk. 4,522.00	Tk. 4,503.00	Tk. 4,451.00	Tk. 4,451.00

26.13	Supplying of country made glazed vitreous W/H wash basin of any color including pedestal. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position with heavy type C.I. Brackets. 30 mm dia PVC waste water pipe with brass coupling (not exceeding 750 mm in length), basin waste with chain plug including making holes in walls and floors and fitting with wall, screws and mending good the damages, finishing etc. all complete approved and accepted by the Engineer-in-charge.					
26.13.1	Approx. 436 x 343 mm, minimum weight 14.8 kg	each	Tk. 3,833.23	Tk. 3,823.00	Tk. 3,787.00	Tk. 3,787.00
26.13.2	Approx. 457 x 406 mm, minimum weight 15.3 kg	each	Tk. 4,080.00	Tk. 4,070.00	Tk. 4,034.00	Tk. 4,034.00
26.13.3	Approx. 521 x 419 mm, minimum weight 17.7 kg	each	Tk. 4,196.00	Tk. 4,187.00	Tk. 4,150.00	Tk. 4,150.00
26.13.4	Approx. 508 x 394 mm, minimum weight 17.3 kg	each	Tk. 4,221.00	Tk. 4,211.00	Tk. 4,175.00	Tk. 4,175.00
26.13.5	Approx. 540 x 450 mm, minimum weight 20.1 kg	each	Tk. 4,270.00	Tk. 4,260.00	Tk. 4,224.00	Tk. 4,224.00
26.13.6	Approx. 495 x 394 mm, minimum weight 17.3 kg	each	Tk. 4,405.00	Tk. 4,395.00	Tk. 4,359.00	Tk. 4,359.00
26.13.7	Approx. 513 x 432 mm, minimum weight 17.8 kg	each	Tk. 4,528.00	Tk. 4,518.00	Tk. 4,482.00	Tk. 4,482.00
26.13.8	Approx. 564 x 419 mm, minimum weight 18.8 kg	each	Tk. 4,650.00	Tk. 4,641.00	Tk. 4,604.00	Tk. 4,604.00
26.13.9	Approx. 584 x 437 mm, minimum weight 20 kg	each	Tk. 4,834.00	Tk. 4,825.00	Tk. 4,788.00	Tk. 4,788.00
26.13.10	Approx. 560 x 450 mm, minimum weight 22.8 kg	each	Tk. 4,902.00	Tk. 4,892.00	Tk. 4,856.00	Tk. 4,856.00
26.13.11	Approx. 572 x 437 mm, minimum weight 20.3 kg	each	Tk. 5,012.00	Tk. 5,003.00	Tk. 4,966.00	Tk. 4,966.00
26.13.12	Approx. 648 x 488 mm, minimum weight 20 kg	each	Tk. 5,295.00	Tk. 5,285.00	Tk. 5,249.00	Tk. 5,249.00
26.13.13	Approx. 605 x 485 mm, minimum weight 22.5 kg	each	Tk. 5,374.00	Tk. 5,365.00	Tk. 5,328.00	Tk. 5,328.00
26.13.14	Approx. 630 x 520 mm, minimum weight 28 kg	each	Tk. 5,632.00	Tk. 5,622.00	Tk. 5,586.00	Tk. 5,586.00
26.13.15	Approx. 559 x 483 mm, minimum weight 24.5 kg	each	Tk. 5,877.00	Tk. 5,868.00	Tk. 5,831.00	Tk. 5,831.00
26.13.16	Approx. 656 x 510 mm, minimum weight 26 kg	each	Tk. 7,055.00	Tk. 7,046.00	Tk. 7,009.00	Tk. 7,009.00
26.13.17	Approx. 500 x 450 mm, minimum weight 25.6 kg	each	Tk. 7,154.00	Tk. 7,144.00	Tk. 7,108.00	Tk. 7,108.00
26.13.18	Approx. 520 x 425 mm, minimum weight 26.2 kg	each	Tk. 7,497.00	Tk. 7,487.00	Tk. 7,451.00	Tk. 7,451.00
26.13.19	Approx. 665 x 490 mm, minimum weight 29.5 kg	each	Tk. 7,626.00	Tk. 7,616.00	Tk. 7,580.00	Tk. 7,580.00
26.13.20	Approx. 660 x 525 mm, minimum weight 31.5 kg	each	Tk. 7,822.00	Tk. 7,812.00	Tk. 7,776.00	Tk. 7,776.00
26.13.21	Approx. 600 x 440 mm, minimum weight 26 kg	each	Tk. 7,902.00	Tk. 7,892.00	Tk. 7,856.00	Tk. 7,856.00
26.13.22	Approx. 525 x 415 mm, minimum weight 27.4 kg	each	Tk. 8,442.00	Tk. 8,432.00	Tk. 8,396.00	Tk. 8,396.00
26.13.23	Approx. 650 x 510 mm, minimum weight 25 kg	each	Tk. 8,516.00	Tk. 8,506.00	Tk. 8,470.00	Tk. 8,470.00

26.13.24	Approx. 550 x 470 mm, minimum weight 34 kg	each	Tk. 10,356.00	Tk. 10,346.00	Tk. 10,310.00	Tk. 10,310.00
26.14	Supplying of country made glazed vitreous W/H wash basin of any color including semi-pedestal/ half pedestal. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position with heavy type C.I. Brackets. 30 mm dia PVC waste water pipe with brass coupling (not exceeding 750 mm in length), basin waste with chain plug including making holes in walls and floors and fitting with wall, screws and mending good the damages, finishing etc. all complete approved and accepted by the Engineer-in-charge.					
26.14.1	Approx. 510X 400 mm size, minimum weight 15 kg	each	Tk. 4,171.89	Tk. 4,162.00	Tk. 4,126.00	Tk. 4,126.00
26.14.2	Approx. 550X 440 mm size, minimum weight 17 kg	each	Tk. 6,405.00	Tk. 6,395.00	Tk. 6,359.00	Tk. 6,359.00
26.14.3	Approx. 600X 500 mm size, minimum weight 18.5 kg	each	Tk. 7,227.00	Tk. 7,217.00	Tk. 7,181.00	Tk. 7,181.00
26.14.4	Approx. 550 X 460 mm size, minimum weigh 20.0 kg	each	Tk. 7,399.00	Tk. 7,389.00	Tk. 7,353.00	Tk. 7,353.00
26.15	Supplying, fitting and fixing basin pedestal of Approx. size of 670 mm x 186 mm x 170 mm, any color with rowel plug and screws etc. all complete approved and accepted by the Engineer- in- charge.	each	Tk. 2,495.77	Tk. 2,490.00	Tk. 2,474.00	Tk. 2,474.00

26.16	Supplying, fitting and fixing of country made porcelain sink of any color, approx. 635 mm x 480 mm x 195 mm size, 19.5 kg of weight. The sanitary ware shall conform BDS1162:2014. The glaze shall be thoroughly fused to body. The minimum thickness of body at any section shall be 5 mm. When assembled together and when examined from a distance of 60 cm, the outer surface shall not show to the unaided eye, blemishes or defects in excess of those listed in BDS standard. The mean value of water absorption shall not be greater than 0.5% of the ware when dry. When tested with chemical solutions (Acetic acid, Citric acid, Detergent, Hydrochloric acid, Sodium hydroxide, Sodium stearate and Sulfuric acid of various strength) as per BDS1162:2014 procedure, none of the test pieces should suffer any loss of reflectivity on the glaze. There shall be no crazing and no stain on the ware. The materials used for making glaze shall not contain lead compound. In case of certain coloring oxides used for making colored glaze, the lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position fitted and fixed with heavy type C.I. brackets, 12 mm dia C.P. bib cock, 40 mm dia C.P. chain plug, 40 mm dia C.P. waste, 32 mm dia PVC waste pipe with brass coupling (750 mm length) including making holes in walls and floors and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer- in- charge.	each	Tk. 3,850.00	Tk. 3,842.00	Tk. 3,819.00	Tk. 3,819.00
26.17	Supplying, fitting and fixing of best quality single bowl stainless steel sink with heavy type CI steel brackets, 40 mm dia C.P. chain plug, 40 mm dia C.P. waste, 40 mm dia PVC waste pipe with brass coupling (750 mm length) including making hole in walls and floors and mending good the damages, finishing with 1:4 cement mortar etc. all complete approved and accepted by the Engineer-in-charge.					
26.17.1	Approx. 508 X 406 mm size single bowl without sink tray	each	Tk. 3,587.00	Tk. 3,579.00	Tk. 3,556.00	Tk. 3,556.00
26.17.2	Approx. 508 X 457 mm size single bowl without sink tray	each	Tk. 3,697.00	Tk. 3,690.00	Tk. 3,666.00	Tk. 3,666.00
26.17.3	Approx. 610 X 457 mm size single bowl without sink tray	each	Tk. 4,089.00	Tk. 4,082.00	Tk. 4,058.00	Tk. 4,058.00
26.17.4	Approx. 762 X 457 mm size single bowl with sink tray	each	Tk. 5,298.00	Tk. 5,290.00	Tk. 5,267.00	Tk. 5,267.00
26.17.5	Approx. 914 X 457 mm size single bowl with sink tray	each	Tk. 5,420.00	Tk. 5,413.00	Tk. 5,390.00	Tk. 5,390.00
26.17.6	Approx. 1016 X 508 mm size single bowl with sink tray	each	Tk. 6,392.00	Tk. 6,384.00	Tk. 6,361.00	Tk. 6,361.00
26.18	Supplying, fitting and fixing of best quality double bowl stainless steel sink with heavy type CI or steel brackets, 12 mm dia C.P. chain plug, 40 mm dia C.P. waste, 40 mm dia PVC waste pipe with brass coupling (750 mm length) including making hole in walls and floors and mending good the damages, finishing etc. all complete approved and accepted by the Engineer-in-charge.	each				
26.18.1	Approx. 1160 x 500 mm size double bowl with sink tray (Standard)	each	Tk. 8,486.00	Tk. 8,477.00	Tk. 8,446.00	Tk. 8,446.00
26.18.2	Approx. 1160 x 500 mm size double bowl with sink tray (Super)	each	Tk. 12,167.00	Tk. 12,158.00	Tk. 12,127.00	Tk. 12,127.00
26.18.3	Approx. 1160 x 500 mm size double bowl with sink tray (Special)	each	Tk. 17,689.00	Tk. 17,679.00	Tk. 17,649.00	Tk. 17,649.00

26.19	Supplying, fitting and fixing of 5 mm thick unframed super quality double coated mirror with all necessary fitting including making holes in walls and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer-in-charge.					
26.19.1	Local Mirror, 5 mm thickness	sqm	Tk. 2,590.00	Tk. 2,572.00	Tk. 2,510.00	Tk. 2,510.00
26.19.2	Belgium Mirror, 5 mm thickness	sqm	Tk. 1,724.00	Tk. 1,719.00	Tk. 1,703.00	Tk. 1,703.00
26.20	Supplying, fitting and fixing of super quality 600 mm x 125 mm coloured glass (plate) shelf having 5 mm thickness with fancy C.P. brackets, screws and frames including making holes in walls and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer- in- charge.	each	Tk. 1,829.00	Tk. 1,825.00	Tk. 1,808.00	Tk. 1,808.00
26.21	Supplying, fitting and fixing of super quality 600 x 125 mm white glass (plate) shelf having 5 mm thickness with fancy C.P. brackets, screws and frames including making holes in walls and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer- in- charge.	each	Tk. 1,032.00	Tk. 1,027.00	Tk. 1,010.00	Tk. 1,010.00
26.22	Supplying, fitting and fixing of country made best quality towel rail/ring/shelf minimum 20 mm in dia with C.P. holder including making holes in walls and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer- in- charge.					
26.22.1	C.P. towel ring/ 450 mm long single towel rail	each	Tk. 718.00	Tk. 717.00	Tk. 714.00	Tk. 714.00
26.22.2	600 mm long C.P. single towel rail	each	Tk. 718.00	Tk. 717.00	Tk. 714.00	Tk. 714.00
26.22.3	600 mm long C.P. double towel rail	each	Tk. 1,264.00	Tk. 1,263.00	Tk. 1,260.00	Tk. 1,260.00
26.22.4	600 mm long C.P. towel shelf	each	Tk. 1,646.00	Tk. 1,645.00	Tk. 1,641.00	Tk. 1,641.00
26.23	Supplying, fitting and fixing of best quality toilet paper holder of standard size including making holes in walls and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer- in- charge.					
26.23.1	PVC toilet paper holder	each	Tk. 293.00	Tk. 292.00	Tk. 288.00	Tk. 288.00
26.23.2	C.P iron bracket toilet paper holder	each	Tk. 372.00	Tk. 372.00	Tk. 368.00	Tk. 368.00
26.23.3	Porceain toilet paper holder (any color)	each	Tk. 808.00	Tk. 807.00	Tk. 804.00	Tk. 804.00
26.23.4	Stainless steel toilet paper holder	each	Tk. 563.00	Tk. 562.00	Tk. 558.00	Tk. 558.00
26.24	Supplying, fitting and fixing of standard size country made soap tray including making holes in walls and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer- in- charge.					
26.24.1	PVC soap tray	each	Tk. 150.00	Tk. 149.00	Tk. 145.00	Tk. 145.00
26.24.2	C.P. soap tray	each	Tk. 236.00	Tk. 235.00	Tk. 231.00	Tk. 231.00
26.24.3	Porcelain soap tray (any color)	each	Tk. 420.00	Tk. 419.00	Tk. 415.00	Tk. 415.00
26.24.4	Stainless steel soap tray	each	Tk. 628.00	Tk. 627.00	Tk. 624.00	Tk. 624.00

26.25	Supplying, fitting and fixing of Automatic sensor liquid soap dispenser to hold and dispense up to min 600 ml of liquid and lotion soaps, synthetic detergents and antibacterial soaps. Dispenser shall be powered by batteries and shall operate automatically from IR sensor that shall activate soap dispensing when hands are placed in sensor zone. Sensor detection range shall be adjustable by owner. Dispenser shall pump a drop volume of 0.8 ml to 1.0 ml of soap during each activation. A tamper resistant soap level-viewing window shall be provided. Unit shall have a swing open body secured with side-mounted lock that is opened with special key provided, all complete and accepted by the Engineer-in-charge.					
26.25.1	Automatic liquid soap dispenser (local): standard quality	each	Tk. 1,633.00	Tk. 1,626.00	Tk. 1,611.00	Tk. 1,611.00
26.25.2	Automatic liquid soap dispenser (local): super quality	each	Tk. 3,835.00	Tk. 3,828.00	Tk. 3,814.00	Tk. 3,814.00
26.25.3	Automatic liquid soap dispenser (foreign): special quality	each	Tk. 7,795.00	Tk. 7,788.00	Tk. 7,773.00	Tk. 7,773.00
26.26	Supplying, fitting and fixing of Push-Up type liquid soap dispenser to dispense 16 oz (.47 liters) of soap, detergent or lotion. Unit shall be fabricated of tamper resistant, chrome plated plastic, a translucent white polyethylene soap container, and a corrosion resistant dispensing valve which can provide no more than 5 lbs (22.2 Newtons) pressure and unit can be reloaded by unscrewing base from bracket collar, inverting unit, and unscrewing soap container in which soap will continue to feed properly until empty with a continuously visible soap level etc. all complete approved and accepted by the Engineer-in-charge.					
26.26.1	Push-Up type liquid soap dispenser (local): standard quality	each	Tk. 1,347.00	Tk. 1,340.00	Tk. 1,325.00	Tk. 1,325.00
26.26.2	Push-Up type liquid soap dispenser (local): super quality	each	Tk. 2,052.00	Tk. 2,045.00	Tk. 2,031.00	Tk. 2,031.00
26.26.3	Push-Up type liquid soap dispenser (foreign): superior quality	each	Tk. 6,077.00	Tk. 6,070.00	Tk. 6,055.00	Tk. 6,055.00
26.26.4	Push-Up type liquid soap dispenser (foreign): special quality	each	Tk. 8,825.00	Tk. 8,819.00	Tk. 8,804.00	Tk. 8,804.00
26.27	Supplying, fitting and fixing of floor grating net in traps or in drains including making holes in walls/floors and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer-in-charge. of following type					
26.27.1	Approx. 50~75 mm dia stainless steel floor grating net	each	Tk. 185.00	Tk. 183.00	Tk. 180.00	Tk. 180.00
26.27.2	Approx. 100~125 mm dia stainless steel floor grating net	each	Tk. 326.00	Tk. 325.00	Tk. 321.00	Tk. 321.00
26.27.3	Approx. 50 mm x 50 mm stainless steel floor grating net	each	Tk. 185.00	Tk. 183.00	Tk. 180.00	Tk. 180.00
26.27.4	Approx. 100 mm x 100 mm stainless steel floor grating net	each	Tk. 209.00	Tk. 208.00	Tk. 205.00	Tk. 205.00
26.27.5	Approx. 125 mm x 125 mm stainless steel floor grating net	each	Tk. 988.00	Tk. 987.00	Tk. 984.00	Tk. 984.00
26.27.6	125 mm dia convex cast iron grating on rain water down pipe (roof top)	each	Tk. 553.00	Tk. 552.00	Tk. 550.00	Tk. 550.00
26.28	Supplying, fitting and fixing of 100 mm floor drain with vertical outlet in traps including making holes in walls/floors and mending good the damages with cement mortar (1:4) etc. all complete approved and accepted by the Engineer-in-charge. of following type.	each	Tk. 1,369.00	Tk. 1,368.00	Tk. 1,364.00	Tk. 1,364.00

26.29	Supplying, fitting and fixing of best quality country made 12 mm bib cock made from copper or copper alloy, chromium plated. The faucet conforms BDS EN 200:2009. The faucet tap shall be free from any leakage, permeation and other abnormalities. The water hammer value shall be 1.47 Mpa or under. The faucet after it has been operated 100000 times it shall be free from a seat- leakage and shall satisfy the reverse flow preventing performance. The inside and outside surfaces of the faucet shall be smooth and be free from blowholes, fissures, remarkable flows or injurious defects. The chromium plating on the faucet shall be of class 2 grade 1. The final coating should be done with chromium of minimum 0.1 micron. The faucet shall be made leak proof and fixing in position with selected tape etc. , all complete approved and accepted by the Engineer- in- charge.					
26.29.1	Round headed CP Conceal Bib Cock short body (Essco/coaster/star/diamond/delux/deco/orbed/delin/triple/master/classic headed or similar head)	each	Tk. 1,172.00	Tk. 1,169.00	Tk. 1,161.00	Tk. 1,161.00
26.29.2	Lever CP Conceal Bib Cock long body with aerator (Oliver/spoon/delta/decent lever or similar)	each	Tk. 1,172.00	Tk. 1,169.00	Tk. 1,161.00	Tk. 1,161.00
26.29.3	12 mm CP open Bib Cock	each	Tk. 981.00	Tk. 979.00	Tk. 971.00	Tk. 971.00
26.29.4	12 mm CP Two in One Bib Cock	each	Tk. 1,846.00	Tk. 1,844.00	Tk. 1,836.00	Tk. 1,836.00
26.29.5	Elbow action CP Bib Cock	each	Tk. 3,270.00	Tk. 3,267.00	Tk. 3,259.00	Tk. 3,259.00
26.29.6	CP Push Cock	each	Tk. 1,012.00	Tk. 1,010.00	Tk. 1,002.00	Tk. 1,002.00
26.29.7	CP Leaf Cock	each	Tk. 816.00	Tk. 813.00	Tk. 805.00	Tk. 805.00
26.30	Supplying, fitting and fixing of best quality surface mounted or conceal stop cock/angle stop cock (lever or round headed) with wall flange chromium plated. The faucet conforms BDS EN 200:2009. The faucet tap shall be free from any leakage, permeation and other abnormalities. The water hammer value shall be 1.47 Mpa or under. The faucet after it has been operated 100000 times it shall be free from a seat- leakage and shall satisfy the reverse flow preventing performance. The inside and outside surfaces of the faucet shall be smooth and be free from blowholes, fissures, remarkable flows or injurious defects. The chromium plating on the faucet shall be of class 2 grade 1. The final coating should be done with chromium of minimum 0.1 micron. The faucet shall be made leak proof and fixing in position with selected tape etc. , all complete approved and accepted by the Engineer- in- charge.					
26.30.1	12 mm dia CP Concealed Stop Cock	each	Tk. 1,153.00	Tk. 1,151.00	Tk. 1,143.00	Tk. 1,143.00
26.30.2	20 mm dia CP Concealed Stop Cock	each	Tk. 1,202.00	Tk. 1,200.00	Tk. 1,192.00	Tk. 1,192.00
26.30.3	25 mm dia CP Concealed Stop Cock	each	Tk. 1,356.00	Tk. 1,353.00	Tk. 1,345.00	Tk. 1,345.00
26.30.4	CP angle stop cock	each	Tk. 988.00	Tk. 985.00	Tk. 977.00	Tk. 977.00
26.30.5	CP Two in One angle stop cock	each	Tk. 1,264.00	Tk. 1,261.00	Tk. 1,253.00	Tk. 1,253.00

26.31	Supplying, fitting and fixing of best quality pillar cock made from copper or copper alloy, chromium plated materials. The faucet conforms BDS EN 200:2009. The faucet tap shall be free from any leakage, permeation and other abnormalities. The water hammer value shall be 1.47 Mpa or under. The faucet after it has been operated 100000 times it shall be free from a seat- leakage and shall satisfy the reverse flow preventing performance. The inside and outside surfaces of the faucet shall be smooth and be free from blowholes, fissures, remarkable flows or injurious defects. The chromium plating on the faucet shall be of class 2 grade 1. The final coating should be done with chromium of minimum 0.1 micron. The faucet shall be made leak proof and fixing in position with selected tape etc. all complete approved and accepted by the Engineer- in- charge.					
26.31.1	CP Round headed fixed Pillar cock	each	Tk. 1,540.00	Tk. 1,537.00	Tk. 1,529.00	Tk. 1,529.00
26.31.2	CP Single lever pillar cock	each	Tk. 1,540.00	Tk. 1,537.00	Tk. 1,529.00	Tk. 1,529.00
26.31.3	CP Single lever pillar cock (High top 8")	each	Tk. 3,069.00	Tk. 3,066.00	Tk. 3,058.00	Tk. 3,058.00
26.31.4	CP Single lever pillar cock (High top 12")	each	Tk. 4,050.00	Tk. 4,048.00	Tk. 4,040.00	Tk. 4,040.00
26.31.5	CP Single lever movig pillar cock	each	Tk. 1,891.00	Tk. 1,888.00	Tk. 1,880.00	Tk. 1,880.00
26.31.6	CP Elbow action Pillar cock	each	Tk. 3,270.00	Tk. 3,267.00	Tk. 3,259.00	Tk. 3,259.00
26.31.7	CP One way Swan Cock	each	Tk. 5,920.00	Tk. 5,918.00	Tk. 5,910.00	Tk. 5,910.00
26.31.8	CP Two way Swan Cock	each	Tk. 6,902.00	Tk. 6,899.00	Tk. 6,891.00	Tk. 6,891.00
26.31.9	CP Three way Swan Cock	each	Tk. 8,178.00	Tk. 8,176.00	Tk. 8,168.00	Tk. 8,168.00
26.32	Supplying, fitting and fixing of best quality CP self closing pillar cock without mixer (SensorType /Push Type) with necessary hardware etc. complete approved and accepted by the Engineer- in- charge.					
26.32.1	Sensor Type self closing pillar cock	each	Tk. 9,224.00	Tk. 9,220.00	Tk. 9,208.00	Tk. 9,208.00
26.32.2	Push Type Self closing pillar cock	each	Tk. 1,366.00	Tk. 1,362.00	Tk. 1,350.00	Tk. 1,350.00
26.33	Supplying, fitting and fixing of best quality sink cock made from copper or copper alloy, chromium plated. The faucet conforms BDS EN 200:2009 . The faucet tap shall be free from any leakage, permeation and other abnormalities. The water hammer value shall be 1.47 Mpa or under. The faucet after it has been operated 100000 times it shall be free from a seat- leakage and shall satisfy the reverse flow preventing performance. The inside and outside surfaces of the faucet shall be smooth and be free from blowholes, fissures, remarkable flows or injurious defects. The chromium plating on the faucet shall be of class 2 grade 1. The final coating should be done with chromium of minimum 0.1 micron. The faucet shall be made leak proof and fixing in position with selected tape etc. , all complete approved and accepted by the Engineer- in- charge.					
26.33.1	12 mm CP concealed fixed sink cock	each	Tk. 1,202.00	Tk. 1,200.00	Tk. 1,192.00	Tk. 1,192.00
26.33.2	12 mm CP concealed super quality moving sink cock	each	Tk. 1,685.00	Tk. 1,682.00	Tk. 1,674.00	Tk. 1,674.00
26.33.3	12 mm CP concealed special moving sink cock	each	Tk. 1,891.00	Tk. 1,888.00	Tk. 1,880.00	Tk. 1,880.00

26.34	Supplying, fitting and fixing of best quality CP basin/sink mixer The faucet conforms BDS EN 200:2009. The faucet tap shall be free from any leakage, permeation and other abnormalities. The water hammer value shall be 1.47 Mpa or under. The faucet after it has been operated 100000 times it shall be free from a seat- leakage and shall satisfy the reverse flow preventing performance. The inside and outside surfaces of the faucet shall be smooth and be free from blowholes, fissures, remarkable flows or injurious defects. The chromium plating on the faucet shall be of class 2 grade 1. The final coating should be done with chromium of minimum 0.1 micron. The faucet shall be made leak proof and fixing in position with selected tape etc. , all complete approved and accepted by the Engineer- in- charge.					
26.34.1	Single handle/lever basin mixer	each	Tk. 3,067.00	Tk. 3,065.00	Tk. 3,057.00	Tk. 3,057.00
26.34.2	Fixed basin mixer	each	Tk. 3,258.00	Tk. 3,255.00	Tk. 3,247.00	Tk. 3,247.00
26.34.3	Moving basin mixer	each	Tk. 3,319.00	Tk. 3,317.00	Tk. 3,308.00	Tk. 3,308.00
26.34.4	Moving sink mixer	each	Tk. 1,891.00	Tk. 1,888.00	Tk. 1,880.00	Tk. 1,880.00
26.35	Supplying, fitting and fixing of best quality 20 mm CP shower mixer. The faucet conforms BDS EN 200:2009. The faucet tap shall be free from any leakage, permeation and other abnormalities. The water hammer value shall be 1.47 Mpa or under. The faucet after it has been operated 100000 times it shall be free from a seat- leakage and shall satisfy the reverse flow preventing performance. The inside and outside surfaces of the faucet shall be smooth and be free from blowholes, fissures, remarkable flows or injurious defects. The chromium plating on the faucet shall be of class 2 grade 1. The final coating should be done with chromium of minimum 0.1 micron. The faucet shall be made leak proof and fixing in position with selected tape etc. , all complete approved and accepted by the Engineer- in- charge.					
26.35.1	CP shower mixture including shower head only	each	Tk. 5,763.00	Tk. 5,761.00	Tk. 5,753.00	Tk. 5,753.00
26.35.2	CP shower mixture including shower head and faucet	each	Tk. 7,123.00	Tk. 7,120.00	Tk. 7,112.00	Tk. 7,112.00
26.36	Supplying, fitting and fixing of best quality Bathtub mixer. The faucet conforms BDS EN 200:2009. The faucet tap shall be free from any leakage, permeation and other abnormalities. The water hammer value shall be 1.47 Mpa or under. The faucet after it has been operated 100000 times it shall be free from a seat- leakage and shall satisfy the reverse flow preventing performance. The inside and outside surfaces of the faucet shall be smooth and be free from blowholes, fissures, remarkable flows or injurious defects. The chromium plating on the faucet shall be of class 2 grade 1. The final coating should be done with chromium of minimum 0.1 micron. The faucet shall be made leak proof and fixing in position with selected tape etc. , all complete approved and accepted by the Engineer- in- charge.					
26.36.1	Bathtub mixer without hand shower	each	Tk. 4,973.00	Tk. 4,970.00	Tk. 4,932.00	Tk. 4,932.00
26.36.2	Bathtub mixer including hand shower	each	Tk. 5,421.00	Tk. 5,418.00	Tk. 5,379.00	Tk. 5,379.00

26.37	Supplying, fitting and fixing of best quality Single lever angle stop mixer . The faucet conforms BDS EN 200:2009. The faucet tap shall be free from any leakage, permeation and other abnormalities. The water hammer value shall be 1.47 Mpa or under. The faucet after it has been operated 100000 times it shall be free from a seat- leakage and shall satisfy the reverse flow preventing performance. The inside and outside surfaces of the faucet shall be smooth and be free from blowholes, fissures, remarkable flows or injurious defects. The chromium plating on the faucet shall be of class 2 grade 1. The final coating should be done with chromium of minimum 0.1 micron. The faucet shall be made leak proof and fixing in position with selected tape etc. , all complete approved and accepted by the Engineer- in- charge.	each	Tk. 2,283.00	Tk. 2,281.00	Tk. 2,273.00	Tk. 2,273.00
26.38	Supplying and fitting-fixing Chromium plated (CP) Shower head . all complete approved and accepted by the Engineer- in- charge.					
26.38.1	Moving type 100 mm CP shower head	each	Tk. 1,251.00	Tk. 1,249.00	Tk. 1,241.00	Tk. 1,241.00
26.38.2	Moving type 150 mm CP shower head	each	Tk. 2,276.00	Tk. 2,274.00	Tk. 2,266.00	Tk. 2,266.00
26.38.3	Moving type 200 mm CP shower head	each	Tk. 2,534.00	Tk. 2,531.00	Tk. 2,523.00	Tk. 2,523.00
26.38.4	Moving type 250 mm CP shower head	each	Tk. 2,828.00	Tk. 2,826.00	Tk. 2,818.00	Tk. 2,818.00
26.39	Supplying and fitting-fixing stainless steel hand / push shower for ablution including holder etc. all complete approved and accepted by the Engineer- in- charge.					
26.39.1	Moving type hand shower	each	Tk. 1,454.00			
26.39.2	Moving type push shower	each	Tk. 1,049.00	Tk. 1,047.00	Tk. 1,039.00	Tk. 1,039.00
PIPE AND PIPE FITTING ACCESSORIES						
PIPE ITEMS						
26.40	Supplying, fitting and fixing of 100 mm inside diameter & wall thickness 2.7 mm - 3.4 mm best quality uPVC rain water down pipe fitting, fixed in position with head and shoes, bends, min.20 mm width F.I. Bar clamp and nails, and including all accessories such as round grating/domed roof grating, bands, sockets approved and accepted by the Engineer- in- charge.	meter	Tk. 993.28	Tk. 988.00	Tk. 970.00	Tk. 970.00
26.41	Supplying, laying, fitting and fixing of best quality HDPE soil, waste and ventilation pipe having PE-100 (Hydrostatic Strength for PE-100 is 5.0 MPa & 12.4 Mpa at 80°C & 20°C respectively according to ISO 4427:2007(E)), PN-6 (nominal pressure for PN-6 is 0.6 MPa according to ISO 4427:2007(E)) & SDR 26 for soil, waste and ventilation pressure pipe having density 0.940 – 0.970 gm/cm ³ and other physical, chemical, thermal, fire resistivity properties etc. as per BSTI approved manufacturer standards or ASTM/BS/ISO/IS standards ,fitted and fixed in position with necessary joints and fittings with all accesories complete approved and accepted by the Engineer- in- charge.					
26.41.1	For underground including earth cutting & preperation of bed					
26.41.1.1	HDPE pipe of 110 mm outside dia with wall thickness 4.2 mm - 4.8 mm	meter	Tk. 1,512.50	Tk. 1,509.00	Tk. 1,486.00	Tk. 1,486.00
26.41.1.2	HDPE pipe of 160 mm outside dia with wall thickness 6.2 mm - 7.0 mm	meter	Tk. 2,382.19	Tk. 2,376.00	Tk. 2,348.00	Tk. 2,348.00
26.41.1.3	HDPE pipe of 200 mm outside dia with wall thickness 7.7 mm - 8.6 mm	meter	Tk. 3,952.79	Tk. 3,947.00	Tk. 3,916.00	Tk. 3,916.00
26.41.1.4	HDPE pipe of 250 mm outside dia with wall thickness 9.6 mm -10.7 mm	meter	Tk. 6,065.34	Tk. 6,046.00	Tk. 5,973.00	Tk. 5,973.00
26.41.1.5	HDPE pipe of 280 mm outside dia with wall thickness 10.7 mm -11.9 mm	meter	Tk. 7,892.58	Tk. 7,873.00	Tk. 7,801.00	Tk. 7,801.00
26.41.1.6	HDPE pipe of 315 mm outside dia with wall thickness 12.1 mm -13.5 mm	meter	Tk. 9,671.62	Tk. 9,652.00	Tk. 9,580.00	Tk. 9,580.00

26.41.1.7	HDPE pipe of 400 mm outside dia with wall thickness 15.3 mm -17.0 mm	meter	Tk. 15,400.84	Tk. 15,383.00	Tk. 15,301.00	Tk. 15,301.00
26.41.1.8	HDPE pipe of 630 mm outside dia with wall thickness 24.2 mm - 26.7 mm	meter	Tk. 35,799.40	Tk. 35,776.00	Tk. 35,670.00	Tk. 35,670.00
26.41.1.9	HDPE pipe of 710 mm outside dia with wall thickness 27.2 mm - 30.1 mm	meter	Tk. 46,015.57	Tk. 45,992.00	Tk. 45,886.00	Tk. 45,886.00
26.41.1.10	HDPE pipe of 900 mm outside dia with wall thickness 34.4 mm - 38.3 mm	meter	Tk. 68,549.90	Tk. 68,526.00	Tk. 68,420.00	Tk. 68,420.00
26.41.2	For in house system					
26.41.2.1	HDPE pipe 110 mm outside dia with wall thickness 4.2 mm - 4.8 mm	meter	Tk. 2,012.74	Tk. 2,009.00	Tk. 1,995.00	Tk. 1,995.00
26.41.2.2	HDPE pipe 160 mm outside dia with wall thickness 6.2 mm -7.0 mm	meter	Tk. 3,276.26	Tk. 3,270.00	Tk. 3,251.00	Tk. 3,251.00
26.41.2.3	HDPE pipe 200 mm outside dia with wall thickness 7.7 mm -8.6 mm	meter	Tk. 5,229.81	Tk. 5,224.00	Tk. 5,205.00	Tk. 5,205.00
26.41.2.4	HDPE pipe 250 mm outside dia with wall thickness 9.6 mm -10.7 mm	meter	Tk. 8,190.84	Tk. 8,185.00	Tk. 8,166.00	Tk. 8,166.00
26.42	Supplying, fitting and fixing of G.I. pipe with all special fittings, such as bends, elbows, sockets, reducing sockets, Tee, unions, jam-nuts etc. including cutting trenches where necessary and fitting the same with earth duly rammed and fixing in walls with holder bats and making hole in floors, walls and consequent mending good the damages etc. all complete in all respects approved and accepted by the Engineer- in- charge.					
26.42.1	12.5 mm dia G.I. pipe with wall thickness 2.65 mm, outside diameter min 17.8 mm, weight 1.31 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 440.72	Tk. 439.00	Tk. 432.00	Tk. 432.00
26.42.2	20 mm dia G.I. pipe with wall thickness 2.65 mm, outside diameter min 25.3 mm, weight 1.7 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 549.01	Tk. 547.00	Tk. 540.00	Tk. 540.00
26.42.3	25 mm dia G.I. pipe with wall thickness 3.35 mm, outside diameter min 31.7 mm, weight 2.59 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 700.48	Tk. 698.00	Tk. 692.00	Tk. 692.00
26.42.4	32 mm dia G.I. pipe with wall thickness 3.25 mm, outside diameter min 38.5 mm, weight 3.33 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 884.16	Tk. 881.00	Tk. 873.00	Tk. 873.00
26.42.5	40 mm dia G.I. pipe with wall thickness 3.25 mm, outside diameter min 46.5 mm, weight 3.83 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 1,145.88	Tk. 1,142.00	Tk. 1,130.00	Tk. 1,130.00
26.42.6	50 mm dia G.I. pipe with wall thickness 3.65 mm, outside diameter min 57.3 mm, weight 5.38 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 1,503.95	Tk. 1,498.00	Tk. 1,479.00	Tk. 1,479.00
26.42.7	62-65 mm dia G.I. pipe with wall thickness 3.25 mm, outside diameter min 71.5 mm, weight 6.15 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 2,396.30	Tk. 2,386.00	Tk. 2,356.00	Tk. 2,356.00
26.42.8	75 -80 mm dia G.I. pipe with wall thickness 3.25 mm, outside diameter min 86.5 mm, weight 7.22 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 2,607.46	Tk. 2,597.00	Tk. 2,565.00	Tk. 2,565.00
26.42.9	100 mm dia G.I. pipe with wall thickness 3.65 mm, outside diameter min 107.3 mm, weight 10.42 kg/m, can withstand min 50 kg/cm ² hydraulic pressure.	meter	Tk. 3,721.07	Tk. 3,709.00	Tk. 3,671.00	Tk. 3,671.00
26.43	Supplying different inside dia best quality uPVC soil, waste and ventilation pipe having specific gravity 1.35 - 1.45, wall thickness 2.5 mm - 3.0 mm, and other physical, chemical, thermal, fire resistivity properties etc. as per BSTI approved manufacturer standards or ASTM, BS/ISO/IS standards fitting and fixing in position with sockets, bends, of uPVC Pipe with all accessories such as Round grating /domed roof grating bands, sockets etc. approved and accepted by the Engineer- in- charge.					
26.43.1	50 mm inside dia wall thickness 2.5 mm - 3.0 mm	meter	Tk. 564.43	Tk. 559.00	Tk. 541.00	Tk. 541.00
26.43.2	100 mm inside dia wall thickness 3.4- 4.0 mm	meter	Tk. 881.24	Tk. 877.00	Tk. 864.00	Tk. 864.00
26.43.3	150 mm inside dia wall thickness 4.5 mm - 5.2 mm	meter	Tk. 1,598.68	Tk. 1,593.00	Tk. 1,574.00	Tk. 1,574.00

26.44	Supplying and laying uPVC pipe for for underground including fitting, fixing etc. all complete approved and accepted by the Engineer- in- charge.					
26.44.1	150 mm dia with wall thickness 4.5 mm - 6.6 mm uPVC pipe	meter	Tk. 1,415.53	Tk. 1,413.00	Tk. 1,394.00	Tk. 1,394.00
26.44.2	200 mm dia with wall thickness 5.3 mm - 7.8 mm uPVC pipe	meter	Tk. 2,104.41	Tk. 2,102.00	Tk. 2,078.00	Tk. 2,078.00
26.44.3	250 mm dia with wall thickness 6.6 mm - 7.6 mm uPVC pipe	meter	Tk. 3,077.26	Tk. 3,074.00	Tk. 3,053.00	Tk. 3,053.00
26.44.4	280 mm dia with wall thickness 6.9 mm - 8.6 mm uPVC pipe	meter	Tk. 3,442.61	Tk. 3,439.00	Tk. 3,413.00	Tk. 3,413.00
26.44.5	315 mm dia with wall thickness 7.8 mm - 9.0 mm uPVC pipe	meter	Tk. 4,332.33	Tk. 4,328.00	Tk. 4,303.00	Tk. 4,303.00
26.45	Supplying different inside dia best quality uPVC pressure pipe for water supply having specific gravity 1.35 - 1.45, and other physical, chemical, thermal, fire resistivity properties etc. as per BSI approved manufacturer standards or ASTM, BS/ISO/IS standards fitted and fixed in position with sockets, bends, with all accessories such as Round grating/domed roof grating, bends, sockets etc. approved and accepted by the Engineer-in-charge.					
26.45.1	12 mm dia wall thickness 1.7 mm - 2.8 mm	meter	Tk. 167.53	Tk. 166.00	Tk. 159.00	Tk. 159.00
26.45.2	19 mm dia wall thickness 1.9 mm - 2.9 mm	meter	Tk. 215.99	Tk. 214.00	Tk. 207.00	Tk. 207.00
26.45.3	25 mm dia wall thickness 2.2 mm - 3.4 mm	meter	Tk. 278.84	Tk. 276.00	Tk. 266.00	Tk. 266.00
26.45.4	32 mm dia wall thickness 2.7 mm - 3.6 mm	meter	Tk. 284.78	Tk. 283.00	Tk. 277.00	Tk. 277.00
26.45.5	37 mm dia wall thickness 2.5 mm - 3.1 mm	meter	Tk. 298.82	Tk. 297.00	Tk. 291.00	Tk. 291.00
26.45.6	60 mm dia wall thickness 3.0 mm - 3.9 mm	meter	Tk. 570.72	Tk. 568.00	Tk. 559.00	Tk. 559.00
26.45.7	75 mm dia wall thickness 2.9 mm - 3.5 mm	meter	Tk. 638.00	Tk. 636.00	Tk. 626.00	Tk. 626.00
26.45.8	88 mm dia wall thickness 3.5 mm - 4.1 mm	meter	Tk. 734.32	Tk. 732.00	Tk. 721.00	Tk. 721.00
26.45.9	114 mm dia wall thickness 4.5 mm - 5.2 mm	meter	Tk. 899.18	Tk. 897.00	Tk. 886.00	Tk. 886.00
26.46	Supplying different inside dia best quality CPVC pressure pipe for water supply having specific gravity 1.35 - 1.45, and other physical, chemical, thermal, fire resistivity properties etc. as per BSI approved manufacturer standards or ASTM, BS/ISO/IS standards fitting and fixing in position with sockets, bends, with all accessories such as round grating/domed roof grating, bends, sockets etc. approved and accepted by the Engineer-in-charge.					
26.46.1	12 mm dia wall thickness 2.8 mm - 3.3 mm	meter	Tk. 265.71	Tk. 264.00	Tk. 257.00	Tk. 257.00
26.46.2	19 mm dia wall thickness 2.9 mm - 3.4 mm	meter	Tk. 386.74	Tk. 385.00	Tk. 378.00	Tk. 378.00
26.46.3	25 mm dia wall thickness 3.4 mm - 4.0 mm	meter	Tk. 453.57	Tk. 451.00	Tk. 441.00	Tk. 441.00
26.46.4	32 mm dia wall thickness 3.6 mm - 4.2 mm	meter	Tk. 624.97	Tk. 623.00	Tk. 617.00	Tk. 617.00
26.46.5	37 mm dia wall thickness 3.7 mm - 4.3 mm	meter	Tk. 867.88	Tk. 866.00	Tk. 860.00	Tk. 860.00
26.46.6	50 mm dia wall thickness 3.9 mm - 4.5 mm	meter	Tk. 1,484.59	Tk. 1,482.00	Tk. 1,473.00	Tk. 1,473.00
26.47	Supplying, fitting and fixing of PP-R Pipe for water supply sustaining minimum water pressure PN 20, fulfilling corresponding standard dimension ratio SDR meeting the performance & long-term strength requirements as per DIN 8077 /78 or ASTM F 2983 including nipple, elbow, reducers, Tee, end cap, plug, socket etc as per manufacturer instruction. all complete accepted by the Engineer- in- charge.					
26.47.1	12 mm dia, minimum wall thickness 2.8 mm	meter	Tk. 322.80	Tk. 320.00	Tk. 307.00	Tk. 307.00
26.47.2	19 mm dia, minimum wall thickness 3.5 mm	meter	Tk. 363.12	Tk. 357.00	Tk. 343.00	Tk. 343.00
26.47.3	25 mm dia, minimum wall thickness 4.4 mm	meter	Tk. 424.13	Tk. 418.00	Tk. 405.00	Tk. 405.00
26.47.4	32 mm dia, minimum wall thickness 5.5 mm	meter	Tk. 538.16	Tk. 524.00	Tk. 510.00	Tk. 510.00

26.47.5	38 mm dia, minimum wall thickness 6.9 mm	meter	Tk. 660.24	Tk. 623.00	Tk. 609.00	Tk. 609.00
26.47.6	50 mm dia, minimum wall thickness 8.6 mm	meter	Tk. 888.91	Tk. 863.00	Tk. 850.00	Tk. 850.00
26.48	Supplying, fitting and laying of Pipes up to depth of invert dia R.C.C pipe having 1.83 meter in length made of stone chips with 40 mm cement joint including all fittings and specials like plain bend, Tees, reducing sockets, junctions, door bends cowls, anti-siphon, gasket and cement joints, making holes in walls and mending good the damages etc. all complete approved and accepted by the Engineer- in-charge.					
26.48.1	R.C.C pipe of 150 mm dia with thickness 40mm	meter	Tk. 1,092.73	Tk. 1,088.00	Tk. 1,072.00	Tk. 1,072.00
26.48.2	R.C.C pipe of 225 mm dia with thickness 40mm	meter	Tk. 1,731.70	Tk. 1,725.00	Tk. 1,696.00	Tk. 1,696.00
26.48.3	R.C.C pipe of 300 mm dia with thickness 50mm	meter	Tk. 2,844.69	Tk. 2,835.00	Tk. 2,795.00	Tk. 2,795.00
26.48.4	R.C.C pipe of 375 mm dia with thickness 50mm	meter	Tk. 3,608.46	Tk. 3,596.00	Tk. 3,549.00	Tk. 3,549.00
	Pipe fitting Accessories (union, gate valve, globe valve, check valve, float valve, clean out)					
26.49	Supplying, fitting and fixing of PP-R Union meeting the performance & long-tem strength requirements of ASTM F 2389 in accordance with the manufacturer's specifications all complete accepted by the Engineer- in- charge.					
26.49.1	12 mm PP-R Union	each	Tk. 317.80	Tk. 258.00	Tk. 251.00	Tk. 251.00
26.49.2	19 mm PP-R Union	each	Tk. 363.20	Tk. 281.00	Tk. 273.00	Tk. 273.00
26.49.3	25 mm PP-R Union	each	Tk. 407.99	Tk. 358.00	Tk. 348.00	Tk. 348.00
26.49.4	32 mm PP-R Union	each	Tk. 642.96	Tk. 515.00	Tk. 504.00	Tk. 504.00
26.49.5	37 mm PP-R Union	each	Tk. 1,040.52	Tk. 735.00	Tk. 717.00	Tk. 717.00
26.49.6	50 mm PP-R Union	each	Tk. 1,189.60	Tk. 836.00	Tk. 820.00	Tk. 820.00
26.50	Supplying, fitting and fixing of PP-R Gate-Valve meeting the performance & long-tem strength requirements of ASTM F 2389 in accordance with the manufacturer's specifications all complete accepted by the Engineer- in- charge.					
26.50.1	12 mm PP-R Gate-Valve	each	Tk. 418.42	Tk. 494.00	Tk. 486.00	Tk. 486.00
26.50.2	19 mm PP-R Gate-Valve	each	Tk. 547.25	Tk. 640.00	Tk. 632.00	Tk. 632.00
26.50.3	25 mm PP-R Gate-Valve	each	Tk. 596.34	Tk. 731.00	Tk. 723.00	Tk. 723.00
26.50.4	32 mm PP-R Gate-Valve	each	Tk. 964.44	Tk. 832.00	Tk. 824.00	Tk. 824.00
26.50.5	37 mm PP-R Gate-Valve	each	Tk. 1,025.79	Tk. 936.00	Tk. 928.00	Tk. 928.00
26.50.6	50 mm PP-R Gate-Valve	each	Tk. 1,271.20	Tk. 1,374.00	Tk. 1,366.00	Tk. 1,366.00
26.51	Supplying, fitting and fixing of G.I. Union with sealant etc. complete in all respects approved and accepted by the Engineer- in- charge.					
26.51.1	12 mm dia G.I. Union	each	Tk. 176.70	Tk. 174.00	Tk. 168.00	Tk. 168.00
26.51.2	20 mm dia G.I. Union	each	Tk. 227.74	Tk. 225.00	Tk. 217.00	Tk. 217.00
26.51.3	25 mm dia G.I. Union	each	Tk. 274.06	Tk. 271.00	Tk. 262.00	Tk. 262.00
26.51.4	32 mm dia G.I. Union	each	Tk. 352.28	Tk. 349.00	Tk. 337.00	Tk. 337.00
26.51.5	40 mm dia G.I. Union	each	Tk. 490.64	Tk. 485.00	Tk. 468.00	Tk. 468.00
26.51.6	50 mm dia G.I. Union	each	Tk. 525.60	Tk. 521.00	Tk. 505.00	Tk. 505.00
26.51.7	62 mm dia G.I. Union	each	Tk. 865.10	Tk. 860.00	Tk. 841.00	Tk. 841.00

26.51.8	75 mm dia G.I. Union	each	Tk. 1,157.34	Tk. 1,151.00	Tk. 1,128.00	Tk. 1,128.00
26.51.9	100 mm dia G.I. Union	each	Tk. 1,569.96	Tk. 1,563.00	Tk. 1,537.00	Tk. 1,537.00
26.52	Supplying, fitting and fixing of best quality G.I. gate valve with sealant etc. complete approved and accepted by the Engineer- in- charge.					
26.52.1	12 mm brass gate valve	each	Tk. 458.92	Tk. 456.00	Tk. 450.00	Tk. 450.00
26.52.2	20 mm brass gate valve	each	Tk. 632.66	Tk. 630.00	Tk. 622.00	Tk. 622.00
26.52.3	25 mm brass gate valve	each	Tk. 746.46	Tk. 743.00	Tk. 734.00	Tk. 734.00
26.52.4	32 mm brass gate valve	each	Tk. 1,045.97	Tk. 1,042.00	Tk. 1,031.00	Tk. 1,031.00
26.52.5	40 mm brass gate valve	each	Tk. 1,220.72	Tk. 1,216.00	Tk. 1,198.00	Tk. 1,198.00
26.52.6	50 mm brass gate valve	each	Tk. 1,875.33	Tk. 1,871.00	Tk. 1,854.00	Tk. 1,854.00
26.52.7	62 mm brass gate valve	each	Tk. 3,785.43	Tk. 3,780.00	Tk. 3,761.00	Tk. 3,761.00
26.52.8	75 mm brass gate valve	each	Tk. 5,139.05	Tk. 5,132.00	Tk. 5,110.00	Tk. 5,110.00
26.52.9	100 mm brass gate valve	each	Tk. 8,085.47	Tk. 8,078.00	Tk. 8,053.00	Tk. 8,053.00
26.53	Supplying, fitting and fixing of best quality G.I. globe valve , threaded joint with sealant etc. complete approved and accepted by the Engineer- in- charge.					
26.53.1	12 mm dia brass globe valve	each	Tk. 508.00	Tk. 506.00	Tk. 499.00	Tk. 499.00
26.53.2	20 mm dia brass globe valve	each	Tk. 589.71	Tk. 587.00	Tk. 579.00	Tk. 579.00
26.53.3	25 mm dia brass globe valve	each	Tk. 850.76	Tk. 848.00	Tk. 839.00	Tk. 839.00
26.53.4	32 mm dia brass globe valve	each	Tk. 1,168.67	Tk. 1,165.00	Tk. 1,153.00	Tk. 1,153.00
26.53.5	40 mm dia brass globe valve	each	Tk. 1,398.64	Tk. 1,393.00	Tk. 1,376.00	Tk. 1,376.00
26.53.6	50 mm dia brass globe valve	each	Tk. 2,133.00	Tk. 2,128.00	Tk. 2,112.00	Tk. 2,112.00
26.53.7	62 mm dia brass globe valve	each	Tk. 4,460.29	Tk. 4,455.00	Tk. 4,436.00	Tk. 4,436.00
26.53.8	75 mm dia brass globe valve	each	Tk. 5,666.67	Tk. 5,660.00	Tk. 5,638.00	Tk. 5,638.00
26.53.9	100 mm dia brass globe valve	each	Tk. 8,944.39	Tk. 8,937.00	Tk. 8,912.00	Tk. 8,912.00
26.54	Supplying, fitting and fixing of best quality G.I. check valve , threaded joint with sealant etc. complete approved and accepted by the Engineer- in- charge.					
26.54.1	12 mm brass check valve	each	Tk. 465.06	Tk. 463.00	Tk. 456.00	Tk. 456.00
26.54.2	20 mm brass check valve	each	Tk. 559.03	Tk. 556.00	Tk. 549.00	Tk. 549.00
26.54.3	25 mm brass check valve	each	Tk. 758.73	Tk. 756.00	Tk. 746.00	Tk. 746.00
26.54.4	32 mm brass check valve	each	Tk. 1,009.16	Tk. 1,005.00	Tk. 994.00	Tk. 994.00
26.54.5	40 mm brass check valve	each	Tk. 1,269.80	Tk. 1,265.00	Tk. 1,247.00	Tk. 1,247.00
26.54.6	50 mm brass check valve	each	Tk. 1,856.92	Tk. 1,852.00	Tk. 1,836.00	Tk. 1,836.00
26.54.7	62 mm brass check valve	each	Tk. 3,184.18	Tk. 3,179.00	Tk. 3,160.00	Tk. 3,160.00
26.54.8	75 mm brass check valve	each	Tk. 4,329.21	Tk. 4,322.00	Tk. 4,300.00	Tk. 4,300.00
26.54.9	100 mm brass check valve	each	Tk. 7,103.85	Tk. 7,096.00	Tk. 7,071.00	Tk. 7,071.00
26.55	Supplying, fitting and fixing of best quality G.I. float valve , threaded joint Pit valve etc. complete approved and accepted by the Engineer- in- charge.					
26.55.1	12 mm brass float valve	each	Tk. 422.82	Tk. 420.00	Tk. 414.00	Tk. 414.00
26.55.2	20 mm brass float valve	each	Tk. 560.25	Tk. 558.00	Tk. 550.00	Tk. 550.00
26.55.3	25 mm brass float valve	each	Tk. 625.89	Tk. 623.00	Tk. 614.00	Tk. 614.00

26.55.4	32 mm brass float valve	each	Tk. 1,024.06	Tk. 1,020.00	Tk. 1,009.00	Tk. 1,009.00
26.55.5	40 mm brass float valve	each	Tk. 1,316.10	Tk. 1,311.00	Tk. 1,293.00	Tk. 1,293.00
26.56	Supplying different inside dia best quality uPVC clean out / trap having specific gravity 1.35 -1.45 and other physical, chemical, thermal, fire resistivity properties etc. as per BSTI approved manufacturer standards or ASTM, BS/ISO/IS standards, fitting and fixing in position etc. complete approved and accepted by the Engineer- in- charge.					
26.56.1	50 mm dia uPVC Y or T - Y Cleanout	each	Tk. 353.69	Tk. 351.00	Tk. 343.00	Tk. 343.00
26.56.2	50 mm dia uPVC 'P' or 'S' trap	each	Tk. 693.33	Tk. 690.00	Tk. 677.00	Tk. 677.00
26.56.3	100 mm dia uPVC Y or T - Y Cleanout	each	Tk. 671.18	Tk. 667.00	Tk. 653.00	Tk. 653.00
26.56.4	100 mm dia uPVC 'P' or 'S' trap	each	Tk. 439.58	Tk. 437.00	Tk. 429.00	Tk. 429.00
	Pressure reducing valve					
26.57	Supply, fitting and fixing of Water pressure reducing valve to reduce static and flowing pressures in water distribution systems and it shall be a diaphragm/balanced piston type with integral stainless steel strainer and a built-in bypass to relieve pressure build-up downstream of the assembly. The valve shall be spring loaded to open and shall be diaphragm actuated to close under pressure conditions. The valve shall be suitable for supply pressure up to 400 psi and water temperatures from 33° to 180°F. The valve's internal parts such as the cartridge assembly, strainer, diaphragm, and spring shall be accessible for inspection, repair or replacement without disconnecting the device from the pipeline. In addition, the valve shall have a clean-out feature to allow for removal of the strainer and seat disc without disturbing the pressure setting etc. all complete accepted by the Engineer- in- charge.					
26.57.1	12 mm dia water pressure reducing valve (MS)	each	Tk. 2,535.00	Tk. 2,532.00	Tk. 2,524.00	Tk. 2,524.00
26.57.2	19 mm dia water pressure reducing valve (MS)	each	Tk. 3,357.00	Tk. 3,354.00	Tk. 3,346.00	Tk. 3,346.00
26.57.3	25 mm dia water pressure reducing valve (MS)	each	Tk. 4,191.00	Tk. 4,188.00	Tk. 4,181.00	Tk. 4,181.00
26.57.4	32 mm dia water pressure reducing valve (MS)	each	Tk. 6,277.00	Tk. 6,274.00	Tk. 6,267.00	Tk. 6,267.00
26.57.5	38 mm dia water pressure reducing valve (MS)	each	Tk. 7,749.00	Tk. 7,747.00	Tk. 7,739.00	Tk. 7,739.00
26.57.6	50 mm dia water pressure reducing valve (MS)	each	Tk. 11,430.00	Tk. 11,428.00	Tk. 11,420.00	Tk. 11,420.00
26.57.7	62 mm dia water pressure reducing valve (MS)	each	Tk. 19,774.00	Tk. 19,772.00	Tk. 19,764.00	Tk. 19,764.00
26.57.8	75 mm dia water pressure reducing valve (MS)	each	Tk. 28,363.00	Tk. 28,361.00	Tk. 28,353.00	Tk. 28,353.00
26.57.9	19 mm dia water pressure reducing valve (Brass double net)	each	Tk. 4,039.00	Tk. 4,036.00	Tk. 4,028.00	Tk. 4,028.00
26.57.10	25 mm dia water pressure reducing valve (Brass double net)	each	Tk. 5,195.00	Tk. 5,192.00	Tk. 5,184.00	Tk. 5,184.00
26.57.11	38 mm dia water pressure reducing valve (Brass double net)	each	Tk. 9,525.00	Tk. 9,522.00	Tk. 9,514.00	Tk. 9,514.00
26.57.12	50 mm dia water pressure reducing valve (Brass double net)	each	Tk. 13,134.00	Tk. 13,131.00	Tk. 13,123.00	Tk. 13,123.00
	WATER TANK, CONNECTION PIPE, WATER PURIFIER & DISPENSER, BOOSTER PUMP, HAND DRIER					
26.58	Supplying, fitting and fixing of food graded plastic overhead water reservoir /tank including all necessary fittings, hardware and consumables etc. all complete approved and accepted by the Engineer- in- charge.					
26.58.1	1000 liter capacity	each	Tk. 11,998.00	Tk. 11,961.00	Tk. 11,814.00	Tk. 11,814.00
26.58.2	1500 liter capacity	each	Tk. 16,477.00	Tk. 16,440.00	Tk. 16,292.00	Tk. 16,292.00
26.58.3	2000 liter capacity	each	Tk. 20,955.00	Tk. 20,918.00	Tk. 20,771.00	Tk. 20,771.00

26.58.4	3000 liter capacity	each	Tk. 29,912.00	Tk. 29,876.00	Tk. 29,728.00	Tk. 29,728.00
26.58.5	5000 liter capacity	each	Tk. 47,827.00	Tk. 47,790.00	Tk. 47,643.00	Tk. 47,643.00
26.59	Supply and installation of food-graded plastic internal mini water tank for the use in kitchen, bathroom and toilet for emergency storage and supply of water manufactured from liner low density polyethylene (ILDPE) roto-grade (ultra violet) stabilized which complies FDA (Federal Department of Agriculture, USA) regulations 21 CFR 1277. 152, having food grade quality where no recycled material is used carrying, lifting, fitting, fixing in position including supply of necessary hardware, consumables, fittings etc. all complete approved and accepted by the Engineer- in- charge.					
26.59.1	150 liter capacity	each	Tk. 4,133.00	Tk. 4,120.00	Tk. 4,071.00	Tk. 4,071.00
26.59.2	300 liter capacity	each	Tk. 5,666.00	Tk. 5,654.00	Tk. 5,605.00	Tk. 5,605.00
26.59.3	500 liter capacity	each	Tk. 5,850.00	Tk. 5,838.00	Tk. 5,789.00	Tk. 5,789.00
26.60	Supply and installation of 400 gallon capacity ferro-cement water tank (HBRI made or equivalent) including supply of hardware, consumables, fittings etc. all complete approved and accepted by the Engineer- in- Charge.	each	Tk. 9,448.00	Tk. 9,448.00	Tk. 9,448.00	Tk. 9,448.00
26.61	Supplying, fitting and fixing of 12 mm dia connection pipe approved and accepted by the Engineer- in-charge.					
26.61.1	450 mm long plastic flexible lead (connection) pipe	each	Tk. 112.00	Tk. 111.00	Tk. 108.00	Tk. 108.00
26.61.2	600 mm long plastic flexible lead (connection) pipe	each	Tk. 124.00	Tk. 124.00	Tk. 120.00	Tk. 120.00
26.61.3	450 mm long metal cover connection pipe	each	Tk. 188.00	Tk. 187.00	Tk. 184.00	Tk. 184.00
26.61.4	600 mm long metal cover connection pipe	each	Tk. 222.00	Tk. 222.00	Tk. 218.00	Tk. 218.00
26.62	Supplying, fitting, fixing of foreign made best quality Reverse Osmosis Water Purifier (Kent/Unilever/Vapor water purifier or equivalent brand) for purification of water by reverse osmosis system to have potable water conforming WHO standard and Bangladesh concentration parameters including installation charge, carriage, sundries etc. complete approved and accepted by the Engineer- in- charge.					
26.62.1	Water purification capacity: 10 L/hr, storage 6 litre; RO+MF	each	Tk. 18,577.00	Tk. 18,577.00	Tk. 18,577.00	Tk. 18,577.00
26.62.2	Water purification capacity: 10~12 L/hr, digital display, storage minimum 9 litre, 7 stage purification; Ro+UV+MF	each	Tk. 32,528.00	Tk. 32,528.00	Tk. 32,528.00	Tk. 32,528.00
26.62.3	Water purification capacity: 15 L/hr; storage 8 litre, RO+UF	each	Tk. 22,516.00	Tk. 22,516.00	Tk. 22,516.00	Tk. 22,516.00
26.62.4	Water purification capacity: 20 L/hr, minimum storage 8 litre, 6 stage purification; RO+UV+MF	each	Tk. 29,400.00	Tk. 29,400.00	Tk. 29,400.00	Tk. 29,400.00
26.62.5	Water purification capacity: 60 L/hr, at least 5 stage filtration RO purifier ;	each	Tk. 94,481.00	Tk. 94,481.00	Tk. 94,481.00	Tk. 94,481.00
26.63	Supply, installation and commissioning of 50 liter capacity best quality 'Water dispenser' with container of approved brand/origin.	each	Tk. 4,601.00	Tk. 4,601.00	Tk. 4,601.00	Tk. 4,601.00

26.64	Supply and Installation of Automatic Pressure and Flow Controller having Maximum flow rate-80 Liter/min (minimum), Starting Flow rate- 2-2.5 Liter/min, Maximum Switching current: 12 A. Maximum total pressure 10 Bar and operating pressure should be 1-3.5 Bar. Supported Motor voltage 230 V $\pm$ 5% 50 Hz AC supply. Having pressure gauge of minimum 40mm dia (0-12 Bar) and inlet and outlet size shall be 1 inch minimum and inline position threaded and protection: IP-65 (minimum). Manufactured in EU/G7 countries as per sample accepted / approved by the Engineer-in-charge. (For Centrifugal Pump motor upto 2 HP)					
26.64.1	Maximum flow rate-80 Liter/min	each	Tk. 11,442.00	Tk. 11,436.00	Tk. 11,381.00	Tk. 11,381.00
26.64.2	Maximum flow rate-200 Liter/min	each	Tk. 17,454.00	Tk. 17,448.00	Tk. 17,393.00	Tk. 17,393.00
26.65	Supplying, fitting and installation of automatic hand drier of high speed, energy efficient, rapid drying, automatic sensor, adjustable speed and sound control, adjustable heat control, electric hand dryer; surface mounted or semi-recessed which can eliminate over 99% of airborne bacteria and particles as small as 0.3 microns with the removable Super Filter System and it shall be powered by 220v AC supply/ batteries and shall operate automatically from IR sensor that shall activate it when hands are placed in sensor zone etc. all complete accepted by the Engineer- in- charge.					
26.65.1	Automatic Hand drier Local (Standard quality)	each	Tk. 5,804.00	Tk. 5,796.00	Tk. 5,785.00	Tk. 5,785.00
26.65.2	Automatic Hand drier Local (Super quality)	each	Tk. 7,031.00	Tk. 7,024.00	Tk. 7,012.00	Tk. 7,012.00
26.65.3	Automatic Hand drier Local (Superior quality)	each	Tk. 8,871.00	Tk. 8,864.00	Tk. 8,853.00	Tk. 8,853.00
26.65.4	Automatic Hand drier (Special quality)	each	Tk. 14,025.00	Tk. 14,018.00	Tk. 14,007.00	Tk. 14,007.00
MISCELLANEOUS ITEMS						
26.66	Manufacturing, supplying, fitting & fixing of adjustable hanger made of 38 mm x 38 mm x 6 mm m.s. angle with 600 mm long 12 mm dia nut-bolt for having pvc pipe from ceiling as per direction of the Engineer-in-charge.	meter	Tk. 868.92	Tk. 865.00	Tk. 835.00	Tk. 835.00
26.67	Groove cutting in brick work and R.C.C work, including cost for concealing of water distribution pipe work (12 mm and 20 mm dia) in brick wall by cutting groove in wall, lintel, beam etc. by any means carefully without damaging the structure and filling the grooves with C.C (4:2:1) after laying of pipe including cost of scaffolding, finishing, curing etc. all complete approved and accepted by the Engineer-in- charge.					
26.67.1	Groove size 40 mm x 40 mm (For laying 12 mm to 20 mm dia pipes)	meter	Tk. 322.00	Tk. 316.00	Tk. 296.00	Tk. 296.00
26.67.2	Groove size 75 mm x 75 mm (For 25 mm to 40 mm dia pipes)	meter	Tk. 523.00	Tk. 515.00	Tk. 483.00	Tk. 483.00
26.68	Punching or cutting hole of any diameter in RCC floor for sanitary works.	each hole	Tk. 202.00	Tk. 198.00	Tk. 185.00	Tk. 185.00
26.69	Supply, fitting & fixing of 600 mm dia C.I. man hole cover etc. all complete approved by the Engineer-in- charge.	each	Tk. 1,819.00	Tk. 1,817.00	Tk. 1,810.00	Tk. 1,810.00
INSPECTION PIT & SOAK WELL						
26.70	Construction of masonry inspection pit with 250 mm thick brick work in cement mortar (1:4) including necessary earth work, side filling and one layer brick flat soling, 75 mm thick (1:3:6) base concrete for making invert channel and 12 mm thick (1:2) cement plaster with neat finishing etc. all complete up to a depth of 700 mm approved and accepted by the Engineer- in- charge.					

26.70.1	Clear 450 mm x 450 mm and depth 450 mm to 600 mm, average 525 mm for single 150 mm dia R.C.C. pipes and 400 mm PVC pipe.	each	Tk. 4,021.59	Tk. 4,001.00	Tk. 3,867.00	Tk. 3,825.00
26.70.2	Clear 525 mm x 525 mm and depth 675 mm to 825 mm, average 750 mm for single 225 mm dia R.C.C. pipes and 400 mm PVC pipe.	each	Tk. 5,934.69	Tk. 5,904.00	Tk. 5,711.00	Tk. 5,647.00
26.70.3.	Clear 600 x 600 mm and depth 750 mm to 900 mm, average 825 mm for single 300 mm dia R.C.C. pipes and 400 mm PVC pipe.	each	Tk. 8,259.34	Tk. 8,219.00	Tk. 7,953.00	Tk. 7,861.00
26.70.4	Master-pit of Size: Clear 600 mm x 500 mm and average 750 mm depth for Septic tank.	each	Tk. 3,931.88	Tk. 3,915.00	Tk. 3,796.00	Tk. 3,758.00
26.71	Construction and placement M.H. cover with locking/unlocking arrangement including necessary earth work, side filling, shuttering, curing, cement plaster (1:4) with neat finishing on edges and top etc. all complete approved and accepted by the Engineer- in- charge.					
26.71.1	900 mm x 900 mm x 75 mm R.C.C. pit cover	each	Tk. 1,143.44	Tk. 1,140.00	Tk. 1,126.00	Tk. 1,120.00
26.71.2	1025 mm x 1025 mm x 75 mm R.C.C. pit cover	each	Tk. 1,319.14	Tk. 1,315.00	Tk. 1,299.00	Tk. 1,293.00
26.71.3	1100 mm x 1100 mm x 75 mm R.C.C. pit cover	each	Tk. 1,511.73	Tk. 1,507.00	Tk. 1,490.00	Tk. 1,483.00
26.72	Construction and placing of R.C.C inspection pit cover (slab) with supplying and provisions for placing, fitting, fixing 450 mm dia C.I. man-hole cover with locking/un-locking arrangement including concrete (1:2:4) with approx. 1% re-enforcement, necessary earth cutting, or cleaning, side filling, curing etc. with minimum 12 mm cement plaster (1:4) and neat cement finishing on edges and top all complete and accepted by the Engineer- in- charge.					
26.72.1	950 mm x 950 mm x 75 mm R.C.C. pit cover with 450 mm dia C.I. manhole cover.	each	Tk. 2,072.06	Tk. 2,069.00	Tk. 2,056.00	Tk. 2,052.00
26.72.2	1025 mm x 1025 mm x 75 mm R.C.C. pit cover with 450 mm dia C.I. manhole cover.	each	Tk. 2,247.16	Tk. 2,244.00	Tk. 2,229.00	Tk. 2,224.00
26.72.3	1100 mm x 1100 mm x 75 mm R.C.C. pit cover with 450 mm dia C.I. manhole cover.	each	Tk. 2,440.12	Tk. 2,436.00	Tk. 2,420.00	Tk. 2,414.00
26.73	Construction of masonry inspection pits up to a depth of 700 mm with 250 mm thick brick work in cement mortar (1:4) including 100 mm thick R.C.C. top slab (1:2:4) with 1% reinforcement, man-hole cover with locking arrangement including necessary earth work, side filling and one brick flat soling. 75 mm thick (1:3:6) base concrete for making invert channel and 12 mm thick (1:2) cement plaster with neat finishing etc. all complete approved and accepted by the Engineer- in- charge.					
26.73.1	Clear 450 mm x 450 mm and depth 450 mm to 600 mm, average 525 mm for single 150 mm dia R.C.C. pipes and 400 mm PVC pipe with pit cover and 450 mm dia C.I. man-hole cover.	each	Tk. 6,093.66	Tk. 6,070.00	Tk. 5,923.00	Tk. 5,877.00
26.73.2	Clear 525 mm x 525 mm and depth 675 mm to 825 mm, average 750 mm for single 225 mm dia R.C.C. pipes and 400 mm PVC pipe with pit cover and 450 mm dia C.I. man-hole cover.	each	Tk. 8,181.86	Tk. 8,148.00	Tk. 7,940.00	Tk. 7,871.00
26.73.3	Clear 600 mm x 600 mm and depth 750 mm to 900 mm, average 825 mm for single 300 mm dia R.C.C. pipes and 375 mm and 400 mm PVC pipe with pit cover and 450 mm dia C.I. man-hole cover.	each	Tk. 10,699.46	Tk. 10,655.00	Tk. 10,373.00	Tk. 10,275.00

26.74	Construction of septic tank of different sizes with walls of brick work in cement mortar (1:6) having a lining of minimum 125 mm R.C.C cast against the walls as per approved type plan over a brick flat soling and 150 mm thick reinforced cement concrete flooring (1:2:4) with 125 mm thick walls in partition and 12 mm thick cement plaster (1:4) with N.C.F. to insides of walls on floor and all around outside walls by 450 mm height at top including supplying, fitting and fixing of two R.C.C. Tees and providing 450 mm dia water sealed heavy type C.I. manhole cover with locking/unlocking arrangement and 100 mm thick R.C.C (1:2:4) top slab, including centering, shuttering, fabricating, casting and curing etc. complete up to required depth including necessary earth work in excavation and shoring, bailing out water and side filling including the cost of all materials, operations and incidental charges. etc. all complete as per type plan approved and accepted by the Engineer-in-charge (Rate is including cost of reinforcement of the weight of the glaze materials. Applia					
26.74.1	For 200 users	each	Tk. 281,691.99	Tk. 280,592.00	Tk. 275,259.00	Tk. 273,206.00
26.74.2	For 100 users	each	Tk. 221,765.70	Tk. 220,916.00	Tk. 216,561.00	Tk. 214,852.00
26.74.3	For 50 users	each	Tk. 139,537.06	Tk. 139,019.00	Tk. 136,453.00	Tk. 135,441.00
26.74.4	For 30 users	each	Tk. 102,247.13	Tk. 101,875.00	Tk. 99,944.00	Tk. 99,179.00
26.74.5	For 20 users	each	Tk. 95,696.10	Tk. 95,343.00	Tk. 93,533.00	Tk. 92,813.00
26.74.6	For 10 users	each	Tk. 56,654.06	Tk. 56,461.00	Tk. 55,502.00	Tk. 55,116.00
26.75	Construction of non electric Eco STP of different sizes (as per detail drawing attached in annexure) with 250 mm walls of brick work in cement mortar (1:4) having a brick flat soling and 125 mm thick reinforced cement concrete flooring (1:2:4) with 125 mm thick walls (1:4) in partition and flush pointing (1:2) on inside wall surface and 12 mm thick cement plaster (1:4) with N.C.F. on floor and slab surface including supplying, fitting and fixing of upvc pipes & Tees and providing 450 mm dia water sealed heavy type C.I. manhole cover with locking/unlocking arrangement and 100 mm thick R.C.C (1:2:4) top slab, including centering, shuttering, fabricating, casting and curing etc. complete up to required depth including necessary earth work in excavation and shoring, bailing out water and side filling, dressing, inside cleaning, cow dung or old septic tank liquied for charging/start up including the cost of all materials, operations and incidental charges. etc. all complete as per type plan approved and accepted lead content, if any, shall not exceed 5 percent of the weight of the glaze materials. Appliances shall be cl					
26.75.1	500 users	each	Tk. 608,660.00	Tk. 608,660.00	Tk. 608,660.00	Tk. 608,660.00
26.75.2	300 users	each	Tk. 466,520.00	Tk. 466,520.00	Tk. 466,520.00	Tk. 466,520.00
26.76	Construction of soak well of different sizes (as per detail drawing attached in annexure) (medium and large sizes) with 250 mm thick solid brick work and 250 mm honey comb brick work with cement mortar (1:6) as per design over R.C.C. (1:2:4) well curb with 1% reinforcement up to the depth as per drawing with 450 mm dia water sealed heavy type. C.I. manhole cover with locking arrangement, filling the well up to the required depth with graded khoa including supplying and fabricating M.S. rod, casting, curing including necessary earth work in excavation, side filling and bailing out water including cost of all materials etc. all complete as per drawing, design approved and accepted by the (Rate is including cost of reinforcement and its fabrication, binding and placing) Engineer- in- charge.					

26.76.1	200 users	each	Tk. 130,097.00	Tk. 130,097.00	Tk. 130,097.00	Tk. 130,097.00
26.76.2	100 users	each	Tk. 116,983.00	Tk. 116,983.00	Tk. 116,983.00	Tk. 116,983.00
26.76.3	50 users	each	Tk. 104,808.00	Tk. 104,808.00	Tk. 104,808.00	Tk. 104,808.00
26.77	Purification of supplied raw water using Mobile Water Desalination Plant which includes mobilization, demobilization, necessary maintenance of all associated electromechanical machines, tank, membrane, micron filter, sand tube, gravel tube, manganese tube, activated carbon tube, resin tube, anti-scalant, salt tube and any other related instruments as and when required. This rate also includes cost of energy, laboratory testing to conform following specification and as per the direction of Engineer in charge.					
	Water in outlet must conform the following indicators and levels:					
	A. 1) pH- 6.5 to 8.5, 2) Iron as Fe $\leq$ 0.3 mg/l, 3) Arsenic $\leq$ 0.05mg/l, 4) Total Hardness as CaCO ₃ - 5 - 25 mg/l, 5) Chloride as CL ⁻ 500 mg/l, 6) Sulfate as SO ₄ ²⁻ $\leq$ 2000mg/l, 7) Alkali equivalent sodium oxide $\leq$ 1500 mg/l, 8) Sugars $\leq$ 100mg/l, 9)Phosphates as P ₂ O ₅ $\leq$ 100mg/l, 10)Nitrates as NO ₃ ⁻ $\leq$ 500mg/l, 11) Lead as Pb ⁺ $\leq$ 100mg/l, 12) Zinc as Zn ²⁺ $\leq$ 100 mg/l, 13) Salinity $\leq$ 1000ppm/l,					
	B. Setting time and strength test: The initial setting time obtained on paste samples made with the water shall not be less than 1 hr and not differ by more than 25% from the initial setting time obtained on specimens made with distilled or de-ionized water. The final setting time obtained on paste samples made with the water shall not be less than 10 hr and not differ by more than 25% from the final setting time obtained on specimens made with distilled or de-ionized water. The mean compressive strength at 7 days and 28 days of the concrete or mortar specimens prepared with water shall be at least 90% of the mean compressive strength of corresponding specimens prepared with distilled or de-ionized water.					
	C. To determine the value of aforementioned indicators, test shall be carried out following ISO 29581-1, ISO 7890-1, BDS ISO 12439:2011 (ISO 12439:2010).					
	D. Supplier shall be responsible for all kind of maintenance and providing spare parts untill the completion of service.					
	E. The authority or engineer in charge reserves the right to select the materials/goods/ machineries to be used in the plant					
26.77.1	Consumption Quantity: 50,000 to 100,000 liter	per liter	Tk. 2.45	Tk. 2.45	Tk. 2.45	Tk. 2.45
26.77.2	Consumption Quantity: 100,000 to 500,000 liter	per liter	Tk. 2.15	Tk. 2.15	Tk. 2.15	Tk. 2.15
26.77.3	Consumption Quantity: 500,000 to 10,000,000 liter	per liter	Tk. 1.84	Tk. 1.84	Tk. 1.84	Tk. 1.84
26.77.4	Consumption Quantity: 10,00,000 to 50,000,000 liter	per liter	Tk. 1.53	Tk. 1.54	Tk. 1.54	Tk. 1.54
26.77.5	Consumption Quantity: 50,000,000 liter and above	per liter	Tk. 1.23	Tk. 1.23	Tk. 1.23	Tk. 1.23

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CHAPTER 27: DEEP TUBE-WELL

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
	Note: Tender for test & observation well must be called before hand. Tender for main production well will then be called on the basis of required data obtained from the test well. Requirements of water must be the guiding factor for dimensioning the main well.					
	SINKING OF 38 mm DIA TEST AND OBSERVATION WELL					
27.1	Mobilization of test boring or observation well: Transportation of materials, equipments, tools and plants and boring rig to work site. Construction of derrick and dismantling the same. Cleaning the site after completion of the work accepted by the Engineer-in-charge.	per set	Tk. 7,852.97	Tk. 7,853.00	Tk. 7,853.00	Tk. 7,853.00
27.2	Boring by using 100 mm dia cutter and 40 mm dia heavy type G.I pipe (wall thickness 2.9 mm, outside diameter min 47.8 mm, weight 3.517 kg/m, capable to withstand 50 kg/cm ² pressure) and other equipments capable of drilling up to a depth of 500 meter by water jet system through all sorts of soil strata providing protection of caving by supplying necessary casing pipe. Collection of soil samples at every 3 meter interval and at every change of soil strata and preserving them in a controlled environment for analysis and laboratory test. Finally, withdrawal of boring and casing pipes etc. complete and accepted by the Engineer-in-charge.					
27.2.1	From 0.0 m to 50 m = 50 m	meter	Tk. 196.32	Tk. 196.00	Tk. 196.00	Tk. 196.00
27.2.2	Above 50 m to 100 m = 50 m	meter	Tk. 225.77	Tk. 226.00	Tk. 226.00	Tk. 226.00
27.2.3	Above 100 m to 152 m = 52 m	meter	Tk. 259.64	Tk. 260.00	Tk. 260.00	Tk. 260.00
27.2.4	Above 152 m to 200 m = 48 m	meter	Tk. 298.58	Tk. 299.00	Tk. 299.00	Tk. 299.00
27.2.5	Above 200 m to 250 m = 50 m	meter	Tk. 343.37	Tk. 343.00	Tk. 343.00	Tk. 343.00
27.2.6	Beyond 250 m	meter	Tk. 394.88	Tk. 395.00	Tk. 395.00	Tk. 395.00
27.3	Installation of tube-well fixture for observation well: Supplying and lowering of 40/38 mm inside dia best quality uPVC pressure tube well pipe (specific gravity 1.35 - 1.45, wall thickness 2.5 mm - 3.0 mm capable to withstand a minimum hydrostatic pressure of 170 psi for 1 hr). Fitting and fixing in position with all accessories, uPVC sand trap of length 3.00m with uPVC strainer of desired slot installing in or at mid depth of the most suitable water bearing strata, 38 mm dia, best quality G.I. pipe of length 1.52 m with M.S. welded flat bar welded on each side to prevent from rotation up to desired depth, fitting, fixing the best quality no. 6 C.I. head hand-pump etc including the cost of solvent cement, socket adapter, filling medium sand up to 18.00 m above strainer and the remaining portion with available soil from boring etc. complete as per standard specification and accepted by the Engineer-in-charge.					
27.3.1	Fitting & fixing of no. 6 Hand-pump set including all necessary accessories etc	per set	Tk. 3,460.22	Tk. 3,455.00	Tk. 3,436.00	Tk. 3,436.00

27.3.2	Supplying and fitting 38/40 mm dia heavy type G.I pipe conforming to standard BS-1387 & BDS-1031, having chemical composition (in%) carbon 0.17-0.25, manganese 0.95-1.20, sulphur 0.060 max, phosphorous 0.060 max, tensile properties: minimum yield strength 188 MPa, minimum tensile strength 313 MPa and marking for a) manufacturer b) material c) wall thickness d) nominal outside dia e) intended use etc. on the body of the pipe having wall thickness 2.9 mm, weight 3.517 kg/m capable to withstand 50 kg/cm ² hydraulic pressure	meter	Tk. 1,016.90	Tk. 1,013.00	Tk. 997.00	Tk. 997.00
27.3.3	38/40 mm dia water grade uPVC pipe having wall thickness 2.5 mm -3.00 mm	meter	Tk. 220.25	Tk. 219.00	Tk. 215.00	Tk. 215.00
27.3.4	38/40 mm dia water grade uPVC strainer having wall thickness 2.5 mm -3.00 mm	meter	Tk. 290.81	Tk. 289.00	Tk. 282.00	Tk. 282.00
27.3.5	38/40 mm dia uPVC socket adapter	each	Tk. 131.91	Tk. 131.00	Tk. 129.00	Tk. 129.00
27.3.6	Best quality 38 mm dia uPVC cap	each	Tk. 107.36	Tk. 107.00	Tk. 104.00	Tk. 104.00
27.3.7	Supplying, fitting & fixing tube-well log describing technical information of the tube-well	each	Tk. 5,006.27	Tk. 5,006.00	Tk. 5,006.00	Tk. 5,006.00
27.3.8	Filling the top 0.6 meter of the bore hole around the top pipe with sanitary seal of 1:2:4 (Cement: 1.2 FM sand: 20 mm downgraded brick chips) concrete	each	Tk. 60.08	Tk. 60.00	Tk. 59.00	Tk. 58.00
27.4	Complete development of the tube well to obtain sand and turbidity free water at a satisfactory yield confirmed by tests as per specification and accepted by the Engineer-in-charge. Water sample must be taken from 3 separate aquifers from three tube-wells					
27.4.1	Wells up to 200 meter depth					
27.4.1.1	For 1 No. test tube-well	each	Tk. 8,834.59	Tk. 8,835.00	Tk. 8,835.00	Tk. 8,835.00
27.4.1.2	For 2 Nos. test tube-wells	each	Tk. 13,251.89	Tk. 13,253.00	Tk. 13,253.00	Tk. 13,253.00
27.4.1.3	For 3 Nos. test tube-wells	each	Tk. 17,669.19	Tk. 17,670.00	Tk. 17,670.00	Tk. 17,670.00
27.4.2	Wells beyond 200 meter depth					
27.4.2.1	For 1 No. test tube-well	meter	Tk. 50.80	Tk. 51.00	Tk. 51.00	Tk. 51.00
27.4.2.2	For 2 Nos. test tube-wells	meter	Tk. 76.20	Tk. 76.00	Tk. 76.00	Tk. 76.00
27.4.2.3	For 3 Nos. test tube-wells	meter	Tk. 101.60	Tk. 102.00	Tk. 102.00	Tk. 102.00
27.5	"Furnishing design of a production well by testing of water and soil samples from test bores up to 350 m depth collecting samples of water from the best suitable aquifer, filling in sterilized air-tight glass bottles having no air space inside; collecting soil (sand, solids etc) at every 3 m interval and at each change of soil strata from the bores, arranging them in the cell of a box in an orderly manner. Carrying out test of water for (1) pH, (2) CO ₂ (3) Chloride, (4) Fluoride, (5) Iron, (6) Manganese, (7) Nitrate-Nitrogen, (8) Total alkalinity, (9) Total hardness, (10) Arsenic, (11) Turbidity, followed by test reports along with Bangladesh Standards of potable water and (12) sieve analysis on soils throughout a bore, leading to ascertain the best suitable sustained aquifers, depths and slot sizes of strainer, types and dia of tube well for specific yield, thickness of shrouding around strainer, fineness modulus of shrouding materials, mechanical units (pumps, compressor) for testing and operation of the tubent of the weight of the glaze materials. Appliances shall be clearly and indelibly marked at a prominent place, visible even after the appliances are installed with the following: a) manufacturer's name and/or registered trademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixture should be placed in position with heavy type C.I. Brackets. 30 mm dia PVC waste water pipe with brass coupling (not exceeding 750 mm in length), basin waste with chain plug including making holes in walls and floors and fitt					
	Total No. of test on water: 11 types of test x 1 test bore x one set sample (3 specimen) = 11 Nos. test (each specimen 1.5 liter of water)					

	Sieve analysis: min 9 Nos. Sample x one test x One bore = 9 Nos. test (Samples are of 1 kg dry wt. each)					
	There are about 55 (fifty-five) tests for potable water quality standard according to Bangladesh Environment Conservation Rules-12 (Schedule-3), 1997, under ECA 1995 of Ministry of Environment and Forest. Therefore, additional test/ tests on water or sieve analysis of soil if required, other than the above specified 12 (twelve) tests, may be carried out for public interest as per direction of the design engineer on actual payment including profit, VAT and all other charges.					
27.5.1	Furnishing design considering 1 No. test bore	LS	Tk. 37,669.73	Tk. 37,670.00	Tk. 37,670.00	Tk. 37,670.00
27.5.2	Furnishing design considering 2 Nos. test bores	LS	Tk. 73,267.14	Tk. 73,267.00	Tk. 73,267.00	Tk. 73,267.00
27.5.3	Furnishing design considering 3 Nos. test bores	LS	Tk. 109,900.71	Tk. 109,901.00	Tk. 109,901.00	Tk. 109,901.00
27.6	Disinfections of the well by using minimum 3 kg of bleaching powder of 33% strength or any other disinfectant as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 454.00	Tk. 454.00	Tk. 454.00	Tk. 454.00
	PRODUCTION WELL					
27.7	Mobilization: Preparation for installation of tube-well and transportation of complete boring equipments, labour and materials to the work site and remove the materials after completion of the works. Installation of drilling rig and accessories, making arrangements for electricity, water, labour shed, site office and stores, digging mud pit etc, leveling, dressing and clearing of the site as per original condition and a testing for discharge using minimum 1000 liter water tub etc. all complete as per specifications and accepted by the Engineer-in-charge.					
27.7.1	For Tube-wells up to 100 meter depth	per set	Tk. 47,976.75	Tk. 47,977.00	Tk. 47,977.00	Tk. 47,977.00
27.7.2	For Tube-wells up to 200 meter depth	per set	Tk. 51,403.66	Tk. 51,404.00	Tk. 51,404.00	Tk. 51,404.00
27.7.3	For Tube-wells beyond 200 meter depth	per set	Tk. 54,830.58	Tk. 54,831.00	Tk. 54,831.00	Tk. 54,831.00
27.8	Preparation and making gravel pack around the tube well fixtures with supply of designed graded pea gravels free from any substance harmful to pipe and health confirmed by test (Passing through #10 mesh and retained on #40 mesh) including sieving, washing etc. complete as per standard specification and accepted by the Engineer-in-charge. (The rate is inclusive of the cost of extra amount required for caving and extra depth below the tube well fixtures, shrouding should not be more than 15 m above the strainer and measurement should be allowed from bottom of the tube-well fixture to top of the gravel pack)	cum	Tk. 5,483.77	Tk. 5,468.00	Tk. 4,987.00	Tk. 4,987.00
27.9	Making a sanitary seal up to a minimum depth of 6 meter from G.L. or as specified including the supply of approved quality of cement, sand and khoa (12 mm down graded) in a ratio of (1:1:5:3) as per standard specification and accepted by the Engineer-in-charge.	cum	Tk. 9,526.95	Tk. 9,479.00	Tk. 9,366.00	Tk. 9,294.00
	INSTALLATION OF 50 mm x 125 mm DIA SHROUDED					
	PRODUCTION WELL WITH PVC PIPE					

27.10	Boring: Execution of 325 mm dia boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter). Boring for 325 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval or as required and at every change in soil stratum including sinking and with drawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from E.G.L. to bottom of the blind pipe and the rate is inclusive of the cost of extra depth or of further boring required due to caving and siltation)					
27.10.1	From 0.0 m to 50 m = 50 m	meter	Tk. 957.08	Tk. 957.00	Tk. 957.00	Tk. 957.00
27.10.2	From 50 m to 100 m = 50 m	meter	Tk. 1,100.64	Tk. 1,101.00	Tk. 1,101.00	Tk. 1,101.00
27.10.3	From 100 m to 152 m = 52 m	meter	Tk. 1,265.74	Tk. 1,266.00	Tk. 1,266.00	Tk. 1,266.00
27.10.4	From 152 m to 200 m = 48 m	meter	Tk. 1,455.60	Tk. 1,456.00	Tk. 1,456.00	Tk. 1,456.00
27.10.5	From 200 m to 250 m = 50 m	meter	Tk. 1,673.94	Tk. 1,674.00	Tk. 1,674.00	Tk. 1,674.00
27.10.6	Beyond 250 m	meter	Tk. 1,925.03	Tk. 1,925.00	Tk. 1,925.00	Tk. 1,925.00
27.11	Supplying the following tube well fixtures and pipes having density 1350 kg/m ³ , opacity 0.2% of visible light as per ISO 7686, minimum required strength 25 MPa conforming to BDS-1:2013, 4.4.1, 4.4.2 at working site as per standard practice and accepted by the Engineer-in-charge. Note : Pipe size (Wall Thickness) will be chosen as per table page no:)					
27.11.1	Best quality 50 mm dia uPVC cap	each	Tk. 176.69	Tk. 176.00	Tk. 172.00	Tk. 172.00
27.11.2.1	50 mm dia water grade (Class D) uPVC blind pipe/sand trap of wall thickness 3.1 mm to 3.7 mm	meter	Tk. 263.40	Tk. 263.00	Tk. 259.00	Tk. 259.00
27.11.2.2	50 mm dia water grade (Class E) uPVC blind pipe/sand trap of wall thickness 3.9 mm to 4.5 mm	meter	Tk. 330.89	Tk. 330.00	Tk. 327.00	Tk. 327.00
27.11.3.1	50 mm dia strainer of any slot as per design made with grade(Class D) uPVC pipe of wall thickness 3.1 mm to 3.7 mm	meter	Tk. 353.38	Tk. 351.00	Tk. 341.00	Tk. 341.00
27.11.3.2	50 mm dia strainer of any slot as per design made with grade(Class E) uPVC pipe of wall thickness 3.9 mm to 4.5 mm	meter	Tk. 420.87	Tk. 418.00	Tk. 409.00	Tk. 409.00
27.11.4	Best quality 50 mm x 125 mm dia uPVC Reducer	meter	Tk. 1,095.74	Tk. 1,095.00	Tk. 1,091.00	Tk. 1,091.00
27.11.5.1	125mm dia water grade (Class D) uPVC housing pipe of wall thickness 7.3 to 8.4mm	meter	Tk. 1,283.47	Tk. 1,282.00	Tk. 1,279.00	Tk. 1,279.00
27.11.5.2	125mm dia water grade (Class E) uPVC housing pipe of wall thickness 9.0 to 10.4mm	meter	Tk. 1,516.61	Tk. 1,516.00	Tk. 1,512.00	Tk. 1,512.00
27.11.6	Supplying centralizer made with 12 mm dia M.S. rod of approved design	each	Tk. 499.58	Tk. 497.00	Tk. 487.00	Tk. 487.00
27.12	Complete installation of 50 mm dia uPVC tube well fixtures such as expanded top or furnishing pipe, reducer, tube well pipes, strainer, blind pipe, pipe cap etc. fitting and fixing with best quality PVC solution, iron screws and PVC tape, sockets and placing in position with proper size M.S. centralizer @ 3.0 meter c/c in strainer portion including blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge.	meter	Tk. 105.05	Tk. 103.00	Tk. 99.00	Tk. 99.00

27.13	Complete development of the tube well by air compressor and pump of suitable capacity by alternate surging and pumping or any specified method until satisfactory yield is achieved in sand and turbid free water confirmed by laboratory test using necessary quantity of cologne and conducting step drawn-down test before and after the development as per specification and accepted by the Engineer-in-charge. (Development should be started within 24 hours of completion of shrouding)					
27.13.1	Wells up to 200 m depth	each	Tk. 31,289.19	Tk. 31,289.00	Tk. 31,289.00	Tk. 31,289.00
27.13.2	Wells beyond 200 m depth	each	Tk. 39,632.97	Tk. 39,633.00	Tk. 39,633.00	Tk. 39,633.00
27.14	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer-in-charge.	per hour	Tk. 873.32	Tk. 871.00	Tk. 859.00	Tk. 859.00
27.15	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) to obtain chlorinated water having 150 ppm free chlorine in the well complete as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 1,644.22	Tk. 1,644.00	Tk. 1,644.00	Tk. 1,644.00
27.16	Filling of the borehole around the tube well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or by local clay free from any substances harmful to pipes and health as per standard specification and accepted by the Engineer-in-charge.	meter	Tk. 44.94	Tk. 44.00	Tk. 41.00	Tk. 41.00
27.17	Protective top cover: Supplying and fitting fixing of best quality 150 mm dia 4 mm thick MS. Pipe 1.0 m (3'-4") long one end blocked with 200 mm dia 4 mm thick M.S. plate and other end minimum 300 mm (1'-0") embedded in sanitary seal for protection of uPVC Housing Pipe as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 5,325.30	Tk. 5,313.00	Tk. 5,264.00	Tk. 5,264.00
	INSTALLATION OF 75 mm x 150 mm DIA SHROUDED					
	PRODUCTION WELL WITH PVC PIPE					
27.18	Boring: Execution of 350 mm dia boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter). Boring for 350 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval or as required and at every change in soil stratum including sinking and withdrawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from E.G.L. to bottom of the blind pipe and the rate is inclusive of the cost of extra depth or of further boring required due to caving and siltation)					
27.18.1	From 0.0 m to 50 m = 50 m	meter	Tk. 1,042.97	Tk. 1,043.00	Tk. 1,043.00	Tk. 1,043.00
27.18.2	From 50 m to 100 m = 50 m	meter	Tk. 1,199.42	Tk. 1,199.00	Tk. 1,199.00	Tk. 1,199.00
27.18.3	From 100 m to 152 m = 52 m	meter	Tk. 1,379.33	Tk. 1,379.00	Tk. 1,379.00	Tk. 1,379.00
27.18.4	From 152 m to 200 m = 48 m	meter	Tk. 1,586.23	Tk. 1,586.00	Tk. 1,586.00	Tk. 1,586.00
27.18.5	From 200 m to 250 m = 50 m	meter	Tk. 1,824.17	Tk. 1,824.00	Tk. 1,824.00	Tk. 1,824.00
27.18.6	Beyond 250 m	meter	Tk. 2,097.79	Tk. 2,098.00	Tk. 2,098.00	Tk. 2,098.00

27.19	Supplying the following tube well fixtures and pipes having density 1350 kg/m ³ , opacity 0.2% of visible light as per ISO 7686, minimum required strength 25 MPa conforming to BDS-1:2013, 4.4.1, 4.4.2 at working site as per standard practice and accepted by the Engineer-in-charge.					
27.19.1	Best quality 75 mm dia uPVC cap	each	Tk. 238.04	Tk. 237.00	Tk. 233.00	Tk. 233.00
27.19.2.1	75 mm dia water grade(Class D) uPVC blind pipe/sand trap of wall thickness 4.6 mm to 5.3 mm	meter	Tk. 522.71	Tk. 522.00	Tk. 518.00	Tk. 518.00
27.19.2.2	75 mm dia water grade(Class E) uPVC blind pipe/sand trap of wall thickness 5.7 mm to 6.6 mm	meter	Tk. 658.91	Tk. 658.00	Tk. 654.00	Tk. 654.00
27.19.3.1	75 mm dia strainer of any slot as per design made with water grade (Class D) uPVC pipe of wall thickness 4.6 mm to 5.3 mm	meter	Tk. 603.70	Tk. 601.00	Tk. 591.00	Tk. 591.00
27.19.3.2	75 mm dia strainer of any slot as per design made with water grade (Class E) uPVC pipe of wall thickness 5.7 mm to 6.6 mm	meter	Tk. 739.90	Tk. 737.00	Tk. 728.00	Tk. 728.00
27.19.4	Best quality 75 mm x 150 mm dia uPVC Reducer	each	Tk. 1,570.59	Tk. 1,570.00	Tk. 1,566.00	Tk. 1,566.00
27.19.5.1	150 mm dia water grade(Class D) uPVC housing pipe having wall thickness 8.8 mm to 10.2 mm	meter	Tk. 2,114.17	Tk. 2,113.00	Tk. 2,109.00	Tk. 2,109.00
27.19.5.2	150 mm dia water grade(Class E) uPVC housing pipe having wall thickness 10.8 mm to 12.5 mm	meter	Tk. 2,575.53	Tk. 2,575.00	Tk. 2,571.00	Tk. 2,571.00
27.19.6	Supplying of centralizer made with 12 mm dia M.S. rod of approved design	meter	Tk. 499.58	Tk. 497.00	Tk. 487.00	Tk. 487.00
27.20	Complete installation of 75 mm dia uPVC tube-well fixtures such as expanded top pipes, reducer, tube-well pipes, strainer, blind pipe, pipe cap etc fitting and fixing with best quality uPVC solution, iron screws and uPVC tape and placing in position with proper size M.S. centralizer @ 3 meter c/c .in strainer portion including blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge.	meter	Tk. 112.02	Tk. 110.00	Tk. 107.00	Tk. 107.00
27.21	Complete development of the well by suitable air compressor and pump of suitable capacity by alternate surging and pumping untill satisfactory yield, sand free water and the turbidity condition is reached using necessary quantity of calgon and conducting step draw down test before and after the development as per specification and accepted by the Engineer-in-charge. (Development should be started within 24 hours of completion of shrouding)					
27.21.1	Wells up to 200 m depth	each	Tk. 41,473.51	Tk. 41,474.00	Tk. 41,474.00	Tk. 41,474.00
27.21.2	Wells beyond 200 m depth	each	Tk. 49,768.21	Tk. 49,768.00	Tk. 49,768.00	Tk. 49,768.00
27.22	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer.	hour	Tk. 1,069.62	Tk. 1,068.00	Tk. 1,055.00	Tk. 1,055.00
27.23	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) chlorinated water having 150 ppm available free chlorine complete as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 1,996.55	Tk. 1,997.00	Tk. 1,997.00	Tk. 1,997.00

27.24	Filling of the bore-hole around the tube well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or local clay free from any substances harmful to pipes and health as per standard specification and accepted by the Engineer-in-charge.	meter	Tk. 53.99	Tk. 53.00	Tk. 50.00	Tk. 50.00
27.25	Protective top cover: Supplying and fitting fixing best quality 200 mm dia 4 mm thick M.S. pipe 1.0 m (3'-4") long, one end blocked with 250 mm dia 4 mm thick M.S. plate and other end minimum 300 mm (1'-0) embedded in sanitary seal for protection of uPVC housing pipe as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 7,147.43	Tk. 7,129.00	Tk. 7,055.00	Tk. 7,055.00
	INSTALLATION OF 100 mm x 200 mm DIA SHROUDED					
	PRODUCTION WELL WITH PVC PIPE					
27.26	Boring: Execution of 400 mm dia of boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter) Boring for 400 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval or as required and at every change in soil stratum including sinking and withdrawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from E.G.L. to bottom of the blind pipe and the rate is inclusive of the cost of extra depth or of further boring required due to caving and siltation)					
27.26.1	From 0.0 m to 50 m = 50 m	meter	Tk. 1,092.05	Tk. 1,092.00	Tk. 1,092.00	Tk. 1,092.00
27.26.2	From 50 m to 100 m = 50 m	meter	Tk. 1,255.86	Tk. 1,256.00	Tk. 1,256.00	Tk. 1,256.00
27.26.3	From 100 m to 152 m = 52 m	meter	Tk. 1,444.24	Tk. 1,444.00	Tk. 1,444.00	Tk. 1,444.00
27.26.4	From 152 m to 200 m = 48 m	meter	Tk. 1,660.88	Tk. 1,661.00	Tk. 1,661.00	Tk. 1,661.00
27.26.5	From 200 m to 250 m = 50 m	meter	Tk. 1,910.01	Tk. 1,910.00	Tk. 1,910.00	Tk. 1,910.00
27.26.6	Beyond 250 m	meter	Tk. 2,196.51	Tk. 2,197.00	Tk. 2,197.00	Tk. 2,197.00
27.27	Supplying the following tube well fixtures and pipes having density 1350 kg/m ³ , opacity 0.2% of visible light as per ISO 7686, minimum required strength 25 MPa conforming to BDS-1878:2014, 4.4.1, 4.4.2 at working site as per standard practice and accepted by the Engineer-in-charge.					
27.27.1	Best quality 100 mm dia uPVC cap	each	Tk. 544.80	Tk. 544.00	Tk. 540.00	Tk. 540.00
27.27.2.1	100 mm dia water grade (Class D) uPVC blind pipe/ sand trap having wall thickness 6.0 mm to 6.9 mm	meter	Tk. 705.54	Tk. 705.00	Tk. 701.00	Tk. 701.00
27.27.2.2	100 mm dia water grade (Class E) uPVC blind pipe/ sand trap having wall thickness 7.3 mm to 8.4 mm	meter	Tk. 1,024.57	Tk. 1,024.00	Tk. 1,020.00	Tk. 1,020.00
27.27.3.1	100 mm dia strainer of any slot as per design made with water grade(Class D) uPVC pipe having wall thickness 6.0 mm to 6.9 mm	meter	Tk. 786.52	Tk. 784.00	Tk. 774.00	Tk. 774.00
27.27.3.2	100 mm dia strainer of any slot as per design made with water grade(Class E) uPVC pipe having wall thickness 7.3 mm to 8.4 mm	meter	Tk. 1,105.55	Tk. 1,103.00	Tk. 1,093.00	Tk. 1,093.00
27.27.4	Best quality 100 mm x 200 mm dia uPVC reducer	each	Tk. 2,323.99	Tk. 2,323.00	Tk. 2,319.00	Tk. 2,319.00
27.27.5.1	200 mm dia water grade (Class D) uPVC housing pipe having wall thickness 10.3 mm to 11.9 mm	meter	Tk. 3,180.45	Tk. 3,179.00	Tk. 3,176.00	Tk. 3,176.00

27.27.5.2	200 mm dia water grade(Class E) uPVC housing pipe having wall thickness 12.6 mm to 14.8 mm	meter	Tk. 3,825.87	Tk. 3,825.00	Tk. 3,821.00	Tk. 3,821.00
27.27.6	Supplying centralizer made with 12 mm dia M.S. rod of approved design	each	Tk. 499.58	Tk. 497.00	Tk. 487.00	Tk. 487.00
27.28	Complete installation of 100 mm dia uPVC tube-well fixtures such as expanded top pipes, reducer, tube-well pipes, strainer, blind pipe, pipe cap etc. fitting and fixing with best quality PVC solution, iron screws and uPVC tape and placing in position with proper size M.S. centralizer @ 3 meter c/c. in strainer portion including blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge.	meter	Tk. 133.34	Tk. 131.00	Tk. 126.00	Tk. 126.00
27.29	Complete development of the tube-well by air compressor and pump of suitable capacity by alternate surging and pumping until satisfactory yield of sand free water and the turbidity condition is reached using necessary quantity of calgon and conducting step draw down test before and after the development as per specification and accepted by the Engineer-in-charge. (Development should be started within 24 hours of completion of shrouding)					
27.29.1	Wells up to 200 m depth	each	Tk. 50,921.62	Tk. 50,922.00	Tk. 50,922.00	Tk. 50,922.00
27.29.2	Wells beyond 200 m depth	each	Tk. 61,105.94	Tk. 61,106.00	Tk. 61,106.00	Tk. 61,106.00
27.30	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer-in-charge.	hour	Tk. 1,122.52	Tk. 1,120.00	Tk. 1,107.00	Tk. 1,107.00
27.31	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) chlorinated water having 150 ppm available free chlorine etc. complete as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 2,348.88	Tk. 2,349.00	Tk. 2,349.00	Tk. 2,349.00
27.32	Filling of the borehole around the tube-well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or local clay free from any substances harmful to pipes and health as per standard specification and accepted by the Engineer-in-charge.	meter	Tk. 61.35	Tk. 60.00	Tk. 56.00	Tk. 56.00
27.33	Protective top cover: Supplying and fitting fixing best quality 250 mm dia 4 mm thick M.S. pipe, 1.0 m (3'-4") long one, end blocked with 300 mm dia 4 mm thick M.S. plate and other end minimum 300 (1'-0") embedded in sanitary seal for protection of uPVC Housing pipe as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 11,741.42	Tk. 11,711.00	Tk. 11,588.00	Tk. 11,588.00
	INSTALLATION OF 150 mm x 200 mm & 150 mm x 250 mm DIA SHROUDED					
	PRODUCTION WELL WITH uPVC PIPE					

27.34	Boring: Execution of 600 mm dia boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter). Boring for 600 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval or as required and at every change in soil stratum including sinking and withdrawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from E.G.L. to bottom of the blind pipe and the rate is inclusive of the cost of extra depth or of further boring required due to caving and siltation)					
27.34.1	From 0.0 m to 50 m = 50 m	meter	Tk. 1,374.27	Tk. 1,374.00	Tk. 1,374.00	Tk. 1,374.00
27.34.2	From 50 m to 100 m = 50 m	meter	Tk. 1,580.41	Tk. 1,580.00	Tk. 1,580.00	Tk. 1,580.00
27.34.3	From 100 m to 152 m = 52 m	meter	Tk. 1,817.47	Tk. 1,817.00	Tk. 1,817.00	Tk. 1,817.00
27.34.4	From 152 m to 200 m = 48 m	meter	Tk. 2,090.09	Tk. 2,090.00	Tk. 2,090.00	Tk. 2,090.00
27.34.5	From 200 m to 250 m = 50 m	meter	Tk. 2,403.61	Tk. 2,404.00	Tk. 2,404.00	Tk. 2,404.00
27.34.6	Beyond 250 m	meter	Tk. 2,764.15	Tk. 2,764.00	Tk. 2,764.00	Tk. 2,764.00
27.35	Supplying the following tube well fixtures and pipes having density 1350 kg/m ³ , opacity 0.2% of visible light as per ISO 7686, minimum required strength 25 MPa conforming to BDS-1878:2014, 4.4.1, 4.4.2 at working site as per standard practice and accepted by the Engineer-in-charge.					
27.35.1	Best quality 150 mm dia uPVC cap for blind pipe/ sand trap.	each	Tk. 1,143.59	Tk. 1,143.00	Tk. 1,139.00	Tk. 1,139.00
27.35.2.1	150 mm dia water grade(Class D) uPVC blind pipe/ sand trap having wall thickness 8.8 mm to 10.2 mm	meter	Tk. 1,816.00	Tk. 1,815.00	Tk. 1,811.00	Tk. 1,811.00
27.35.2.2	150 mm dia water grade(Class E) uPVC blind pipe/ sand trap having wall thickness 10.8 mm to 12.5 mm	meter	Tk. 2,212.33	Tk. 2,211.00	Tk. 2,207.00	Tk. 2,207.00
27.35.3.1	150 mm dia strainer of any slot as per design made with water grade(Class D) uPVC pipe of wall thickness 8.8 to 10.2 mm	meter	Tk. 1,917.23	Tk. 1,914.00	Tk. 1,903.00	Tk. 1,903.00
27.35.3.2	150 mm dia strainer of any slot as per design made with water grade(Class E) uPVC pipe of wall thickness 10.8 to 12.5 mm	meter	Tk. 2,313.56	Tk. 2,311.00	Tk. 2,299.00	Tk. 2,299.00
27.35.4.1	Best quality 150 mm x 200 mm dia uPVC Reducer	each	Tk. 2,531.36	Tk. 2,530.00	Tk. 2,525.00	Tk. 2,525.00
27.35.4.2	Best quality 150 mm x 250 mm dia uPVC Reducer	each	Tk. 3,748.57	Tk. 3,747.00	Tk. 3,742.00	Tk. 3,742.00
27.35.5.1	200 mm dia water grade(Class D) uPVC housing pipe of wall thickness 10.3 mm to 11.9 mm	meter	Tk. 3,180.45	Tk. 3,179.00	Tk. 3,176.00	Tk. 3,176.00
27.35.5.2	200 mm dia water grade(Class E) uPVC housing pipe of wall thickness 12.6 mm to 14.8 mm	meter	Tk. 3,825.87	Tk. 3,825.00	Tk. 3,821.00	Tk. 3,821.00
27.35.5.3	250 mm dia water grade(Class D) uPVC housing pipe of wall thickness 12.8 mm to 14.8 mm	meter	Tk. 4,409.93	Tk. 4,409.00	Tk. 4,405.00	Tk. 4,405.00
27.35.5.4	250 mm dia water grade(Class E) uPVC housing pipe of wall thickness 15.7 mm to 18.1 mm	meter	Tk. 5,870.10	Tk. 5,869.00	Tk. 5,865.00	Tk. 5,865.00
27.35.6	Supplying of centralizer made with 12 mm dia M.S. rod of approved design	meter	Tk. 499.58	Tk. 497.00	Tk. 487.00	Tk. 487.00
27.36	Complete installation of 150 mm dia uPVC tube-well fixtures such as expanded top pipes, reducer, tube-well pipes, strainer, blind pipe, pipe cap etc. fitting and fixing with best quality uPVC solution, iron screws and uPVC tape and placing in position with proper size M.S. centralizer @ 3 meter c/c in strainer portion including blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge.	meter	Tk. 129.29	Tk. 127.00	Tk. 124.00	Tk. 124.00

27.37	Complete development of the tube well by suitable air compressor and pump of suitable capacity by alternate surging and pumping until satisfactory yield, sand free water and the turbidity condition is reached using necessary quantity of calgon and conducting step draw down test before and after the development as per specification and accepted by the Engineer-in-charge. (Development should be started within 24 hours of completion of shrouding)					
27.37.1	Wells up to 200 m depth	each	Tk. 60,860.54	Tk. 60,861.00	Tk. 60,861.00	Tk. 60,861.00
27.37.2	Wells beyond 200 m depth	each	Tk. 68,975.27	Tk. 68,975.00	Tk. 68,975.00	Tk. 68,975.00
27.38	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer-in-charge.	hour	Tk. 1,274.92	Tk. 1,272.00	Tk. 1,259.00	Tk. 1,259.00
27.39	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) chlorinated water having 150 ppm available free chlorine etc. complete as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 2,818.66	Tk. 2,819.00	Tk. 2,819.00	Tk. 2,819.00
27.40	Filling of the borehole around the tube well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or local clay as free from any substances harmful to pipes and health per standard specification and accepted by the Engineer-in-charge.	meter	Tk. 85.12	Tk. 83.00	Tk. 78.00	Tk. 78.00
27.41	Protective top cover: Supplying fitting and fixing of best quality 250 mm dia 4 mm thick M.S. pipe 1.0 m (3'-4") long one end blocked with 300 dia 4 mm thick M.S. plate and other end minimum 300 (1'-0") embedded in sanitary seal for protection of uPVC Housing pipe as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 10,956.12	Tk. 10,925.00	Tk. 10,803.00	Tk. 10,803.00
	INSTALLATION OF 150 mm x 300 mm DIA SHROUDED					
	PRODUCTION WELL WITH G.I. PIPE					
27.42	Boring: Execution of 600 mm dia boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter) for boring 600 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval and at every change of formation including sinking and withdrawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from G.L. to bottom of the blind pipe and the rate is inclusive of the cost of extra depth of boring required for caving and siltation)					
27.42.1	From 0.0 m to 50 m = 50 m	meter	Tk. 1,374.27	Tk. 1,374.00	Tk. 1,374.00	Tk. 1,374.00
27.42.2	From 50 m to 100 m = 50 m	meter	Tk. 1,580.41	Tk. 1,580.00	Tk. 1,580.00	Tk. 1,580.00
27.42.3	From 100 m to 152 m = 52 m	meter	Tk. 1,817.47	Tk. 1,817.00	Tk. 1,817.00	Tk. 1,817.00
27.42.4	From 152 m to 200 m = 48 m	meter	Tk. 2,090.09	Tk. 2,090.00	Tk. 2,090.00	Tk. 2,090.00
27.42.5	From 200 m to 250 m = 50 m	meter	Tk. 2,403.61	Tk. 2,404.00	Tk. 2,404.00	Tk. 2,404.00
27.42.6	Beyond 250 m	meter	Tk. 2,764.15	Tk. 2,764.00	Tk. 2,764.00	Tk. 2,764.00

27.43	Supplying the following tube well fixtures and pipes conforming to standard BS-1387 & BDS-1031, having chemical composition (in%) carbon 0.17-0.25, manganese 0.95-1.20, sulphur 0.060 max, phosphorous 0.060 max, tensile properties: minimum yield strength 188 MPa, minimum tensile strength 313 MPa and marking for a) manufacturer b) material c) wall thickness d) nominal outside dia e) intended use etc. on the body of the pipe at working site as per standard practice and accepted by the Engineer-in-charge.					
27.43.1	150 mm dia M.S. cap 6 mm thick for blind pipe/sand trap	each	Tk. 1,550.96	Tk. 1,550.00	Tk. 1,546.00	Tk. 1,546.00
27.43.2	150 mm dia G.I. Pipe having wall thickness 5.4 mm, weight 21.2 kg/m can withstand 50 kg/ cm ² hydraulic pressure for blind pipe/sand trap	meter	Tk. 3,788.45	Tk. 3,785.00	Tk. 3,773.00	Tk. 3,773.00
27.43.3	150 mm dia strainer of slot opening as per design including tagging 4 Nos. 12 mm dia M.S. rod continuously.					
27.43.3.1	Stainless steel strainer of continuous slot having weight 28.84 kg per meter.	meter	Tk. 18,358.03	Tk. 18,356.00	Tk. 18,346.00	Tk. 18,346.00
27.43.3.2	2.50 mm thick stainless steel Bridge type strainer.	meter	Tk. 13,946.87	Tk. 13,944.00	Tk. 13,935.00	Tk. 13,935.00
27.43.4	150 mm x 300 mm dia M.S. reducer of minimum thickness of 6 mm.	each	Tk. 3,306.84	Tk. 3,301.00	Tk. 3,276.00	Tk. 3,276.00
27.43.5	300 mm dia 6 mm minimum thick M.S. housing pipe (Seamless)	meter	Tk. 10,460.40	Tk. 10,454.00	Tk. 10,430.00	Tk. 10,430.00
27.43.6	Centralizer made with 12 mm dia M.S. rod of approved design.	each	Tk. 499.58	Tk. 497.00	Tk. 487.00	Tk. 487.00
27.43.7	350 mm dia M.S. cap having 6 mm thick for housing pipe covering.	each	Tk. 3,509.30	Tk. 3,495.00	Tk. 3,436.00	Tk. 3,436.00
27.44	Complete installation of 150 mm dia deep tube-well fixtures such as expanded top pipes, reducer, tube-well pipes, strainer, blind pipe, pipe cap etc. fitting and fixing socket, welding and additional welding with fibre and placing in position with proper size M.S. centralizer @ 3 meter c/c in strainer portion including blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge.	meter	Tk. 804.80	Tk. 790.00	Tk. 732.00	Tk. 732.00
27.45	Complete development of the tube well by suitable air compressor and pump of suitable capacity by alternate surging and pumping until satisfactory yield, sand free water and the turbidity condition is reached using necessary quantity of calgon and conducting step draw down test before and after the development as per specification and accepted by the Engineer-in-charge. (Development should be started within 24 hours of completion of shrouding)					
27.45.1	Wells up to 200 m depth	each	Tk. 60,860.54	Tk. 60,861.00	Tk. 60,861.00	Tk. 60,861.00
27.45.2	Wells beyond 200 m depth	each	Tk. 68,975.27	Tk. 68,975.00	Tk. 68,975.00	Tk. 68,975.00
27.46	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer-in-charge.	hour	Tk. 1,315.82	Tk. 1,313.00	Tk. 1,300.00	Tk. 1,300.00
27.47	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) chlorinated water having 150 ppm available free chlorine etc. complete as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 2,877.43	Tk. 2,877.00	Tk. 2,877.00	Tk. 2,877.00
27.48	Filling of the borehole around the tube well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or local clay free from any substances harmful to pipes and health as per standard specification and accepted by the Engineer-in-charge.	meter	Tk. 85.12	Tk. 83.00	Tk. 78.00	Tk. 78.00

	INSTALLATION OF 200 mm x 400 mm DIA SHROUDED					
	PRODUCTION WELL WITH G.I. PIPE					
27.49	Boring: Execution of 800 mm dia boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter). For boring 800 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval and at every change of formation including sinking and withdrawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from G.L. to bottom of the blind pipe and the rate is required for caving and siltation)					
27.49.1	From 0.0 m to 50 m = 50 m	meter	Tk. 1,546.05	Tk. 1,546.00	Tk. 1,546.00	Tk. 1,546.00
27.49.2	From 50 m to 100 m = 50 m	meter	Tk. 1,777.96	Tk. 1,778.00	Tk. 1,778.00	Tk. 1,778.00
27.49.3	From 100 m to 152 m = 52 m	meter	Tk. 2,044.66	Tk. 2,045.00	Tk. 2,045.00	Tk. 2,045.00
27.49.4	From 152 m to 200 m = 48 m	meter	Tk. 2,351.35	Tk. 2,351.00	Tk. 2,351.00	Tk. 2,351.00
27.49.5	From 200 m to 250 m = 50 m	meter	Tk. 2,704.06	Tk. 2,704.00	Tk. 2,704.00	Tk. 2,704.00
27.49.6	Beyond 250 m	meter	Tk. 3,109.67	Tk. 3,110.00	Tk. 3,110.00	Tk. 3,110.00
27.50	Supplying the following tube-well fixtures and pipes conforming to standard BS-1387 & BDS-1031, having chemical composition (in%) carbon 0.17-0.25, manganese 0.95-1.20, sulphur 0.060 max, phosphorous 0.060 max, tensile properties: minimum yield strength 188 MPa, minimum tensile strength 313 MPa and marking for a) manufacturer b) material c) wall thickness d) nominal outside dia e) intended use etc. on the body of the pipe at working site as per standard practice and accepted by the Engineer-in-charge.					
27.50.1	200 mm dia M.S. cap 6mm thick for blind pipe/ sand strap	each	Tk. 1,968.15	Tk. 1,967.00	Tk. 1,963.00	Tk. 1,963.00
27.50.2	200 mm dia 6 mm thick tube well pipe and blank G.I. pipe of wt. 33.45 kg per meter can withstand minimum 700 psi hydraulic pressure.	meter	Tk. 7,776.28	Tk. 7,773.00	Tk. 7,761.00	Tk. 7,761.00
27.50.3	200 mm dia strainer of slot opening as per design including tagging 4 Nos. 12 mm dia M.S. rod continuously.	meter				
27.50.3.1	200 mm dia stainless steel strainer of continuous slot having 38.36 kg per meter.	meter	Tk. 21,948.15	Tk. 21,945.00	Tk. 21,933.00	Tk. 21,933.00
27.50.3.2	200 mm dia 2.5 mm thick stainless steel bridge type strainer	meter	Tk. 14,587.44	Tk. 14,584.00	Tk. 14,572.00	Tk. 14,572.00
27.50.4	200 mm x 400 mm dia 6 mm thick M.S. reducer.	each	Tk. 3,374.32	Tk. 3,369.00	Tk. 3,350.00	Tk. 3,350.00
27.50.5	400 mm dia 6 mm thick M.S. housing pipe (Seamless)	meter	Tk. 14,509.59	Tk. 14,503.00	Tk. 14,479.00	Tk. 14,479.00
27.50.6	Centralizer made with 12 mm dia M.S. rod of approved design	each	Tk. 499.58	Tk. 497.00	Tk. 487.00	Tk. 487.00
27.50.7	450 mm dia M.S. cap of 6 mm thick for housing pipe covering.	each	Tk. 785.30	Tk. 773.00	Tk. 724.00	Tk. 724.00
27.51	Complete installation of 200 mm dia deep tube well fixtures such as expanded top pipes, reducer, and tube well pipes, strainer, blind pipe. pipe cap etc. Fitting and fixing socket, welding and additional welding with fibre and placing in position with proper size M.S. centralizer @ 3 meter c/c in strainer position including blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge.	meter	Tk. 879.27	Tk. 863.00	Tk. 799.00	Tk. 799.00

27.52	Complete development of the tube well by suitable air compressor and pump of suitable capacity by alternate surging and pumping until satisfactory yield, sand free water and the turbidity condition is reached using necessary quantity of calgon and conducting step draw down test before and after the development as per specification and accepted by the Engineer-in-charge. (Development should be started within 24 hours of completion of shrouding).					
27.52.1	Wells up to 200 m depth	each	Tk. 57,792.97	Tk. 57,793.00	Tk. 57,793.00	Tk. 57,793.00
27.52.2	Wells beyond 200 m depth	each	Tk. 64,592.14	Tk. 64,592.00	Tk. 64,592.00	Tk. 64,592.00
27.53	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer-in-charge.	hour	Tk. 1,361.32	Tk. 1,359.00	Tk. 1,346.00	Tk. 1,346.00
27.54	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) chlorinated water having 150 ppm available free chlorine complete as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 3,288.43	Tk. 3,288.00	Tk. 3,288.00	Tk. 3,288.00
27.55	Filling of the borehole around the tube well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or local clay free from any substances harmful to pipes and health as per standard specification and accepted by the Engineer-in-charge.	meter	Tk. 85.12	Tk. 83.00	Tk. 78.00	Tk. 78.00
	INSTALLATION OF 50 mm x 150 mm DIA SHROUDED					
	PRODUCTION WELL WITH G.I. PIPE					
27.56	Boring: Execution of 325 mm dia boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter) for boring 325 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval and at every change of formation including sinking and withdrawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from G.L. to bottom of the blind pipe and the rate is inclusive of the cost of extra depth of boring required for caving and siltation)					
27.56.1	From 0.0 m to 50 m = 50 m	meter	Tk. 957.08	Tk. 957.00	Tk. 957.00	Tk. 957.00
27.56.2	From 50 m to 100 m = 50 m	meter	Tk. 1,100.64	Tk. 1,101.00	Tk. 1,101.00	Tk. 1,101.00
27.56.3	From 100 m to 152 m = 52 m	meter	Tk. 1,265.74	Tk. 1,266.00	Tk. 1,266.00	Tk. 1,266.00
27.56.4	From 152 m to 200 m = 48 m	meter	Tk. 1,455.60	Tk. 1,456.00	Tk. 1,456.00	Tk. 1,456.00
27.56.5	From 200 m to 250 m = 50 m	meter	Tk. 1,673.94	Tk. 1,674.00	Tk. 1,674.00	Tk. 1,674.00
27.56.6	Beyond 250 m	meter	Tk. 1,925.03	Tk. 1,925.00	Tk. 1,925.00	Tk. 1,925.00
27.57	Supplying the following tube-well fixtures and pipes conforming to standard BS-1387 & BDS-1031, having chemical composition (in%) carbon 0.17-0.25, manganese 0.95-1.20, sulphur 0.060 max, phosphorous 0.060 max, tensile properties: minimum yield strength 188 MPa, minimum tensile strength 313 MPa and marking for a) manufacturer b) material c) wall thickness d) nominal outside dia e) intended use etc. on the body of the pipe at working site as per standard practice and accepted by the Engineer-in-charge.					
27.57.1	150 mm dia 6mm thick M.S. cap for blind pipe/ sand trap.	each	Tk. 1,550.96	Tk. 1,550.00	Tk. 1,546.00	Tk. 1,546.00
27.57.2	50 mm dia 4.85 mm thick tube well pipe and blank G.I. pipe	meter	Tk. 1,175.19	Tk. 1,172.00	Tk. 1,160.00	Tk. 1,160.00

27.57.3	50 mm dia strainer of slot opening as per design including tagging 4 Nos. 12 mm dia M.S. rod continuously.	meter				
27.57.3.1	50 mm dia stainless steel strainer of continuous slot having 28.84 kg per meter.	meter	Tk. 7,388.41	Tk. 7,386.00	Tk. 7,376.00	Tk. 7,376.00
27.57.3.2	50 mm dia 2.5 mm thick stainless steel bridge type strainer	meter	Tk. 7,388.41	Tk. 7,386.00	Tk. 7,376.00	Tk. 7,376.00
27.57.4	50 mm x 150 mm dia 6 mm thick M.S. reducer.	each	Tk. 1,349.73	Tk. 1,344.00	Tk. 1,319.00	Tk. 1,319.00
27.57.5	150 mm dia 6 mm thick M.S. housing pipe (Seamless).	meter	Tk. 5,061.49	Tk. 5,055.00	Tk. 5,031.00	Tk. 5,031.00
27.57.6	Centralizer made with 12 mm dia M.S. rod of approved design.	each	Tk. 499.58	Tk. 497.00	Tk. 487.00	Tk. 487.00
27.57.7	150 mm dia M.S. cap of 6 mm thick for housing pipe covering.	each	Tk. 2,306.81	Tk. 2,292.00	Tk. 2,233.00	Tk. 2,233.00
27.58	Complete installation of 50 mm dia deep tube-well fixtures such as expanded top pipes, reducer, tube-well pipes, strainer, blind pipe, pipe cap etc. fitting and fixing socket, welding and additional welding with fibre and placing in position with proper size M.S. centralizer @ 3 meter c/c in strainer portion including. blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge	meter	Tk. 804.80	Tk. 790.00	Tk. 732.00	Tk. 732.00
27.59	Compete development of the tube well by suitable air compressor and pump of suitable capacity by alternate surging and pumping until satisfactory yield, sand free water and the turbidity condition is reached using necessary quantity of calgon and conducting step draw down test before and after the development as per specification and accepted by the Engineer-in-charge (Development should be started within 24 hours of completion of shrouding)					
27.59.1	Wells up to 200 m depth	each	Tk. 31,289.19	Tk. 31,289.00	Tk. 31,289.00	Tk. 31,289.00
27.59.2	Wells beyond 200 m depth	each	Tk. 35,461.08	Tk. 35,461.00	Tk. 35,461.00	Tk. 35,461.00
27.60	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer-in-charge.	hour	Tk. 1,078.08	Tk. 1,076.00	Tk. 1,064.00	Tk. 1,064.00
27.61	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) chlorinated water having 150 ppm available free chlorine etc. complete as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 1,644.22	Tk. 1,644.00	Tk. 1,644.00	Tk. 1,644.00
27.62	Filling of the borehole around the tube well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or local clay free from any substances harmful to pipes and health as per standard specification and accepted by the Engineer-in-charge.	meter	Tk. 48.31	Tk. 47.00	Tk. 44.00	Tk. 44.00
	INSTALLATION OF 75 mm x 150 mm DIA SHROUDED					
	PRODUCTION WELL WITH G.I. PIPE					

27.63	Boring: Execution of 350 mm dia boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter) for boring 350 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval and at every change of formation including sinking and withdrawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from G.L. to bottom of the blind pipe and the rate is inclusive of the cost of extra depth of boring required for caving and siltation)					
27.63.1	From 0.0 m to 50 m = 50 m	meter	Tk. 1,042.97	Tk. 1,043.00	Tk. 1,043.00	Tk. 1,043.00
27.63.2	From 50 m to 100 m = 50 m	meter	Tk. 1,199.42	Tk. 1,199.00	Tk. 1,199.00	Tk. 1,199.00
27.63.3	From 100 m to 152 m = 52 m	meter	Tk. 1,379.33	Tk. 1,379.00	Tk. 1,379.00	Tk. 1,379.00
27.63.4	From 152 m to 200 m = 48 m	meter	Tk. 1,586.23	Tk. 1,586.00	Tk. 1,586.00	Tk. 1,586.00
27.63.5	From 200 m to 250 m = 50 m	meter	Tk. 1,824.17	Tk. 1,824.00	Tk. 1,824.00	Tk. 1,824.00
27.63.6	Beyond 250 m	meter	Tk. 2,097.79	Tk. 2,098.00	Tk. 2,098.00	Tk. 2,098.00
27.64	Supplying the following tube-well fixtures and pipes conforming to standard BS-1387 & BDS-1031, having chemical composition (in%) carbon 0.17-0.25, manganese 0.95-1.20, sulphur 0.060 max, phosphorous 0.060 max, tensile properties: minimum yield strength 188 MPa, minimum tensile strength 313 MPa and marking for a) manufacturer b) material c) wall thickness d) nominal outside dia e) intended use etc. on the body of the pipe at working site as per standard practice and accepted by the Engineer-in-charge.					
27.64.1	150 mm dia 6mm thick M.S. cap for blind pipe/ sand trap.	each	Tk. 1,398.16	Tk. 1,397.00	Tk. 1,394.00	Tk. 1,394.00
27.64.2	75 mm dia 4.85 mm thick tube well pipe and blank G.I. pipe	meter	Tk. 1,728.75	Tk. 1,728.00	Tk. 1,724.00	Tk. 1,724.00
27.64.3	75 mm dia strainer of slot opening as per design including tagging 4 Nos. 12 mm dia M.S. rod continuously.	meter				
27.64.3.1	75 mm dia stainless steel strainer of continuous slot having 28.84 kg per meter.	meter	Tk. 9,966.09	Tk. 9,963.00	Tk. 9,951.00	Tk. 9,951.00
27.64.3.2	75 mm dia 2.5 mm thick stainless steel bridge type strainer	meter	Tk. 9,900.19	Tk. 9,897.00	Tk. 9,885.00	Tk. 9,885.00
27.64.4	75 mm x 150 mm dia 6 mm thick M.S. reducer.	meter	Tk. 1,349.73	Tk. 1,344.00	Tk. 1,319.00	Tk. 1,319.00
27.64.6	150 mm dia 6 mm thick M.S. housing pipe (Seamless)	meter	Tk. 5,061.49	Tk. 5,055.00	Tk. 5,031.00	Tk. 5,031.00
27.64.7	Centralizer made with 12 mm dia M.S. rod of approved design	each	Tk. 499.58	Tk. 497.00	Tk. 487.00	Tk. 487.00
27.64.8	150 mm dia M.S. cap of 6 mm thick for housing pipe covering.	meter	Tk. 2,306.81	Tk. 2,292.00	Tk. 2,233.00	Tk. 2,233.00
27.65	Complete installation of 75 mm dia deep tube-well fixtures such as expanded top pipes, reducer, tube-well pipes, strainer, blind pipe, pipe cap etc. fitting and fixing socket, welding and additional welding with fibre and placing in position with proper size M.S. centralizer @ 3 meter c/c in strainer portion including blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge.	meter	Tk. 771.26	Tk. 758.00	Tk. 702.00	Tk. 702.00

27.66	Complete development of the tube well by suitable air compressor and pump of suitable capacity by alternate surging and pumping until satisfactory yield, sand free water and the turbidity condition is reached using necessary quantity of calgon and conducting step draw down test before and after the development as per specification and accepted by the Engineer-in-charge(Development should be started within 24 hours of completion of shrouding).					
27.66.1	Wells up to 200 m depth	each	Tk. 41,473.51	Tk. 41,474.00	Tk. 41,474.00	Tk. 41,474.00
27.66.2	Wells beyond 200 m depth	each	Tk. 47,003.31	Tk. 47,003.00	Tk. 47,003.00	Tk. 47,003.00
27.67	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer-in-charge.	hour	Tk. 1,202.06	Tk. 1,199.00	Tk. 1,186.00	Tk. 1,186.00
27.68	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) chlorinated water having 150 ppm available free chlorine etc. complete as per standard specification and accepted by the Engineer-in-charge.	each	Tk. 1,996.55	Tk. 1,997.00	Tk. 1,997.00	Tk. 1,997.00
27.69	Filling of the borehole around the tube well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or local clay free from any substances harmful to pipes and health as per standard specification and accepted by the Engineer-in-charge.	meter	Tk. 55.68	Tk. 54.00	Tk. 51.00	Tk. 51.00
	INSTALLATION OF 100 mm x 200 mm DIA SHROUDED					
	PRODUCTION WELL WITH G.I. PIPE					
27.70	Boring: Execution of 400 mm dia boring through all sorts of strata by reverse circulation or any other approved method (Method other than reverse circulation is to be applied only for wells deeper than 152 meter) for boring 400 mm dia vertical hole up to designed depth and collection of soil samples at every 3 meter interval and at every change of formation including sinking and withdrawing necessary casing pipe and all necessary arrangements complete as per specification and accepted by the Engineer-in-charge. (Measurement will be allowed from G.L. to bottom of the blind pipe and the rate is inclusive of the cost of extra depth of boring required for caving and siltation)					
27.70.1	From 0.0 m to 50 m = 50 m	meter	Tk. 1,092.05	Tk. 1,092.00	Tk. 1,092.00	Tk. 1,092.00
27.70.2	From 50 m to 100 m = 50 m	meter	Tk. 1,255.86	Tk. 1,256.00	Tk. 1,256.00	Tk. 1,256.00
27.70.3	From 100 m to 152 m = 52 m	meter	Tk. 1,444.24	Tk. 1,444.00	Tk. 1,444.00	Tk. 1,444.00
27.70.4	From 152 m to 200 m = 48 m	meter	Tk. 1,660.88	Tk. 1,661.00	Tk. 1,661.00	Tk. 1,661.00
27.70.5	From 200 m to 250 m = 50 m	meter	Tk. 1,910.01	Tk. 1,910.00	Tk. 1,910.00	Tk. 1,910.00
27.70.6	Beyond 250 m	meter	Tk. 2,196.51	Tk. 2,197.00	Tk. 2,197.00	Tk. 2,197.00
27.71	Supplying the following tube-well fixtures and pipes conforming to standard BS-1387 & BDS-1031, having chemical composition (in%) carbon 0.17-0.25, manganese 0.95-1.20, sulphur 0.060 max, phosphorous 0.060 max, tensile properties: minimum yield strength 188 MPa, minimum tensile strength 313 MPa and marking for a) manufacturer b) material c) wall thickness d) nominal outside dia e) intended use etc. on the body of the pipe at working site as per standard practice and accepted by the Engineer-in-charge.					

27.71.1	Supplying 150 mm dia 6mm thick M.S. cap for blind pipe/ sand trap.	each	Tk. 1,550.96	Tk. 1,550.00	Tk. 1,546.00	Tk. 1,546.00
27.71.2	Supplying 100 mm dia 6 mm thick tube well pipe and blank G.I. pipe	meter	Tk. 2,596.32	Tk. 2,593.00	Tk. 2,581.00	Tk. 2,581.00
27.71.3	100 mm dia strainer of slot opening as per design including tagging 4 Nos. 12 mm dia M.S. rod continuously.	meter				
27.71.3.1	Supplying 100 mm dia stainless steel strainer of continuous slot having 28.84 kg per meter.	meter	Tk. 11,042.27	Tk. 11,042.00	Tk. 11,041.00	Tk. 11,041.00
27.71.3.2	Supplying 100 mm dia 2.5 mm thick stainless steel bridge type strainer	meter	Tk. 8,550.46	Tk. 8,547.00	Tk. 8,535.00	Tk. 8,535.00
27.71.4	Supplying 100 mm x 200 mm dia 6 mm thick M.S. reducer.	each	Tk. 1,619.68	Tk. 1,615.00	Tk. 1,595.00	Tk. 1,595.00
27.71.5	Supplying 200 mm dia 6 mm thick M.S. housing pipe (Seamless)	meter	Tk. 5,257.81	Tk. 5,252.00	Tk. 5,227.00	Tk. 5,227.00
27.71.6	Supplying centralizer made with 12 mm dia M.S. rod of approved design	each	Tk. 315.75	Tk. 313.00	Tk. 303.00	Tk. 303.00
27.71.7	Supplying 200 mm dia M.S. cap of 6 mm thick for housing pipe covering.	each	Tk. 2,773.08	Tk. 2,773.00	Tk. 2,712.00	Tk. 2,712.00
27.72	Complete installation of 100 mm dia deep tube-well fixtures such as expanded top pipes, reducer, and tube-well pipes, strainer, blind pipe, pipe cap etc. fitting and fixing socket, welding and additional welding with fibre and placing in position with proper size M.S. centralizer @ 3 meter c/c in strainer portion including blind pipe etc. complete as per standard practice and accepted by the Engineer-in-charge	meter	Tk. 742.12	Tk. 732.00	Tk. 692.00	Tk. 692.00
27.73	Complete development of the tube well by suitable air compressor and pump of suitable capacity by alternate surging and pumping until satisfactory yield, sand free water and the turbidity condition is reached using necessary quantity of calgon and conducting step draw down test before and after the development as per specification and accepted by the Engineer-in-charge (Development should be started within 24 hours of completion of shrouding)					
27.73.1	Wells up to 200 m depth	each	Tk. 50,921.62	Tk. 50,922.00	Tk. 50,922.00	Tk. 50,922.00
27.73.2	Wells beyond 200 m depth	each	Tk. 57,711.17	Tk. 57,711.00	Tk. 57,711.00	Tk. 57,711.00
27.74	Pumping test of the well by designed turbine/submersible pump with supply of testing equipments for pressure/ velocity/ discharge etc. and all materials for 72 hours including yield drawn and recovery test etc. complete as per specification and accepted by the Engineer-in-charge	hour	Tk. 1,247.56	Tk. 1,245.00	Tk. 1,232.00	Tk. 1,232.00
27.75	Disinfections of the well including supply of sufficient quantity of bleaching powder (33% strength) chlorinated water having 150 ppm available free chlorine etc. complete as per standard specification and accepted by the Engineer-in-charge	each	Tk. 2,348.88	Tk. 2,349.00	Tk. 2,349.00	Tk. 2,349.00
27.76	Filling of the borehole around the tube well from top of the shrouding up to the bottom of sanitary seal by withdrawn selected soil or local clay free from any substances harmful to pipes and health as per standard specification and accepted by the Engineer-in-charge	meter	Tk. 63.04	Tk. 61.00	Tk. 58.00	Tk. 58.00
	INSTALLATION OF 40 MM DIA uPVC HAND TUBE WEL					
27.77	Supplying and sinking of 40 mm dia water grade uPVC tube well pipe having wall thickness 2.5 mm to 3.0 mm including trial boring and with all necessary fittings as required etc. all complete and accepted by the Engineer-in-charge.					
27.77.1	40 mm dia pipe up to 60 m depth	meter	Tk. 277.79	Tk. 274.00	Tk. 258.00	Tk. 258.00

27.77.2	40 mm dia pipe beyond 60 m depth	meter	Tk. 493.17	Tk. 487.00	Tk. 463.00	Tk. 463.00
27.78	Supplying and sinking of 3 m long uPVC strainer with all necessary fittings as required etc. complete and accepted by the Engineer-in-charge.	each	Tk. 701.71	Tk. 698.00	Tk. 683.00	Tk. 683.00
27.79	Supplying, fitting and fixing of No. 6 pump for 40 mm dia tube well with all necessary fittings including 0.61 m long 40 mm dia G.I. Pipe etc. all complete as per accepted by the Engineer-in-charge.	each	Tk. 4,166.98	Tk. 4,167.00	Tk. 4,167.00	Tk. 4,167.00
27.80	Construction of platform of size 1.4 m x 1.0 m and outlet drain up to 1 meter long, as per drawing with 75 mm thick cement concrete of f _c min 17 MPa with brick chips and cement content relates to 1:2:4, over a brick flat soling and providing 125 mm x 75mm brick band in 1:6 all around, including finished with 12 mm thick cement plaster in 1:4 followed by neat cement finishing all complete and accepted by the Engineer-in-charge.	each	Tk. 6,159.93	Tk. 6,139.00	Tk. 5,944.00	Tk. 5,894.00
DEEP SET HAND OPERATED TUBE WELL						
27.81	Supplying and sinking uPVC, tube well pipe, uPVC strainer including trial boring with all necessary fittings as required etc. all complete and accepted by the Engineer-in-charge.					
27.81.1	50 mm dia water grade uPVC pipe having wall thickness 2.5 mm to 3 mm	meter	Tk. 583.75	Tk. 580.00	Tk. 564.00	Tk. 564.00
27.81.2	40 mm dia water grade uPVC pipe having wall thickness 2.5 mm to 3 mm	meter	Tk. 382.46	Tk. 378.00	Tk. 362.00	Tk. 362.00
27.81.3	40 mm dia uPVC strainer 3.0 m long	each	Tk. 701.71	Tk. 698.00	Tk. 683.00	Tk. 683.00
27.82	Supplying, fitting and fixing of deep set tube well pump (Hand set) with all necessary fittings as required such as No. 6 pump, 50 mm dia G.I. pipe 0.61 m long, 50 mm x 40 mm reducing socket, 40 mm dia pipe Nipple 1.22 m long cylinder, tie foot ball, tie plunger, 3 Nos. 10 mm dia sockets, 4 Nos. 10mm dia nuts, 5 Nos. end threaded 10 mm dia M.S. rod 3 m long, necessary quantity of solution and tape etc. complete and accepted by the Engineer-in-charge.	each	Tk. 8,248.33	Tk. 8,224.00	Tk. 8,126.00	Tk. 8,126.00

..... **END OF CHAPTER TWENTY SEVEN**

CHAPTER 28 : GAS CONNECTION WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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28.1	Pipe line installation: Installation of 50 psi, API, 5L, Grade-B, ERW Natural pipe line for supplying Titas Gas including, supplying, fitting, fixing and laying different sizes M.S. pipes with necessary tape and primer including cutting trenches (450 mm x 910 mm) in all kinds of soil, cleaning, cutting, applying coal-tar, wrapping the pipes, marking the pipeline with 200mm X 200mm demarcation plate (C.C./ M.S. plate) @ 15m interval and back filling the trenches, excavating earth including. leveling, dressing and removing excess earth in all respect as per specification of Titas Gas T and D Co. Ltd. and accepted by the Engineer-in-charge. [Rate is excluding the cost of welding and back filling with sand at the bottom by 300 mm in thickness (compacted)] (Pipes manufactured by National Tubes Limited under Steel & Engineering Corporation, Ministry of Industries, Government of Bangladesh shall be given priority)					
28.1.1	100 mm ND (114.30 mm O.D) M.S pipe, weight of min 14.4 kg/m, standard test pressure 176 kg/ m ² , wall thickness of min 5.4 mm	meter	Tk. 4,135.18	Tk. 4,129.00	Tk. 4,113.00	Tk. 4,113.00
28.1.2	75 mm N.D (88.90 mm O.D) M.S pipe, weight of min 10.1 kg/m, standard test pressure 176 kg/ m ² , wall thickness of min 4.85 mm	meter	Tk. 2,356.46	Tk. 2,351.00	Tk. 2,334.00	Tk. 2,334.00
28.1.3	50 mm N.D. (60.30 mm O.D) M.S pipe, weight of min 6.17 kg/m, standard test pressure 176 kg/ m ² , wall thickness of min 4.47 mm	meter	Tk. 1,422.29	Tk. 1,416.00	Tk. 1,400.00	Tk. 1,400.00
28.1.4	25 mm N.D. (33.80 mm O.D) M.S pipe weight of min 3.84 kg/m, standard test pressure 50 kg/ m ² , wall thickness of min 4.05 mm	meter	Tk. 849.31	Tk. 843.00	Tk. 827.00	Tk. 827.00
28.1.5	20 mm N.D. (27.20 mm O.D) M.S pipe weight of min 1.90 kg/m, standard test pressure 50 kg/ m ² , wall thickness of min 3.25 mm	meter	Tk. 666.49	Tk. 661.00	Tk. 644.00	Tk. 644.00
28.2	Providing Butt welded joints with pipes to Tees, Elbows, Saddles, end caps etc above or below ground including. cutting and V-leveling in each end of the pipe, grinding by machine etc. complete as per specification of Titas Gas T and D C Ltd. and accepted by the Engineer-in-charge.					
28.2.1	75 mm N.D (88.90 mm O.D)-6.95 mm	each	Tk. 280.38	Tk. 280.00	Tk. 280.00	Tk. 280.00
28.2.2	50 mm N.D (60.30 mm O.D)-5.15 mm	each	Tk. 240.32	Tk. 240.00	Tk. 240.00	Tk. 240.00
28.2.3	25 mm N.D (33.40 mm O.D)-4.2mm	each	Tk. 186.92	Tk. 187.00	Tk. 187.00	Tk. 187.00
28.2.4	20 mm N.D (26.70 mm O.D)-3.35mm	each	Tk. 160.22	Tk. 160.00	Tk. 160.00	Tk. 160.00
28.3	Supplying, fitting and fixing the following materials including carriage to the work site from Titas Gas Store as per specification of Titas Gas (T and D) Co. Ltd. and accepted by the Engineer-in-charge. (Rate is excluding the cost of welding)					
28.3.1	50 mm x 50 mm x 50 mm (N.D) equal tee	each	Tk. 314.55	Tk. 314.00	Tk. 314.00	Tk. 314.00
28.3.2	50 mm x 50 mm x 25 mm (N.D) reducing tee	each	Tk. 512.84	Tk. 513.00	Tk. 512.00	Tk. 512.00
28.3.3	25 mm x 25 mm x 25 mm (N.D) equal tee	each	Tk. 181.58	Tk. 181.00	Tk. 180.00	Tk. 180.00
28.3.4	50 mm x 90° elbow	each	Tk. 255.01	Tk. 255.00	Tk. 254.00	Tk. 254.00
28.3.5	25 mm x 90° elbow	each	Tk. 214.96	Tk. 215.00	Tk. 214.00	Tk. 214.00
28.3.6	50 mm x 25 mm 45° elbow	each	Tk. 117.49	Tk. 117.00	Tk. 116.00	Tk. 116.00
28.3.7	50 mm x 25 mm reducer.	each	Tk. 134.85	Tk. 135.00	Tk. 134.00	Tk. 134.00
28.3.8	20 mm plug, socket	each	Tk. 528.71	Tk. 528.00	Tk. 527.00	Tk. 527.00
28.3.9	50mm X 20 mm saddle	each	Tk. 435.25	Tk. 435.00	Tk. 434.00	Tk. 434.00
28.3.10	50 mm dia end cap	each	Tk. 104.14	Tk. 104.00	Tk. 103.00	Tk. 103.00

28.3.11	25 mm dia end cap	each	Tk. 113.49	Tk. 113.00	Tk. 112.00	Tk. 112.00
28.4	Supplying, welding, fitting and fixing of different size valve tee/ service tee to connect the pipe line with the Titas Gas Main pipe line etc. all complete in all respect as per specification of Titas Gas T and D Co. Ltd. approved and accepted by the Engineer-in-charge.					
28.4.1	50 mm x 50 mm Valve Tee (50 mm and above 50 mm low pressure line) Sch 40	each	Tk. 16,799.34	Tk. 16,737.00	Tk. 16,489.00	Tk. 16,489.00
28.4.2	50 mm x 25 mm Valve Tee	each	Tk. 3,070.81	Tk. 3,024.00	Tk. 2,837.00	Tk. 2,837.00
28.4.3	25 mm x 25 mm Service Tee	each	Tk. 6,976.08	Tk. 6,936.00	Tk. 6,776.00	Tk. 6,776.00
28.4.4	50 mm x 25 mm Service Tee	each	Tk. 13,618.38	Tk. 13,565.00	Tk. 13,351.00	Tk. 13,351.00
28.5	Supplying, fitting and fixing in position different dia I/F (Insulated Flange) etc. all complete in all respect as per specification of Titas Gas T and D Co. Ltd. approved and accepted by the Engineer-in-charge. 50 mm N.D. (60.30 mm O.D) Insulation Flange.	each	Tk. 7,189.70	Tk. 7,130.00	Tk. 6,889.00	Tk. 6,889.00
28.6	Construction of 1520 mm x 1520 mm x 1370 mm (depth) water-tight valve pit , enclosed by 250 mm thick brick wall in 1:6, plastered on each face in 1:4 with net cement finishing, having a 75 mm thick C.C (1:3:6) bed upon a layer of brick flat soling, over laid by 38 mm thick patent stone flooring including standard M.S. sheet valve pit cover as per drawing and specification of Titas Gas T and D Co. Ltd. etc. all complete in all respect and accepted by the Engineer-in-charge.	each	Tk. 121,986.15	Tk. 121,708.00	Tk. 119,912.00	Tk. 119,730.00
28.7	Supplying, fitting and fixing of ball valve class 150 RF including supplying of 2 Nos. flange along with necessary nuts and bolts, welding the flange etc. all complete in all respect as per specification of Titas Gas T and D Co. Ltd. approved and accepted by the Engineer-in-charge.					
28.7.1	50 mm ball valve class 150 RF. 2" dia	each	Tk. 18,712.59	Tk. 18,616.00	Tk. 18,229.00	Tk. 18,229.00
28.7.2	25 mm ball valve class 150 RF. 1" dia	each	Tk. 14,478.87	Tk. 14,382.00	Tk. 13,995.00	Tk. 13,995.00
	Part - B (Domestic & Commercial line)					
28.8	Riser Pipe line installation: Installation of 2 psi, API, 5L, Grade-B, ERW Natural pipe line for supplying Titas Gas including, supplying, fitting, fixing and laying different sizes M.S. pipes with necessary tape and painting in all respect as per specification of Titas Gas T and D Co. Ltd. and accepted by the Engineer-in-charge. [Rate is excluding the cost of welding and tee, elbow socket etc fittings] (Pipes manufactured by National Tubes Limited under Steel & Engineering Corporation, Ministry of Industries, Government of Bangladesh shall be given priority)					
28.8.1	12 mm N.D. with wall thickness 2 mm	each	Tk. 319.40	Tk. 318.00	Tk. 312.00	Tk. 312.00
28.8.2	20 mm N.D. with wall thickness 2.35 mm	each	Tk. 386.65	Tk. 385.00	Tk. 379.00	Tk. 379.00
28.9	Supplying, fitting and fixing in position different dia one-way brass stop cock (best quality and heavy type) etc. complete in all respect as per specification of Titas Gas T and D Co. Ltd. approved and accepted by the Engineer-in-charge.					
28.9.1	12 mm x 6 mm N.D.	each	Tk. 403.88	Tk. 403.00	Tk. 401.00	Tk. 401.00
28.9.2	20 mm x 20 mm N.D.	each	Tk. 728.76	Tk. 728.00	Tk. 723.00	Tk. 723.00
28.10	Cutting of permanent pucca (rigid/flexible) road and mending good the damages etc. all complete in all respect and accepted by the Engineer-in-charge.					
28.10.1	Bituminous Road	sqm	Tk. 1,355.58	Tk. 1,331.00	Tk. 1,232.00	Tk. 1,232.00
28.10.2	R.C.C. Road	sqm	Tk. 2,027.44	Tk. 1,991.00	Tk. 1,843.00	Tk. 1,843.00

28.11	Testing and purging of the house line properly as per specifications Titas Gas T and D Co. Ltd. and accepted by the Engineer-in-charge.	meter	Tk. 29.37	Tk. 29.00	Tk. 29.00	Tk. 29.00
28.12	Supplying and installation the complete LP Gas Reticulated System for highrise buildings (Upto 40 unit 20 storied) with Storage facility (45 KG POL Valve cylinders/Bulk Tank) with isolation and safety valve, pressure gauges, LPG transfer hose, Seamless MS pipe network, LPG Regulator (1st Stage and 2nd Stage), Household LP Gas meter, Safety fittings & accessories, system design, documentation, supervision and commissioning etc. (as per standard of department of Explosive and NFPA-58) all complete and accepted by engineer in charge.					
28.12.1	Cylinder heading line for multistoried Building with 40 unit apartment, 2X10 Nos cylinder Heading line, 45 Kg LP Gas Cylinder Bank Vapor Off Take (VOT) System with MS Seamless SCH 40 pipes. (min. dia 25 mm), Flanze Type MS 150 class ball valve, Pressure Gauge (100 mm dia) liquid type (range: 0-20 Bar), Bleed Valve, Safety Relief Valve (LPG Type, Relief at 14.6 Bar)), LP Gas pigtail, Ball valve and necessary fittings suitable for LP Gas as per standard of department of Explosive and NFPA-58) , Supply, Installation of Medium pressure main distribution pipeline, MS Seamless (Sch-40) 50 mm dia pipe upto 100 meter length , Flanze Type MS Valve 150 Class Dia 50 mm, Sch 40 MS 50 mm TEE, Sch 40 MS 50 mm Elbow, 150 Class 50 mm MS Flanze, Painting, Pipe support with MS Clamp, First Stage Regulator; Brand: REGO / FISHER / CAVAGNA or similar from USA or EU, MS Seamless SCH 40 pipes, Flanze Type MS 150 class ball valve, Pressure Gauge (100 mm dia) liquid type (range: 0-20 Bar), Bleed Valve, Safety Relief Valve (LPG Type, Relief at 14.6 Bar) etc. as per requirement all complete, approved and accepted by the Engineer-in-charge. (All the welding joints must be TIG welded) (Rate is excluding the cost of shed, perimeter fencing and RCC floor).	set	Tk. 1,159,540.00	Tk. 1,159,540.00	Tk. 1,159,540.00	Tk. 1,159,540.00
28.12.2	Reduction cost per unit below 40 unit connection for all necessary arrangement including heading line.	each	Tk. 10,000.00	Tk. 10,000.00	Tk. 10,000.00	Tk. 10,000.00
28.12.3	Additional cost per apartment upto 100 apartment for all necessary arrangement including heading line.	each	Tk. 9,202.70	Tk. 9,203.00	Tk. 9,203.00	Tk. 9,203.00
28.12.4	Supplying and installation of 45 Kg LP Gas empty cylinder [Water capacity: 108 litre; design code: DOT, Service pressure 17 Kgs/cm ² , test pressure 34 Kgs/cm ² , Steel grade SG295 or equiv.], certified by Department of Explosives.	each	Tk. 7,362.00	Tk. 7,362.00	Tk. 7,362.00	Tk. 7,362.00
28.13	Supplying and instalation of LP Gas Prepaid meter with smart recharging system. (Smart card for recharge gas, Meter connection, Second Stage Regulator, Gas meter header with individual valves (MS 150 class) in both ends of meter and other fittings etc. all complete and accepted by the Engineer-in-charge.	each	Tk. 24,541.00	Tk. 24,541.00	Tk. 24,541.00	Tk. 24,541.00

..... END OF CHAPTER TWENTY EIGHT

CHAPTER 29 : TERMITE TREATMENT WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
29.1	Supplying anti-termite chemicals named DURS BAN 20 EC / equivalent and mixing the same with pure water in required proportion (For DURSBAN 20 EC 1 Liter in 19 liters of pure water) and spraying the emulsified mixture @ 5 liters per square meter of soil surface by chemical sprayer or by any other means to the sides and bottom of the foundation trench and over plinth filling or floor bed accepted by the Engineer-in-charge.	sqm	Tk. 337.12	Tk. 337.00	Tk. 336.00	Tk. 336.00
29.2	Supplying anti termite chemicals named DURS BAN 20 EC / equivalent and mixing the same with pure water in required proportion (For DURSBAN 20 EC 1 Liter in 19 liters of pure water) and using the same emulsified water as mixing water in cement-sand mortar of brick works in foundation and up to plinth and accepted by the Engineer-in-charge.	cum	Tk. 2,490.04	Tk. 2,485.00	Tk. 2,478.00	Tk. 2,478.00
29.3	Supplying anti-termite chemicals named DURS BAN 20 EC / equivalent and mixing the same with pure water in required proportion (For DURSBAN 20 EC 1 Liter in 19 liters of pure water) and spraying the emulsified mixture with hand sprayer over both outside and inside back fill in foundation trenches @ 7.50 liters per sqm of the vertical surface area of the foundation wall accepted by the Engineer-in-charge. (The maximum depth of the back fill to be treated is 500 mm. and the measurement for depth should not be given more than 500 mm)	sqm	Tk. 503.80	Tk. 503.00	Tk. 502.00	Tk. 502.00
29.4	Anti-termite treatment in existing super structures by making downwardly slanting holes at a required depth of 9" from the existing finished F.L/G.L inside and outside the building and applying the emulsion of anti termite chemicals, named DURS BAN 20 EC into the holes preparing chemical solution mixing with water at the ratio of chemical: water = 1:19 maintaining a dosage rate of 250 ml per hole including supplying all materials and repairing the holes all complete and accepted by the Engineer-in-charge.	per hole	Tk. 75.10	Tk. 74.00	Tk. 71.00	Tk. 71.00
29.5	Trench treatment for protection against incoming termites to the establishment through underground by making trench around existing building /structures at a required depth of 750 mm from existing level including supplying anti-termite chemicals named DURS BAN 20 EC / equivalent preparing the solution to be mixed with water in the ratio of chemical: water = 1:19, applying the solution into trench with the help of a chemical sprayer in three layers and each layer follows a 250 mm thick back-fill in trench etc all complete and accepted by the Engineer-in-charge.	meter	Tk. 343.90	Tk. 341.00	Tk. 334.00	Tk. 334.00

29.6	Fumigation treatment (Pest Control) against mosquitoes, flies, bees, cockroaches, spiders, silver fishes, lizards and other common pests and insects in wooden walls, bricks walls, floors, doors and windows, furniture fanlight, carpets, records, curtains, electrical conduits etc. with the application of fumigants such as Syngenta or NEOCIDOL-60 EC or similar chemically equivalent govt. registered pest control products preparing chemical solution mixing with water in the ratio of chemical: water = 1:100, applying with the help of electrically/mechanically operated jet fogging machine, swing fog, dyna fog, pneumatic hand sprayer, knap sac hand sprayer in air-tight position, after fumigation the surface will be kept remain at air-tight condition at least for 3 (three) hours etc. complete and accepted by the Engineer-in-charge. (Volume of the room will be considered towards measurement.)	cum	Tk. 47.12	Tk. 47.00	Tk. 45.00	Tk. 45.00
29.7	Anti termite treatment of all wooden doorframes, window frames, wood paneling etc. by using DURS BAN 20 EC / equivalent emulsion in the ratio of 1 liter. DURS BAN 20 EC + 19 liters of Kerosene oil suitable for wood, plywood, particle board, hard board etc. on both surfaces by brush on both surfaces by brush-on-method etc. complete and accepted by the Engineer-in-charge.	sqm	Tk. 193.76	Tk. 194.00	Tk. 191.00	Tk. 191.00
29.8	Anti-termite treatment of electric telephone wiring system by inserting a plugging materials (cotton, foam etc.) soaked in anti-termite chemical DURS BAN 20 EC / equivalent to an electric switch board point for all floors and spraying the emulsion (to be mixed at the ratio of 1 liters DURS BAN 20 EC + 19 liter of water) to the casings and grooving of electric wirings, telephone cables utility pipes etc. for all floors etc. complete and accepted by the Engineer-in-charge.	per board	Tk. 325.55	Tk. 322.00	Tk. 317.00	Tk. 317.00
29.9	Destruction of termite nest around an infested building, tree stumps in lawn with the treatment of DURS BAN 20 EC / equivalent by excavation of soil in affected area up to a depth of 1.5 meter, mixing the DURS BAN 20 EC / equivalent solution to be mixed with water in 1:19 ratio with the soil and laying the same in same place etc. all complete as per direction including. cost of all materials and chemicals etc. complete and accepted by the Engineer-in-charge.	each nest	Tk. 2,531.64	Tk. 2,511.00	Tk. 2,467.00	Tk. 2,467.00

..... END OF CHAPTER TWENTY NINE

CHAPTER 30 : CONCRETE SOLID AND HOLLOW BLOCKS, FACING AND PAVING STONES ETC.

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
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30.1	240 mm thick Concrete Solid Block wall works having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm , laying with specified mortar of cement and sand in a ratio 1:5 (sand not less than F.M = 1.2) and minimum thickness of 12 mm in foundation and plinth , filling the joints/ interstices fully with mortar, racking out the joints and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by the contractor and accepted by the Engineer-in-charge.	Cum	Tk. 8,859.00	Tk. 8,835.00	Tk. 8,722.00	Tk. 8,722.00
30.2	One layer of Concrete Solid Block Soling having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm in ground floor/ foundation including preparation of bed and filling the interstices with fine sand (F.M.= 0.8), leveling etc. all completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 661.94	Tk. 659.00	Tk. 649.00	Tk. 649.00
30.3	240 mm thick Exterior Wall with Concrete Solid Block having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm , laying with specified mortar of cement and sand in a ratio 1:5 (sand not less than F.M =1.2) and minimum thickness of 12 mm , filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.					
30.3.1	Ground floor	Cum	Tk. 8,967.33	Tk. 8,944.00	Tk. 8,830.00	Tk. 8,830.00
30.3.2	Add for each addl. floor above ground floor	Cum	Tk. 73.00	Tk. 73.00	Tk. 73.00	Tk. 73.00
30.4	114 mm thick Partition Wall with Concrete Solid Block having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm , laying with specified mortar of cement and sand in a ratio 1:5 (sand not less than F.M =1.2) and minimum thickness of 12 mm, filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.					
30.4.1	Ground floor	sqm	Tk. 1,236.24	Tk. 1,231.00	Tk. 1,208.00	Tk. 1,208.00
30.4.2	Add for each addl. floor up above ground floor	sqm	Tk. 36.00	Tk. 36.00	Tk. 36.00	Tk. 36.00

30.5	240 mm thick Exterior Wall with Concrete Hollow Block having minimum compressive strength of 15 MPa of size (390 x 190 x 240) mm, thickness of individual Block Shell should not be less than 32 mm, water absorption not more than 7%, laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) and minimum thickness of 15 mm (9 mm for shell portion and 6 mm for inside hollow portion), filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.					
30.5.1	Ground floor	sqm	Tk. 1,907.26	Tk. 1,903.00	Tk. 1,880.00	Tk. 1,880.00
30.5.2	Add for each addl. floor up above ground floor	sqm	Tk. 36.00	Tk. 36.00	Tk. 36.00	Tk. 36.00
30.6	112 mm thick Partition Wall with Concrete Hollow Block having minimum compressive strength of 15 MPa of size (390 x 190 x 112) mm, thickness of individual Block Shell should not be less than 25 mm, water absorption not more than 7%, laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) minimum thickness of 15 mm (9 mm for shell portion and 6 mm for inside hollow portion), filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.					
30.6.1	Ground floor	sqm	Tk. 1,015.08	Tk. 1,011.00	Tk. 988.00	Tk. 988.00
30.6.2	Add for each addl. floor up above ground floor	sqm	Tk. 36.00	Tk. 36.00	Tk. 36.00	Tk. 36.00
30.7	240 mm thick Exterior Wall with 3 Holes Concrete Hollow Block having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70-3H) mm, laying with specified mortar of cement and sand in a ratio 1:5 (sand not less than F.M =1.2) and minimum thickness of 12 mm, filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.					
30.7.1	Ground floor	cum	Tk. 7,202.07	Tk. 7,189.00	Tk. 7,112.00	Tk. 7,112.00
30.7.2	Add for each addl. floor up above ground floor	cum	Tk. 73.00	Tk. 73.00	Tk. 73.00	Tk. 73.00
30.8	114 mm thick Partition Wall with 3 Holes Concrete Hollow Block having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70-3H) mm, laying with specified mortar of cement and sand in a ratio 1:5 (sand not less than F.M =1.2) and minimum thickness of 12 mm, filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.					
30.8.1	Ground floor	sqm	Tk. 1,115.46	Tk. 1,112.00	Tk. 1,088.00	Tk. 1,088.00

30.8.2	Add for each addl. floor up above ground floor	sqm	Tk. 36.00	Tk. 36.00	Tk. 36.00	Tk. 36.00
30.9	240 mm thick Boundary Wall with Concrete Solid Block having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm , laying with specified mortar of cement and sand in a ratio 1:5 (sand not less than F.M =1.2) and minimum thickness of 12 mm, filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.	Cum	Tk. 8,967.33	Tk. 8,944.00	Tk. 8,830.00	Tk. 8,830.00
30.10	114 mm thick boundary wall with Concrete Solid block having minimum compressive strength of 15 MPa and water absorption not greater than 7% of size (240 x 114 x 70) mm , laying with specified mortar of cement and sand in a ratio 1:5 (sand not less than F.M= 1.2) and minimum thickness of 12 mm, filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,236.24	Tk. 1,231.00	Tk. 1,208.00	Tk. 1,208.00
30.11	240 mm thick Boundary Wall with Concrete Hollow Block having minimum compressive strength of 15 MPa of size (390 x 190 x 240) mm , thickness of individual Block Shell should not be less than 32 mm, water absorption not more than 7%, laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) and minimum thickness of 15 mm (9 mm for shell portion and 6 mm for inside hollow portion), filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,907.26	Tk. 1,903.00	Tk. 1,880.00	Tk. 1,880.00
30.12	112 mm thick Boundary Wall with Concrete Hollow Block having minimum compressive strength of 10 MPa of size (390 x 190 x 112) mm , thickness of individual Block Shell should not be less than 25 mm, water absorption not more than 7%, laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) minimum thickness of 15 mm (9 mm for shell portion and 6 mm for inside hollow portion), filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,015.08	Tk. 1,011.00	Tk. 988.00	Tk. 988.00
30.13	240 mm thick Boundary Wall with 3 Holes Concrete Hollow Block having minimum compressive strength of 15 MPa of size (240 x 114 x 70 -3H) mm , thickness of individual Block Shell should not be less than 32 mm, water absorption not more than 7%, laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) and minimum thickness of 15 mm (9 mm for shell portion and 6 mm for inside hollow portion), filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.	cum	Tk. 7,202.07	Tk. 7,189.00	Tk. 7,112.00	Tk. 7,112.00

30.14	114 mm thick Boundary Wall with 3 Holes Concrete Hollow Block having minimum compressive strength of 10 MPa of size (240 x 114 x 70 - 3H) mm , thickness of individual Block Shell should not be less than 25 mm , water absorption not more than 7%, laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) minimum thickness of 15 mm (9 mm for shell portion and 6 mm for inside hollow portion), filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,115.46	Tk. 1,112.00	Tk. 1,088.00	Tk. 1,088.00
30.15	112 mm thick partition wall with sand cement hollow block wall (SCHB) having compressive strength of not less than 10 MPa, size (390 x 190 x 112) mm, thickness of individual Block shell should not be less than 25 mm, water absorption not more than 7%, laying with specified mortar (with admixture) of cement and sand in a ratio of not less than 1:5 (sand not less than F.M = 1.2), filling the interstices with mortar and making bond with connecting walls including necessary scaffolding and curing for minimum 14 days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 7 days including cost of water, electricity and other charges etc. in all completed by the contractor and accepted by the Engineer-in-charge.					
30.15.1	Ground Floor	sqm	Tk. 861.87	Tk. 858.00	Tk. 835.00	Tk. 835.00
30.15.2	Add for each additional floor above ground floor	sqm	Tk. 36.00	Tk. 36.00	Tk. 36.00	Tk. 36.00
30.16	Minimum 10 mm thick cement plaster in cement : sand = 1:5 in surface of external/internal wall with specified external/internal plaster mix applied in two coats (first coat 6 mm and second coat 4 mm), finishing the corner and edge including washing of sand, cleaning the surface, scaffolding and curing for minimum 3 (three) days on mortar joints by jute/brush in wet, followed by impervious/wet covering for another 4 (four) days etc. all complete etc. all complete and accepted by the Engineer-in-charge. (Rate is excluding the cost of grouting and reinforcement)					
30.16.1	Ground floor	sqm	Tk. 286.15	Tk. 284.00	Tk. 266.00	Tk. 266.00
30.16.2	Add for each additional floor above ground floor	sqm	Tk. 6.00	Tk. 6.00	Tk. 6.00	Tk. 6.00
30.17	Supplying and placing of 60 mm thick grey uni-block paving for walk way having size (222x110x60) mm , water absorption not greater than 4% and minimum compressive strength of 25 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,124.51	Tk. 1,122.00	Tk. 1,106.00	Tk. 1,106.00
30.18	Supplying and placing of 60 mm thick coloured uni-block paving for walk way having size (222x110x60) mm , water absorption not greater than 4% and minimum compressive strength of 25 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,249.98	Tk. 1,248.00	Tk. 1,231.00	Tk. 1,231.00

30.19	Supplying and placing of 70 mm thick grey uni-block paving for walk way having size (222x110x70) mm , water absorption not greater than 4% and minimum compressive strength of 30 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge	sqm	Tk. 1,224.88	Tk. 1,222.00	Tk. 1,206.00	Tk. 1,206.00
30.20	Supplying and placing of 70 mm thick coloured uni-block paving for walk way having size (222x110x70) mm , water absorption not greater than 4% and minimum compressive strength of 30 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge	sqm	Tk. 1,300.17	Tk. 1,298.00	Tk. 1,281.00	Tk. 1,281.00
30.21	Supplying and placing of 80 mm thick grey uni-block for paving foot-path, major walk way, car porch having size (222x110x80) mm , water absorption not greater than 4% and compressive strength of 40 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,500.93	Tk. 1,499.00	Tk. 1,482.00	Tk. 1,482.00
30.22	Supplying and placing of 80 mm thick coloured uni-block for paving foot-path, major walk way, car porch having size (222x110x80) mm , water absorption not greater than 4% and compressive strength of 40 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,651.49	Tk. 1,649.00	Tk. 1,633.00	Tk. 1,633.00
30.23	Supplying and placing of 90 mm grey uni-block for paving heavy traffic areas, roads rounds abuts, industrial yards etc having size (222x110x90) mm , water absorption not greater than 4% and compressive strength of 40 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,551.11	Tk. 1,549.00	Tk. 1,532.00	Tk. 1,532.00
30.24	Supplying and placing of 90 mm thick coloured uni-block for paving heavy traffic areas, road, industrial yard etc having size (222x110x90) mm , water absorption not greater than 4% and compressive strength of 40 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,701.68	Tk. 1,699.00	Tk. 1,683.00	Tk. 1,683.00
30.25	Supplying and placing of 100 mm grey uni-block for paving heavy traffic areas, roads rounds abuts, industrial yards etc having size (222x110x100) mm , water absorption not greater than 4% and compressive strength of 40 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,852.25	Tk. 1,850.00	Tk. 1,834.00	Tk. 1,834.00
30.26	Supplying and placing of 100 mm thick coloured uni-block for paving heavy traffic areas, road, industrial yard etc having size (222x110x100) mm , water absorption not greater than 4% and compressive strength of 40 MPa on compacted sand bed of 50 mm on stabilized soil base, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,952.63	Tk. 1,950.00	Tk. 1,934.00	Tk. 1,934.00

30.27	Supplying and placing of grey 200 mm x 100 mm x 60 mm rectangular paving block having water absorption not greater than 4% and compressive strength of 30 MPa on 50 mm thick compacted sand bed, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,226.73	Tk. 1,224.00	Tk. 1,208.00	Tk. 1,208.00
30.28	Supplying and placing of coloured 200 mm x 100 mm x 60 mm rectangular paving block having water absorption not greater than 4% and compressive strength of 30 MPa on 50 mm thick compacted sand bed, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,471.87	Tk. 1,470.00	Tk. 1,453.00	Tk. 1,453.00
30.29	Supplying and placing of grey 200 mm x 100 mm x 80 mm rectangular paving block having water absorption not greater than 4% and compressive strength of 30 MPa on 50 mm thick compacted sand bed, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,655.72	Tk. 1,653.00	Tk. 1,637.00	Tk. 1,637.00
30.30	Supplying and placing of coloured 200 mm x 100 mm x 80 mm rectangular paving block having water absorption not greater than 4% and compressive strength of 30 MPa on 50 mm thick compacted sand bed, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,778.29	Tk. 1,776.00	Tk. 1,759.00	Tk. 1,759.00
30.31	Supplying and placing of grey 200 mm x 100 mm x 100 mm rectangular paving block having water absorption not greater than 4% and compressive strength of 30 MPa on 50 mm thick compacted sand bed, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,839.57	Tk. 1,837.00	Tk. 1,821.00	Tk. 1,821.00
30.32	Supplying and placing of coloured 200 mm x 100 mm x 100 mm rectangular paving block having water absorption not greater than 4% and compressive strength of 30 MPa on 50 mm thick compacted sand bed, and filling all interstices with sand, cleaning etc. completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 1,962.14	Tk. 1,960.00	Tk. 1,943.00	Tk. 1,943.00
30.33	Supplying, carrying, placing, providing of Concrete Kerb Stone having minimum compressive strength 15 MPa of size 600 mm x 300 mm x 100/60 mm all completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 236.91	Tk. 236.00	Tk. 234.00	Tk. 234.00
30.34	Supplying, carrying, placing, providing of Concrete Kerb Stone having minimum compressive strength 15 MPa of size 500 mm x 350 mm x 100 mm all completed by the contractor and accepted by the Engineer-in-charge.	sqm	Tk. 243.04	Tk. 242.00	Tk. 240.00	Tk. 240.00
30.35.1	Supplying and placing of grey engraved cement concrete tiles (250 mm x 250 mm in size) having minimum thickness of 24 mm over 12 mm sand cement mortar (4:1) each at a distance of 6 mm apart and filling the joints with cement sand mortar (1:4) including raking out joints and cutting the tiles to required sizes wherever necessary with high class recessed pointing, cleaning and washing of sand, curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days etc. all complete and accepted by the Engineer-in-charge.					

30.35.1.1	Ground floor	sqm	Tk. 957.56	Tk. 955.00	Tk. 941.00	Tk. 941.00
30.35.1.2	Add for each additional floor above ground floor	sqm	Tk. 10.00	Tk. 10.00	Tk. 10.00	Tk. 10.00
30.35.2	Supplying and placing of coloured engraved cement concrete tiles (250 mm x 250 mm in size) having minimum thickness of 24 mm over 12 mm sand cement mortar (4:1) each at a distance of 6 mm apart and filling the joints with cement sand mortar (1:4) including raking out joints and cutting the tiles to required sizes wherever necessary with high class recessed pointing, cleaning and washing of sand, curing for min 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days etc. all complete and accepted by the Engineer-in-charge.					
30.35.2.1	Ground floor	sqm	Tk. 1,096.24	Tk. 1,094.00	Tk. 1,079.00	Tk. 1,079.00
30.35.2.2	Add for each additional floor above ground floor	sqm	Tk. 10.00	Tk. 10.00	Tk. 10.00	Tk. 10.00
30.36.1	38 mm thick grey cladding work in facing on superstructure with 200 mm x 90 mm x 38 mm size concrete clad pested with 20 mm thick cement mortar (1:4) to be placed at 6 mm apart, filling the interstices with same mortar, raking out joints and cutting the clad to required sizes wherever necessary with high class recessed pointing, cleaning and washing of sand, necessary scaffolding, curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days etc. all complete and accepted by the Engineer-in-charge.					
30.36.1.1	Ground floor	sqm	Tk. 1,579.87	Tk. 1,571.00	Tk. 1,535.00	Tk. 1,535.00
30.36.1.2	Add for each additional floor above ground floor	sqm	Tk. 18.00	Tk. 18.00	Tk. 18.00	Tk. 18.00
30.36.2	38 mm thick coloured cladding work in facing on superstructure with 200 mm x 92 mm x 38 mm size concrete clad pested with 20 mm thick cement mortar (1:4) to be placed at 6 mm apart, filling the interstices with same mortar, raking out joints and cutting the clad to required sizes wherever necessary with high class recessed pointing, cleaning and washing of sand, necessary scaffolding, curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days etc. all complete and accepted by the Engineer-in-charge.					
30.36.2.1	Ground floor	sqm	Tk. 1,706.67	Tk. 1,698.00	Tk. 1,662.00	Tk. 1,662.00
30.36.2.2	Add for each additional floor above ground floor	sqm	Tk. 18.00	Tk. 18.00	Tk. 18.00	Tk. 18.00
30.36.3	50 mm thick grey cladding work in facing on superstructure with 400 mm x 100 mm x 50 mm size concrete clad pested with 20 mm thick cement mortar (1:4) to be placed at 6 mm apart, filling the interstices with same mortar, raking out joints and cutting the clad to required sizes wherever necessary with high class recessed pointing, cleaning and washing of sand, necessary scaffolding, curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days etc. all complete and accepted by the Engineer-in-charge.					
30.36.3.1	Ground floor	sqm	Tk. 1,337.38	Tk. 1,327.00	Tk. 1,282.00	Tk. 1,282.00
30.36.3.2	Add for each additional floor above ground floor	sqm	Tk. 18.00	Tk. 18.00	Tk. 18.00	Tk. 18.00

30.36.4	50 mm thick coloured cladding work in facing on superstructure with 400 mm x 100 mm x 50 mm size concrete clad pested with 20 mm thick cement mortar (1:4) to be placed at 6 mm apart, filling the interstices with same mortar, raking out joints and cutting the clad to required sizes wherever necessary with high class recessed pointing, cleaning and washing of sand, necessary scaffolding, curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days etc. all complete and accepted by the Engineer-in-charge.					
30.36.4.1	Ground floor	sqm	Tk. 1,482.67	Tk. 1,472.00	Tk. 1,427.00	Tk. 1,427.00
30.36.4.2	Add for each additional floor above ground floor	sqm	Tk. 18.00	Tk. 18.00	Tk. 18.00	Tk. 18.00
30.36.5	15 mm thick grey/coloured cladding work in facing on superstructure with 420 mm x 330 mm x 15 mm size concrete clad pested with 20 mm thick cement mortar (1:4) to be placed at 6 mm apart, filling the interstices with same mortar, raking out joints and cutting the clad to required sizes wherever necessary with high class recessed pointing, cleaning and washing of sand, necessary scaffolding, curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days etc. all complete and accepted by the Engineer-in-charge					
30.36.5.1	Ground floor	sqm	Tk. 987.08	Tk. 979.00	Tk. 943.00	Tk. 943.00
30.36.5.2	Add for each additional floor above ground floor	sqm	Tk. 18.00	Tk. 18.00	Tk. 18.00	Tk. 18.00
30.37.1	240 mm thick wall in super structure with Autoclaved Aerated Concrete (AAC) Block of size (600 x 200 x 240 : L x H x W) mm having Compressive Strength not less than 6 MPa, Nominal Dry Density 850 kg/m ³ and Drying Shrinkage not exceeding 0.02% conforming BNBC-2020 (ASTM C1386), laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) and minimum thickness of 10 mm, (or other binding materials which comply with BNBC-2020) filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by the contractor and accepted by the Engineer-in-charge.					
30.37.1.1	Ground floor	cum	Tk. 5,962.24	Tk. 5,943.00	Tk. 5,853.00	Tk. 5,853.00
30.37.1.2	Add for each additional floor above ground floor	cum	Tk. 73.00	Tk. 73.00	Tk. 73.00	Tk. 73.00
30.37.2	200 mm thick wall in super structure with Autoclaved Aerated Concrete (AAC) Block of size (600 x 200 x 200 : L x H x W) mm having Compressive Strength not less than 6 MPa, Nominal Dry Density 850 kg/m ³ and Drying Shrinkage not exceeding 0.02% conforming BNBC-2020 (ASTM C1386), laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) and minimum thickness of 10 mm, (or other binding materials which comply with BNBC-2020) filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by the contractor and accepted by the Engineer-in-charge.					
30.37.2.1	Ground floor	cum	Tk. 6,324.91	Tk. 6,305.00	Tk. 6,215.00	Tk. 6,215.00
30.37.2.2	Add for each additional floor above ground floor	cum	Tk. 73.00	Tk. 73.00	Tk. 73.00	Tk. 73.00

30.37.3	150 mm thick wall in super structure with Autoclaved Aerated Concrete (AAC) Block of size (600 x 200 x 150 : L x H x W) mm having Compressive Strength not less than 6 MPa, Nominal Dry Density 850 kg/m ³ and Drying Shrinkage not exceeding 0.02% conforming BNBC-2020 (ASTM C1386), laying with specified mortar of cement and sand in a ratio 1:4 (sand not less than F.M =1.2) and minimum thickness of 10 mm, (or other binding materials which comply with BNBC-2020) filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding and curing for minimum 3 (three) days on mortar joints by jute /brush in wet, followed by impervious /wet covering for another 4 (four) days curing etc. all completed by the contractor and accepted by the Engineer-in-charge.					
30.37.3.1	Ground floor	cum	Tk. 7,180.19	Tk. 7,161.00	Tk. 7,070.00	Tk. 7,070.00
30.37.3.2	Add for each additional floor above ground floor	cum	Tk. 73.00	Tk. 73.00	Tk. 73.00	Tk. 73.00
30.38.1	240 mm thick wall in super structure with Autoclaved Aerated Concrete (AAC) Block of size (600 x 200 x 240 : L x H x W) mm having Compressive Strength not less than 6 MPa, Nominal Dry Density 850 kg/m ³ and Drying Shrinkage not exceeding 0.02% conforming BNBC-2020 (ASTM C1386), laying with approved polymer modified adhesive mortar having water retention not less than 75% & average compressive strength at 28 days, min 5.2 MPa conforming ASTM C270, Splitting Tensile Strength not less than (0.40 – 0.49 MPa) conforming ASTM C1660, maintaining W/p ratio 25% - 30% and jointing adhesive mortar layer thickness of (3 mm -4 mm), (or other binding materials which comply with BNBC-2020) filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding, cutting and sizing AAC blocks etc. all completed by the contractor and accepted by the Engineer-in-charge.					
30.38.1.1	Ground floor	cum	Tk. 6,124.14	Tk. 6,109.00	Tk. 6,041.00	Tk. 6,041.00
30.38.1.2	Add for each additional floor above ground floor	cum	Tk. 73.00	Tk. 73.00	Tk. 73.00	Tk. 73.00
30.38.2	200 mm thick wall in super structure with Autoclaved Aerated Concrete (AAC) Block of size (600 x 200 x 200 : L x H x W) mm having Compressive Strength not less than 6 MPa, Nominal Dry Density 850 kg/m ³ and Drying Shrinkage not exceeding 0.02% conforming BNBC-2020 (ASTM C1386), laying with approved polymer modified adhesive mortar having water retention not less than 75% & average compressive strength at 28 days, min 5.2 MPa conforming ASTM C270, Splitting Tensile Strength not less than (0.40 – 0.49 MPa) conforming ASTM C1660, maintaining W/p ratio 25% - 30% and jointing adhesive mortar layer thickness of (3 mm -4 mm), (or other binding materials which comply with BNBC-2020) filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding, cutting and sizing AAC blocks etc. all completed by the contractor and accepted by the Engineer-in-charge.					
30.38.2.1	Ground floor	cum	Tk. 6,576.10	Tk. 6,561.00	Tk. 6,492.00	Tk. 6,492.00
30.38.2.2	Add for each additional floor above ground floor	cum	Tk. 73.00	Tk. 73.00	Tk. 73.00	Tk. 73.00

30.38.3	150 mm thick wall in super structure with Autoclaved Aerated Concrete (AAC) Block of size (600 x 200 x 150 : L x H x W) mm having Compressive Strength not less than 6 MPa, Nominal Dry Density 850 kg/m ³ and Drying Shrinkage not exceeding 0.02% conforming BNBC-2020 (ASTM C1386), laying with approved polymer modified adhesive mortar having water retention not less than 75% & average compressive strength at 28 days, min 5.2 MPa conforming ASTM C270, Splitting Tensile Strength not less than (0.40 – 0.49 MPa) conforming ASTM C1660, maintaining W/p ratio 25% - 30% and jointing adhesive mortar layer thickness of (3 mm -4 mm), (or other binding materials which comply with BNBC-2020) filling the interstices with mortar and making bond with connecting walls /frames including necessary scaffolding, cutting and sizing AAC blocks etc. all completed by the contractor and accepted by the Engineer-in-charge.					
30.38.3.1	Ground floor	cum	Tk. 7,508.61	Tk. 7,493.00	Tk. 7,422.00	Tk. 7,422.00
30.38.3.2	Add for each additional floor above ground floor	cum	Tk. 73.00	Tk. 73.00	Tk. 73.00	Tk. 73.00

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CHAPTER 31 : SUB-SOIL INVESTIGATION WORKS & DIGITAL SURVEY

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
	Percussion Method of Drilling					
31.1	Mobilization and demobilization of boring equipment and man-power: at site (drilling rig comprising drilling pipe, drop hammer, tripod, pulley, chain, wrangle, sample collection devices etc tools and plants; tripod for temporary camp, necessary work-force etc) (Once for one site)	per site	Tk. 14,701.59	Tk. 14,585.00	Tk. 14,252.00	Tk. 14,252.00
31.2	Sub - Soil investigation by 100 mm dia percussion wash boring including collecting disturbed and undisturbed soil samples in numbers as required for classification of soil, conducting SPT using auto trip hammer, stratification of layers, analysing physical parameters of soils like Atterberg limits, specific gravity, grain size distribution (by wet sieve, hydrometer if required), ground water table location, direct shear test, unconfined compression test, unit weight (dry/weight), natural moisture content; C - ϕ values and other strength parameters to ascertain bearing capacity, skin friction, end bearings etc at every 1.5m interval as per respective national/international standards and entering all these data & information in necessary tables & graphs and finally furnishing them in the form of standard sub-soil investigation report with CD containing video of sub-soil investigation operation of concerned site duly signed by competent engineer & exploratory office.					
31.2.1	Bore hole depth from 0 to 20 m	per bore hole	Tk. 33,879.95	Tk. 33,722.00	Tk. 33,259.00	Tk. 33,259.00
31.2.2	Additional charge for bore hole depth above 20.0 m and upto 30.0 m	meter	Tk. 495.30	Tk. 491.00	Tk. 480.00	Tk. 480.00
31.2.3	Additional charge for bore hole depth above 30.0 m and upto 40.0 m	meter	Tk. 540.61	Tk. 535.00	Tk. 523.00	Tk. 523.00

31.2.4	Additional charge for bore hole depth above 40.0 m and upto 50.0 m	meter	Tk. 570.57	Tk. 565.00	Tk. 553.00	Tk. 553.00
31.3	Sub-soil investigation by 100 mm dia wash boring including collecting disturbed and undisturbed soil samples in numbers as required for classification of soil, conducting SPT using auto trip hammer, stratification of layers, and entering all these data & information in necessary tables & graphs, furnishing them in the form of standard sub-soil investigation report duly signed by competent engineer & exploratory office.(Only after written advice from concerned design division/circle)					
31.3.1	Bore hole depth from 0 to 20m	per bore hole	Tk. 9,308.81	Tk. 9,226.00	Tk. 9,042.00	Tk. 9,042.00
31.3.2	Additional charge for bore hole depth above 20.0 m and upto 30.0 m	meter	Tk. 408.63	Tk. 404.00	Tk. 394.00	Tk. 394.00
31.3.3	Additional charge for bore hole depth above 30.0 m and upto 40.0 m	meter	Tk. 453.19	Tk. 448.00	Tk. 437.00	Tk. 437.00
31.3.4	Additional charge for bore hole depth above 40.0 m and upto 50.0 m	meter	Tk. 482.54	Tk. 477.00	Tk. 466.00	Tk. 466.00
31.4	Added cost per bore hole, if the same is under water by more than 1.5 m in depth	per bore hole	Tk. 6,954.11	Tk. 6,903.00	Tk. 6,643.00	Tk. 6,643.00
	Rotary Method of Drilling					
31.5	Mobilization and demobilization of boring equipment and man-power: at site (rotary hydraulic drilling rig comprising mud pump, drilling pipe, auto trip hammer, pulley, chain, wrangle, sample collection devices (shelby tube, split spoon sampler) etc tools and plants; tripol for temporary camp, necessary work-force etc) (Once for one site) (Only after written advice from concerned design division/circle)	per site	Tk. 25,784.53	Tk. 25,785.00	Tk. 25,785.00	Tk. 25,785.00
31.6	Sub - Soil investigation by 100 mm dia rotary wash boring and including collecting disturbed and undisturbed soil samples in numbers as required for classification of soil, conducting SPT using auto trip hammer, stratification of layers, analysing physical parameters of soils like Atterberg limits, specific gravity, grain size distribution(by wet seive, hydrometer if required) , ground water table location, direct shear test, unconfined compression test, unit weight(dry/weight), natural moisture content ; C - ϕ values and other strength parameters to ascertain bearing capacity , skin friction, end bearings etc at every 1.5m interval as per respective national/international standards and entering all these data & information in necessary tables & graphs and finally furnishing them in the form of standard sub-soil investigation report with CD containing video of sub-soil investigation operation of concerned site duly signed by competent engineer & exploratory office. (Only after written advice from concerned lead content, if any, sha					
31.6.1	Bore hole depth from 0 to 20 m	per bore hole	Tk. 52,846.35	Tk. 52,846.00	Tk. 52,846.00	Tk. 52,846.00
31.6.2	Additional charge for bore hole depth above 20.0 m and upto 30.0 m	meter	Tk. 899.88	Tk. 900.00	Tk. 900.00	Tk. 900.00
31.6.3	Additional charge for bore hole depth above 30.0 m and upto 40.0 m	meter	Tk. 1,027.94	Tk. 1,028.00	Tk. 1,028.00	Tk. 1,028.00
31.6.4	Additional charge for bore hole depth above 40.0 m and upto 50.0 m	meter	Tk. 1,159.29	Tk. 1,159.00	Tk. 1,159.00	Tk. 1,159.00
	Downhole Siesmic Testing					

31.7	Mobilization of downhole seismic testing equipments such as a hammer weighing 10lb, a wooden Beam of L= 2.40m and B & t = 150mm, Receiver borehole pick with 3D geophones to measure the velocity and travel time of P and S wave inside the borehole and 2nos. 1D Geophone to measure surface velocity of P and S wave. Trigger: Sending signal to control unit to record First Arrival Time (FAT) and velocity of P and S wave. Control and Processing Unit with Laptop (Only after written advice from concerned design division/circle)	per test	Tk. 9,734.29	Tk. 9,734.00	Tk. 9,734.00	Tk. 9,734.00
31.8	Downhole Seismic Testing- The test is to be conducted in accordance to ASTM D7400-08 to determine the interval velocities (VS & VP) from arrival times of compression (P) and shear (S) seismic waves which are generated near surface and travel down to an array of vertically installed seismic sensors. This item includes the necessary cost for entering all these data & information in necessary tables & graphs and finally furnishing them in the form of standard sub-soil investigation report with CD containing video of sub-soil investigation operation of concerned site duly signed by competent engineer & exploratory office. (Only after written advice from concerned design division/circle)					
31.8.1	For first test	per test	Tk. 125,540.52	Tk. 125,541.00	Tk. 125,541.00	Tk. 125,541.00
31.8.2	For each consecutive test after first test	per test	Tk. 113,282.52	Tk. 113,283.00	Tk. 113,283.00	Tk. 113,283.00
31.9	Conducting plate load test as per ASTM D1194 or equivalent standard providing required arrangement of three circular steel plates with minimum thickness of 25 mm and varying in diameter from 305 mm to 762 mm respectively or square steel bearing plates of equivalent area for minimum 50 tons loading capacity of hydraulic or mechanical jack assembly and applying load in 2 cycles with maintaining time interval of each loading increment of not less than 15 minutes. Continue each test until a peak load is reached or until the ratio of load increment to settlement increment reaches a minimum, steady magnitude. If sufficient load is available, continue the test until the total settlement reaches at least 10 percent of the plate diameter, unless a well defined failure load is observed and all other required materials, excavation, protection against collapse, preparation of test pit, calibration of hydraulic jacks with pressure gauge (with an error not exceeding $\pm 2\%$ of the load increment used) & dial gauges (an accpe, Relief at 14.6 Bar) etc. as per requirement all complete, approved and accepted by the Engineer-in-charge. (All the welding joints must be TIG welded)(Rate is excluding the cost of shed, perimeter fencing and RCC floor).ademark, b) the number of Bangladesh standard and c) country of origin. Each product shall also be marked with the BSTI Certification Mark. The fixure should be placed in position with heavy type C.I. Brackets. 30 mm dia PVC waste water pipe with brass coupling (not exceeding 750 mm in length), basin waste with chain plug including making holes in walls and floors and fitting with wall, screws and mending good the damages, finishing etc. all complete approved and accepted by the Engineer-in-charge.e.paring the base with cement conc	per test	Tk. 90,549.17	Tk. 90,288.00	Tk. 88,908.00	Tk. 88,908.00
31.9.1	Per test rate for conducting additional plate load test at the same site after first 03(Three) tests.	per test	Tk. 28,520.49	Tk. 28,222.00	Tk. 27,101.00	Tk. 27,101.00

31.9.2	Additional charge per km (beyond 25 Km) outside Dhaka city (two ways)/any city for mobilization & demobilization	per km	Tk. 193.35	Tk. 193.00	Tk. 193.00	Tk. 193.00
31.10	Carrying out different tests on soil samples in BUET laboratory collected from the sub-soil investigation site accepted by the Engineer-in-Charge. (Only after written advice from concern design division/circle)					
31.10.1	Void ratio	per set of test	Tk. 7,700.00	Tk. 7,700.00	Tk. 7,700.00	Tk. 7,700.00
31.10.2	Organic matter content	per set of test	Tk. 4,500.00	Tk. 4,500.00	Tk. 4,500.00	Tk. 4,500.00
31.10.3	Permeability for cohesive soil	per set of test	Tk. 24,000.00	Tk. 24,000.00	Tk. 24,000.00	Tk. 24,000.00
31.10.4	Permeability non-cohesive soil	per set of test	Tk. 9,500.00	Tk. 9,500.00	Tk. 9,500.00	Tk. 9,500.00
31.10.5	Consolidation test for cohesive soil	per set of test	Tk. 24,000.00	Tk. 24,000.00	Tk. 24,000.00	Tk. 24,000.00
31.10.6	Triaxial shear Test					
31.10.6.1	Consolidated undrained triaxial test with pore pressure for cohesive and non co-hesive soil	per set of test	Tk. 52,000.00	Tk. 52,000.00	Tk. 52,000.00	Tk. 52,000.00
31.10.6.2	Consolidated undrained triaxial test without pore pressure for cohesive soil	per set of test	Tk. 46,000.00	Tk. 46,000.00	Tk. 46,000.00	Tk. 46,000.00
31.10.6.3	Consolidated drained triaxial test	per set of test	Tk. 52,000.00	Tk. 52,000.00	Tk. 52,000.00	Tk. 52,000.00
31.10.6.4	Unconsolidated undrained triaxial test without pore pressure	per set of test	Tk. 24,000.00	Tk. 24,000.00	Tk. 24,000.00	Tk. 24,000.00
31.10.7	One dimensional swelling / settlement potential of cohesive soil.	per set of test	Tk. 10,000.00	Tk. 10,000.00	Tk. 10,000.00	Tk. 10,000.00
31.10.8	Modified proctor compaction test	per set of test	Tk. 20,000.00	Tk. 20,000.00	Tk. 20,000.00	Tk. 20,000.00
	Detail digital topographic & physical feature survey					
31.11.1	Mobilization and demobilization of survey equipment and man-power: survey equipment comprising Total Station, RTK-GPS, level machine, Prism with stand, Tripod and others necessary tools & plants, triple for temporary camp and necessary work-force.	per site	Tk. 11,390.06	Tk. 11,390.00	Tk. 11,257.00	Tk. 11,257.00
31.11.2	Detail digital topographic & physical feature survey of project area. The survey will cover the location and alignment of all topographic features inside the project area such as: Boundary, roads, exposed surface drains, walk ways, existing physical features such as trees, structures, exposed utility network etc. Spot land levels of the project area will be surveyed for contouring@ 5 m C/C. GPS should be used for establishing co-ordinate of global axis-Latitude/Longitude. Total station should be used for survey of spot level, detailed physical features etc. Field data shall be processed in GIS or Auto CAD Software. Finally layout map with all topographic and physical features shall be prepared using Auto CAD Software. All data & maps should be Submitted both in hard and soft format in 5(five) copy.	per Acre	Tk. 14,955.29	Tk. 16,291.00	Tk. 16,157.00	Tk. 16,157.00

..... END OF CHAPTER THIRTY ONE.....

CHAPTER 32: REPAIR WORKS

Item No.	Description of Item	Unit	Unit Rate (Dhaka, Mymensingh)	Unit Rate (Chattogram, Sylhet)	Unit Rate (Khulna, Barisal, Gopalganj)	Unit Rate (Rajshahi, Rangpur)
32.1	Stripping of unserviceable plaster including racking out joints, cleaning etc.	sqm	Tk. 30.92	Tk. 30.00	Tk. 28.00	Tk. 28.00

32.2.1	Picking-up of unserviceable patent stone from floor and dado etc. and removal of debris to a safe distance.	sqm	Tk. 46.69	Tk. 45.00	Tk. 43.00	Tk. 43.00
32.2.2	Picking-up of unserviceable tiles including bonding mortar from floor, walls and dado etc. and removal of debris to a safe distance.	sqm	Tk. 61.41	Tk. 60.00	Tk. 56.00	Tk. 56.00
32.3	Picking up of damaged/unserviceable terraced (lime concrete) floor or roof etc. and removal of debris to a safe distance.	sqm	Tk. 64.91	Tk. 63.00	Tk. 59.00	Tk. 59.00
32.4	Dismantling of unserviceable/damaged terraced roof including beam, column, barga & tiles etc. and removal of debris to a safe distance.	sqm	Tk. 164.30	Tk. 161.00	Tk. 153.00	Tk. 153.00
32.5.1.1	Dismantling of unserviceable/damaged brick works (with cement or lime mortar) of thickness 250 mm in foundation and superstructure and removal of debris to a safe distance.	cum	Tk. 537.85	Tk. 524.00	Tk. 491.00	Tk. 491.00
32.5.1.2	Dismantling of unserviceable/damaged brick works (with cement or lime mortar) of thickness 75/125 mm in foundation and superstructure and removal of debris to a safe distance.	sqm	Tk. 62.27	Tk. 61.00	Tk. 57.00	Tk. 57.00
32.5.2	Dismantling of plain concrete made of lime or cement and removal of debris to a safe distance.	cum	Tk. 1,647.28	Tk. 1,606.00	Tk. 1,503.00	Tk. 1,503.00
32.5.3	Dismantling of brick flat soling and removal of debris to a safe distance and removal of debris to a safe distance.	sqm	Tk. 17.77	Tk. 17.00	Tk. 16.00	Tk. 16.00
32.5.4	Dismantling of H.B.B. Brick soling and removal of debris to a safe distance and removal of debris to a safe distance.	sqm	Tk. 23.62	Tk. 23.00	Tk. 21.00	Tk. 21.00
32.6.1.1	Dismantling of R.C.C. beam, lintel, column, drop wall, sun shade from any height and removal of debris to a safe distance.	cum	Tk. 3,928.33	Tk. 3,841.00	Tk. 3,612.00	Tk. 3,612.00
32.6.1.2	Dismantling of R.C.C. Railing of any kind, jally brickwork, and hollow block wall from any height and removal of debris to a safe distance.	sqm	Tk. 217.98	Tk. 214.00	Tk. 202.00	Tk. 202.00
32.6.2	Dismantling of R.C.C. in roof slab and removal of debris to a safe distance.	cum	Tk. 3,319.48	Tk. 3,244.00	Tk. 3,065.00	Tk. 3,065.00
32.6.3	Recovery of reinforcing bars and metals during dismantling of RCC including cleaning, making bundle, stacking or storing as per direction of engineer in charge and removal of debris to a safe distance.	quintal	Tk. 552.16	Tk. 540.00	Tk. 503.00	Tk. 503.00
32.6.4	Recovery of bricks during dismantling of brick works including cleaning, stacking or storing etc. all complete as per direction of engineer in charge and removal of debris to a safe distance.	% 0 nos	Tk. 1,484.70	Tk. 1,458.00	Tk. 1,350.00	Tk. 1,350.00
32.6.5	Recovery of bricks during dismantling of soling including cleaning, stacking or storing all complete as per direction of engineer in charge and removal of debris to a safe distance.	% 0 nos	Tk. 998.80	Tk. 981.00	Tk. 908.00	Tk. 908.00
32.6.6	Recovery of brick bats during dismantling of soling, brick works, HBB etc. including cleaning, stacking or storing as per direction of engineer in charge and removal of debris to a safe distance.	cum	Tk. 249.70	Tk. 245.00	Tk. 227.00	Tk. 227.00
32.7	Repairs to minor/fine (non-structural affecting aesthetic) cracks in brick walls, floor and roof etc.	meter	Tk. 41.71	Tk. 41.00	Tk. 38.00	Tk. 38.00
32.8.1	Cutting and polishing of old mosaic on wall with pumice stone etc.	sqm	Tk. 99.73	Tk. 98.00	Tk. 92.00	Tk. 92.00
32.8.2	Cutting and polishing of old mosaic on floor with pumice stone and mosaic cutting machine etc.	sqm	Tk. 111.69	Tk. 110.00	Tk. 106.00	Tk. 106.00
32.8.3	Cleaning and washing the mosaic floor etc.	sqm	Tk. 19.63	Tk. 19.00	Tk. 18.00	Tk. 18.00
32.8.4	Cleaning and refilling of wornout joints between glazed tiles, marble etc. in floor or walls.	sqm	Tk. 71.20	Tk. 69.00	Tk. 65.00	Tk. 65.00
32.9	Repairing of terraced roof cracks by cutting in 'traphozoided' shape	meter	Tk. 539.23	Tk. 531.00	Tk. 501.00	Tk. 500.00

32.10	Taking out and relaying of old brick soling including replacement of 10% brick	sqm	Tk. 128.22	Tk. 127.00	Tk. 120.00	Tk. 119.00
32.11	Taking out and relaying of brick soling after redressing the surfaces	sqm	Tk. 84.91	Tk. 84.00	Tk. 78.00	Tk. 78.00
32.12.1	Taking out of old brick on edge pavement and resetting (Herring bone bond) including replacement of 10% brick	sqm	Tk. 209.86	Tk. 208.00	Tk. 197.00	Tk. 196.00
31.12.2	Taking out of old brick on edge pavement and resting Herring bone bond and redressing the surface	sqm	Tk. 117.30	Tk. 116.00	Tk. 107.00	Tk. 107.00
32.13	Taking out of old roof tiles and ridges	sqm	Tk. 66.69	Tk. 65.00	Tk. 61.00	Tk. 61.00
32.14	Supplying and fitting-fixing of wind tie with M.S. Flat bar clamp	meter	Tk. 565.42	Tk. 562.00	Tk. 550.00	Tk. 550.00
32.15	Supplying and fitting-fixing of J-hook including nuts, washers	each	Tk. 48.89	Tk. 49.00	Tk. 48.00	Tk. 48.00
32.16	Supplying and fitting-fixing of M.S. U-hook including nuts, washers	each	Tk. 64.41	Tk. 64.00	Tk. 63.00	Tk. 63.00
32.17	Taking out of old & unserviceable wooden chowkat /door or window frame	each	Tk. 328.23	Tk. 318.00	Tk. 301.00	Tk. 301.00
32.18	Taking out of old wooden chowkat/door or window frame and refitting and refixing after minor repair (excluding the cost of wood and M.S. clamp)	each	Tk. 1,058.00	Tk. 1,042.00	Tk. 972.00	Tk. 972.00
32.19	Taking out and refitting of wooden shutter after easing excluding the cost of wood	sqm	Tk. 449.64	Tk. 445.00	Tk. 413.00	Tk. 413.00
32.20.1	Thorough repairs to wooden door and window Pannel shutter excluding the cost of wood	sqm	Tk. 729.96	Tk. 723.00	Tk. 672.00	Tk. 672.00
32.20.2	Thorough repairs to wooden door and window Batten shutter excluding the cost of wood	sqm	Tk. 677.27	Tk. 671.00	Tk. 626.00	Tk. 626.00
32.20.3	Thorough repairs to wooden door and window Venetian shutter excluding the cost of wood	sqm	Tk. 657.14	Tk. 651.00	Tk. 605.00	Tk. 605.00
32.20.4	Thorough repairs to wooden door and window Glazed shutter excluding the cost of glass & wood	sqm	Tk. 627.23	Tk. 622.00	Tk. 578.00	Tk. 578.00
32.21.1	Renewing of 3 mm glass panes in wooden frame including supply of glass	sqm	Tk. 1,024.42	Tk. 1,020.00	Tk. 1,000.00	Tk. 1,000.00
32.21.2	Renewing of 3 mm glass panes in steel frame including supply of glass	sqm	Tk. 1,130.01	Tk. 1,125.00	Tk. 1,103.00	Tk. 1,103.00
32.22	Renewing of 5 mm glass panes including supply of glass	sqm	Tk. 1,347.72	Tk. 1,344.00	Tk. 1,329.00	Tk. 1,329.00
32.23	Cleaning and washing of glass panes including supply of necessary fresh putty, pins etc.	per pane	Tk. 3.53	Tk. 3.00	Tk. 3.00	Tk. 3.00
32.24.1	Replacing / renewing of fly proof wire net (with all new materials) the excluding cost of wood	sqm	Tk. 767.81	Tk. 767.00	Tk. 726.00	Tk. 726.00
32.24.2	Replacing / renewing of synthetic fly proof net (with all new materials) the excluding cost of wood	sqm	Tk. 795.42	Tk. 795.00	Tk. 753.00	Tk. 753.00
32.25	Renewing of fly proof wire netting (with old and new materials) the excluding cost of wood	sqm	Tk. 737.24	Tk. 736.00	Tk. 695.00	Tk. 695.00
32.26	Renewing of 12 x 12 mm mesh wire netting the excluding cost of wood	sqm	Tk. 629.87	Tk. 629.00	Tk. 595.00	Tk. 595.00
32.27	Taking out of hard board ceiling, walling	sqm	Tk. 21.57	Tk. 21.00	Tk. 20.00	Tk. 20.00
32.28	Renewing of hard board ceiling or walling excluding cost of wood	sqm	Tk. 656.64	Tk. 656.00	Tk. 628.00	Tk. 628.00
32.29	Replacing of wooden cleat, making, finishing and fixing with chowkat by means of 40 mm hinge	each	Tk. 79.78	Tk. 79.00	Tk. 79.00	Tk. 79.00
32.30	Replacing of cleat with brass hinge	each	Tk. 88.93	Tk. 88.00	Tk. 87.00	Tk. 87.00
32.31	Replacing of wooden buffer block	each	Tk. 59.63	Tk. 59.00	Tk. 59.00	Tk. 59.00
32.32	Replacig of wooden catcher block	each	Tk. 88.15	Tk. 87.00	Tk. 88.00	Tk. 88.00
32.33.1	Replacing of 75 mm long galvanized iron catch hook	each	Tk. 16.87	Tk. 17.00	Tk. 16.00	Tk. 16.00
32.33.2	Replacing of 100 mm long galvanized iron catch hook	each	Tk. 19.33	Tk. 19.00	Tk. 19.00	Tk. 19.00
32.33.3	Replacing of 150 mm long galvanized iron catch hook	each	Tk. 21.78	Tk. 22.00	Tk. 21.00	Tk. 21.00
32.33.4	Replacing of 200 mm long galvanized iron catch hook	each	Tk. 24.23	Tk. 24.00	Tk. 23.00	Tk. 23.00
32.33.5	Replacing of 225 mm long galvanized iron catch hook	each	Tk. 31.60	Tk. 32.00	Tk. 31.00	Tk. 31.00
32.34	Replacing of iron socket bolt : 50 mm dia, 100 mm long	each	Tk. 29.99	Tk. 30.00	Tk. 29.00	Tk. 29.00
32.35.1	Replacing of SS socket bolt : 50 mm dia, 100 mm long (SS)	each	Tk. 65.57	Tk. 66.00	Tk. 65.00	Tk. 65.00

32.35.2	Replacing of iron socket bolt : 50 mm dia, 150 mm long	each	Tk. 36.16	Tk. 36.00	Tk. 35.00	Tk. 35.00
32.35.3	Replacing of SS socket bolt : 50 mm dia, 150 mm long(SS)	each	Tk. 94.44	Tk. 94.00	Tk. 93.00	Tk. 93.00
32.35.4	Replacing of iron socket bolt : 50 mm dia, 200 mm long	each	Tk. 51.46	Tk. 51.00	Tk. 51.00	Tk. 51.00
32.35.5	Replacing of SS socket bolt : 50 mm dia, 200 mm long(SS)	each	Tk. 117.72	Tk. 118.00	Tk. 117.00	Tk. 117.00
32.36.1	Replacing of brass socket bolt: 50 mm dia, 150 mm long	each	Tk. 170.48	Tk. 170.00	Tk. 170.00	Tk. 170.00
32.36.2	Replacing of brass socket bolt : 50 mm dia, 200 mm long	each	Tk. 216.76	Tk. 217.00	Tk. 216.00	Tk. 216.00
32.36.3	Replacing of brass socket bolt : 50 mm dia, 250 mm long	each	Tk. 266.19	Tk. 266.00	Tk. 265.00	Tk. 265.00
32.37.1	Replacing of 75 mm long iron hinge	each	Tk. 52.07	Tk. 52.00	Tk. 51.00	Tk. 51.00
32.37.2	Replacing of 100 mm long iron hinge	each	Tk. 82.17	Tk. 82.00	Tk. 81.00	Tk. 81.00
32.38.1	Replacing of 75 mm long brass hinge	each	Tk. 164.34	Tk. 164.00	Tk. 163.00	Tk. 163.00
32.38.2	Replacing of 100 mm long brass hinge	each	Tk. 216.76	Tk. 217.00	Tk. 216.00	Tk. 216.00
32.39.1	Replacing of 50 mm dia iron ring	each	Tk. 31.83	Tk. 32.00	Tk. 31.00	Tk. 31.00
32.39.2	Replacing of 50 mm dia brass ring	each	Tk. 64.96	Tk. 65.00	Tk. 64.00	Tk. 64.00
32.40.1	Supplying, fitting and fixing of 25 x 25 mm expanded mesh metal net of wooden or steel frame (excluding the cost of wood or steel sections and painting)	sqm	Tk. 1,107.62	Tk. 1,107.00	Tk. 1,042.00	Tk. 1,042.00
32.40.2	Supplying, fitting and fixing of 25 x 25 mm expanded mesh metal net in F.I. Bar frame (including the cost of F.I. Bar)	sqm	Tk. 2,082.51	Tk. 2,082.00	Tk. 2,000.00	Tk. 2,000.00
32.41.1	Renewing of white washing: 1 coat	sqm	Tk. 14.90	Tk. 15.00	Tk. 14.00	Tk. 14.00
32.41.2	Renewing of white washing: 2 coats	sqm	Tk. 21.84	Tk. 22.00	Tk. 20.00	Tk. 20.00
32.42.1	Renewing of colour washing with yellow ochre: 1 coat	sqm	Tk. 23.08	Tk. 23.00	Tk. 22.00	Tk. 22.00
32.42.2	Renewing of colour washing with yellow ochre: 2 coats	sqm	Tk. 26.06	Tk. 26.00	Tk. 24.00	Tk. 24.00
32.43.1	Renewing of Exterior standard acrylic emulsion paint of approved brand and quality	sqm	Tk. 153.38	Tk. 152.00	Tk. 144.00	Tk. 144.00
32.43.2	Renewing of Exterior premium acrylic emulsion paint of approved brand and quality	sqm	Tk. 176.69	Tk. 175.00	Tk. 166.00	Tk. 166.00
32.44.1	Renewing of standard synthetic enamel paint to door-windows frame and shutter:1coats	sqm	Tk. 118.65	Tk. 118.00	Tk. 111.00	Tk. 111.00
32.44.2	Renewing of premium synthetic enamel paint to door-windows frame and shutter: 1coats	sqm	Tk. 122.95	Tk. 122.00	Tk. 115.00	Tk. 115.00
32.45.1	Renewing of standard synthetic enamel paint to door-windows frame and shutter: 2 coat	sqm	Tk. 182.70	Tk. 181.00	Tk. 170.00	Tk. 170.00
32.45.2	Renewing of premium synthetic enamel paint to door-windows frame and shutter: 2coat	sqm	Tk. 190.50	Tk. 189.00	Tk. 178.00	Tk. 178.00
32.46.1	Renewing of synthetic polyvinyl distemper : 1 coats	sqm	Tk. 95.57	Tk. 95.00	Tk. 89.00	Tk. 89.00
32.46.2	Renewing of synthetic polyvinyl distemper : 2 coat	sqm	Tk. 137.67	Tk. 137.00	Tk. 128.00	Tk. 128.00
32.47.1	Renewing of Interior standard acrylic emulsion paint (Plastic) to walls and ceiling: 1 coat	sqm	Tk. 107.86	Tk. 107.00	Tk. 101.00	Tk. 101.00
32.47.2	Renewing of Interior Premium acrylic emulsion painting (silky finish)to walls and ceiling: 1 coat	sqm	Tk. 124.35	Tk. 124.00	Tk. 117.00	Tk. 117.00
32.47.3	Renewing of Interior super premium acrylic emulsion painting (odorless) to walls and ceiling: 1 coat	sqm	Tk. 144.27	Tk. 143.00	Tk. 137.00	Tk. 137.00
32.48.1	Renewing of Interior standard acrylic emulsion paint (Plastic) to walls and ceiling: 2 coat	sqm	Tk. 155.22	Tk. 154.00	Tk. 145.00	Tk. 145.00
32.48.2	Renewing of Interior Premium acrylic emulsion painting (silky finish)to walls and ceiling: 2 coat	sqm	Tk. 178.78	Tk. 178.00	Tk. 169.00	Tk. 169.00
32.48.3	Renewing of Interior super premium acrylic emulsion painting (odorless) to walls and ceiling 2 coat	sqm	Tk. 198.66	Tk. 198.00	Tk. 189.00	Tk. 189.00
32.49	Washing of painted wall surface with suitable soap or any other detergent	sqm	Tk. 35.85	Tk. 35.00	Tk. 33.00	Tk. 33.00
32.50.1	Renewing of chlorinated rubber paint: 1 coats	sqm	Tk. 190.68	Tk. 190.00	Tk. 181.00	Tk. 181.00
32.50.2	Renewing of chlorinated rubber paint: 2 coat	sqm	Tk. 273.97	Tk. 272.00	Tk. 259.00	Tk. 259.00

32.51.1	Renewing of french polishing: 1 coats	sqm	Tk. 130.13	Tk. 129.00	Tk. 123.00	Tk. 123.00
32.51.2	Renewing of french polishing: 2 coat	sqm	Tk. 223.26	Tk. 222.00	Tk. 212.00	Tk. 212.00
32.51.3	Providing of 2 coats laque spray painting with spray machine to high decorative Teak or similar wooden furniture	sqm	Tk. 181.14	Tk. 180.00	Tk. 174.00	Tk. 174.00
32.52.1	Renewing of painting with synthetic black enamel Paint to plinth and skirting :1 coats	sqm	Tk. 117.24	Tk. 116.00	Tk. 110.00	Tk. 110.00
32.52.2	Renewing of painting with synthetic black enamel Paint to plinth and skirting :2 coat	sqm	Tk. 167.61	Tk. 166.00	Tk. 157.00	Tk. 157.00
32.53	Renewing of painting to C.I. Roof with red corrugal paint: 2 coats	sqm	Tk. 162.03	Tk. 161.00	Tk. 152.00	Tk. 152.00
32.54.1	Renewing earth oiling with solignum or equivalent materials: 1 coat	sqm	Tk. 96.94	Tk. 96.00	Tk. 90.00	Tk. 90.00
32.54.2	Renewing earth oiling with solignum or equivalent materials: 2 coat	sqm	Tk. 145.40	Tk. 144.00	Tk. 136.00	Tk. 136.00
32.55.1	Renewing of wood varnishing with varnish (yacht)	sqm	Tk. 116.38	Tk. 116.00	Tk. 110.00	Tk. 110.00
32.55.2	Renewing of wood varnishing with varnish (egg-shell)	sqm	Tk. 123.30	Tk. 122.00	Tk. 116.00	Tk. 116.00
32.55.3	Renewing of wood varnishing with varnish (matt)	sqm	Tk. 124.44	Tk. 124.00	Tk. 117.00	Tk. 117.00
32.56	Scrapping of old paint from wood	sqm	Tk. 47.24	Tk. 46.00	Tk. 43.00	Tk. 43.00
32.57	Scrapping of old mosses from wall and roof	sqm	Tk. 34.42	Tk. 34.00	Tk. 31.00	Tk. 31.00
32.58	Renewing of painting to old 100 mm dia rain-water pipe	meter	Tk. 38.61	Tk. 38.00	Tk. 36.00	Tk. 36.00
32.59	Renewing of painting to old 50mm dia G.I. Pipe: 1 coat	meter	Tk. 19.32	Tk. 20.00	Tk. 18.00	Tk. 18.00
32.60	Stopping of roof leakage of C.I. sheet roof with putty	sqm	Tk. 31.63	Tk. 31.00	Tk. 29.00	Tk. 29.00
32.61	Labour for taking out C.I. sheet roofing	sqm	Tk. 39.69	Tk. 39.00	Tk. 37.00	Tk. 37.00
32.62	Labour for taking out and refitting of old G.I. Ridge/valley	meter	Tk. 160.05	Tk. 158.00	Tk. 147.00	Tk. 147.00
32.63	Labour for C.I. Sheet roofing including supply of limpet, washer, screws, putty etc.	sqm	Tk. 126.54	Tk. 124.00	Tk. 115.00	Tk. 115.00
32.64	Labour for C.I. Sheet walling including supply of limpet, washer, screws, putty etc.	sqm	Tk. 94.90	Tk. 93.00	Tk. 86.00	Tk. 86.00
32.65	Labour for taking out barbed wire fencing	meter	Tk. 9.22	Tk. 9.00	Tk. 8.00	Tk. 8.00
32.66	Labour charge for refitting of barbed wire fencing	meter	Tk. 19.43	Tk. 19.00	Tk. 18.00	Tk. 18.00
32.67	Labour charge for tightening of old loose barbed wire fencing	meter	Tk. 2.41	Tk. 2.00	Tk. 2.00	Tk. 2.00
32.68	Renewing, supplying and fitting-fixing of 300 mm long F.I. Clamp in CC (1:2:4)	each	Tk. 107.12	Tk. 107.00	Tk. 104.00	Tk. 103.00
32.69	Renewing, supplying and fitting-fixing of 150 mm long F.I. Clamp in CC (1:2:4)	each	Tk. 74.42	Tk. 74.00	Tk. 71.00	Tk. 71.00
32.70	Cutting jungles and shrub including removing and cleaning the spot/site	sqm	Tk. 6.64	Tk. 7.00	Tk. 6.00	Tk. 6.00
32.71	Supplying of the following items related to ARBORICULTURE work approved by the officer-in-charge					
32.71.1	200mm size earthen tub	each	Tk. 51.54	Tk. 52.00	Tk. 52.00	Tk. 52.00
32.71.2	250mm size earthen tub	each	Tk. 61.35	Tk. 61.00	Tk. 61.00	Tk. 61.00
32.71.3	300mm size earthen tub	each	Tk. 85.89	Tk. 86.00	Tk. 86.00	Tk. 86.00
32.71.4	350mm size earthen tub	each	Tk. 116.57	Tk. 117.00	Tk. 117.00	Tk. 117.00
32.71.5	400mm size earthen tub	each	Tk. 202.46	Tk. 202.00	Tk. 202.00	Tk. 202.00
32.71.6	300 mm x 300 mm size earthen plate	each	Tk. 36.81	Tk. 37.00	Tk. 37.00	Tk. 37.00
32.71.7	Supply of mustard oil-cake	kg	Tk. 56.44	Tk. 56.00	Tk. 56.00	Tk. 56.00
32.71.8	Supply of Bone mele	kg	Tk. 55.22	Tk. 55.00	Tk. 55.00	Tk. 55.00
32.71.9	Supply of steel spade	each	Tk. 552.16	Tk. 552.00	Tk. 552.00	Tk. 552.00
32.71.10	Supply of Urea fertilizer	kg	Tk. 30.68	Tk. 31.00	Tk. 31.00	Tk. 31.00
32.71.11	Supply of Murate of potash	kg	Tk. 26.99	Tk. 27.00	Tk. 27.00	Tk. 27.00
32.71.12	Supply of T. S. P.	kg	Tk. 36.81	Tk. 37.00	Tk. 37.00	Tk. 37.00
32.71.13	Supply of 5 liter capacity spray machine (plaotic)	each	Tk. 2,208.65	Tk. 2,209.00	Tk. 2,209.00	Tk. 2,209.00

32.71.14	Supply of 450 mm size Steel funna with wooden handle	each	Tk. 454.00	Tk. 454.00	Tk. 454.00	Tk. 454.00
32.71.15	Supply of 225 mm long hand hoe with wooden handle	each	Tk. 122.70	Tk. 123.00	Tk. 123.00	Tk. 123.00
32.71.16	Supply of standard steel hand trolley as per direction of officer-in-charge	each	Tk. 7,607.57	Tk. 7,608.00	Tk. 7,608.00	Tk. 7,608.00
32.71.17	Supply of China made or equivalent Hedge Scissors	each	Tk. 1,472.43	Tk. 1,472.00	Tk. 1,472.00	Tk. 1,472.00
32.71.18	Supply of 250 mm size China made or equivalent Secature	each	Tk. 797.57	Tk. 798.00	Tk. 798.00	Tk. 798.00
32.71.19	Supply of 2.5 gallon capacity 22 BWG G.I. Water cane	each	Tk. 429.46	Tk. 429.00	Tk. 429.00	Tk. 429.00
32.71.20	Supply of 350 mm size G.I. bucket	each	Tk. 343.57	Tk. 344.00	Tk. 344.00	Tk. 344.00
32.71.21	Supply of best quality 19 mm dia plastic hose pipe	each	Tk. 22.09	Tk. 22.00	Tk. 22.00	Tk. 22.00
32.71.22	Supply of insecticide perfection 40 EC 1 pic, 100ml bottle	each	Tk. 300.62	Tk. 301.00	Tk. 301.00	Tk. 301.00
32.71.23	Supply of Redomil gold 68WG (metalexil+mencojeb) 1 pack, 500gm pack	each	Tk. 1,012.30	Tk. 1,012.00	Tk. 1,012.00	Tk. 1,012.00
32.71.24	Supply of Ripcord 400EC, 1 pic, 100ml bottle	each	Tk. 269.95	Tk. 270.00	Tk. 270.00	Tk. 270.00
32.71.25	Supply of Proclaim 5SG(amectin benzoet) 1 box, (10 pack, each pack 10gm)	each	Tk. 779.16	Tk. 779.00	Tk. 779.00	Tk. 779.00
32.71.26	Supply of Vertimec 018ECG (ebamectin) 1 pic, 50ml bottle	each	Tk. 147.24	Tk. 147.00	Tk. 147.00	Tk. 147.00
32.72	Painting to earthen tub: 2 coats over a prime coat					
32.72.1	200mm and 250mm size tub	each	Tk. 21.77	Tk. 22.00	Tk. 20.00	Tk. 20.00
32.72.2	300 mm, 350mm and 400 mm size tub	each	Tk. 34.55	Tk. 34.00	Tk. 33.00	Tk. 33.00
32.73.1	Levelling & dressing the compound or plant/flower bed as per direction of Engineer-in-charge.	sqm	Tk. 9.33	Tk. 9.00	Tk. 8.00	Tk. 8.00
32.73.2	Lawn grass cutting, dressing, brooming and removing debris etc. as per direction of Officer in charge	sqm	Tk. 16.66	Tk. 16.00	Tk. 15.00	Tk. 15.00
32.73.3	Spading, pulverizing soil lump etc. as per direction of Officer in charge	sqm	Tk. 5.87	Tk. 6.00	Tk. 5.00	Tk. 5.00
32.74.1	Labour for picking-up and cleaning damaged bituminous road surface	sqm	Tk. 77.00	Tk. 76.00	Tk. 70.00	Tk. 70.00
32.74.2	Repairing the potholes of the road by picking up and removing damaged materials with stone chips	cum	Tk. 30,553.33	Tk. 30,410.00	Tk. 29,068.00	Tk. 28,681.00
31.75	Applying necessary coat of bituminous aluminium paint seal-coat on top surface of newly laid roofing felt	sqm	Tk. 105.66	Tk. 105.00	Tk. 101.00	Tk. 101.00
32.76.1	Supplying timber (unfinished) for repairing in wood works: Garjan/Jam/local Sal	cum	Tk. 95,331.40	Tk. 90,998.00	Tk. 97,498.00	Tk. 97,498.00
32.76.2	Supplying timber (unfinished) for repairing in wood works: Jarul	cum	Tk. 77,998.42	Tk. 77,998.00	Tk. 77,998.00	Tk. 77,998.00
32.76.3	Supplying timber (unfinished) for repairing in wood works: Shilkorai/Chickrashi	cum	Tk. 108,331.14	Tk. 108,331.00	Tk. 108,331.00	Tk. 108,331.00
32.76.4	Supplying timber (unfinished) for repairing in wood works: Teak Chambal	cum	Tk. 160,330.09	Tk. 155,997.00	Tk. 162,497.00	Tk. 162,497.00
32.76.5	Supplying timber (unfinished) for repairing in wood works: Chattogram Teak	cum	Tk. 190,662.81	Tk. 186,330.00	Tk. 199,286.00	Tk. 199,286.00
32.76.6	Supplying timber (unfinished) for repairing in wood works: Kathal	cum	Tk. 103,997.89	Tk. 103,998.00	Tk. 103,998.00	Tk. 103,998.00
32.76.7	Supplying timber (unfinished) for repairing in wood works: Gamari	cum	Tk. 82,331.67	Tk. 77,998.00	Tk. 86,665.00	Tk. 86,665.00
32.76.8	Supplying timber (unfinished) for repairing in wood works: Chapalish	cum	Tk. 99,664.65	Tk. 95,331.00	Tk. 103,998.00	Tk. 103,998.00
32.76.9	Supplying timber (unfinished) for repairing in wood works: Loha/Pyankado	cum	Tk. 116,997.63	Tk. 116,998.00	Tk. 116,998.00	Tk. 116,998.00
32.76.10	Supplying timber (unfinished) for repairing in wood works: Myanmar(Burma) teak	cum	Tk. 307,660.44	Tk. 303,327.00	Tk. 311,994.00	Tk. 311,994.00
32.76.11	Supplying timber (unfinished) for repairing in wood works: Ordinary timber (Mehogoni)	cum	Tk. 86,664.91	Tk. 86,665.00	Tk. 86,665.00	Tk. 86,665.00
32.77	Labour charge for cleaning, washing, brooming from all heights and stacking of debris at a place beyond plinth area of the site during execution of repair works as per direction of the engineer-in-charge	cum	Tk. 1,450.96	Tk. 1,425.00	Tk. 1,319.00	Tk. 1,319.00

32.78	Cleaning, washing, brooming and taking away of debris from the site during and or after the completion of the repair works in all heights by truck or by any other means as per direction of the engineer-in-charge	cum	Tk. 807.79	Tk. 798.00	Tk. 757.00	Tk. 757.00
32.79	Repairing of steel grill/window frame and shutter/stair railing, veranda grill, rolling shutter excluding the cost of steel materials.	sqm	Tk. 915.51	Tk. 902.00	Tk. 839.00	Tk. 839.00
32.80	Supply of steel materials for repairing work:					
32.80.1	40-grade deform bar	kg	Tk. 79.76	Tk. 80.00	Tk. 80.00	Tk. 80.00
32.80.2	60-grade deform bar	kg	Tk. 88.35	Tk. 88.00	Tk. 88.00	Tk. 88.00
32.80.3	M.S. angle, T, and Z - section, Channel etc.	kg	Tk. 85.89	Tk. 86.00	Tk. 86.00	Tk. 86.00
32.80.4	M.S. plain bar, F.I. Bar	kg	Tk. 85.89	Tk. 86.00	Tk. 86.00	Tk. 86.00
32.80.5	M.S. sheet, plate used in repairing work	kg	Tk. 94.48	Tk. 94.00	Tk. 94.00	Tk. 94.00
32.80.6	38mm dia G.I. Pipe used in repairing work	meter	Tk. 704.53	Tk. 705.00	Tk. 705.00	Tk. 705.00
32.80.7	25mm dia G.I. Pipe used in repairing work	meter	Tk. 322.07	Tk. 322.00	Tk. 322.00	Tk. 322.00
32.80.8	20 mm dia G.I. Pipe used in repairing work	meter	Tk. 330.12	Tk. 330.00	Tk. 330.00	Tk. 330.00
32.80.9	12mm dia G.I. Pipe used in repairing work	meter	Tk. 245.58	Tk. 246.00	Tk. 246.00	Tk. 246.00
32.80.10	Spring for rolling shutter used in repairing work	meter	Tk. 72.47	Tk. 72.00	Tk. 72.00	Tk. 72.00
32.80.11	Spring box for rolling shutter used in repairing work	each	Tk. 159.51	Tk. 160.00	Tk. 160.00	Tk. 160.00
32.81	Cleaning and mending good the damages surrounding the mouth of spout/rain water down pipe	each	Tk. 172.44	Tk. 171.00	Tk. 158.00	Tk. 158.00
32.82	Cleaning rain water down pipe	meter	Tk. 6.30	Tk. 6.00	Tk. 6.00	Tk. 6.00
32.83	Cleaning, renewing of drain and apron with cement concrete or cement mortar etc as per direction of Engineer-in-charge.	meter	Tk. 231.62	Tk. 231.00	Tk. 220.00	Tk. 219.00
32.84.1	Reexcavation of kutchra drain of 0.30 m width and average 0.6 m deep	meter	Tk. 30.68	Tk. 30.00	Tk. 28.00	Tk. 28.00
32.84.2	Reexcavation of kutchra drain of 0.61 m width and average 1.22 m deep	meter	Tk. 125.44	Tk. 123.00	Tk. 114.00	Tk. 114.00
32.85	Clearing choked up masonry surface drain	meter	Tk. 30.68	Tk. 31.00	Tk. 29.00	Tk. 29.00
32.86	Clearing silt from soak well up to 1.4 m dia	each	Tk. 2,098.22	Tk. 2,096.00	Tk. 1,963.00	Tk. 1,963.00
32.87	Clearing sludges & night soil from septic tank removing to a safe place/distance without affecting environment					
32.87.1	10 user septic tank	each	Tk. 2,392.70	Tk. 2,380.00	Tk. 2,270.00	Tk. 2,270.00
32.87.2	20 user septic tank	each	Tk. 3,374.32	Tk. 3,362.00	Tk. 3,190.00	Tk. 3,190.00
32.87.3	30 user septic tank	each	Tk. 3,865.13	Tk. 3,853.00	Tk. 3,650.00	Tk. 3,650.00
32.87.4	50 user septic tank	each	Tk. 6,073.78	Tk. 6,062.00	Tk. 5,767.00	Tk. 5,767.00
32.87.5	100 user septic tank	each	Tk. 10,491.08	Tk. 10,479.00	Tk. 10,000.00	Tk. 10,000.00
32.87.6	200 user septic tank	each	Tk. 18,344.05	Tk. 18,332.00	Tk. 17,546.00	Tk. 17,546.00
32.87.7	Clearing sewage or similar hazardous substances & removing to a safe place/ distances	cum	Tk. 1,687.16	Tk. 1,681.00	Tk. 1,595.00	Tk. 1,595.00
32.88	Clearing sludges & night soil from Inspection pit	each	Tk. 524.55	Tk. 524.00	Tk. 491.00	Tk. 491.00
32.89	Clearing sludges & night soil from RCC soil pipe & sewer line	meter	Tk. 110.84	Tk. 111.00	Tk. 105.00	Tk. 105.00
32.90	Supplying, fitting-fixing of 400 gallon capacity ferro-cement water tank	each	Tk. 9,509.46	Tk. 9,497.00	Tk. 9,448.00	Tk. 9,448.00
32.91	Taking out 40 mm dia tube-well pipe and strainer up to 70 m depth	meter	Tk. 120.25	Tk. 119.00	Tk. 110.00	Tk. 110.00
32.92	Labour for re-sinking 40 mm dia tube-well pipe	meter	Tk. 327.21	Tk. 323.00	Tk. 300.00	Tk. 300.00
32.93	Renewing Check Valve	each	Tk. 1,000.03	Tk. 996.00	Tk. 982.00	Tk. 982.00
32.94	Renewing of 40 mm dia best quality G.I. Socket	each	Tk. 269.95	Tk. 267.00	Tk. 258.00	Tk. 258.00

32.95	Taking out broken Long Pan	each	Tk. 217.80	Tk. 211.00	Tk. 199.00	Tk. 199.00
32.96	Taking out broken European type Commode	each	Tk. 201.23	Tk. 195.00	Tk. 184.00	Tk. 184.00
32.97	Providing 37 mm dia and 600 mm long PVC spout	each	Tk. 252.50	Tk. 250.00	Tk. 239.00	Tk. 239.00
32.98	Providing 25 mm dia and 450 mm long PVC spout	each	Tk. 250.89	Tk. 247.00	Tk. 232.00	Tk. 232.00
32.99	Supplying and fitting-fixing of plastic low-down including necessary internal and external fittings	each	Tk. 1,898.82	Tk. 1,893.00	Tk. 1,841.00	Tk. 1,841.00
32.100	Supplying and fitting fixing of syphone for plastic low-down	each	Tk. 806.16	Tk. 805.00	Tk. 786.00	Tk. 786.00
32.101	250mm and above thick brick work (1:6) in old/unserviceable septic tank	cum	Tk. 8,403.90	Tk. 8,362.00	Tk. 8,077.00	Tk. 7,982.00
32.102	125mm thick brick work (1:6) in old/unserviceable septic tank	sqm	Tk. 1,332.10	Tk. 1,324.00	Tk. 1,274.00	Tk. 1,261.00
32.103	12 mm plaster (1:4) with neat cement finishing in old/dilapidated septic tank	sqm	Tk. 479.11	Tk. 471.00	Tk. 446.00	Tk. 446.00
32.104	Hire charge of welding set including welder cum fabricator and helper	per day	Tk. 1,745.45	Tk. 1,729.00	Tk. 1,604.00	Tk. 1,604.00
32.105	Hire charge of suspended scaffolding (gondula) made of rope	per day	Tk. 429.46	Tk. 427.00	Tk. 417.00	Tk. 417.00
32.106	Per day rate for Aluminium fabricator (labour rate)	per day	Tk. 1,472.43	Tk. 1,472.00	Tk. 1,472.00	Tk. 1,472.00
32.107	Supply of essential hard wares and consumables for repairing aluminium door-window frame and shutter, partition etc.					
32.107.1	Swing door clouser	each	Tk. 3,926.49	Tk. 3,926.00	Tk. 3,926.00	Tk. 3,926.00
32.107.2	Swing door lock	each	Tk. 920.27	Tk. 920.00	Tk. 920.00	Tk. 920.00
32.107.3	Swing door Mohiar	meter	Tk. 20.13	Tk. 20.00	Tk. 20.00	Tk. 20.00
32.107.4	Fixed Neoprene	meter	Tk. 12.08	Tk. 12.00	Tk. 12.00	Tk. 12.00
32.107.5	Flush bolt	each	Tk. 392.65	Tk. 393.00	Tk. 393.00	Tk. 393.00
32.107.6	Sliding door /window key lock	each	Tk. 245.41	Tk. 245.00	Tk. 245.00	Tk. 245.00
32.107.7	Sliding door /window wheel	each	Tk. 98.16	Tk. 98.00	Tk. 98.00	Tk. 98.00
32.107.8	Sliding door /window Mohiar	meter	Tk. 6.04	Tk. 6.00	Tk. 6.00	Tk. 6.00
32.107.9	Sliding Neoprene	meter	Tk. 11.27	Tk. 11.00	Tk. 11.00	Tk. 11.00
32.107.10	Sealant	gm	Tk. 3.80	Tk. 4.00	Tk. 4.00	Tk. 4.00
32.107.11	Sliding door key lock (for window)	each	Tk. 110.43	Tk. 110.00	Tk. 110.00	Tk. 110.00
32.107.12	4-bar hinge	each	Tk. 920.27	Tk. 920.00	Tk. 920.00	Tk. 920.00
32.107.13	Casement handle	each	Tk. 309.21	Tk. 309.00	Tk. 309.00	Tk. 309.00
32.107.14	Rivet for Aluminium work	each	Tk. 6.14	Tk. 6.00	Tk. 6.00	Tk. 6.00
32.107.15	Screws for Aluminium work	each	Tk. 1.50	Tk. 1.50	Tk. 1.50	Tk. 1.50
32.108	Cleaning over head water reservior and removing the waste to safe place/ distance from building site without affecting the environment etc. All completed as per direction of Engineer- in charge.	per gallon	Tk. 1.00	Tk. 0.98	Tk. 0.94	Tk. 0.94
32.109	Cleaning under ground water reservior and removing the waste to safe place/ distance from building site without affecting the environment etc. All completed as per direction of Engineer- in charge. (If water reservior tank capacity is more then 10,000 gallon then unit price for additional gallon will be reduced by @ 25% of unit price)	per gallon	Tk. 0.86	Tk. 0.85	Tk. 0.83	Tk. 0.83
32.110	Cleaning and removing inside rust from existing old GI pipe of different dia meter etc. All completed as per direction of Engineer- in charge.	per meter	Tk. 4.20	Tk. 4.10	Tk. 3.90	Tk. 3.90
32.111	Taking out old unserviceable MS window shutter/grill etc. All completed as per direction of Engineer- in charge.	each	Tk. 202.46	Tk. 199.00	Tk. 190.00	Tk. 190.00

32.112	Taking out unserviceable old MS collapsible gate and refitting after minor repair (excluding cost of material) etc. All completed as per direction of Engineer- in charge.	each	Tk. 2,587.36	Tk. 2,550.00	Tk. 2,432.00	Tk. 2,430.00
32.113	Picking & uprooting unwanted tree, plants etc from building /structure and removing to safe distance from building/structure and using chemicals to prevent further growing up. All completed as per direction of Engineer- in charge.	each	Tk. 242.34	Tk. 239.00	Tk. 227.00	Tk. 227.00
32.114	Sanitary blockage removal of Combi-closet, commode, Long Pan etc. All completed as per direction of Engineer- in charge.	each	Tk. 368.11	Tk. 368.00	Tk. 353.00	Tk. 353.00
32.115	Sanitary blockage removal of Basin, Sink, waste water down pipe etc. All completed as per direction of Engineer- in charge.	each	Tk. 159.51	Tk. 160.00	Tk. 153.00	Tk. 153.00
32.116	Writing & providing of name plate or Flat/Room numbering on Acrylic transparent sheet including fitting fixing etc. All completed as per direction of Engineer- in charge.	each	Tk. 1,496.97	Tk. 1,497.00	Tk. 1,491.00	Tk. 1,491.00
32.117	Hire charge of mechanical gondola for repair work of high rise building height above 20 meter etc. All completed as per direction of Engineer- in charge.	per day	Tk. 2,454.05	Tk. 2,454.00	Tk. 2,393.00	Tk. 2,393.00
32.118	Taking out old unserviceable Aluminium Door/window frame and shutter etc. All completed as per direction of Engineer- in charge.	each	Tk. 245.41	Tk. 240.00	Tk. 227.00	Tk. 227.00

..... END OF CHAPTER THIRTY TWO

CHAPTER 33: ADDED RATE FOR INACCESSIBLE AREAS

Category	Upazilla/ area	District	Added % rate
A (Accessibility with moderate difficulty)	Lama	Bandarban	5%
	Ali Kadam		
	Naikhongchori		
B (Accessibility with high difficulty)	Roangchori	Bandarban	10%
	Ruma		
	Thanchi		
	Barkal	Rangamati	
	Juraichari		
	Bilaichari		
	Longodu		
	Baghaichari		
	Hatia	Noakhali	
	Nijhum Dwip		

[illegible]